AI Applications and Technological Development



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Foreword

- In today's rapidly advancing technological era, AI has become a key force driving social and economic transformations. With continuous technological progress, large AI models not only offer unprecedented opportunities for businesses but also have a profound impact on various industries such as education, healthcare, and entertainment. By introducing the applications of these advanced technologies, we aim to explore how they help us better understand and improve our world.

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Objectives

After completing this course, you will be able to:

- Describe the current application status of large models
- List the applications of large models in NLP, Speech, Vision, Multimodal, and other fields
- Describe AI industry applications and AI solutions

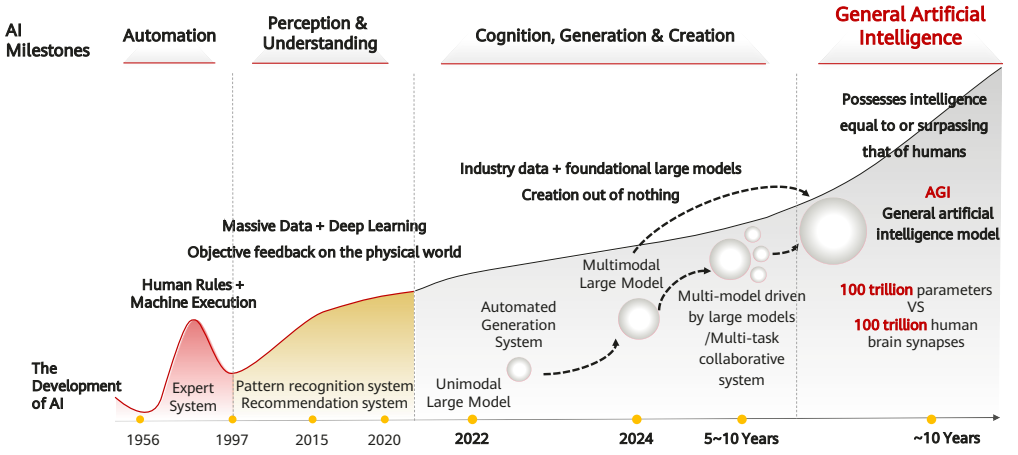
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   - Development of AI Applications
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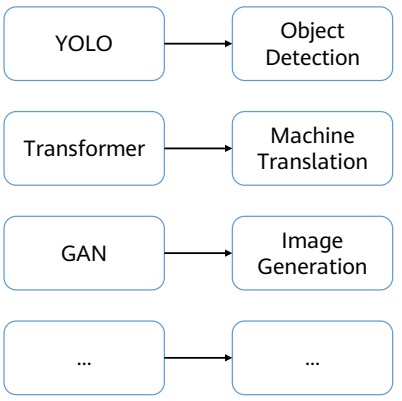
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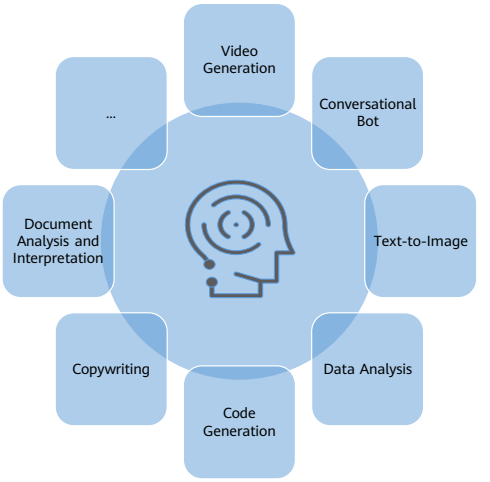
Development of AI Applications



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From Single-Task to Multi-Task





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AI Applications in Computer Vision

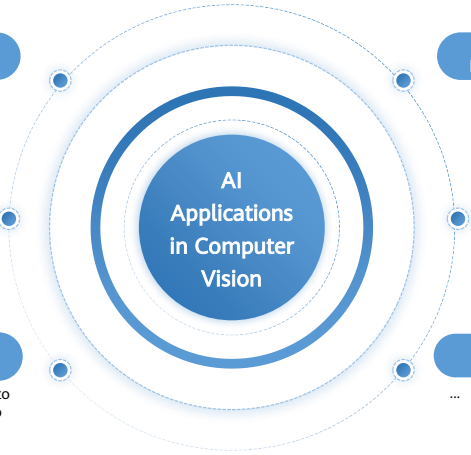


Image Classification

Utilizing massive amounts of unlabeled data to enhance the representational capabilities of training models for application in various general scenarios.

Object Detection

Identifying and locating various objects in images or videos, providing support for automated analysis and decision-making.

Semantic Segmentation

Assigning each pixel in an image to a predefined semantic category to achieve pixel-level image understanding.

Video Restoration

Effectively restoring and enhancing video quality using deep learning techniques.

Autonomous vehicles

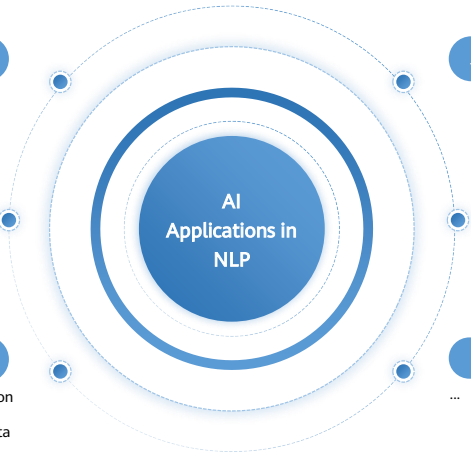
Enabling vehicles to perceive surroundings, make decisions, and perform driving operations like human drivers.

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AI Applications in NLP



Text Classification

Using algorithms to automatically categorize text into predefined classes, enabling efficient information management and precise decision support through the analysis of text features.

Text Summarization

Automatically extracting the core information of a text and generating a simplified version, achieving efficient information compression by capturing semantic relationships and key elements.

Information Extraction

Automatically extracting key information such as entities and relationships from massive amounts of text to support data analysis and decision-making.

Machine Translation

Using algorithmic models to automatically convert one language into another (e.g., Chinese to English, French to Japanese).

Intelligent Writing

Assisting in generating high-quality text, such as news reports, advertising copy, etc., to improve content creation efficiency.

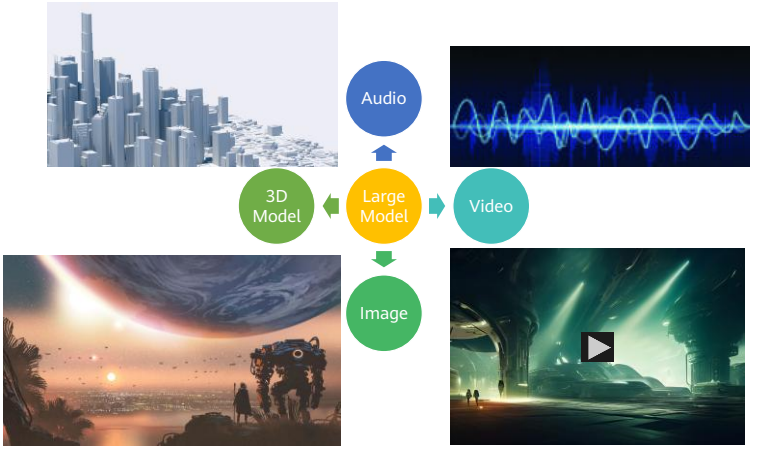
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Multimodal AI Applications

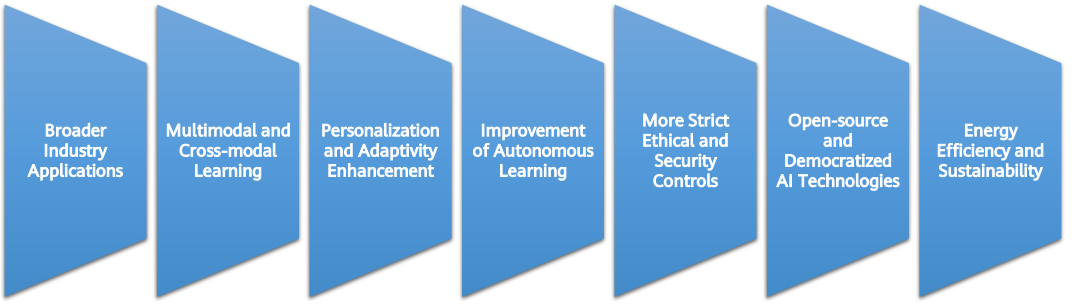
- Integrating linguistic and visual cross-modal information to enable applications such as image generation, image understanding, 3D generation, and video generation.



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The Development Trends of AI Applications

- Future trends in AI applications suggest broader technological integration, deeper industry integration, and ongoing ethical and safety considerations.



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- **Broader Industry Applications**: As the technology matures and becomes more adaptable, it is expected that large models will be applied to more traditional industries, such as manufacturing, energy, and logistics, providing efficient problem-solving solutions and decision support.
- **Multimodal and Cross-modal Learning**: Future large models will not be limited to single domains like text or images, but will be able to process and integrate multiple types of data (such as text, images, videos, and audio), offering a more comprehensive AI application experience.
- **Enhanced Personalization and Adaptability**: Large models will better adapt to the specific needs of individual users, providing customized services. For example, in education and healthcare, models will be able to offer personalized recommendations based on an individual's learning progress or health status.
- **Improved Autonomous Learning Capabilities**: Future large models will have stronger self-learning abilities, enabling them to learn new patterns and knowledge through continuous interaction and feedback, reducing reliance on large-scale labeled data.
- **Stricter Ethical and Safety Controls**: As large models become more widespread, related ethical, privacy, and security issues will receive more attention. Future development will need to focus on ensuring the transparency, explainability, and fairness of models, while also strengthening data protection.
- **Open Source and Democratic AI Technologies**: The development and application of big models may become more democratic. More research institutions and companies may open their models and technologies to promote technology

sharing and innovation around the world.

- **Energy Efficiency and Sustainability**: Given the significant energy consumption required to train large models, future research will focus on improving the energy efficiency of models and reducing their carbon footprint in support of the Sustainable Development Goals.

Growing Market

AI Servers

The global AI server market size is expected to reach $125.1 billion in 2024, increase to $158.7 billion in 2025, and is projected to reach $222.7 billion by 2028. Among these, generative AI servers are expected to account for 29.6% of the market in 2025, rising to 37.7% by 2028.

AI Industry Chain

The global AI industry chain is expected to reach a total scale of approximately $842 billion by 2025, with a Compound Annual Growth Rate (CAGR) of 21.5%.

Artificial Intelligence Enterprises

The number of AI companies is growing rapidly; in the first half of 2024 alone, the number of AI companies in China increased by 35.65% year-on-year, and this figure is expected to continue rising.

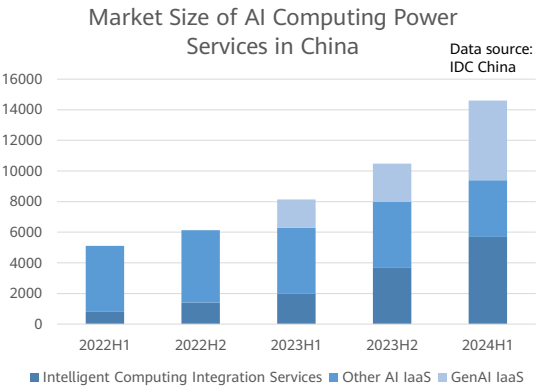
Large Models Solution

In the first half of 2024, the market size of AI large model solutions in China was 1.38 billion yuan. It is projected that the market will experience an average annual compound growth rate of 56.2% from 2024 to 2028, reaching a market size of 21.1 billion yuan by 2028.

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Artificial Intelligence Computing Power Market

- In 2024, the AI computing power market in China reached $19 billion, an increase of 86.9% year-over-year. It is expected to reach $25.9 billion in 2025, representing a year-over-year growth of 36.2%.
- In 2024, China's general-purpose computing power reached 71.5 EFLOPS (exaflops per second), an increase of 20.6 percent year-over-year; AI computing power reached 725.3 EFLOPS, an increase of 74.1 percent, AI computing power's growth rate was more than three times that of general-purpose computing power. By 2025, China's general-purpose computing power is expected to reach 85.8 EFLOPS, an increase of 20 percent year-over-year; AI computing power is expected to reach 1037.3 EFLOPS, an increase of 43 percent.



The rapid transition between old and new drivers has made AI computing power expenditure for GenAI the AI computing power main focus of the AI computing power service market.

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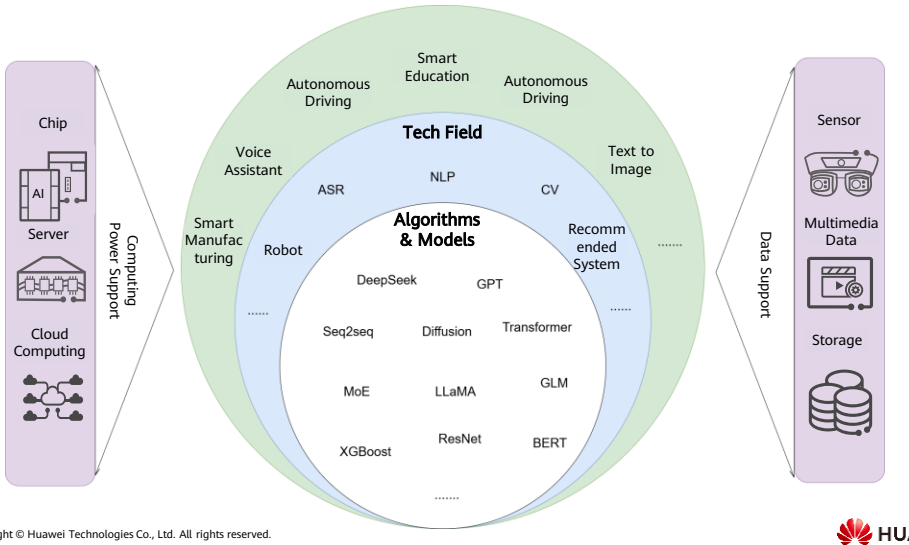
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Overview of AI Technology and Application Areas



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From Small to Large

Algorithms: Parameter count has expanded, with units increasing from millions to billions.

Data: Training data has grown, with individual model training datasets reaching trillions of tokens.

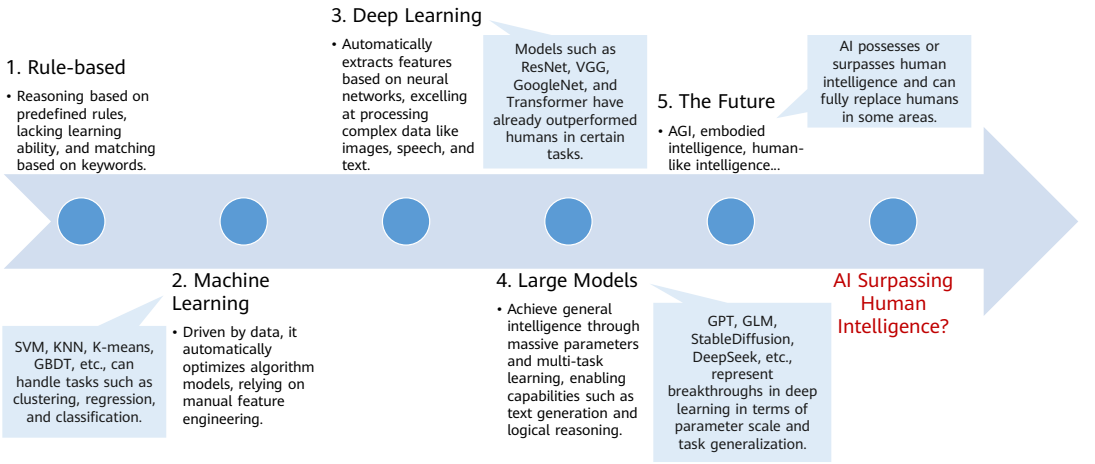
Computing Power: The scale of computing power has increased to EFLOPs.



.....
DeepSeek
GLM
LLaMA
GPT
Transformer
ELMO
ViT
ResNet
VGG
AlexNet

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Learning Algorithms



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Computing Power

- In recent years, the amount of data and parameter scale used by newly launched large language models (LLMs) have shown "exponential" growth, leading to an explosive increase in computing power demand.
- Taking GPT as an example, the GPT-3 model has approximately 174.6 billion parameters, and the total computing power required for one training run is about 3,640 PF-days, which means running at one quadrillion calculations per second for 3,640 days. The subsequent GPT-4 may expand its parameter count to over one trillion, which is (estimated) 10 times that of GPT-3, and the training computing power requirement rises to 68 times that of GPT-3.

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Dataset

	Data Scale	Data Diversity
Small Model	Datasets are typically in the range of millions to tens of millions of samples, making them suitable for resource-constrained scenarios (e.g., mobile devices or edge devices).	Datasets are typically tailored for specific tasks or domains (such as medical image classification and speech keyword recognition), without the need for broad coverage. The data distribution is more focused to optimize model performance in specific scenarios.
Large Model	Large models need to process and learn more complex relationships and patterns, thus requiring large-scale datasets to support their training. For example, the Common Crawl dataset used to train the GPT-3 model contains hundreds of billions of tokens of text data, representing an order of magnitude increase in scale compared to datasets used for smaller models.	The dataset covers multiple fields and topics, including science, technology, culture, art, history, and business. The dataset is no longer limited to a single modality or specific domain, but has instead become multimodal and cross-domain.

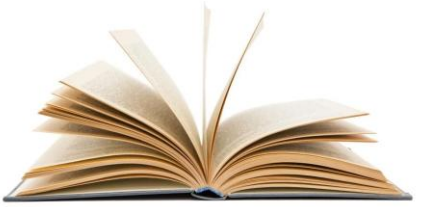
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Large Model vs. Small Model

Small Model

- Low upper limit of learning ability
- Different tasks require different models
- Large amount of training data
- Single-modal data
- Poor few-shot capability

Using AI models to extract knowledge from data



"Knowledge" is better learned

Large Model

- Strong learning ability
- One model solves multiple tasks
- Massive training data
- Multimodal capabilities
- Strong few-shot learning capability

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From Transformer to DeepSeek

Time	Model/Technology	Core Breakthrough	Application Scenarios
2017	Transformer architecture was proposed	Self-attention mechanisms were introduced to replace RNN/LSTM, enabling parallel computation and addressing long-range dependency issues.	It became the foundational architecture for Natural Language Processing (NLP) tasks such as machine translation and text generation.
2018	BERT, GPT-1	BERT: Bidirectional Encoder Representations from Transformers, enhancing contextual understanding; GPT-1: the first pre-trained language model based on Transformer.	NLP tasks such as text classification, question-answering systems, and text generation.
2020	Vision Transformer (ViT)	Applying Transformers to the field of Computer Vision (CV) for the first time, using pure Transformers to handle image tasks.	CV tasks such as image classification and object detection have broken the dominance of CNNs.
2021–2024	GPT-3, GPT-4	GPT-3: 175 billion parameters, advancing few-shot learning; GPT-4: enhanced multimodal capabilities, supporting both text and image inputs.	Chatbots, content generation, intelligent customer service, etc., have initiated the trend of large-scale Transformer pre-training.
Feb 2023 - Apr 2024	LLaMA-1 to LLaMA-3	Open-source large language models with parameter scales ranging from 7B to 65B, providing cost-effective and high-performance model options for academia and developers. Upgraded LLaMA models feature even larger parameter scales (up to 70B), further improved performance, and support for longer context windows.	Academic research and enterprise applications, lowering the barrier to using large models, supporting multiple languages and tasks.
Dec 2024	DeepSeek-V3	The number of parameters has increased to 671 billion, with a training cost of only $5.57 million, offering high cost-effectiveness.	Capable of handling complex tasks such as scientific computing and big data analysis.
Jan 20, 2025	DeepSeek-R1 Officially Released	Achieves performance on par with OpenAI's top models in tasks such as mathematics, coding, and logical reasoning, at a lower cost.	Provides efficient and cost-effective solutions for areas like intelligent customer service, content creation, and code generation.

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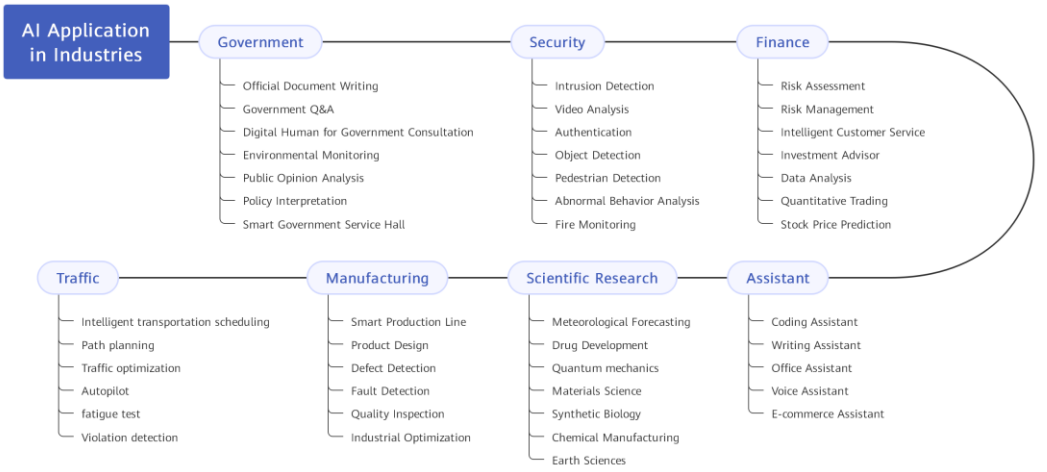
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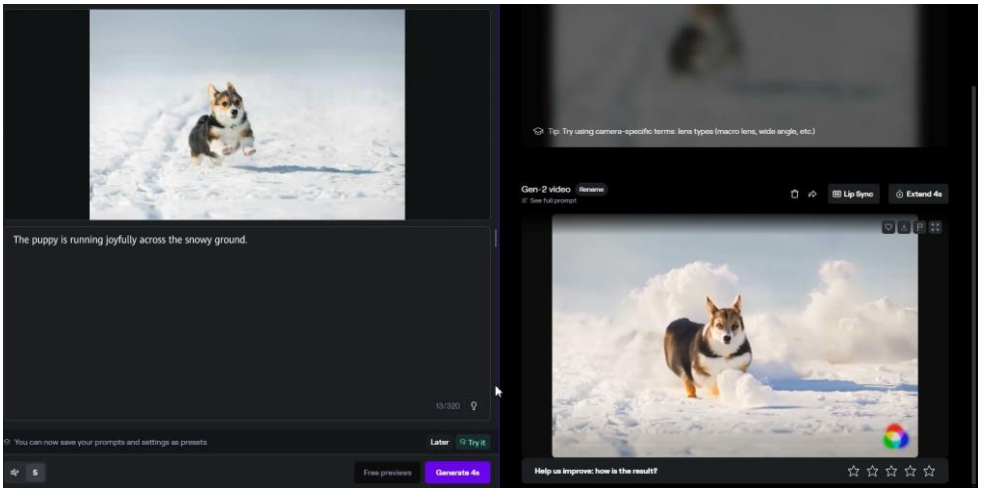
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Current Status of AI Applications Across Industries



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AI Assistant - Video Creation



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Smart Finance

Scenario: 2 Types of Content Generation Output Models

Intelligent Summarization

Stock Exchange

Annual Report Summary Writing

Full Annual Report → Analyze Key Information → Identify Key Points → Summary

Other Scenarios: Meeting Summary; Call Summary

Summary key elements generation accuracy rate **of 98%+**, already launched and in use.

Document Writing

Bank

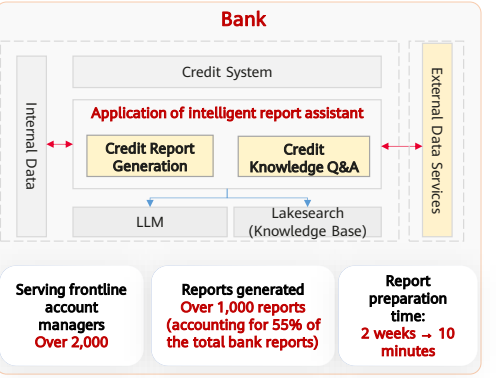
Requirement Document Preparation

Input Original Requirement → Template Auto-Selection → Existing Documents Comparison and Rewriting → Format Correction

Other Scenarios: Credit Report Generation; Official Document Writing

The accuracy rate for generating key elements of summaries **has increased to 85%+,** reducing the time **from 2 weeks to 30 seconds.**

Case & Customer Value: One-click generation of Due Diligence/Credit Reports



Serving frontline account managers Over 2,000

Reports generated Over 1,000 reports (accounting for 55% of the total bank reports)

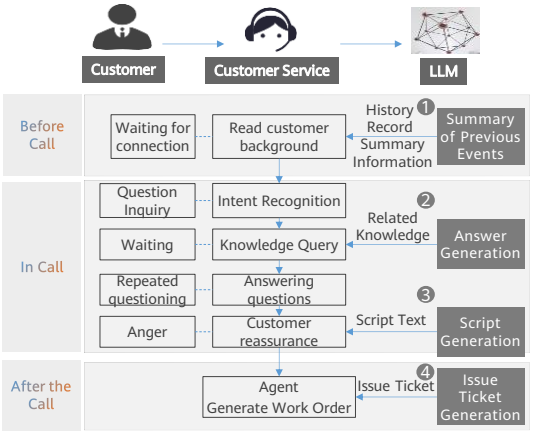
Report preparation time: 2 weeks → 10 minutes

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Smart Finance - Customer Service Assistant

Scenario: The role of large models in the four phases of intelligent customer service.



Case & Customer Value: Customer Service Assistant, Promoted Across All Branches

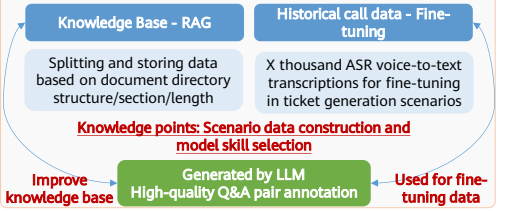
A State-owned Commercial Bank



Insufficient adaptability to complex business scenarios
Leading to poor customer services

Empowering the entire bank over 10,000+ branches, xx million branch employees

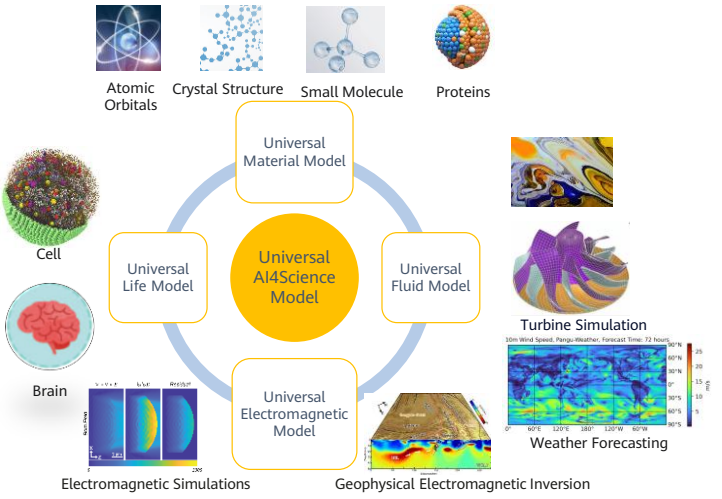
Transaction Processing	Processing Time
5 steps → 1 step	↓ 5 minutes



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Scaling Law in AI for Science

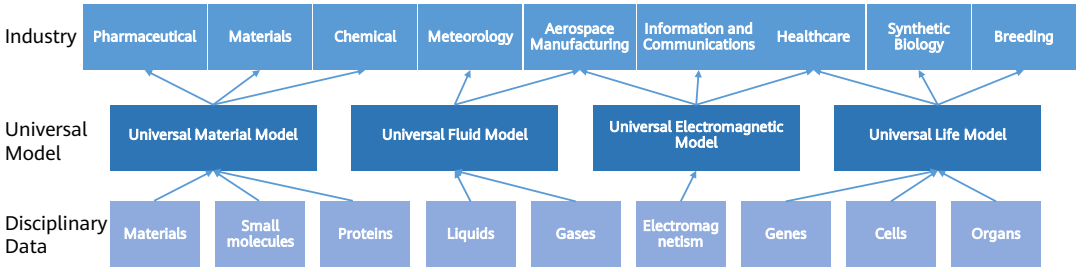
- Integrating interdisciplinary and cross-scale data to train fundamental scientific models in life, matter, fluid, and electromagnetic sciences.



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AI4Science

Unlocking Industry Data for Intelligent Emergence



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AI4Science 2035

Breakthroughs in foundational AI technologies unleash immense industry innovation drivers.



AI for Quantum Mechanics	AI for Molecular Dynamics	AI for Fluid Mechanics	AI for Materials	AI for Small Molecule	AI For Protein	AI for Cells	...

Technology Trends		Industry Trends
Connect scientific big models to industry scenarios, driving the emergence of scientific big models from massive industry scenario data.	**2035**	1. AI cells are widely used in clinical trials and personalized medicine. 2. AI breeding is widely applied. 3. AI accurately predicts extreme weather. 4. AI is extensively used in industrial design, chemical optimization, and synthetic biology.
Interdisciplinary and multi-scale data integration, focusing on building four major scientific models: Matter, Fluid, Field, and Life.	**2030**	1. AI-designed materials have achieved breakthroughs in superconductors, semiconductors, batteries, and special alloys. 2. AI-designed cells have been validated. 3. AI-designed aircraft have been validated.
AI achieves breakthroughs in more disciplines such as Mathematics, Quantum Mechanics, Materials, Fluid Dynamics, Cells, Brain Science, and Earth Science.	**2025**	1. AI drug completes clinical market 2. Global Forecasting in Minutes Based on Large Meteorological Model Based on Satellite 3. AI material design verified, phase diagram prediction achieved breakthrough

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AI4Science - AI + Meteorology

Meteorological Large Model: Discovering and Predicting Natural Laws

80

Approximately 23 typhoons form globally each year.

Weather Large Model



3D Meteorological Model
Hierarchical Temporal Aggregation

More Accurate Predictions



Forecast Accuracy **20%** ↑

Faster and Greener

Computational Power for Predicting Weather Over the Next 10 Days

Traditional Forecasting via Numerical Calculation	Weather Inference via Large Model
3000 servers	1 card
5 hours	10 seconds

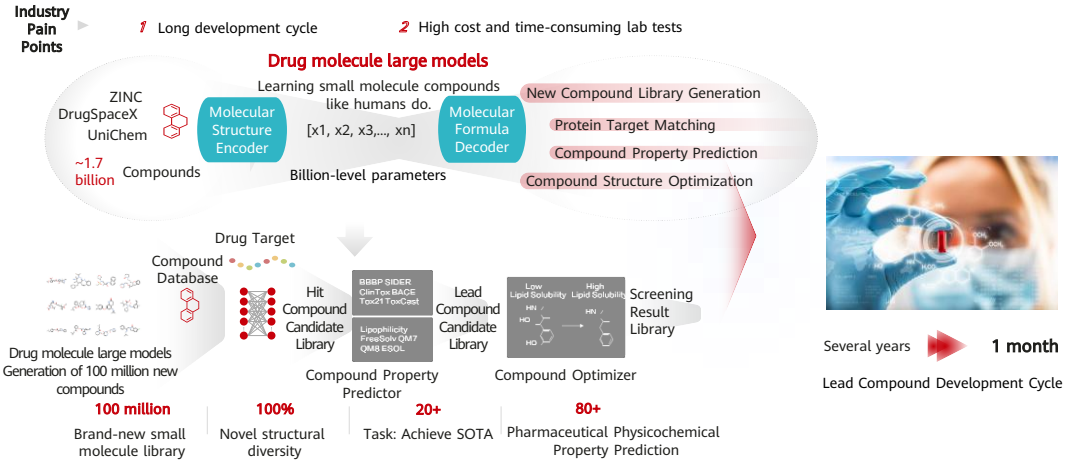
Forecasting Speed **10,000x** ↑

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AI4Science - AI + Pharmaceutical Development

Drug molecule large models significantly enhance pharmaceutical design efficiency.



Several years → **1 month**

Lead Compound Development Cycle

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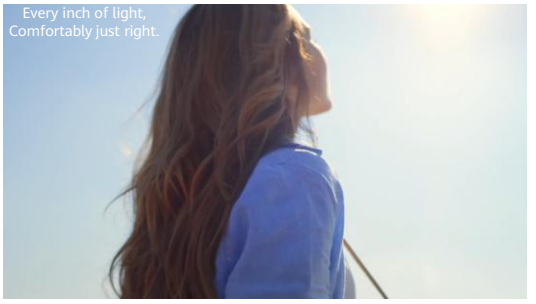
Smart Home

- As a product of the perfect integration of modern technology and daily life, smart homes are transforming our living environments and lifestyles at an unprecedented pace.
- By integrating advanced IoT technologies, artificial intelligence algorithms, and big data analytics, smart homes connect various devices within the home into a highly automated and personalized intelligent ecosystem, offering users an unparalleled experience of convenience, comfort, and security.



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Smart Home - Making Life More Thoughtful



Intelligent sunshade control allows you to adjust the blinds in groups according to the intensity of sunlight, filtering out strong light and creating a natural, soft lighting environment for you.

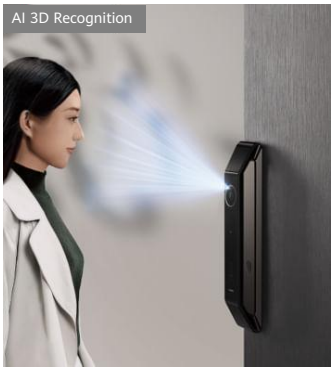
Trigger the home mode, and the zero-wait-to-heat mode will automatically activate. The intelligent system's water temperature adaptation capability allows the water temperature to adjust automatically according to the season, geographical location, and personal preferences.

No matter the season, warm water is always ready.



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Smart Home - Peace of Mind at Home and Away





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Smart Mining

Scenario 1: Electrical and Mechanical Chamber — The electrical and mechanical chamber stores important equipment such as power distribution and communication systems.
Objective: Achieve automatic inspection and unmanned operation.

Challenge 1: The chamber environment needs to be monitored 24 hours a day.



Challenge 2: Manual periodic inspections are required.



Scenario 2: Underground Smart Monitoring

	Challenge
Coal Conveyor Belt	The coal conveyor belt spans over 20 kilometers, making manual inspection time-consuming and unsafe; Multiple sections of the belt need to work in coordination to prevent coal from piling up.
Violations	Personnel violation monitoring: wearing safety helmets, smoking, passing under equipment, walking in the roadway, sitting on the conveyor belt, etc.
Improper Operations	Such as gas extraction, water exploration and drainage, etc.

Scenario 1 Solution: Automated Inspection Robot



Video Surveillance: 24-hour monitoring using **AI** machine vision for anomaly alerts, sensitive area detection, etc.
Sound Detection: AI intelligent recognition to determine whether the machine is operating healthily.

Scenario 2

Conveyor belt tear monitoring



Conveyor belt misalignment monitoring

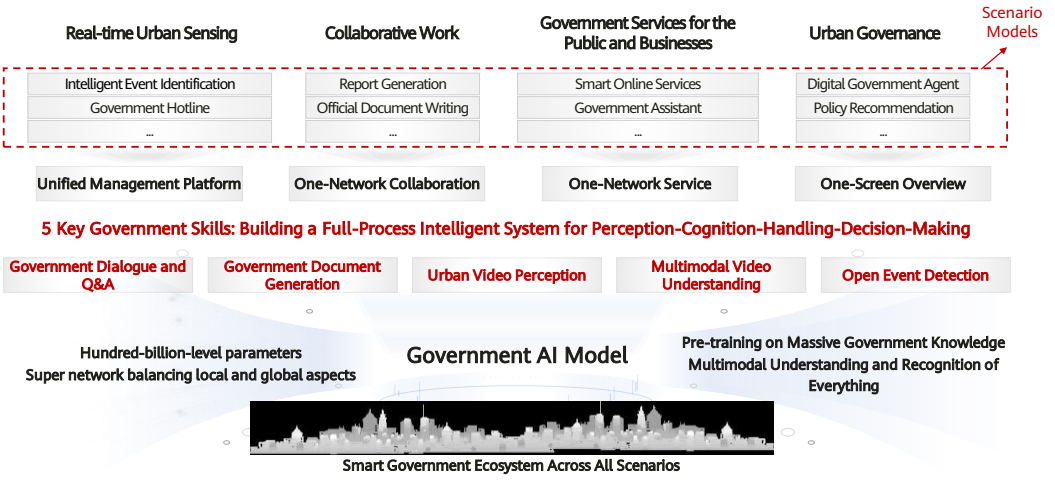


Coal Belt Monitoring



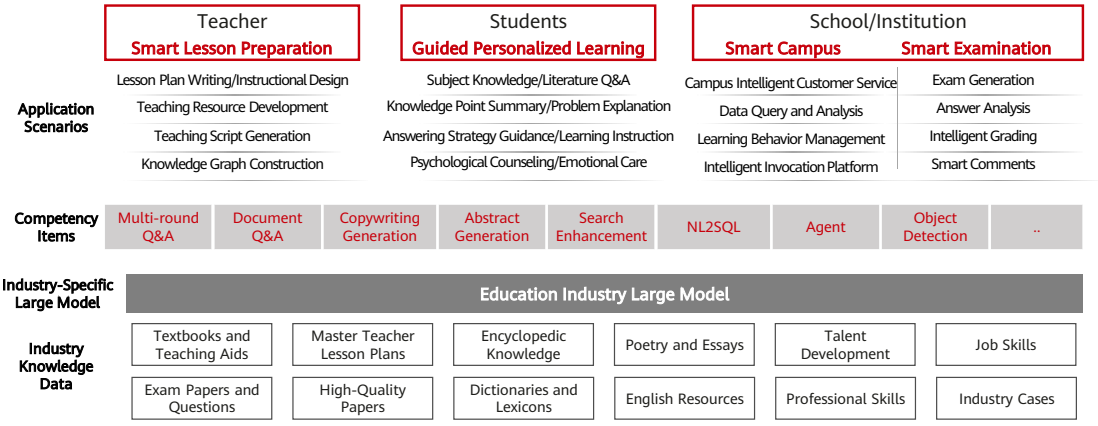
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Government AI Model



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Educational Large Model



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Digital Human

A digital human is a digitized character created using digital technology that closely resembles human appearance, possessing physical features, human-like performance capabilities, and interactive abilities.

Core Values:

- Breaking physical boundaries to provide human-like services.
- Working continuously 24 hours a day, 7 days a week.
- Reduce labor costs and improve service efficiency.
- Achieve the perfect combination of personalization and scale.



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Smart City

- With the advancement of safe city and smart city initiatives, intelligent hardware devices generate massive amounts of data. Currently, high-definition intelligent recognition and analysis have been implemented in most regions across the country. However, the majority of this data remains at the level of viewing and storage. As data continues to grow exponentially, it will lead to a significant increase in storage costs.



The AI long-tail demand scenarios in smart cities are numerous and complex to develop.

AI applications face three major challenges: algorithm development, application development, and model deployment.



Various government departments operate siloed systems, resulting in fragmented data.

Each department and bureau has independently built information systems, leading to fragmented data that prevents the development of unified situational awareness, comprehensive governance, emergency response, and analytical decision-making capabilities.



Innovation development lacks solid foundation.

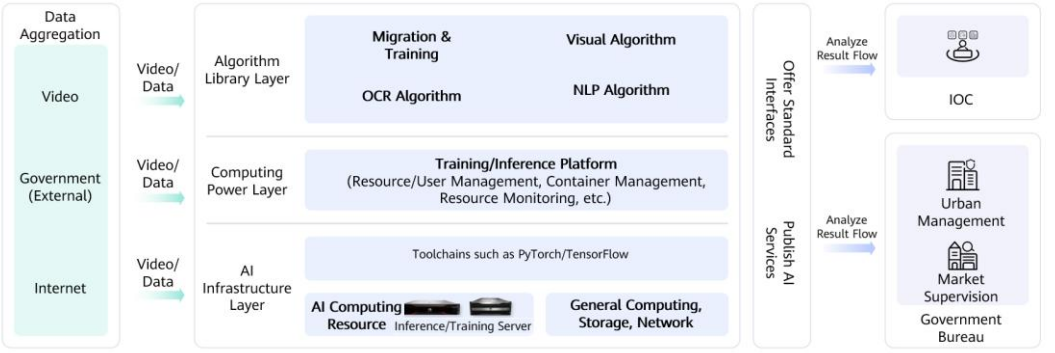
Currently, smart cities largely rely on foreign fundamental software and hardware for AI innovation, posing a constant risk of being constrained by external factors.

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- With the advancement of safe city and smart city initiatives, intelligent hardware devices generate a vast amount of data. Currently, high-definition intelligent recognition and analysis have been implemented in most regions across the country. However, the majority of this data remains at the level of viewing and storage. Moreover, the explosive growth of data leads to a corresponding exponential increase in storage costs.
- The current analytical methods are limited, and there is a lack of capability to mine data information effectively. In addition, various cities have been implementing video intelligent analysis networking and sharing projects for many years, and have now largely achieved the sharing of high-definition data.
- With the development of AI, cloud computing, and big data technologies, the application of safe city initiatives is gradually shifting towards comprehensive models that integrate big data applications, urban public safety operations, and emergency response. Safe city solutions now focus on mining and analyzing data information, providing strong support for urban development.

Smart City Solutions



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- This solution integrates the foundational software and hardware required for underlying AI computing power, a software platform layer for intelligent scheduling capabilities, and various AI software algorithm layers (such as visual algorithms, OCR algorithms, natural language processing algorithms, etc.). Built on this foundation, it provides standard interfaces to various government departments and agencies, offering AI services that can be invoked by these departments to meet diverse urban governance and smart service application needs.

Smart Transportation

- With the continuous increase in the number of vehicles, the current transportation industry faces issues such as worsening urban congestion and frequent traffic accidents.



Massive, real-time, and multi-dimensional data streams exceed the limits of human analysis.



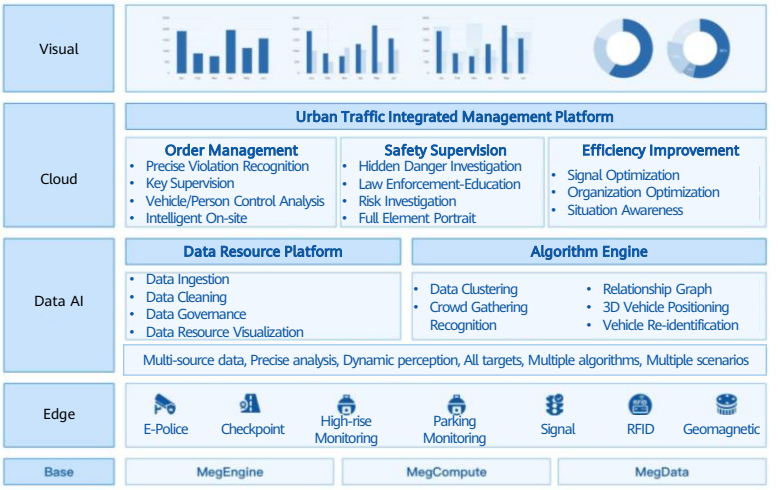
Traffic violations are often concealed, making them difficult to detect and penalize.
Limited enforcement resources make it challenging to achieve comprehensive coverage.



Shift from peak-hour congestion to all-day, normalized congestion.
The phenomenon of tidal traffic congestion is becoming increasingly evident, with a significant increase in passenger flow along axial corridors.

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Smart Transportation Solutions



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Manufacturing

- To enhance core competitiveness and brand value, the manufacturing industry must strictly control product quality and improve customer satisfaction. Under increasingly stringent and variable requirements, how can product quality inspection be completed efficiently and with high quality?

Human Resource Issues

- Manual quality inspection is prone to fatigue, inefficient, and unable to quickly respond to product changes or process upgrades.



Poor adaptability

- When switching product models, traditional manufacturing companies need to continuously invest funds and even modify equipment, making it difficult to efficiently inspect new products in a short time.

Increasingly harsh environments

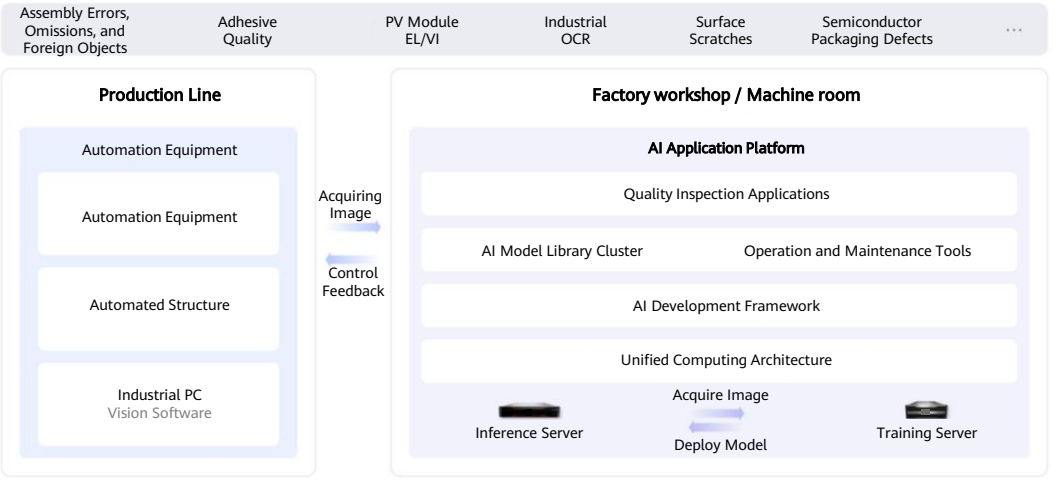
- Different working environments (such as those with specific requirements for temperature, lighting, operational techniques, etc.), the large amount of data that needs to be processed in quality inspection work, and the defects or minor flaws in structurally complex components make manual quality inspection difficult to manage. In addition, some industries may involve working conditions that are harmful to human health, posing risks to workers.

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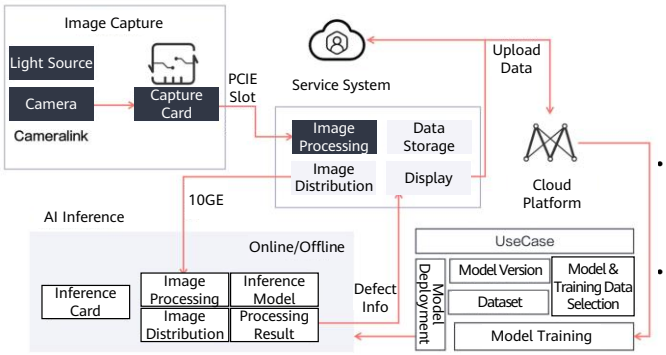
- AI is a key engine for the large-scale development of intelligent manufacturing. It has widespread application needs in various scenarios such as quality inspection, robot vision, human-machine collaboration, production safety, and factory area patrol.
- Quality inspection is a critical part of product manufacturing lines in the manufacturing industry. Traditional optical inspection methods suffer from issues such as insufficient precision and flexibility.
- Based on AI algorithm models, intelligent automatic optical inspection can significantly improve these problems. It holds strong application prospects in quality inspection across many sub-sectors of manufacturing, including electronic products, semiconductor wafers, packaging boxes, batteries, automotive parts, textile prints, camera lens barrels, pipes, and wires.

Intelligent Manufacturing Solutions



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AI Quality Inspection Solution



The detection precision is improved to 95%, and the labor cost is reduced by 30%.

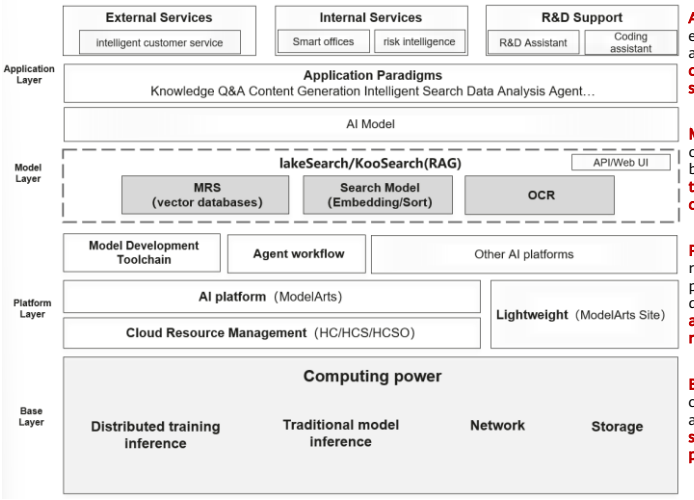
- **Surface defects:** Identify defects on the surface of an object during manufacturing. They are mainly divided into two types:
  - **Appearance problems**, which affect the appearance of delivery, such as scratches.
  - **Quality problems**, affecting product quality, such as weak solder joints and crack detection.
- **Size measurement:** Measure the size of the object to see if it meets the requirements of the design drawing. The accurate specifications are different, such as 1cm, 1mm, and 0.1mm.
- **Process optimization:** Monitor the equipment to optimize the production process through the parameters in the process flow, mainly divided into two types:
  - Reduce the cost, in order to reduce the production cost or save the electric energy.
  - Improve the yield rate of existing products.

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- Quality inspection solutions utilize artificial intelligence for auxiliary defect detection or parameter optimization, and establish a digital management system to provide end-to-end solutions. These solutions include three major modules: hardware, software, and algorithms, addressing customers' specific practical problems, helping them reduce costs and improve efficiency, and enabling intelligent transformation.

Smart Finance Solutions



Application Layer: Various business units within enterprises use AI capabilities to improve their operations according to their specific scenarios, **enabling AI computing power, algorithms, and data to generate strategic, commercial, and operational value.**

Model Layer: Enterprise AI organizations develop AI capabilities that are consumable, callable, and iterative, based on the new generation of AI foundational models, **these models are orchestrated, adjustable, and combinable**.

Platform Layer: Enterprise IT organizations focus on resource scheduling and management of the computing power infrastructure, as well as the production and deployment of AI model capabilities, **enabling the availability of computing power, data, and algorithm resources**.

Base Layer: Enterprise data center organizations build computing power infrastructure centered on products such as computing, networking, and storage, forming **a standardized, deliverable, and sustainable computing power infrastructure**.

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Quiz

1. (Multiple-choice Question) Which of the following tasks can be solved using the large language model DeepSeek V3?
   A. Sentiment analysis
   B. Image caption generation
   C. Video generation
   D. Multilingual translation

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- Answer: AD

Summary

- This course focuses on the development of large model applications and technologies, including the evolution of AI applications, AI industry application scenarios, and AI solutions.

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More Information

Huawei Training App

Enterprise Technical Support App

Huawei Enterprise Business App

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把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

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Artificial Intelligence and Large Model Fundamentals



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Foreword

- This chapter aims to provide an in-depth introduction to the topics related to large models today, covering mainstream large model overviews, large language models, Transformer architectures, GPT and BERT models, Vision Transformer models, Diffusion models, and multimodal large models. Through analysis and discussion, you will have a deeper understanding and mastery of these core elements that drives the development of artificial intelligence.

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Objectives

After completing this course, you will be able to:

- Become familiar with the current status of large models and their impact on AI development
- Master the fundamentals and key technologies of large language models
- Master the Attention mechanism and Transformer architecture
- Become familiar with the development history and architectures of BERT and GPT
- Master the architecture and applications of Vision Transformer
- Master generative learning strategies and the structure of Diffusion models
- Understand the importance and characteristics of multimodal large models
- Become familiar with the significance and training logic of DeepSeek

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1. **Overview of Mainstream Large Models**
2. Definition and Development of Language Models
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4. GPT and BERT Models
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8. DeepSeek Analysis

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What is a Large Model?

- A large language model (LLM) is a language model trained with self-supervised machine learning on a vast amount of text, designed for natural language processing tasks, especially language generation.

  — *Wikipedia*

- Pre-trained models built on massive multi-source data represent a technological upgrade and product iteration of traditional algorithm models. Users can perform zero-shot or few-shot learning using open-source or open APIs/tools to achieve better recognition, understanding, decision-making, and generation outcomes, as well as lower-cost development and deployment solutions.

  —*IDC*

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- International Data Corporation (IDC) is a globally renowned provider of consulting, advisory, and event services specializing in information technology, telecommunications, and consumer technology.

Current Status of Large Models - Across Industries

- **Industrial Production**

Aggregate industry data and use foundational models to build industry-specific large models, addressing the issue of fragmented scenarios and transforming "workshop-style" AI development into "industrialized" production.

- **Equipment Inspection**

Leveraging foundational models integrated with domain-specific knowledge to address key challenges in equipment inspection scenarios—such as limited training samples, a wide variety of fault types, and significant variations in target scales.

- **Process Indicator Prediction**

Prediction of process indicators in industrial production, such as quality prediction for cement rotary kilns and ash content prediction for clean coal in coal washing plants.

- **Smart City**

Utilizing object detection and segmentation to meet the massive long-tail demands of smart cities, efficiently identifying urban events through language interaction.

- **Intelligent Creative Marketing**

Writing tasks in various styles and genres, providing creative commercial copywriting to help products attract more potential customers and unleash boundless creative energy.

- **Government and Enterprise Knowledge Retrieval**

Models are trained based on government and enterprise knowledge, enabling precise acquisition of user needs through conversational interaction, improving the efficiency of knowledge retrieval, and delivering a warm and intelligent AI service experience.

- **Meeting Assistant**

Rewrite meeting minutes into standard agendas and generate text summaries based on meeting content, offering efficient and convenient meeting experiences for organizers and participants.

- **Film and Television Production**

Providing technologies for the industry, including storyboard creation, IP character design, short video generation, scene generation, and video editing.

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- **E-commerce Marketing:** Provides technologies for product design, category packaging, element recognition, material generation, and advertising marketing for various industries.
- **Game Development:** Offers technologies for character creation, concept art design, 3D material generation, figurine creation, and style conversion for the gaming industry.
- **Graphic Design:** Delivers technologies for advertising design, product design, e-commerce/model images, avatar design, and illustration design for the graphic design industry.

Breakthroughs Brought by Large Models to AI - Emergence

- When a complex system is composed of many tiny individuals that interact with each other, and when there are enough of these individuals, they can exhibit special phenomena at the macro level that cannot be explained by the properties of the individuals themselves. This phenomenon is known as "Emergence."
- Currently, large models are believed to have emergence capabilities in two major types of tasks:
  - In-Context Learning (ICL). In-context learning helps algorithms better understand the meanings and relationships between words in a sentence, or helps them better recognize the relationships between different objects in an image.
  - Chain-of-Thought (CoT). Users write out a step-by-step reasoning process and provide it to the large language model, enabling the model to perform relatively complex reasoning tasks.



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- In daily life, there are also Emergences, such as the formation of snowflakes, traffic jams, animal migrations, and vortex formation. Let's take snowflakes as an example to explain this: Snowflakes are composed of water molecules, which are very small. However, when a large number of water molecules interact under certain external temperature conditions, they form a highly regular, symmetrical, and beautiful snowflake at the macroscopic level.
- When the model size is not large enough, it cannot handle various tasks well. But once it crosses a certain critical threshold in model size, the large model suddenly becomes capable of handling these tasks much better.

Breakthroughs Brought by Large Models to AI - Laying Foundation

Traditional Agriculture: Artisanal Farming

Farmer 1 ⇨ Crop 1

Farmer 2 ⇨ Crop 2

⋮ ⋮

- **One model for one scenario**
- **Small parameter size with poor generalization**

Traditional AI: Artisanal Development

Model 1 Training ⇨ Solving Problems 1

Model 2 Training ⇨ Solving Problems 2

⋮ ⋮

Model Generalization Normalization
From Fragmented Models to Single Models

Development cycle
From Monthly to Daily

Model accuracy
Improvement by 2-5%

Each farmer spends a lot of time and effort.

The foundation for achieving large-scale agricultural production is farmland cultivated through large-scale mechanized farming.



Regular Planting

Crop 1 Crop 2 ..... Crop N

Large models have become the fertile ground for artificial intelligence

Enabling large-scale, replicable AI industry applications

Each manufacturer requires a large amount of data and computing power.

The foundation for achieving large-scale AI applications lies in **Large Models** pre-trained on massive industry data.



Generated with minor tuning

Model 1 Model 2 ..... Model N

One model for multiple scenarios

Large number of parameters

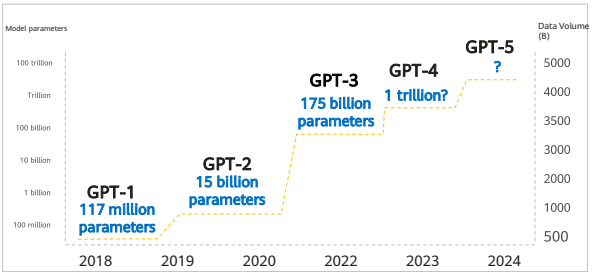
Strong generalization ability

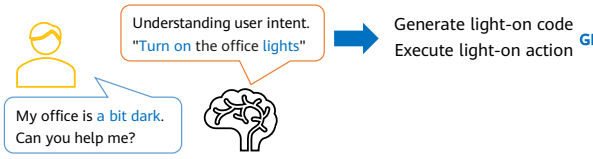
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Breakthroughs Brought by Large Models to AI - Understanding (1)

Model parameters and data volume have expanded, enabling ChatGPT to possess powerful language capabilities for "following instructions."





Application disruption, unleashing strong productivity.

Bing + ChatGPT: Revolutionizing Search, Your Internet Co-Pilot



Office + ChatGPT: Transforming Office Work, Redefining Productivity



GPT has achieved excellent performance in exams such as the GRE and Law.



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Breakthroughs Brought by Large Models to AI - Understanding (2)

PC Era:
Mouse + Graphical User Interface

- Using a mouse to perform most operations, lowered the barrier of using computer, leading to the widespread of PCs.
- Microsoft quickly learned from Apple and captured over 75% of the market share.

Mobile Internet Era:
Touchscreen + Gestures

- Apple's release of the iPhone, with its touchscreen, lowered the barriers to accessing and sharing information, transforming people's lifestyles in consumption, entertainment, and financial scenarios.

Task-Level Interaction Era:
Intelligent Internet

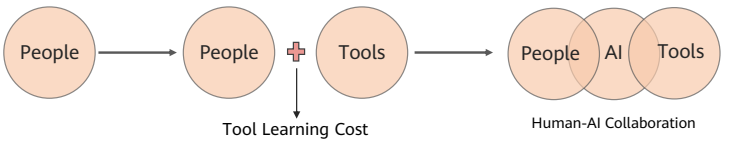
- ChatGPT effectively understands user intent and engages in conversations in a human-like manner, reducing the barriers to using software and hardware.

1990
Windows: Revolutionary Human-Computer Interaction

2007
iPhone: Revolutionary Human-Computer Interaction

2011
Siri: A New Way of Human-Computer Interaction

2022
Revolutionary Intelligent Human-Computer Interaction



Tool Learning Cost

Human-AI Collaboration

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Large Models - Inevitable Trend in AI Development



Before the emergence of the next revolutionary computational model, large models will be the most effective general paradigm in the field of artificial intelligence and will generate significant commercial value.



Research on large models has the potential to inspire the next general-purpose computing model. It could lead to the evolution of statistical learning methods to the next stage, with this breakthrough possibly arising from the integration of large models and knowledge.

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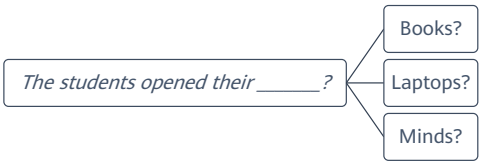
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Language Model Definition

- What is a language model?
  - Given a sequence of words, it predicts what the most likely next word is.



  - A more formal definition: Given a sequence of words, $w_1, w_2, \dots, w_{i-1}$, calculate the probability distribution of the next word w_i:

$$P(w_i|w_1, w_2, \dots, w_{i-1}),\ w_i \text{ could be any word in the vocabulary.}$$

  - It can also be considered that a language model is a model for calculating the probability of a piece of text:

$$P(w_1, w_2, \dots, w_m) = P(w_1)P(w_2|w_1) \dots P(w_i|w_1, w_2, \dots, w_{i-1}) \dots P(w_m|w_1, w_2, \dots, w_{m-1})$$

Different modeling methods

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- Language Model: Calculating the probability of a sentence and predicting what the next word might be.
- A language model is a type of model that calculates the probability distribution of a sequence of words (which can be phrases, sentences, or paragraphs). Its input is a text sentence, and its output is the probability of that sentence. This probability indicates how reasonable the sentence is, i.e., how well it conforms to the rules of human language.

Development of Language Models

- N-gram Language Model
- Neural Network Language Model
- Recurrent Neural Network Language Model
- Transformer Language Model
- Pre-trained Language Models (PLMs)
  - BERT: Bidirectional Encoder Representations from Transformers
  - GPT: Decoder-only Language Model
- Large Language Models (LLMs)
  - GPT-3
  - ChatGPT
  - DeepSeek

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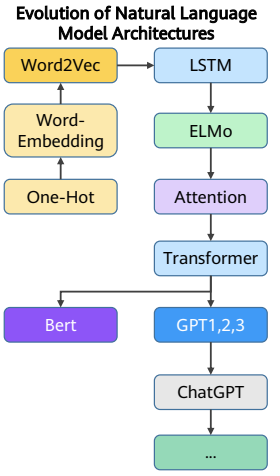
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- N-gram is an algorithm based on statistical language models. Its basic idea is to perform a sliding window operation of size N on the content of a text, treating it as bytes, and thus forming a sequence of byte segments with a length of N.
- Neural network language models are language models implemented using neural networks. By employing neural networks, words are mapped to vectors that serve as inputs to the network model to estimate the joint probability of word sequences.
- Recurrent neural network (RNN) language models aim to address the issue of fixed window size in feedforward neural network models. In addition, feedforward neural network models assume that each input is independent, which is not a reasonable assumption. The structure of RNNs can utilize the contextual sequence relationships of words, enabling better modeling of the relationships between sentences. To some extent, RNNs can also effectively capture contextual connections. However, due to the potential problem of long-term gradient vanishing in RNNs, their ability to remember context is limited.
- The emergence of large-scale pre-trained language models based on the Transformer architecture, represented by GPT and BERT, has ushered in a new era of pre-training and fine-tuning paradigms in natural language processing. When applying pre-trained models to downstream tasks, there is no need to understand too many task-specific details or design specific neural network architectures. Instead, one only needs to "fine-tune" the pre-trained model, i.e., perform supervised training on the pre-trained language model using labeled data for the specific task, to achieve significant performance improvements. These methods are commonly referred to as pre-trained language models (PLMs).
- In 2020, OpenAI released GPT-3 (Generative Pre-trained Transformer 3), a generative large-scale pre-trained language model with 175 billion parameters, marking the beginning of the era of large-scale language models. Due to the enormous number of parameters in these models, fine-tuning them for different

tasks would require significant computational resources, making the traditional pre-training and fine-tuning paradigm no longer suitable for large-scale language models. However, researchers discovered that by using methods such as In-Context Learning (ICL), it was possible to achieve good performance on many tasks with few samples simply by directly utilizing large-scale language models. Following this, researchers proposed various approaches tailored for large-scale language models, including prompt learning methods, Model as a Service (MaaS) paradigms, and instruction tuning, which have achieved excellent results across different tasks. At the same time, companies and research institutions such as Google, Meta, Baidu, and Alibaba have also released their own large language models.

Evolution of Language Algorithms

- Stage 1: Deep learning models driven by labeled data.
  - Static word vectors, such as Word2Vec.
  - Model architecture: RNN architectures, such as LSTM.
- Stage 2: Self-supervised pre-trained large models.
  - Context-based dynamic word vectors, such as BERT.
  - Model architecture: Transformer architecture, specifically Self-Attention.
  - The Transformer can address some of the core limitations of previous RNN architectures:
    - Difficulty in parallelization, as backpropagation requires computing the entire sequence.
    - Insufficient ability to model long-term dependencies.
    - The model scale is difficult to expand.



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- The emergence of deep learning technology marks a new phase in the development of artificial intelligence. We can refer to the wave of AI driven by deep learning that began around 2012 as the era of the new generation of AI (although the application of deep learning in speech recognition dates back even earlier, we consider the year 2012 as the starting point because many subsequent advancements in deep learning originated from the field of computer vision, with the appearance of AlexNet being a key milestone). This period can be regarded as the first stage of the new generation of AI driven by deep learning, where deep learning models trained on labeled data significantly improved model representation capabilities, thereby advancing AI technologies substantially and achieving commercial success in fields such as computer vision and speech recognition. The main limitation of this stage was its heavy reliance on the amount of labeled data. As the number of model parameters increased, solving for these numerous parameters required a large amount of training data as constraints. However, obtaining a large amount of labeled data is very costly, and it becomes increasingly difficult to scale beyond hundreds of millions of data points, thus limiting the effective size of models that can be supported by the available data. From 2012 to 2015, computer vision was the most active area of research, during which various deep network models, including ResNet, were developed.

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Why Introduce Attention?

- Attention:Focus.
- When using CNN models to recognize images, how can we make the model aware of the importance of different local information within the image?
- When using RNN models to handle NLP-related tasks, how can we solve the problem of long-term "memory"?



Encoder: The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-wise fully connected feed-forward network. We employ a residual connection [11] around each of the two sub-layers, followed by layer normalization [1]. That is, the output of each sub-layer is $\mathrm{LayerNorm}(x + \mathrm{Sublayer}(x))$, where $\mathrm{Sublayer}(x)$ is the function implemented by the sub-layer itself. To facilitate these residual connections, all sub-layers in the model, as well as the embedding layers, produce outputs of dimension $d_{\text{model}} = 512$.

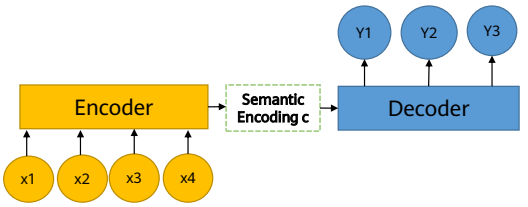
Decoder: The decoder is also composed of a stack of $N = 6$ identical layers. In addition to the two sub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head attention over the output of the encoder stack. Similar to the encoder, we employ residual connections around each of the sub-layers, followed by layer normalization. We also modify the self-attention sub-layer in the decoder stack to prevent positions from attending to subsequent positions. This masking, combined with fact that the output embeddings are offset by one position, ensures that the predictions for position i can depend only on the known outputs at positions less than i.

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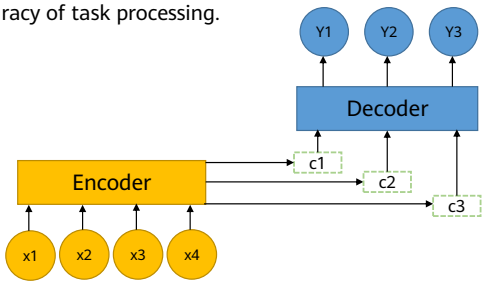
- The visual attention mechanism is an innate ability of the human brain. When we look at an image, we first quickly scan it and then focus on the target areas that require special attention. For example, when observing the image above, our attention is easily drawn to locations such as Peking University. Imagine if we were to process every detail in the image; it would inevitably consume a lot of energy, which would not be beneficial for human survival and evolution. Similarly, introducing a similar mechanism in deep learning networks can simplify models and accelerate computation.
- Attention allows us to focus on key points even when the text is long, enabling us to grasp the main ideas without losing important information. In the figure below, the colored text represents the key points that have been highlighted.

Attention Mechanism

- Attention Mechanism: At each step of the decoder, it uses a direct connection with the encoder to focus on specific parts of the source sequence.
- The attention mechanism focuses on more critical sections in the input data, reducing the focus on less important information and even filtering out irrelevant bits. This approach can solve the problem of information overload and improve the efficiency and accuracy of task processing.



Encoder-Decoder Framework



Encoder-Decoder Framework (with Attention)

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- Attention mechanism added to Encoder-Decoder: The c vector encoded by the Encoder is weighted against each output during the decoding process of the Decoder. This allows the Decoder to adjust the weights at each step to obtain different c vectors.

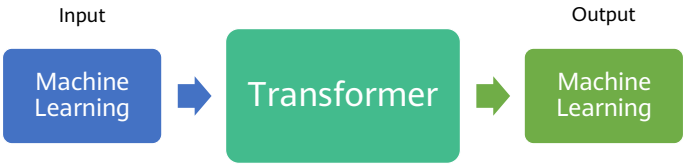
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The Birth of Transformer

- In 2017, the Google Machine Translation team published "Attention is All You Need," in which they introduced the Transformer model. This model completely abandoned traditional network architectures such as Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN), relying solely on the Attention mechanism to perform machine translation tasks. It achieved excellent results, and the Attention mechanism quickly became a hot topic of research.



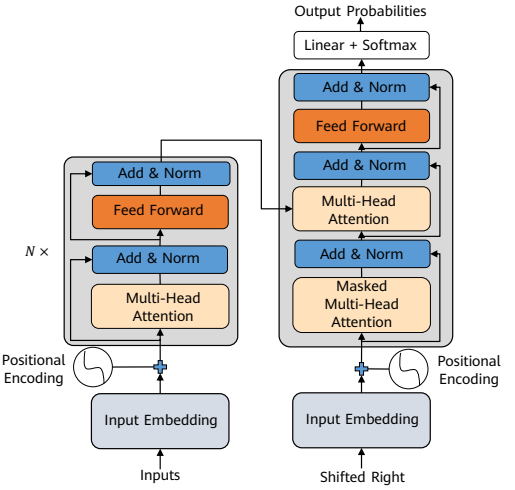
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- The Transformer was proposed by Google in their 2017 paper "Attention is All You Need," which was dedicated to machine translation tasks. This proposal caused a significant stir, and many studies have since demonstrated that the Transformer's ability to extract features is far superior to that of LSTM (Long Short-Term Memory).
- The main problem that the Transformer addresses is capturing textual features, and it achieves this through the use of Attention. This is also why the authors of the Transformer named their paper "Attention is All You Need."
- We know that ELMo (Embeddings from Language Models) uses LSTM instead of the newer Transformer. If ELMo had adopted the Transformer as its feature extractor, it is likely that BERT would not have received the same level of attention and popularity as it has today. Another point is that ELMo's method of feature fusion through bidirectional concatenation might be weaker compared to BERT's integrated feature fusion approach. However, this is merely a theoretical speculation, and there is currently no concrete experimental evidence to support this claim.
- Reference: https://zhuanlan.zhihu.com/p/51679783.

Transformer Model Structure

- The Transformer consists of two parts: Encoder and Decoder.
- The Encoder contains N identical layers, where "layer" refers to the units on the left side of the diagram. The symbol "$N\times$" on the far left indicates N, and if $N=6$, it corresponds to six such units.
- The Decoder also contains $N=6$ identical layers.



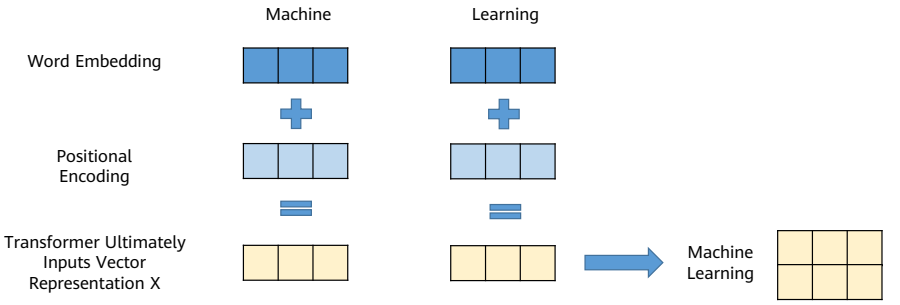
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- The layer structures in the encoder are identical, but they do not share parameters. Each layer can be broken down into two sub-layers. The output dimension of each sub-layer is d_model=512.
- The sentence input to the encoder first passes through a self-attention layer, which helps the encoder focus on other words in the input sentence when encoding each word.
- The output of the self-attention layer is then passed to a feed-forward neural network. The feed-forward neural network corresponding to each position's word is exactly the same (another interpretation is that it is a one-dimensional convolutional neural network with a window size of one word).
- The decoder also has self-attention and feed-forward layers similar to those in the encoder. In addition, there is an attention layer between these two layers, which focuses on relevant parts of the input sentence (similar to the attention mechanism in the seq2seq model).

Transformer Workflow: Obtaining Input

- The input X in the Transformer is obtained by adding word embeddings and positional encodings. Ultimately, each word is embedded into a 512-dimensional vector.



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- Each line represents a word X.

Word Embedding

- There are various ways to obtain word embeddings. For example, they can be pre-trained using algorithms such as Word2Vec and GloVe, or they can be trained within a Transformer model.
- Word2Vec is a statistical method for obtaining word vectors, proposed by Google's Mikolov in 2013 as a set of word embedding techniques. This algorithm has two training modes:
  - Predicting the current word based on its context.
  - Predicting the context based on the current word.
- GloVe is an extension of the Word2Vec method, combining global statistics with Word2Vec's context-based learning.

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- Word2Vec was mainstream until 2018, but with the emergence of BERT and GPT 2.0, this is not the best way to work.
- GloVe is called Global Vectors for Word Representation. GloVe is a word representation tool based on count-based & overall statistics. It can convert a word list into a vector consisting of real numbers. These vectors capture some semantic characteristics between words, such as similarity, analogy, etc. We can calculate the semantic similarity between two words by performing operations on vectors, such as Euclidean distance or cosine similarity.

Positional Encoding

- For the Transformer, since all words in a sentence enter the network simultaneously for processing, the sequential information is lost when the input is fed into the network. Therefore, the Transformer needs additional processing to inform the model about the relative positions of each word. One solution mentioned in the paper is positional encoding, which adds encodings that represent positional information to the input, allowing the network to know the position and order of each word.
- In summary, Positional Encoding is an encoding that represents the relative positions of words in a sentence, enabling the Transformer to retain positional information.

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

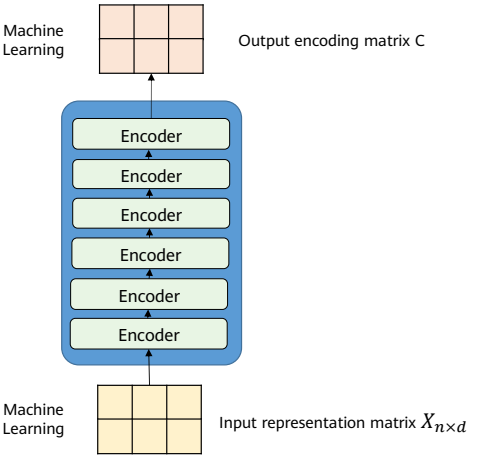
Here, pos represents the position of a word in the sentence, d denotes the dimension of the PE (the same as the word embedding), 2i represents even-numbered dimensions, and 2i+1 represents odd-numbered dimensions (i.e., 2i ≤ d, 2i+1 ≤ d).

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- Using this formula to calculate PE has the following advantages:
  - It allows the PE to adapt to sentences longer than any sentence in the training set.
  - It enables the model to easily compute relative positions; for a fixed interval k, PE(pos+k) can be calculated using PE(pos).

Transformer Workflow: Passing Input into Encoder

- The obtained word representation vector matrix is passed into the Encoder. After passing through 6 Encoder blocks, we can obtain the encoding information matrix C for all words in the sentence.
- The dimension of the matrix output by each Encoder block is exactly the same as the input.



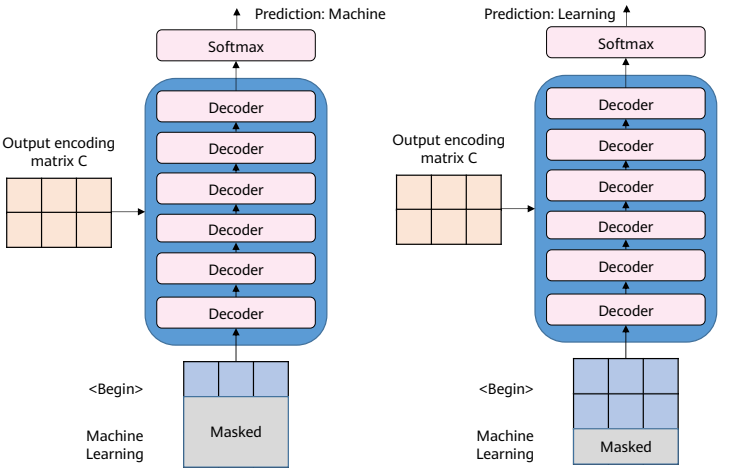
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- $X_{n\times d}$ is input matrix, where n is the number of words in the sentence, and d represents the dimension of the vectors, with d = 512 in the paper.

Transformer Workflow: Passing Encoding Matrix to Decoder

- The encoded information matrix C output by the Encoder is passed to the Decoder. The Decoder then translates the (i+1)th word based on the previously translated words from 1 to i.
- Throughout this process, when translating the (i+1)th word, it is necessary to use a Mask operation to hide the words after the (i+1)th word.



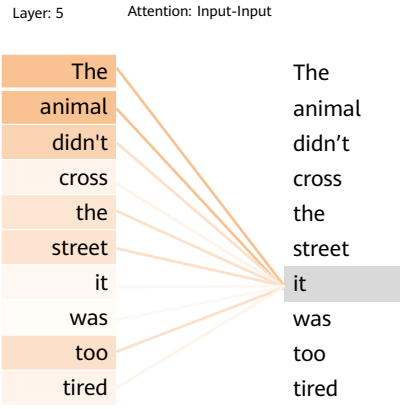
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Self-Attention Mechanism: Self-Attention

- For example, the following sentence is the input sentence we want to translate:
  - *The animal didn't cross the street because it was too tired.*
- What does this "it" refer to in this sentence? Does it refer to "street" or "animal"? This is a simple question for humans, but not for algorithms.
- When the model processes the word "it," the self-attention mechanism allows "it" to establish a connection with "animal."

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The Powerful Role of Self-attention Mechanism



- As the model processes each word in the input sequence, self-attention allows the model to focus on all words in the entire input sequence, helping it better encode the current word.

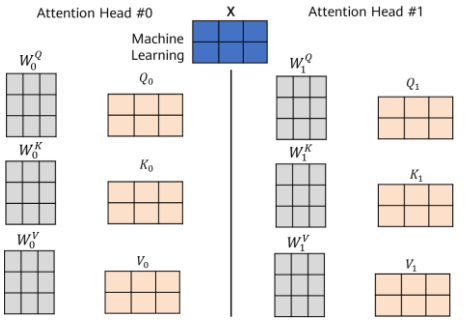
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- If you are familiar with RNNs (Recurrent Neural Networks), recall how they maintain the hidden layer. An RNN combines the representations of all previous words/vectors it has processed with the current word/vector it is processing. In contrast, the self-attention mechanism integrates the understanding of all relevant words into the word we are currently processing.
- When we encode the word "it" in Encoder #5 (the topmost encoder in the stack), the attention mechanism will focus on "The Animal," incorporating part of its representation into the encoding of "it."

Multi-Head Attention Mechanism

- To further improve the self-attention layer, the paper introduces a multi-head attention mechanism (Multi-head Attention).



- This mechanism enhances the performance of the attention layer in two ways: on one hand, it expands the model's ability to focus on different positions; on the other hand, it provides multiple "representation subspaces" for the attention layer.

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- To further improve the self-attention layer, the paper introduces a mechanism called "multi-headed" attention.
- This mechanism can further enhance the performance of the attention layer.
- Multi-headed attention is a combination of multiple sets of self-attention mechanisms. While the self-attention mechanism helps to build word feature representations that include contextual information, multi-headed attention allows the model to learn various types of contextual influences. For example, in the sentence "The weather is nice today, suitable for going out for a run," the correlation between "today," "sunny," and "running" varies depending on different contexts. The more heads there are, the better the model can capture larger and more diverse correlation features, thereby enhancing its expressive power.

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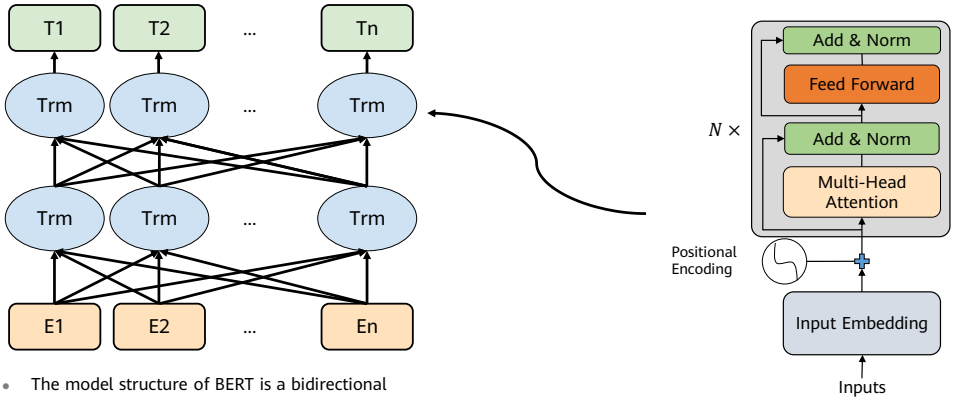
The Birth of BERT

- *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding* is a paper published by the Google team in 2019. Their goal was similar to that of GPT (Generative Pre-Training), which is to learn a general model that can be applied to a wide range of tasks, requiring only minor modifications to adapt to different types of tasks.
- From the experimental results, it is evident that BERT performs better than GPT-1.

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- BERT and GPT-1 both adopt the same two-stage task: first, a pre-training phase where the model is pre-trained on a language modeling task; then, the model is fine-tuned for specific tasks.

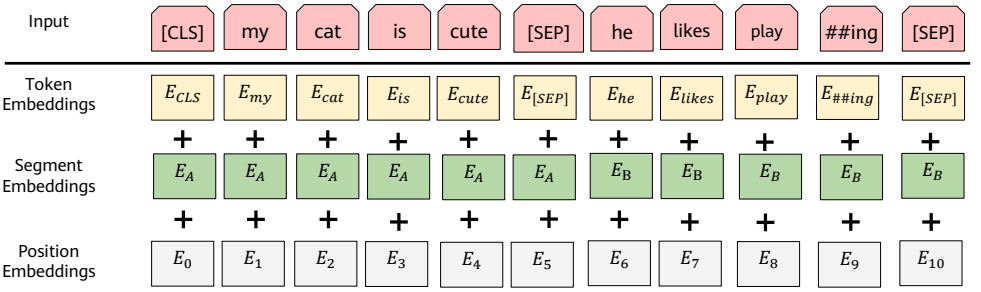
BERT Architecture Diagram



- The model structure of BERT is a bidirectional Transformer encoder. The right figure shows an Encoder of one Transformer.

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Processing of the Input Part



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Processing of Output

```
def get_masked_lm_output(bert_config, input_tensor, output_weights, positions,
                         label_ids, label_weights):
  """Get loss and log probs for the masked LM."""
  input_tensor = gather_indexes(input_tensor, positions)

  with tf.variable_scope("cls/predictions"):
    # We apply one more non-linear transformation before the output layer.
    # This matrix is not used after pre-training.
    with tf.variable_scope("transform"):
      input_tensor = tf.layers.dense(
          input_tensor,
          units=bert_config.hidden_size,
          activation=modeling.get_activation(bert_config.hidden_act),
          kernel_initializer=modeling.create_initializer(
              bert_config.initializer_range))
      input_tensor = modeling.layer_norm(input_tensor)

    output_bias = tf.get_variable(
        "output_bias",
        shape=[bert_config.vocab_size],
        initializer=tf.zeros_initializer())
    logits = tf.matmul(input_tensor, output_weights, transpose_b=True)
    logits = tf.nn.bias_add(logits, output_bias)
    log_probs = tf.nn.log_softmax(logits, axis=-1)

    label_ids = tf.reshape(label_ids, [-1])
    label_weights = tf.reshape(label_weights, [-1])

    one_hot_labels = tf.one_hot(
        label_ids, depth=bert_config.vocab_size, dtype=tf.float32)

    # The `positions` tensor might be zero-padded (if the sequence is too
    # short to have the maximum number of predictions). The `label_weights`
    # tensor has a value of 1.0 for every real prediction and 0.0 for the
    # padding predictions.
    per_example_loss = -tf.reduce_sum(log_probs * one_hot_labels, axis=[-1])
    numerator = tf.reduce_sum(label_weights * per_example_loss)
    denominator = tf.reduce_sum(label_weights) + 1e-5
    loss = numerator / denominator

  return (loss, per_example_loss, log_probs)
```

The first character position CLS corresponds to the classification result output by the Transformer.

Obtain the position information of the masked word, extract the mask, which is of variable length, up to a maximum of 20.

Apply the hidden layer.

Matrix multiplication between the result and the dictionary word matrix.

Apply Softmax to predict the most likely word

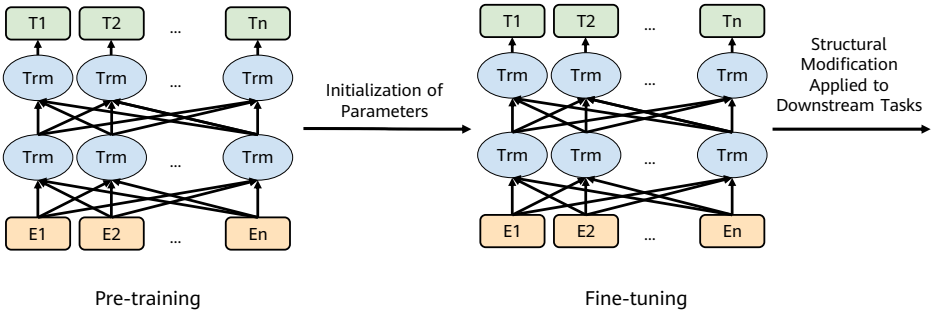
Accumulate loss

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- Code screenshot.

Fine-tuning Phase



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Model Performance

- When BERT was introduced, it indeed caused a significant stir in the NLP field, as its performance was impressive and set new records on many NLP dataset leaderboards.
- However, technological development did not stop there. Later, models like GPT-2, GPT-3, GPT-3.5, and ChatGPT emerged, with increasingly larger parameter sizes. These large models also developed chain-of-thought reasoning capabilities, demonstrating intelligence that surpassed previous models.

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Background of GPT

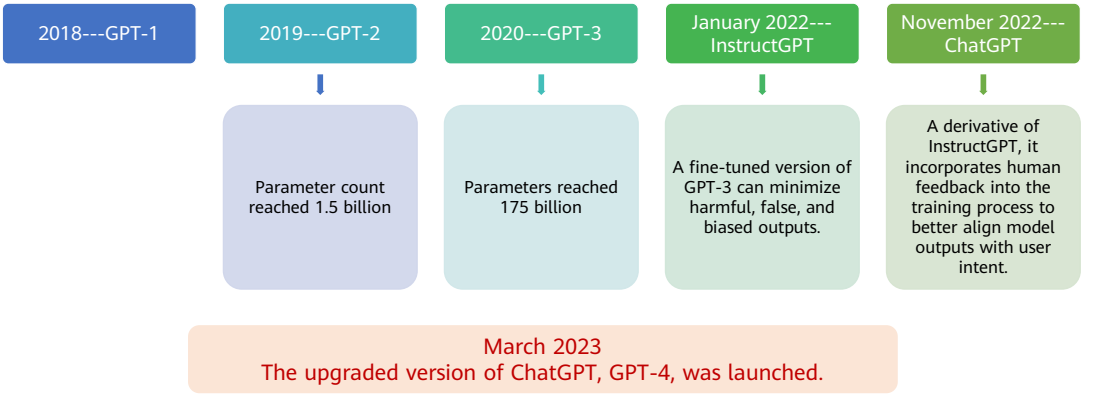
- In the field of Natural Language Processing (NLP), there is only a small portion of labeled data, while a vast amount of data remains unlabeled. To fully utilize this large amount of raw text data, unsupervised learning is needed to extract features from the text. The most classic example of this is word embedding technology.
- Word embeddings can only handle tasks at the word-level (such as synonymy), but they cannot address sentence-level or sentence-pair-level tasks (such as translation or reasoning).
- To solve these problems, OpenAI proposed the GPT (Generative Pre-trained Transformer) framework, which uses a general-purpose large model to perform language understanding tasks.

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- To address the aforementioned issues, OpenAI proposed the GPT framework, which uses a semi-supervised learning approach to accomplish language understanding tasks. The training process of GPT is divided into two stages: pre-training and fine-tuning. The goal is to learn a general representation method that can be easily adapted to different types of tasks with only minor modifications.
- *Improving Language Understanding by Generative Pre-Training* is a paper published by the OpenAI team in 2018. Their objective was to learn a general representation that could be applied to a wide range of tasks, requiring only slight adjustments to adapt to different types of tasks.
- However, NLP is not a new field in AI; there have been many outstanding NLP models developed before. So, what is the innovation of GPT? Previous NLP models were task-specific, such as word segmentation, word vectors, and sentence similarity. Each task had its own model, so every new task required a new model. GPT's approach is to generate a large model that can produce results for different tasks simply by changing the input. This is a very innovative idea.

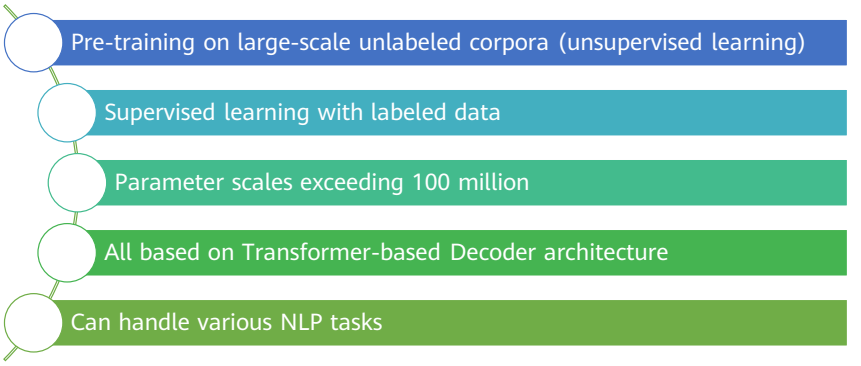
The Development of GPT



March 2023
The upgraded version of ChatGPT, GPT-4, was launched.

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Some Common Features of the GPT Series



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- Various NLP tasks include text classification, text question answering, and more.

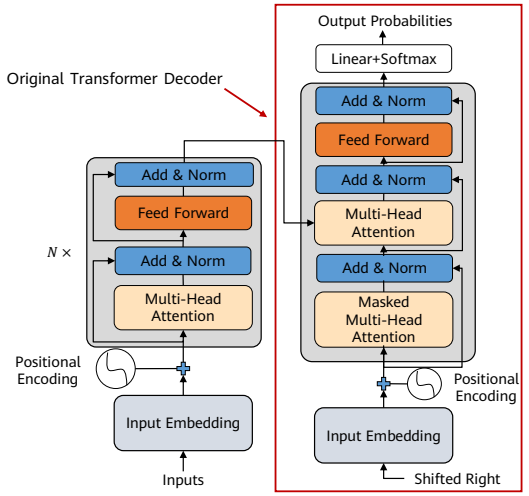
Comparison of Model Parameters in the GPT Series

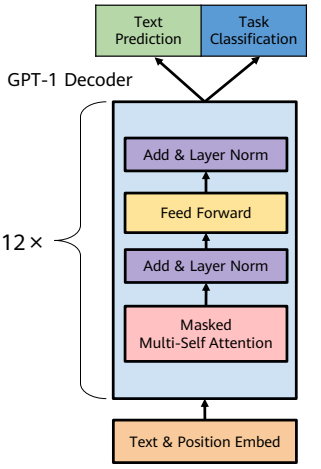
Refers to the number of stacked Transformer Decoders.

Model	Total Number of Parameters	Number of Network Layers	Context Length	Training Data	Hidden Layer Dimensions
GPT-1	117 Million	12	512	10 Billion	768
GPT-2	1.5 Billion	48	1024	10 Billion	1600
GPT-3	1750 Billion	96	2048	Trillion	12288
GPT-4	1.76 Trillion	120	8000	13 Trillion	20k

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GPT-1 Architecture



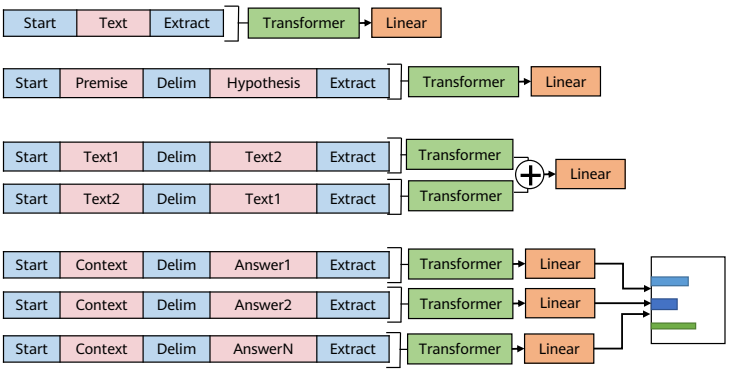


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- OpenAI's first version of GPT was initially called GPT.
- The choice to use Transformers instead of LSTMs was based on the fact that although pre-training helps capture some linguistic information, LSTM models have limitations in their short-term predictive capabilities. In contrast, the choice of Transformers allows us to capture longer-range linguistic structures and generalize better across different tasks.
- GPT-1 uses the Decoder framework of Transformers.
- GPT-1 retained the Masked Multi-Attention layer and Feed Forward layer from the Decoder and expanded the scale of the network. It increased the number of layers to 12, expanded the dimensionality of Attention to 768 (originally 512), increased the number of Attention heads to 12 (originally 8), and increased the hidden layer dimensionality of the Feed Forward layer to 3072 (originally 2048), resulting in a total of 117 million parameters.
- Regarding the positional encoding part, there is actually a significant difference between GPT-1 and the conventional Transformer. In the conventional Transformer, positional encoding is represented using cosine and sine functions. However, in GPT-1, positional encoding is initialized randomly in a manner similar to word embeddings and is updated during training, treating each position as an embedding that needs to be learned.

GPT-1 Task-Specific Input Transformations



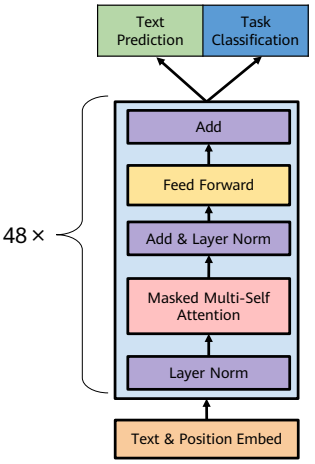
- The figure above illustrates the input transformations for fine-tuning on different tasks. All structured inputs are converted into token sequences, which are then processed by the pre-trained model (Transformer), and finally, a linear and Softmax layer is used to complete specific supervised learning tasks.

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- Input conversion can avoid compatibility issues across different tasks and prevent extensive modifications to the model. All conversions include adding randomly initialized start tokens () and end tokens (<e>).
- The figure above illustrates the input conversion for fine-tuning on different tasks. All structured inputs are converted into token sequences, which are then processed by a pre-trained model (Transformer), and finally, a linear layer and Softmax layer are used to complete specific supervised learning tasks.
- Reference: Improving Language Understanding by Generative Pre-Training.
- Entailment: Text entailment refers to a relationship between two text segments where one text serves as the premise and the other as the hypothesis. If the hypothesis H can be inferred from the premise P, then P entails H. For text entailment, the author concatenates the token sequences of the premise p and the hypothesis h, separated by a delimiter ($). This is essentially a classification problem, with possible labels being "entailment," "contradiction," or "neutral."
- Similarity: For sentence similarity, to eliminate the inherent order between two sentences, the author merges the two sentences in different orders and separates them with delimiters. Each order is then processed independently to obtain representations of the two sentences. These representations are summed element-wise and fed into a Linear layer.

GPT-2

- Differences compared to GPT-1:
  - Emphasizes zero-shot learning.
  - Larger model with 1.5 billion parameters.
  - Larger dataset, WebText dataset contains 40GB of text data, compared to only 5GB in GPT-1.
  - The training parameters were adjusted, with batch_size increased from 64 to 512, and the context window expanded from 512 to 1024.



Compared to GPT-1, GPT-2 has two architectural changes:

- More layers, with 48 stacked Decoder layers (GPT-2 Extra Large).
- Normalization is moved forward, placed before both self-attention and fully connected layers.

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- https://en.wikipedia.org/wiki/GPT-2.
- Zero-shot method:
  - The core idea of GPT-2 is that when the model has a very large capacity and is trained on sufficiently rich data, it can perform other supervised learning tasks simply by learning from the language model, without needing to fine-tune on downstream tasks.
  - For example, these tasks have already been performed during pre-training:
    - Translation: translating English to Chinese, [machine learning], [machine learning];
    - Q&A: answer the question, document, question, answer.
    - Document Summarization: summarize the document, document, summarization.
- Other adjustments related to model architecture are primarily due to the increasing risk of vanishing and exploding gradients as the number of model layers grows. These adjustments help reduce variance changes between layers during pre-training, making the gradients more stable.

GPT-3

- Compared to previous GPT versions, the changes are as follows:
  - Training data: Includes low-quality Common Crawl, high-quality WebText2, Books1, Books2, and Wikipedia, with different weights assigned to data of varying quality.
  - Model architecture: GPT-3 continues to use the GPT model architecture but introduces the sparse attention module from Sparse Transformer.
  - Downstream tasks: Unlike Zero-shot, GPT-3 adopts Few-shot learning.

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- GPT-3 assigns different weights based on the varying quality of the datasets; those with higher weights are more likely to be sampled during training. The final dataset size is approximately 570GB. In comparison, GPT-2's training data included WebText, BooksCorpus, Stories, Wikipedia, and Project Gutenberg, among others.
- The difference between sparse attention and traditional self-attention (referred to as dense attention) lies in:
  - Dense attention: Attention is calculated pairwise between every token, with a complexity of $O(n^2)$.
  - Sparse attention: Each token only calculates attention with a subset of other tokens, with a complexity of $O(n*\log n)$.
  - For details, please refer to https://devpress.csdn.net/aitech/649523aad385f5601380291f.html.
  - However, from a critical perspective, there are certainly drawbacks. In NLP, the content of language has contextual relationships, and such heavy reliance on context will inevitably degrade the performance of long-text modeling.
- In few-shot learning, several examples and task descriptions are provided for the model to learn. One-shot learning provides only one example and task description. Zero-shot learning does not provide any examples; instead, it only offers specific task-related descriptions during testing. The author conducted experiments on these three learning methods separately, and the results showed that the performance of all three methods improved with increasing model

capacity, and the order of effectiveness was: few-shot > one-shot > zero-shot.

GPT-3.5

- Although GPT-3 is very powerful, it does not understand human instructions well enough, which led to the idea behind the creation of GPT-3.5, also known as InstructGPT.
- InstructGPT uses reinforcement learning from human feedback to continuously fine-tune the pre-trained language model, aiming to enable the model to better understand human commands and instructions, such as generating short essays, answering knowledge-based questions, and conducting brainstorming sessions.
- Regarding the technical approach of InstructGPT, the original paper divides it into three steps:
  - Supervised fine-tuning based on GPT-3
  - Reward model training
  - Reinforcement learning training

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ChatGPT

- On November 30, 2022, OpenAI released ChatGPT, which quickly became a sensation. Content and test examples related to ChatGPT were widely shared online with impressive performance. In just two months, it gained 100 million users.
- In terms of the overall technical approach, ChatGPT builds upon the powerful GPT-3.5 Large Language Model (LLM) and introduces "Human-labeled Data + Reinforcement Learning" (RLHF, Reinforcement Learning from Human Feedback, where human feedback essentially refers to human-labeled data) to continuously fine-tune the pre-trained language model.
- Similar to the training process of GPT-3.5, also known as InstructGPT, it consists of three stages:
  - Supervised fine-tuning using the GPT-3.5 model
  - Reward model training
  - Reinforcement learning training

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- The main goal is to enable LLM models to learn how to understand the meanings of human commands (such as generating short essays, answering knowledge-based questions, brainstorming questions, and other types of instructions), and to teach the LLM to judge what kind of answers are high-quality for a given prompt input (user's question) based on various criteria, including being informative, rich in content, helpful to the user, harmless, and free from discriminatory information.
- The improvements mainly lie in the methods of collecting and labeling data; in other aspects, including model architecture and training procedures, it largely follows InstructGPT.
- Compared to InstructGPT, the difference lies in how the data is set up for training (and collection). ChatGPT adds some supervised data from dialogue tasks for fine-tuning.

GPT-4

- On March 14, 2023, OpenAI released GPT-4, a model with trillions of parameters, claiming that GPT-4 provides more reliable and creative responses compared to GPT-3.5 (including ChatGPT), and is capable of handling more nuanced instructions.
- GPT-4 is a multimodal model: it can accept both image and text inputs, enabling it to describe the meaning within specific images, summarize text from screenshots, and answer questions involving charts.

Model	Context Length (Tokens)
GPT-3	2048
GPT-3.5	4096
GPT-4	32768 (8192)

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- https://en.wikipedia.org/wiki/GPT-4.
- OpenAI claims that in their own tests, the model scored 1410 on the SAT (94th percentile), 163 on the LSAT (88th percentile), and 298 on the Uniform Bar Exam (90th percentile). In comparison, GPT-3.5 scored at the 82nd, 40th, and 10th percentiles on the same exams. GPT-4 also passed the Oncology Exam, Engineering Exam, and Plastic Surgery Exam.

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8. DeepSeek Analysis

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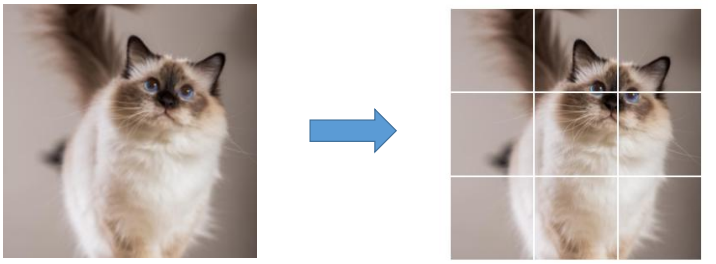
Background of ViT's Emergence

- From 2020 to 2021, convolutional neural networks dominated the field of Computer Vision (CV), while transformers became the standard configuration in Natural Language Processing (NLP). Many studies focused on extending ideas from NLP to CV, which can be mainly divided into two categories:
  - First, combining attention mechanisms with CNNs.
  - Second, replacing certain structures in CNNs with attention mechanisms while keeping the overall architecture unchanged. For example, applying self-attention mechanisms to the semantic segmentation domain led to the development of excellent semantic segmentation models such as Non-local, DANet, and CCNet.
- However, these methods still relied on CNNs to extract image features. Therefore, researchers began to consider whether it was possible to completely eliminate the dependence on CNNs and directly use transformers as encoders to extract image features.
- ViT is thinking about how to extend the Transformer architecture to the field of CV.

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Vision Transformer

- In the Vision Transformer (ViT) algorithm, an attempt is made to directly apply the standard Transformer architecture to images with minimal modifications to the entire image classification process. Specifically, in the ViT algorithm, the entire image is divided into small image patches, and these patches are linearly embedded into sequences that are then fed into the Transformer network. The model is trained for image classification using supervised learning.



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- As one of the most classic Transformer algorithms in the CV field, unlike traditional CNN algorithms, ViT attempts to directly apply the standard Transformer architecture to images with minimal modifications to the entire image classification process.
- To meet the input structure requirements of the Transformer, the entire image is divided into small image patches, and these patches are linearly embedded into sequences before being fed into the network. At the same time, class token is used for classification prediction.

Vision Transformer Variants

- ViT, referencing BERT, has set up three model variants (with an additional Huge variant) as follows. For example, ViT-L/16 represents the Large variant with an input patch size of 16 × 16.

Model	Layers	Hidden Size	MLP Size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

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- The paper provides parameters for three models (Base/Large/Huge). The "Layers" refer to the number of times the Encoder Block is repeatedly stacked in the Transformer Encoder. The "Hidden Size" corresponds to the dimension (length of the vector) of each token after passing through the Embedding layer. The "MLP size" is the number of nodes in the first fully connected layer of the MLP Block in the Transformer Encoder (which is four times the Hidden Size). The "Heads" represent the number of heads in the Multi-Head Attention mechanism within the Transformer.

Vision Transformer Experimental Results

- The algorithm was experimentally validated on medium-scale datasets (e.g., ImageNet) and large-scale datasets (e.g., ImageNet-21K, JFT-300M), and the following findings were observed:
  - Compared to CNN architectures, Transformers lack certain translation invariance and local perception capabilities, making it difficult to achieve comparable performance when data is insufficient. Specifically, Transformers trained on medium-scale ImageNet datasets tend to be a few percentage points lower in accuracy compared to ResNets.
  - However, the results change when there is a large amount of training data. After pre-training on large-scale datasets, applying transfer learning to other datasets can achieve or even surpass current State-of-the-art (SOTA) levels.

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- In machine learning, various assumptions are made about algorithms, and these assumptions are known as inductive biases. For example, convolutional neural networks (CNNs) have strong inductive biases. In the paper, an experiment was conducted showing that when trained on medium-sized datasets, the accuracy of ViTs was slightly inferior to that of ResNets. However, this result was expected because Transformers lack some of the inherent inductive biases of CNNs, such as translational invariance and locality. Yet, it is precisely in this aspect that Transformers excel. Due to their higher computational efficiency, and because their model performance does not saturate with increasing data size—at least not yet—the upper limit of model performance is very high. Therefore, Transformers are well-suited for training on large datasets.
- As shown in the results, ViTs do not perform well when the pre-training dataset is small. However, as the pre-training dataset grows larger, the performance of ViTs improves and eventually surpasses that of ResNets.

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Generating Structured Complex Data



Text

Composed of tokens



Images

Composed of pixels



Speech

16K sampling rate
16,000 samples per second

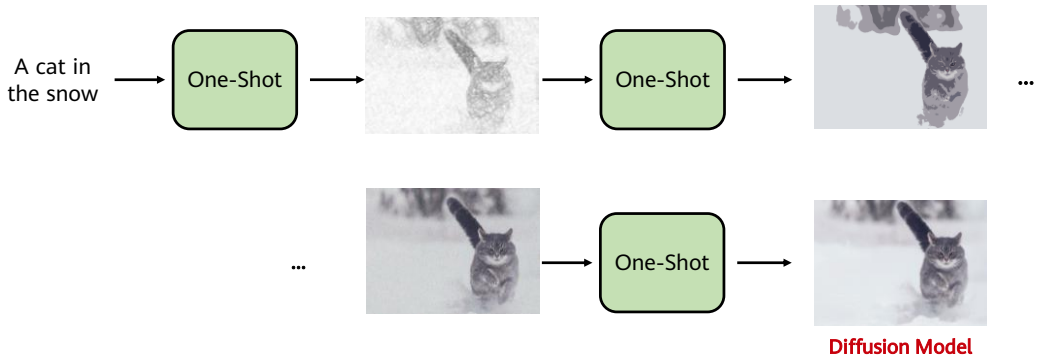
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- In Chinese, the token is the "character."
- In English, the token is a word piece: unbreakable → un break able.
- Step by step, breaking them down one by one, generating characters/words individually.

N “One-Shot”

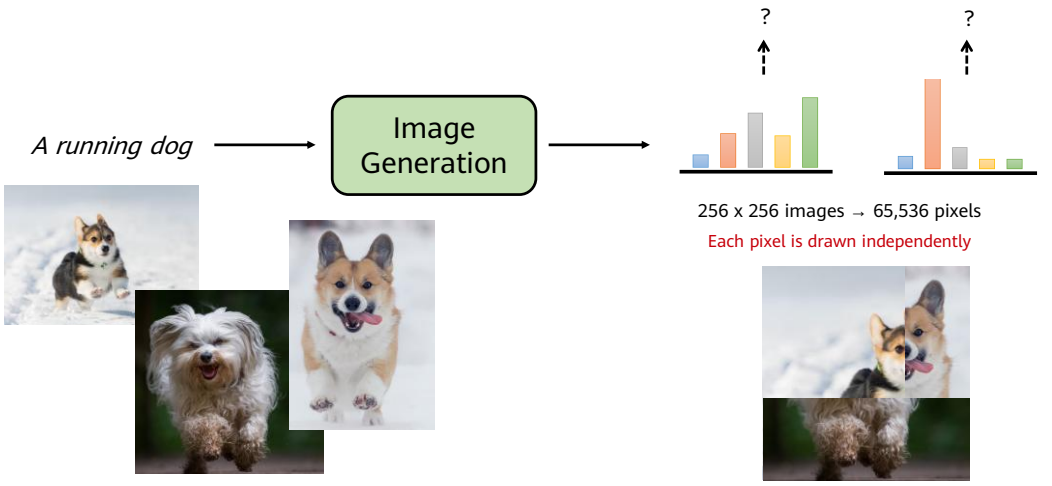
- “One-Shot” changed to “N-Shots”:



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- Initially, it involves inputting text + noise, and in later instances, it will involve inputting images + text.

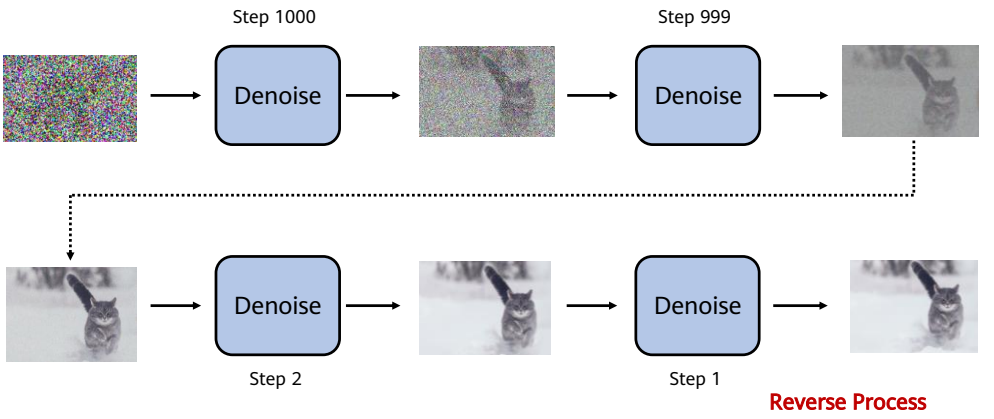
What's wrong with "One-Shot"?



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- Doing it all at once may result in poor outcomes and low quality.

How Diffusion Models Work

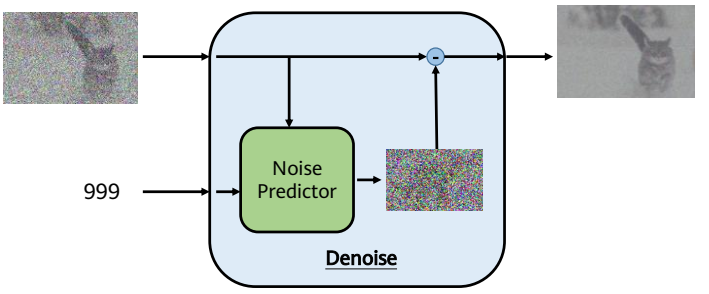


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- How to solve it? N times to get it right.
- First, input random noise and perform Denoise (noise reduction), going through many iterations of Denoise.
- **Reverse Process** Reverse Process.
- Forward Process: Diffusion, adding noise to the original image to generate training data.

What the Denoise module actually does



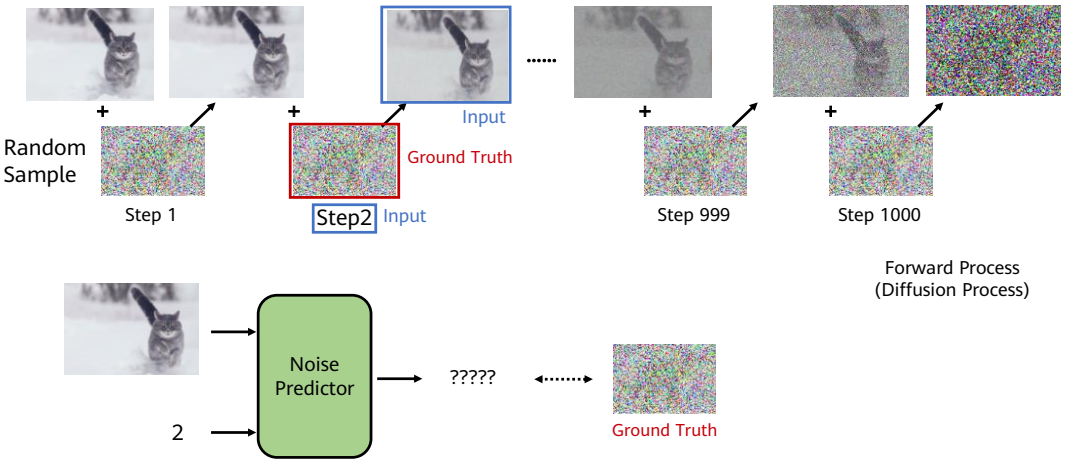
Predicting noise, rather than directly predicting the image

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- A trained noise predictor can denoise a noisy image and also predict the amount of noise added. Since the sampled noise is predictable, subtracting the noise from the image will result in an image that is closer to the one obtained through model training.
- What is predicted is the noise, which is equivalent to predicting the residual; subtracting this residual gives us the desired result.
- When inputting data, we need to inform the model about the current step, e.g., 999, indicating the current stage. This is similar to the process of a painter creating a work of art. When painting, the artist does not work aimlessly; instead, they have a concept in mind, starting with the outline and gradually filling in the details. By telling the model the current stage, it can better understand what needs to be done.

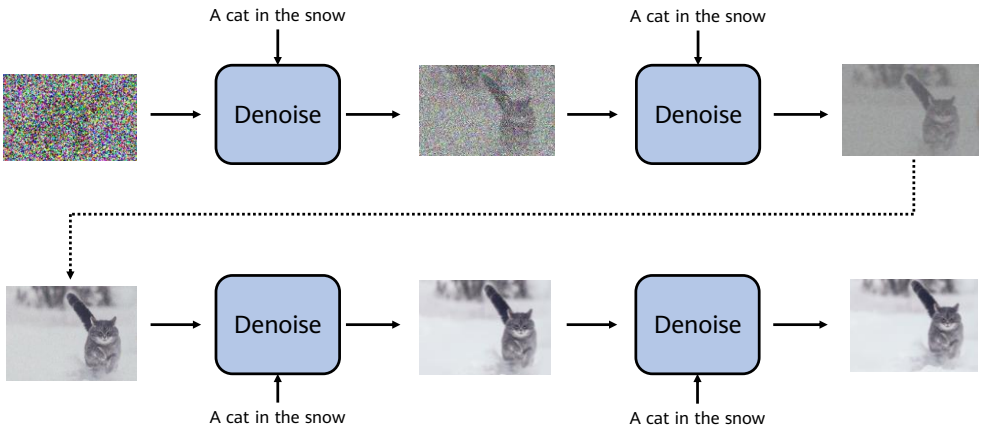
Training Noise Predictor



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- Assuming we already have an image, we can generate some noise and add it to the image, then treat this image as a training sample. By using the same operation, we can generate a large number of training samples to train the core components of an image generation model. Based on the aforementioned dataset, we can train a highly effective noise predictor, with each training step being similar to that of other models. When run under a certain configuration, the noise predictor can generate images.
- GAN and Diffusion are two popular models in the field of generative models.
- Advantages of GAN: High fidelity; however, training is not very stable, requiring alternating training of G and D, which makes it difficult to train, and the diversity of generated images is poor (mode collapse).
- Diffusion: It was not widely popular in the early stages because generating images multiple times was slow. However, it offers better diversity.

Text-to-Image (1)

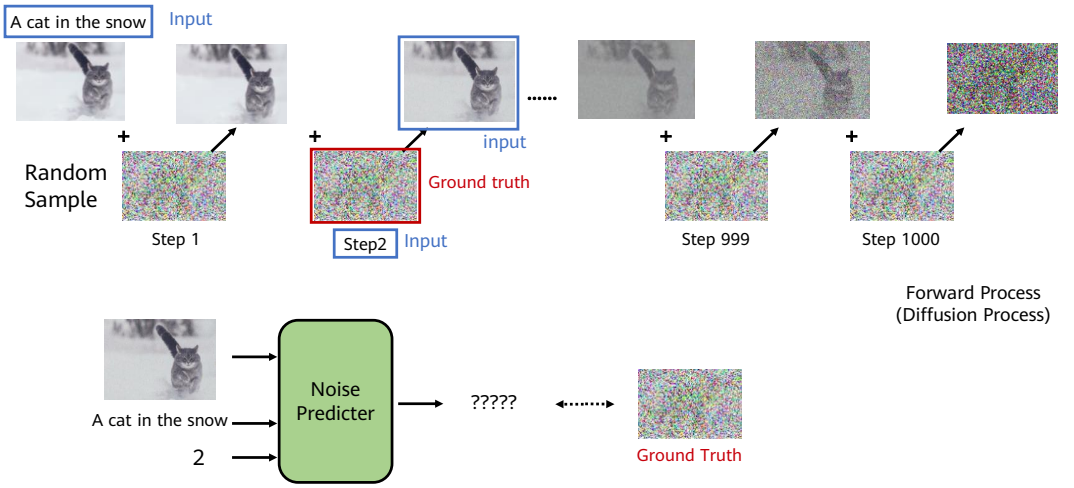


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- It is important to note that the diffusion process described so far has not used any text data to generate images. Therefore, if we were to deploy this model, it would be able to produce aesthetically pleasing images, but users would have no way to control the content being generated. In the following sections, we will describe how to incorporate conditional text into the process to control the type of images the model generates.
- Conditional Image Generation.
- LAION Dataset: 5.85B.

Text-to-Image (2)

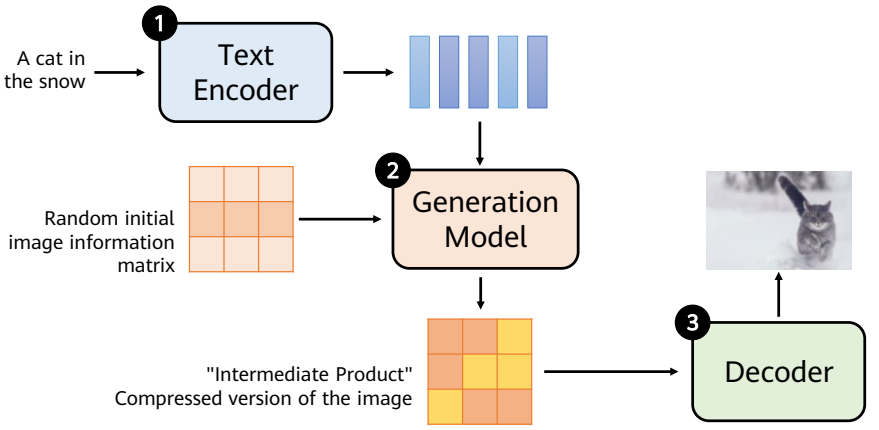


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- To incorporate textual conditions into the image generation process, the input to the noise predictor must be adjusted to include the text.

Stable Diffusion Model Architecture



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- Text Encoder: A text encoder.
- Generation Model: A diffusion model, which is the denoising process.
- Latent Space: Generating in the compressed space allows for faster processing, and the generated image is also a compressed version.
- Through the Decoder, it is decoded back into an image in the original space.
- Functionally, Stable Diffusion mainly includes two aspects: 1) its core function is to generate images based solely on text prompts as input (text2img); 2) you can also use it to modify images according to textual descriptions (i.e., input consists of text + image).
- Stable Diffusion is a system composed of multiple components and models, rather than a single model. When we look inside the model from the perspective of its overall architecture, we can see that it includes a text understanding component that translates textual information into a numeric representation to capture semantic information in the text.
- Then, this information is passed to the image generator, which also consists of multiple internal components. The image generator mainly involves two stages:

1. **Image Information Creator:** This component runs multiple steps to generate image information. These steps are also parameters in the Stable Diffusion interface and library, typically defaulting to 50 or 100. The Image Information Creator operates entirely within the image information space (or latent space), a feature that allows it to run faster than other diffusion models that operate in pixel space.
2. **Image Decoder:** The image decoder paints a picture based on the information obtained from the image information creator. The entire process needs to be run only once to generate the final pixel image. The whole compression process, including subsequent decompression and image rendering, is accomplished through an autoencoder, which compresses the image into latent space and then uses the compressed information to reconstruct the image solely with the

decoder.

- It's not just Stable Diffusion that generates images through denoising; DALL-E 2 and Google's Imagen model do the same.

Text Encoder - A Transformer Language Model

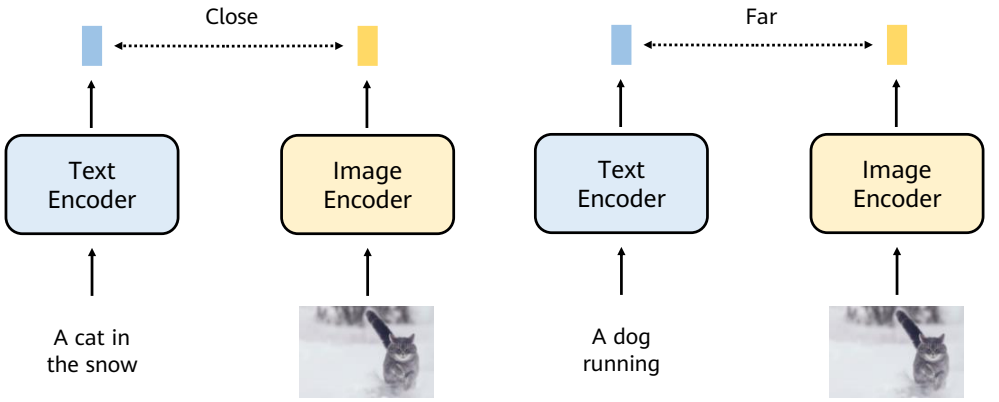
- The language understanding component in the model uses a Transformer-based language model that can convert input text prompts into token embedding vectors.
- Experiments have shown that a larger language model can lead to more significant improvements in image quality compared to choosing a larger image generation component.
- The early Stable Diffusion models used the pre-trained ClipText model released by OpenAI, but in Stable Diffusion V2, they have shifted to the latest and larger variant of the CLIP model, OpenClip.

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- https://arxiv.org/abs/2205.11487。

Contrastive Language-Image Pre-Training (CLIP)



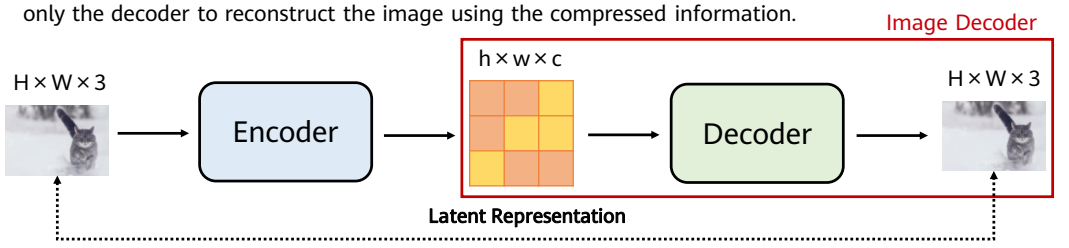
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- The idea of contrastive learning, multimodal models.
- Training uses pairs of text and image data.
- Reference: https://arxiv.org/abs/2103.00020.
- CLIP requires data consisting of images and their captions, with the dataset containing approximately 400 million images and descriptions. The dataset was collected by scraping images from the internet along with their corresponding "alt" tag texts.
- CLIP is a combination of an image encoder and a text encoder. Its training process can be simplified as taking images and their descriptions, encoding them separately using the two encoders. Then, the resulting embeddings are compared using cosine distance. At the beginning of training, even if the text description matches the image, the similarity between them is likely to be very low. As the model continues to update, in later stages, the embeddings obtained by the encoders for images and texts will gradually become more similar.
- By repeating this process across the entire dataset and using an encoder with a large batch size, it is ultimately possible to generate an embedding vector where images of cats and the sentence "a picture of a cat" are similar. Just like in word2vec, the training process also needs to include negative samples of mismatched images and captions, for which the model assigns lower similarity scores.

Image Decoder: Auto-encoder

- To accelerate the image generation process, Stable Diffusion does not run the diffusion process directly on the pixel images themselves, but instead runs it on a compressed version of the images, which is referred to in the paper as "Departure to Latent Space."
- The entire compression process, including subsequent decompression and image rendering, is accomplished through an autoencoder, which compresses the image into latent space and then uses only the decoder to reconstruct the image using the compressed information.

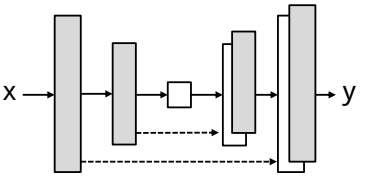


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- Decoder can be trained without labelled data.

Image Information Creator: Diffusion U-Net

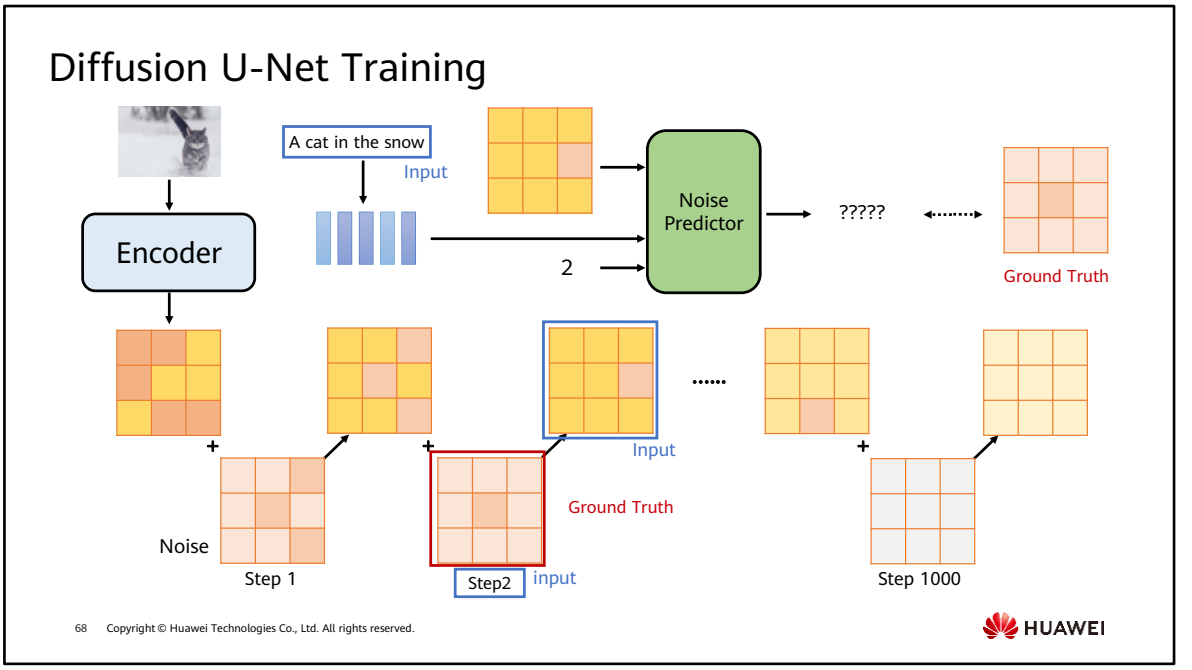
- The Image Information Creator operates entirely within the image information space (or latent space), a feature that allows it to run faster than other Diffusion models that operate in pixel space.
- The entire diffusion process consists of multiple steps, each of which operates on an input latent matrix and generates another latent matrix to better align with the "input text" and the "visual information" obtained from the model's image dataset.



U-Net

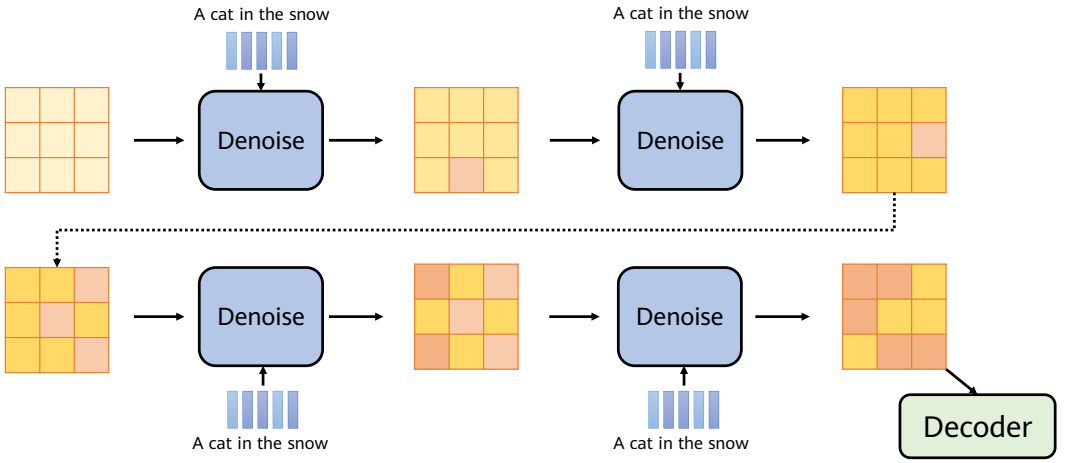
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- The resulting image is not an exact replica of the original image, but rather a distribution—a pixel arrangement of the world, such as the sky usually being blue, people having two eyes, cats having pointed ears, etc. The specific style of the generated image entirely depends on the training dataset.
- 1. The layers in the U-Net model are primarily used to transform latents; 2. Each layer operates on the output of the previous layer; 3. Some outputs (via residual connections) are fed into later processing in the network; 4. Time-step embeddings, which convert time steps into embedding vectors, can be used within the layers.



- All operations are performed in latent space, including the encoded text, input images, and predicted noise.
- The forward diffusion process is completed on compressed latents; noise slices are applied to the latents rather than to pixel images, so the noise predictor is actually trained to predict noise in the compressed representation (latent space).
- The forward process involves training the noise predictor using the encoder of the autoencoder. Once training is complete, images can be generated by running the reverse process (the decoder of the autoencoder).

Stable Diffusion Overview



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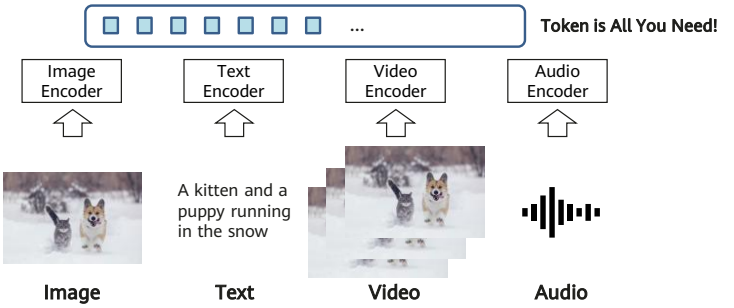
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Multimodal Large Models

- Multimodal AI refers to artificial intelligence that can process and understand multiple types of information, such as text, images, audio, and video. This type of AI is not only capable of handling tasks involving a single data type but also able to establish connections and integrate different data types, thereby achieving a comprehensive and holistic understanding of multimodal information.



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- In the past, our machine learning models generally only handled one type of data, such as text (translation, language modeling), images (object detection, image classification), or audio (speech recognition).
- But think about our human brains. We don't just read and write text; we can also look at pictures, watch videos, listen to music, and recognize various different sounds. Therefore, to make AI more like the real world, it is very important to process multiple types of data.

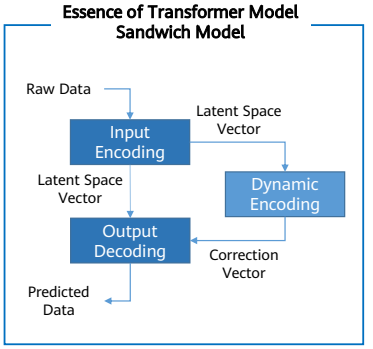
The Importance of Multimodality

- Many application scenarios would be nearly impossible to achieve without multimodal technology support, especially in fields that require processing multiple types of data, such as healthcare, robotics, e-commerce, retail, and gaming.
- By integrating information from various data modalities, the performance of models can be significantly improved. For example, a model capable of processing both text and images simultaneously should perform better than one that can only handle text or images alone.
- Multimodal technology allows for more flexible and diverse ways of interacting with systems, enabling users to choose their preferred interaction methods based on personal preferences or situational needs, such as keyboard input, voice dialogue, or image recognition.

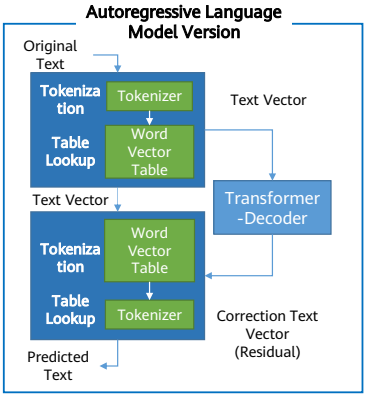
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- With the help of multimodal technology, people with visual impairments can easily browse the internet and even navigate the real world.

Multimodal Model Features (1)



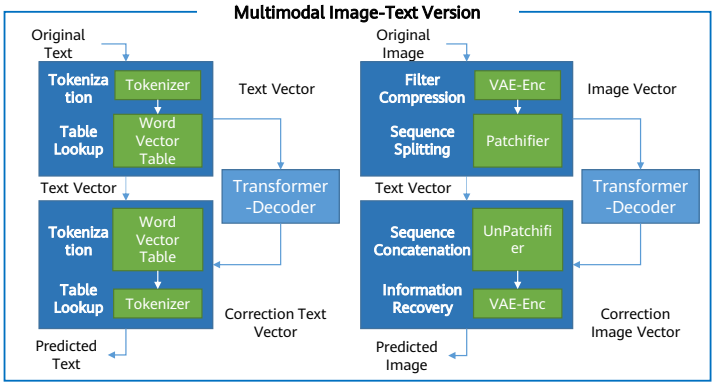
The logic of the Sandwich Model is that **the input and output processes perform encoding and decoding of single data items**. The Transformer architecture is used to **capture the relationships between multiple data points for correction.**



The encoding and decoding process of text-based models uses a tokenization + lookup table mechanism, which is very simple and lightweight, with **extremely low computational cost** during the forward pass**.**

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Multimodal Model Features (2)



Multimodal models emphasize **mapping different modalities to the same latent space** through multi-channel encoding and decoding, enabling unified computation directly. Therefore, the focus shifts to the quality of the encoding and decoding processes.

Due to the large amount of redundant information in images, using **an information compression layer can significantly save computational resources for the main module**, allowing for further segmentation after compression.
Therefore, the diffusion process is actually executed in the Transformer-Decoder part, with the parameters in the dark blue section on the left being fixed. For the text component, more advanced encoders, such as a **pre-trained large language model.**

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- On the left, compression is performed using an Autoencoder structure, followed by diffusion in the compressed space, and then decoding back to the original space.

Development of Multimodal Models

- In recent years, multimodal models have developed rapidly, progressing from Task-specific Models to Pre-trained Models, then to Unified Models with emerging capabilities, and finally to General-purpose Assistants.
- Multimodal models can be categorized based on downstream tasks into **comprehension** tasks (such as visual question answering, visual reasoning, and image-text retrieval) and **generative** tasks (such as text-to-image, image-to-image, image-to-video, and video-to-video generation).

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Multimodal Model Development - Sora Model (1)

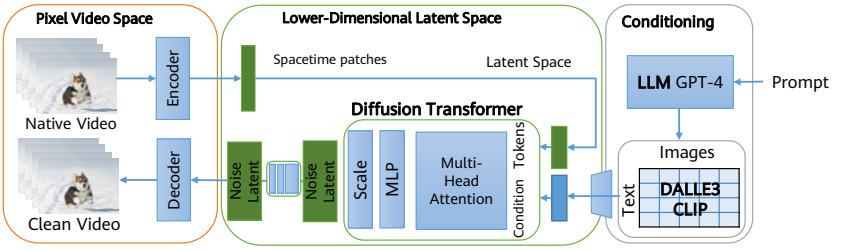
- Content:
  - Supports generating high-fidelity videos up to 60 seconds in 1080p resolution, with extremely high video quality and strong content consistency, featuring cinematic-level shot techniques.
  - Supports short video extensions, editing existing videos, connecting existing videos, and generating images.
- Technology:
  - Compress the video into spacetime patches and use Diffusion Transformer as the backbone network for modeling.
  - Training with the original video allows for direct generation of videos with different sizes, durations, and resolutions.
- Data Engineering:
  - Use DALLE 3's recaptioning technology for video-text annotation.
  - Utilize GPT-4 to expand user-provided short prompts into detailed and complex text descriptions.
- Others:
  - The details of physical interactions still have flaws, such as broken glass, water flow, and the inability to generate footprints in snow.

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- Sora Official Website: https://openai.com/index/sora/.
- https://openai.com/index/video-generation-models-as-world-simulators/.

Multimodal Model Development - Sora Model (2)

- Training Steps:
  - Step 1: Use DALLE 3 recaption technology to generate high-quality text descriptions for the original video.
  - Step 2: Preprocess the video by decoding it into a new format as input, or directly input the video and then decode it.
  - Step 3: Compress the video into a low-dimensional spatial representation using a VAE encoder, and then extract spatiotemporal patches to input into DiT.
  - Step 4: Learn image semantic generation based on Diffusion Transformer to map from text semantics to image semantics.
  - Step 5: The low-dimensional spatial representation generated by DiT is decoded back into pixel-level video data using a VAE decoder.



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1. Overview of Mainstream Large Models
2. Definition and Development of Language Models
3. Transformer Architecture
4. GPT and BERT Models
5. Vision Transformer Model
6. Diffusion Model
7. Multimodal Large Model
8. **DeepSeek Analysis**

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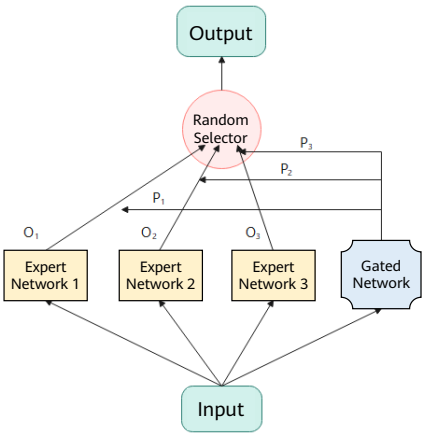
Dense Models and Sparse Models

- Model scale is one of the key factors for improving model performance. As large models based on the Transformer architecture continue to evolve, the number of parameters in these models has been increasing (to enhance model performance). However, if all parameters are involved in computation during training and inference (i.e., most of the matrices involved in computation are dense matrices), the computational power required will also increase significantly. Therefore, such dense models with a large number of parameters have relatively high inference costs.
- So, how can we increase the number of model parameters without significantly increasing the computational demand?
- The Mixture of Experts (MoE) model is an excellent choice! In this model, only a portion of the parameters are involved in computation during the process, meaning that only a small fraction of the parameters are non-zero. This sparsity in matrix computation reduces the demand for computational resources, resulting in lower computational requirements and shorter computation times for both training and inference.

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Mixture of Experts (MoE)

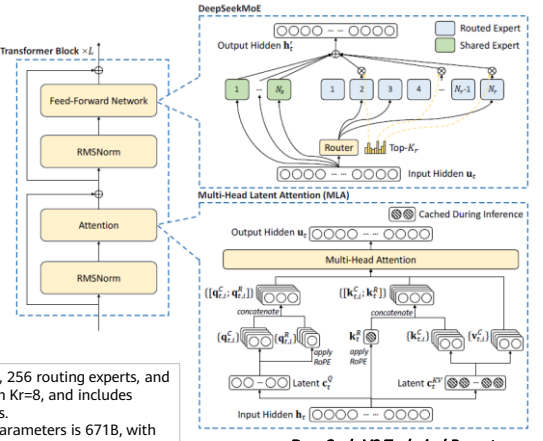
- What exactly is a Mixture of Experts (MoE)? Its overall framework is mainly divided into the following parts:
  - **Expert Networks**: These are used to learn different data. Input data passes through each expert system, and each outputs a value, which in this diagram is labeled as Oi.
  - **Gating Network**: This determines the weight of each expert system's output in the random selector. It assigns a weight Pi to each output. This network can also be trained.
  - **Random Selector**: It takes the current outputs from multiple expert systems and combines them with the assigned weights to produce the final output.



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DeepSeek-V3

- On the evening of December 26, 2024, AI company DeepSeek officially launched the first version of its new series model, DeepSeek-V3, and simultaneously open-sourced it. This model has stunned the AI community with its powerful general AI capabilities (especially in tasks such as knowledge Q&A, text generation, and code writing) and its low training and inference costs.
- The DeepSeek-V3 network adopts a MoE (Mixture of Experts) architecture and has been optimized further.



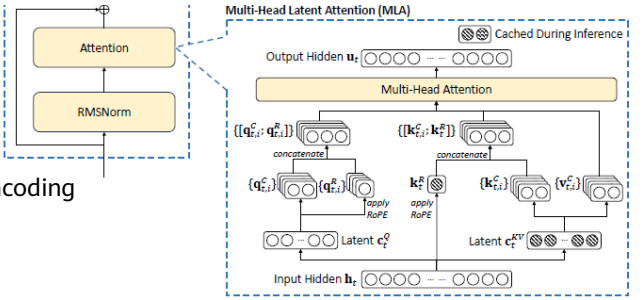
It consists of 61 layers, 256 routing experts, and one shared expert with Kr=8, and includes three initial MLP layers.
The total number of parameters is 671B, with 37B active parameters.

DeepSeek-V3 Technical Report

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DeepSeek-V3 Model Architecture Innovation - MLA

- The primary role of MLA is to optimize KV-Cache to reduce computational card memory usage, thereby improving inference performance.
- Functional Process:
  - Input hidden state ht
  - Compute Q/K/V
  - Fused Rotational Positional Encoding
  - Output Hidden State ut

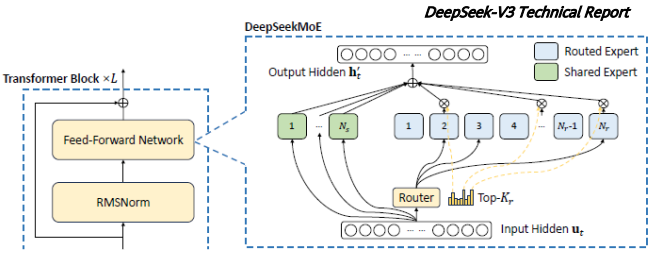


DeepSeek-V3 Technical Report

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DeepSeek-V3 Model Architecture Innovation - DeepSeekMoE

- **Router:**
  - The router determines which experts each part of the input Uc will be routed to.
  - Each token is routed to at most Kr expert nodes, reducing communication overhead.
- **Experts:** There are 256 experts, divided into two categories:
  - Routed Expert: An expert selected by the router.
  - Shared Expert: A shared expert used to capture global features.

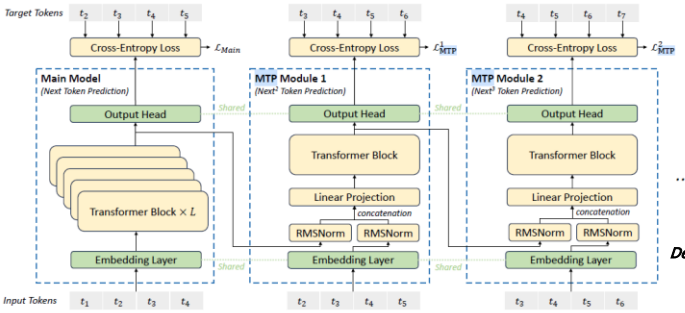


- **Top-Kr Strategy:**
  - Select the Kr experts with the best responses from Nc experts.
  - Limit the scope of cross-node communication to optimize efficiency.
  - Selecting 9 out of 256 experts (8+1)

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DeepSeek-V3 Model Architecture Innovation - MTP

- Traditional prediction methods are slow in inference, generating tokens one by one. Predicting a single token requires extensive interaction with computational card memory, and since the prediction of a single token tends to focus on local optimality, the overall performance may not be satisfactory.
- Core implementation idea: During the decoding phase, the model predicts multiple future tokens at once through multiple sequential modules. Specifically, during the training phase, generating multiple subsequent tokens at once allows the model to learn labels for multiple positions simultaneously, improving the effective utilization of samples and enhancing training efficiency. This technique can also be applied during the inference phase, where multiple tokens are generated at once, resulting in a significant increase in inference speed.



DeepSeek-V3 Technical Report

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DeepSeek-R1

- On January 20, 2025, DeepSeek officially released the DeepSeek-R1 model and simultaneously open-sourced the model weights. DeepSeek-R1 significantly enhances the model's reasoning capabilities with only a minimal amount of labeled data.

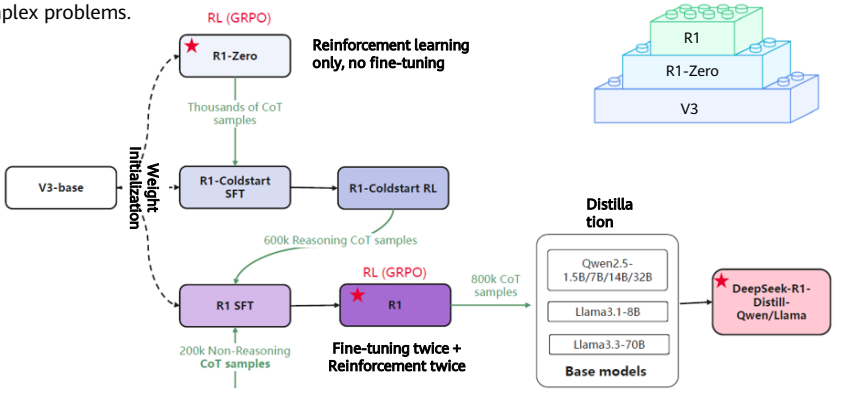


- In a broader sense, DeepSeek-R1 is not a single model. It also includes the initial stage model DeepSeek-R1-Zero and several smaller large models distilled from R1.

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DeepSeek-R1 Training Process

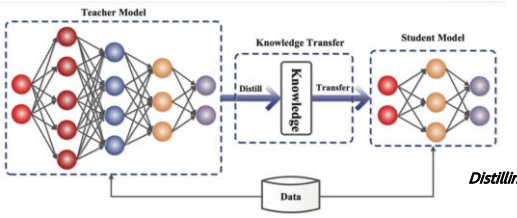
- R1-Zero: Instead of relying on traditional SFT, MCTS, and PRM, it applies the GRPO algorithm on the base model for reinforcement learning, enabling the model to autonomously explore CoT (Chain-of-Thought) reasoning to solve complex problems.



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DeepSeek Model Distillation

- Model Distillation (Knowledge Distillation) is a technique that transfers knowledge from a large, complex model (the teacher model, such as DeepSeek-R1 671B) to a smaller, more efficient model (the student model, such as Qwen 7B).



Distilling the Knowledge in a Neural Network

- DeepSeek-R1 generates 800,000 data points as supervised data.
- Supervised Fine-tuning (SFT) is performed on open-source small models like Qwen and Llama series without using reinforcement learning.
- The performance of the 32B/70B models is close to that of OpenAI's o1-mini. For example, DeepSeek-R1-DistillQwen 32B achieved a 72.6% Pass@1 score on AIME 2024.

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Quiz

1. (Multiple-choice Question) Which of the following technologies are used by Transformer?
   A. Feedforward Neural Network
   B. Normalization
   C. Residual Module
   D. Self-Attention Mechanism

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- Answer: ABCD

Quiz

2. (Multiple-choice Question) Which of the following components are included in the Vision Transformer architecture?

   A. Patch Embedding

   B. Positional Encoding

   C. Transformer Encoder

   D. MLP Head

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- Answer: ABCD

Quiz

3. (Multiple-choice Question) Which modules are included in the Stable Diffusion model?
   A. Text Encoder
   B. Image Encoder
   C. Image Decoder
   D. Image Information Creator

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- Answer: ACD

Summary

- This chapter provides an in-depth introduction to the topics related to large models today, including the current status of large models and their impact on AI development, the fundamentals and key technologies of large language models, detailed explanations of the Attention mechanism and Transformer architecture, the architectures and applications of BERT, GPT, and Vision Transformer, generative learning strategies and the structure of Diffusion models, the importance and characteristics of multimodal large models, an overview of the Sora model, and an analysis of the DeepSeek algorithm, among other core topics.

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Abbreviations

- ACC: Accuracy.
- API: Application Programming Interface.
- BERT: Bidirectional Encoder Representations from Transformers.
- BiLSTM: Bidirectional Long Short Term Memory.
- CLM: Causal Language Modeling.
- CNN: Convolutional Neural Network.
- CLS: Classification.
- CoT: Chain of Thought.
- DDPM: Denoising Diffusion Probabilistic Model.
- ELMo: Embeddings from Language Models.
- EOS: End of Sentence.

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Abbreviations

- GloVe: Global Vectors for Word Representation, it is a word representation tool based on global word frequency statistics. It can represent a word as a vector of real numbers that capture some semantic properties between words.
- GPT: Generative Pre-trained Transformer.
- LLM: Large Language Model.
- LSTM: Long Short-Term Memory, a type of recurrent neural network.
- LTR & No NSP: Left-to-right LM without the next sentence prediction.
- MLM: Masked Language Modeling.
- MLP: Multilayer Perceptron.
- NLEU: Bilingual Evaluation Understudy.
- NLP: Natural Language Processing.

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Abbreviations

- NSP: Next Sentence Prediction.
- PEFT: Parameter Efficient Fine Tuning.
- RLHF: Reinforcement Learning from Human Feedback.
- RM: Reward Model, reward model.
- RNN: Recurrent Neural Network.
- SD: Stable Diffusion.
- SEP: Separate, refers to a delimiter.
- SFT: Supervised Fine Tuning.
- SOTA: State-of-the-Art, the most advanced.

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Business Processes of Large Models



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Foreword

- This chapter mainly discusses the scenario-based application process of large models in enterprises, including requirement analysis, model selection, dataset construction, model training and fine-tuning, and usage. It also provides an in-depth introduction to the commonly used components for model training, optimization, and inference deployment, as well as the requirements for software and hardware.

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Objectives

- On completion of this course, you will be able to:
  - Understand the process of applying large models in enterprise scenarios.
  - Grasp the definitions of foundational, industry, and scenario-specific models.
  - Familiarize with common tools for data preparation, model training, and deployment.

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   - Demand Analysis and Model Selection for Large Models
   - Business Scenarios and Application Scenarios of Large Models
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3. Model Selection
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5. Data Preparation
6. Large Model Training and Optimization
7. Large Model Inference and Deployment

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Scenario Introduction

- To meet the needs of business operations, companies recruit new employees every year. As the company grows, the skill requirements for each position become increasingly specialized, and the number of employees in the same role gradually decreases. Traditional training methods such as uniform instruction and mentorship by experienced employees are no longer sufficient to meet the needs of new employee development.



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Enterprise Pain Points

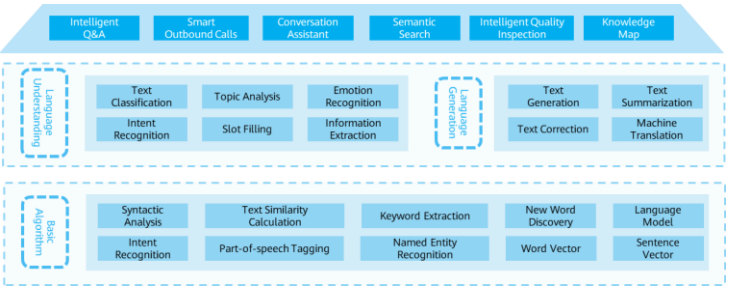
- At the same time, new employees often find it difficult to know where to start when faced with a large amount of product and technical documentation, leading to low self-learning efficiency and difficulty in quickly getting up to speed with their work. Even with mentorship, it still requires a long training period and high training costs.



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Solution: Natural Language Processing Technology

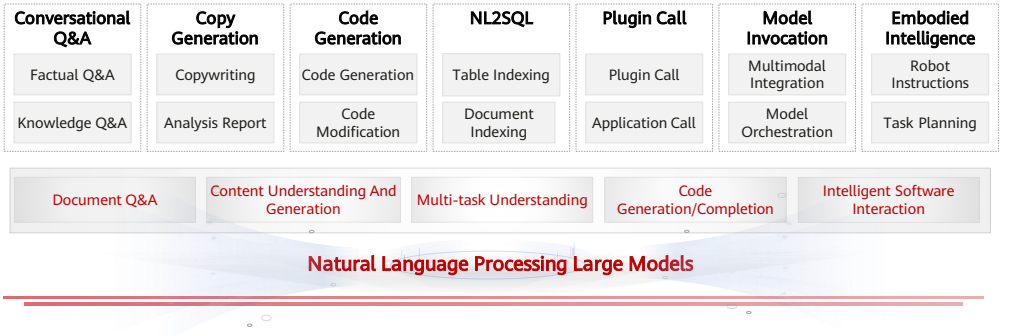
- Natural language processing technology is widely applied in scenarios such as search, dialogue, translation, and recommendation:
  - Intelligent Q&A
  - Conversational Assistant
  - Semantic Search
  - Information Extraction
  - Recommendation Systems
  - Machine Translation
  - Text Creation
  - ...



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Foundation Model: Natural Language Processing Large Models

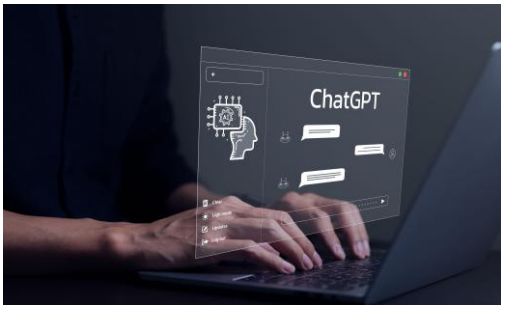
- Natural Language Processing Large Models have demonstrated strong performance across a wide range of NLP tasks. As foundational models, their excellent generalization capabilities significantly reduce the cost of fine-tuning for specific application scenarios.



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Industry Model: Dialogue-based Large Models

- The company's new employee training needs are well matched with the question-answering capabilities of foundational Natural Language Processing (NLP) large models. Therefore, an open-source or proprietary dialogue-based large model (fine-tuned based on foundational NLP large models and general dialogue Q&A datasets) can be selected as the model type to meet this business requirement.



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Scenario Model: New Employee Assistant

- Large dialogue models are primarily fine-tuned based on general dialogue corpora, which gives them strong conversational capabilities but limited understanding of domain-specific professional knowledge related to specific job roles. Therefore, it is possible to collect technical product documents related to the job role, build a dataset, and fine-tune the selected large dialogue model to create a new employee assistant closely associated with the job role. This assistant can answer new employees' questions through dialogue, help them learn the documents, and quickly get up to speed with their work.



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Business and Application Scenarios of Large Models

- Applying the dialogue and Q&A capabilities of large models to the business scenario of training new employees in enterprises, and building a "new employee assistant" scenario model with expanded uses, mainly involves the following aspects:
  - Dialogue and Q&A Model Selection
  - Fine-tuning Dataset Creation
  - Model Usage Expansion





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Dialogue and Q&A Model Selection (1)

- The selection of large models can be divided into two approaches: Closed-source Large Models and Open-source Large Models.
  - **Closed-source Large Models** have the main advantage of being deeply integrated with business needs, offering superior model performance metrics, and having lower costs for fine-tuning, deployment, and post-deployment maintenance. However, they require purchasing services from leading technology companies, which generally come with higher costs.
  - **Open-source Large Models** offer the main advantage of free access to model scripts and weights, but they require users to handle fine-tuning, deployment, and related tasks themselves, which involves a higher technical threshold.
- The primary goal of the new employee assistant is to reduce the cost of training new employees. Therefore, open-source large models should be considered as a priority. One can choose industry-specific models fine-tuned for dialogue and Q&A scenarios, or pay attention to the performance metrics of open-source models related to dialogue and Q&A, or their test results on relevant datasets, to make an informed selection.

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Dialogue and Q&A Model Selection (2)

- Below are open-source models related to dialogue Q&A and commonly used datasets.

Model Name	Description	Parameter Count
DeepSeek-R1	Open-source inference model with MoE architecture, designed to solve complex problems	671B
DeepSeek-V3	Open-source dialogue model, MoE architecture	671B
Baichuan-2	Both 7B and 13B offer chat versions	7B, 13B
ChatGLM3	Open-source bilingual (Chinese-English) dialogue model	6B
Bloom	Language model, open-source with multiple parameter specifications	560M to 176B

Dataset Name	Description	Number of Samples
KdConv	Multi-domain Chinese dialogue dataset	86K Conversations
Zhddline	Chinese dialogue dataset	97k Conversations
NaturalConv	Topic-driven Chinese dialogue dataset	20k Conversations
DuRecDial	Multi-domain and multi-type dialogues	10.2k Dialogues
LCCC	Open domain dialogue dataset	272k Dialogues

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Fine-tuning Dataset Construction

- After selecting a specific dialogue model, it is necessary to construct a fine-tuning dataset based on job-related professional data, including technical product documents, historical equipment parameter tables, programming code, and other textual data. Using large model development tools such as LangChain and Dify, along with libraries for document Q&A and information extraction, operations like text formatting conversion and information extraction are performed to build the fine-tuning dataset.



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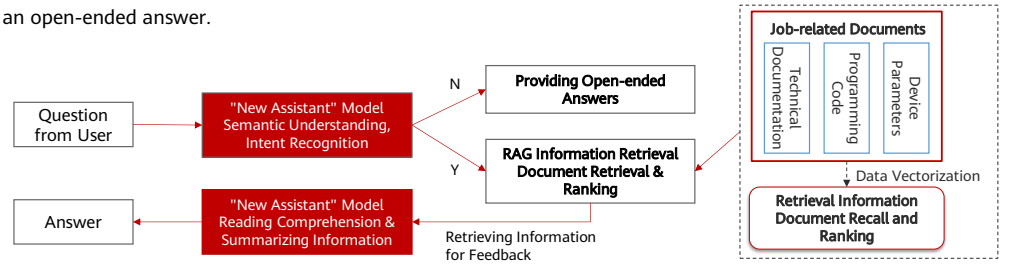
Model Usage

- After selecting an appropriate model and fine-tuning it with a dataset based on job-related technical documents, the "New Employee Assistant" application begins to take shape. It can then provide technical Q&A support for new employees, helping them quickly learn technical documents and get started with their work.
- On the other hand, to further enhance the accuracy and professionalism of the "New Employee Assistant" application's responses, a knowledge base can be integrated. By using Retrieval-Augmented Generation (RAG) technology, the system can first search for relevant information from a large number of documents, and then generate answers or text based on this retrieved information, thereby improving the quality of the responses.

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Model Enhancement: RAG

- To improve the accuracy of the "New Assistant" model in answering professional questions, job-related documents can be vectorized and used as a knowledge base attached to the model. The model primarily identifies the intent behind user queries to determine whether the user is asking a professional question. If so, it extracts semantic information from the question for retrieval, searches for relevant documents in the knowledge base, and uses the model's reading comprehension and summarization capabilities to provide a summary of the main content of the retrieved documents. If the user's question does not involve professional knowledge, the model will directly provide an open-ended answer.



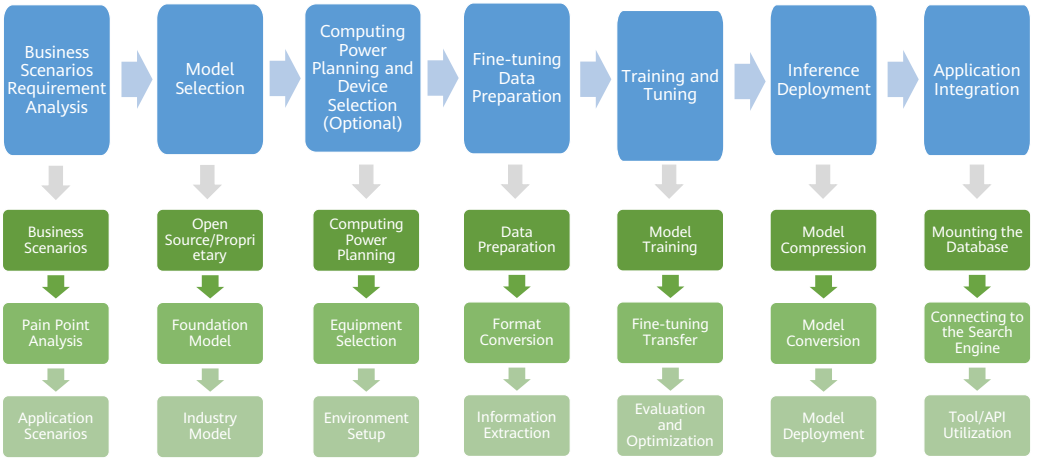
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Overall Process of Large Model Business



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Model Selection (1)

- Model selection primarily considers the overall performance of the model and its performance in specific tasks it is intended for. The performance of a model can be referenced from the papers published by the model developers or from model performance rankings on open-source websites.
- The specific task for the "New Assistant" mentioned above is intelligent dialogue. For this purpose, you can refer to the Chatbot Arena, a large model competition platform released by LMSYS Org. This ranking has become widely recognized as an industry benchmark due to its innovative "Arena" format and the rigor of its testing team. After major companies release their flagship models, they immediately submit them for testing by LMSYS, which speaks to the authority of this ranking.

Ranking	Model	Arena Score	Publishing Company	Agreement
1	Gemini-2.5-Pro-Exp-03-25	1440	Google	Proprietary
2	ChatGPT-4o-latest (2025-03-26)	1406	OpenAI	Proprietary
2	Grok-3-Preview-02-24	1404	xAI	Proprietary
2	GPT-4.5-Preview	1398	OpenAI	Proprietary
5	Gemini-2.0-Flash-Thinking-Exp-01-21	1380	Google	Proprietary
5	Gemini-2.0-Pro-Exp-02-05	1380	Google	Proprietary
5	DeepSeek-V3-0324	1370	DeepSeek	MIT
7	DeepSeek-R1	1359	DeepSeek	MIT

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Model Selection (2)

- In addition, the Big Benchmarks Collection on HuggingFace provides more test benchmarks for selecting models based on different scenario-specific tasks.
- For example, if the specific task is a single task such as code writing and error correction, you can refer to the Big Code Models Leaderboard on HuggingFace, which comprehensively evaluates models' performance in mainstream programming languages like Python, Java, JavaScript, and C++. Models can be selected based on the programming language required for the task.

Ranking	Model	Score	Humaneval-Python	Java	Javascript	Cpp
1	Qwen2.5-Coder-32B-Instruct	59.17	83.2	73.69	76.05	81.95
2	Qwen2.5-Coder-32B	56.67	57.1	65.49	65.07	64.35
3	OpenCodeInterpreter-DS-33B	56.25	75.23	54.8	69.06	64.47
4	Nxcode-CQ-7B-orpo	55.92	87.23	60.91	71.69	68.04
5	CodeQwen1.5-7B-Chat	55.67	87.2	61.04	70.31	67.85

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Model Selection (3)

- In addition to the model's own capabilities in text understanding and generation, if the task also requires integration with search engines, RAG, or other APIs for application purposes, it is necessary to further consider the model's overall performance. You can refer to the Open LLM Leaderboard on HuggingFace and pay special attention to common-sense reasoning metrics such as HellaSwag and Winogrande, as well as logical reasoning metrics like MMLU and TruthfulQA.

Ranking	Model	Average Score	ARC	HellaSwag	MMLU	TruthfulQA	GSM8K
1	Rhea-72b-v0.5	81.22	79.78	91.15	77.95	74.5	76.12
2	MultiVerse_70B	81	78.67	89.77	78.22	75.18	76.65
3	Smaug-72B	80.48	76.02	89.27	77.15	76.67	78.7
4	Alpaca-dragon-72b	79.3	73.89	88.16	77.4	72.69	77.63
5	Mixtral-8x22B-Instruct	79.15	72.7	89.08	77.77	68.14	82.03

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Model Acquisition

- Open-source models are primarily obtained from model maintenance websites or repositories such as GitHub, HuggingFace, etc. The model files obtained mainly include model scripts, weights, and related configuration files, along with instructions for downloading and using the models.
- Closed-source models are generally maintained on the large model service platforms of the respective vendors. These platforms not only maintain different versions of models released by the vendors themselves but also host open-source models trained and published by B2B or B2C users based on the vendor's large model service platform.

Open-source Maintenance Websites
GitHub
Gitee
HuggingFace
ModelScope

Company Name	Enterprise Large Model Platform
Baidu	Qianfan Large Model Platform
Alibaba	Bailian Large Model Platform
Tencent	Industry Cloud Large Model Platform

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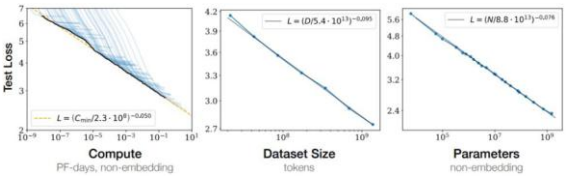
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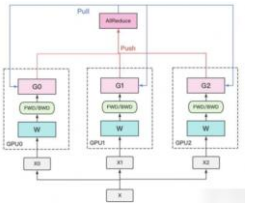
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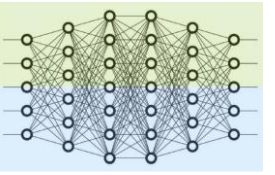
Large Model Training Requirements

- The three key elements of scaling laws: As the model's parameter scale and training data increase, the training performance improves.
- The computing power and storage of a single computational card are insufficient to support the training of large models, thus requiring support for various distributed parallel training strategies:

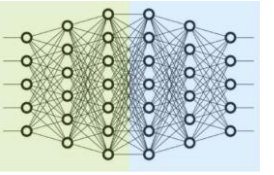




Data parallelism: Splitting the batches of training data.



Tensor parallelism: Splitting the tensors within each layer.



Pipeline Parallelism: Splitting Model Layers

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Computational Resources in Large Model Training

- The coefficient for Forward Pass + Backward Pass + Activation Recomputation = 1 + 2 + 1 = 4. In one iteration using activation recomputation, for each token and each model parameter, 2 * 4 = 8 floating-point operations are required.

$$T_{Training} \approx \frac{8 \times N_{Token} \times N_{Model-Parameter}}{N_{Computing\ Card} \times Computing\ Card\ Max\ FLOPS \times Computing\ Card\ Utilization}$$

- Taking the GPT-3 175B model with 300 billion tokens as an example, the time required for one training run on 1024 AI training cards with 312 TFlops of computing power is:

$$\frac{8 \times (300 \times 10^9) \times (175 \times 10^9)}{1024 \times (312 \times 10^{12}) \times 0.45} \approx 2{,}921{,}340\ Seconds \approx 34\ \text{Days}$$

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Network Bandwidth Requirements of Large Models

- Taking GPT-3 175B as an example, the parallel segmentation methods for large models have the following requirements on communication and latency.

Partitioning Method	Communication Operation	Communication Scale (per card)	Computational Latency Requirements	Cluster Deployment Requirements
Tensor Parallelism	AllReduce	Hundred GB	Second-level	Intra-node
Pipeline parallelism	Send/Recv	100 MB ~ GB	10 milliseconds	Inter-node
Data parallelism	AllReduce	GB	Hundred milliseconds	Intra/Inter-node
Expert parallelism	All2All	Hundred GB	Second-level	High network throughput

- As can be seen from the table above, data parallelism and pipeline parallelism require relatively low bandwidth between nodes. However, Expert Parallelism (MoE) requires higher bandwidth between nodes to be supported.

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Storage Requirements of Large Models

Scenario	NLP (GPT)	CV (ViT, Stable Diffusion)	Autonomous Driving
Model Parameter Count	6B ~ 671B	1B ~ 22B	
CKPT Size	Approximately 16 times the parameter size		
CKPT Saving Frequency	Saved approximately every 500 steps (up to N steps depending on actual conditions)		
CKPT Saving Latency	~Minute-level		
Raw Dataset	10 ~ 100TB	100TB ~ 1PB	1 ~ 10PB
Training Dataset Read Latency	~Minutes level		
Storage IOPS	For 8KB small files, IOPS performance is critical.	For large files, IOPS is not a bottleneck.	

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- Data Processing Requirements
  - Raw Dataset Storage
  - Training Dataset Storage: The raw dataset is processed through data cleaning, filtering, formatting, etc., to obtain the training dataset, which typically results in a compression ratio of more than 10 times.
- Storage Requirements
  - Training dataset fetching: Pull data from storage to memory in one go
  - Checkpoint fetching
  - Periodic checkpoint saving

DeepSeek Series Models Typical Scenario Requirements

Main Scenarios	Scenario Description	Configuration Logic
Deepseek V3 Incremental Pre-training + R1 Reinforcement Learning Training	Based on the training processes of DeepSeek V3 Base and R1, industry-specific capabilities of DeepSeek are enhanced using over TB-level industry data to generate industry-specific R1 models.	DeepSeek V3 itself is trained with PreTraining + PostTraining xx T tokens + xx k CoT data.
R1 Reinforcement Learning Training	Domain-specific models are released based on DeepSeek V3 Base, following the DeepSeek R1 training process, and domain knowledge and capabilities are further enhanced through reinforcement learning to generate domain-specific models.	PP = N EP = X The minimum cluster configuration is EP * PP = NX cards.
R1 Distilled Small Parameter Models	By using DeepSeek R1 distilled small parameter models such as Qwen and LLaMA series, we enhance the chain-of-thought capabilities and performance of small parameter models, and release a series of R1-Distill domain-specific models.	The calculation formula for actual model parameters and expected training time is based on 8 TP / NX. T is the number of tokens, and P is the number of parameters (for example, the number of activated parameters in DeepSeek V3 is 37B)

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Overview of Data Preparation

- Data preparation mainly refers to the dataset used for fine-tuning large models. For a specific task, constructing a high-quality fine-tuning dataset can significantly improve the performance of the fine-tuned model on that task. The main considerations are as follows:
  - **Data Source:** Choosing the right data source is crucial. Data can be collected or scraped from relevant fields. The dataset must include samples related to the specific task. For example, if the task is sentiment analysis, the fine-tuning dataset must contain texts with positive and negative reviews.
  - **Data Quality:** The quality of the dataset determines the upper limit of the model's capabilities. When preparing data, it is important to carefully check and clean the data. Ensure that the dataset does not contain useless samples or other noise, and that each sample in the dataset has clear practical significance.
  - **Data Scale:** A sufficient number of data samples should be ensured to allow the model to accurately learn the format and patterns of the specific task. The number of samples should match the parameter size of the base or industry-specific model.
  - **Similarity:** It should be ensured that the data in the fine-tuning dataset are correlated and easily generalizable, meaning they are representative of the specific task but not repetitive.

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Instruction Tuning Dataset

- The format of the dataset depends on the method used for model fine-tuning. Taking large language models as an example, the most commonly used fine-tuning method is Instruction Tuning, which essentially involves fine-tuning the model on a collection of instances in natural language format. To perform instruction tuning, we first need to collect or create instances in the form of instructions.
- A typical instruction tuning dataset format is: {"instruction": "", "input": "", "output": ""}. If you want to fine-tune a model that has already been instruction-tuned, the format of the fine-tuning dataset should remain consistent. Below is an example of fine-tuning:

Prompt: "Edit the article to make it more engaging for readers. Autonomous robots are computer-controlled machines programmed to perform specific tasks without any human input. Autonomous robots are increasingly being used across various industries, from manufacturing to healthcare to security."

Complete: "Autonomous robots are computer-controlled machines programmed to perform specific tasks without any human input, achieving new levels of efficiency, precision, and reliability. Autonomous robots are increasingly being used across various industries, from manufacturing, where they can assemble complex components with high precision and consistent quality, to healthcare, where they assist in medical testing and procedures, and to security, where they monitor large areas to ensure the safety of people and property. Autonomous robots can also reduce errors and increase safety in hazardous or harmful environments, such as during inspections or maintenance of industrial processes. Due to their versatility, autonomous robots will revolutionize the way we work, making tasks simpler, faster, and ultimately more enjoyable."

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Dialogue Fine-Tuning Dataset

- Dialogue Fine-Tuning is a special type of instruction fine-tuning. Its purpose is to enable large language models to unlock "dialogue" capabilities based on their existing "text completion" abilities.
- The typical data format is {"instruction": "", "input": "", "output": "", "history": [{"", ""}, {"", ""}]}. The specific dialogue format should be consistent with that used during the model's pre-training and fine-tuning. Below is an example of the fine-tuning data format:

Prompt: """[Round 0]
Q: Hello, can you help me answer a question?
A: Of course, what's the question?
[Round 1]
Q: I would like to know about the future development direction of artificial intelligence. Do you have any ideas?
A: The future development direction of artificial intelligence may include more powerful machine learning algorithms, more advanced natural language processing technologies, and even smarter robots.
[Round 2]
Q: That sounds great. What challenges might artificial intelligence face?
A: """

Complete: "The challenges AI faces include issues related to data privacy, security, and ethics, as well as automation that impacts job opportunities."

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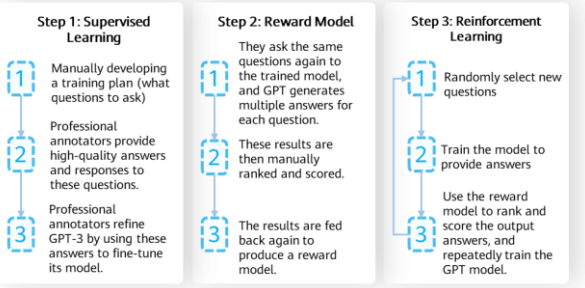
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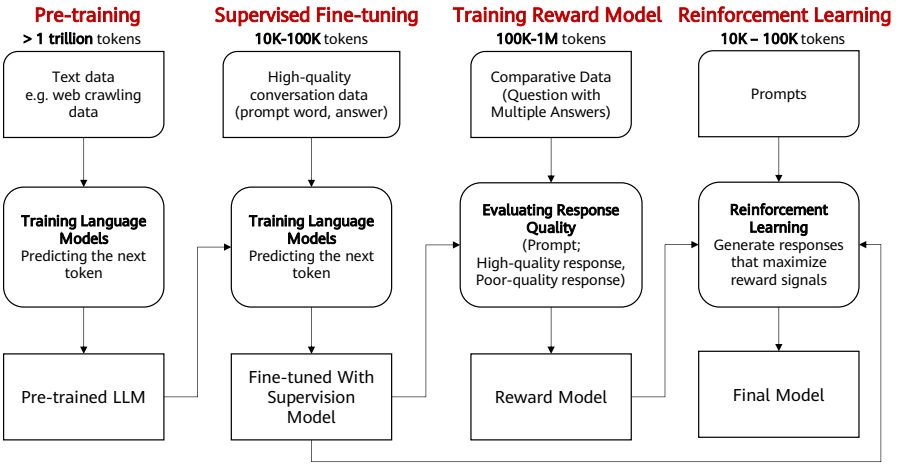
Training and Fine-Tuning of Large Models

- After completing data preparation and model selection, you can proceed with training and fine-tuning the large model.
- Taking large language models as an example, the current mainstream training process mainly follows the approach used by OpenAI based on the GPT-3 foundational model. This involves Supervised Fine-tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) to obtain industry-specific models such as InstructGPT for instruction generation and ChatGPT for dialogue.



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Training and Tuning Process of Large Language Models



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Fine-tuning of Large Models

- General-purpose large models such as ChatGPT, although possessing good general knowledge and capabilities, often fail to achieve top-level performance on specific tasks. They still have a certain gap compared to the actual business requirements (common basic accuracy requirements ≥90%). Open-source LLMs like ChatGLM2 and LLaMA have even larger gaps and cannot be used out-of-the-box. Therefore, it is necessary to fine-tune these large models based on datasets constructed for specific business needs.
- Objectives of fine-tuning large models:
  - **Model Level:** To improve the underfitting issue of large models in specific tasks.
  - **Data Level:** To enable large models to learn information from the corpus of specific scenarios or tasks, while establishing a connection with the pre-trained general corpus, thereby narrowing the gap between pre-training data and business data, and enhancing the model's understanding of specific business domains.
  - **Observable Outcomes:** Improved accuracy in specific scenarios/tasks, reduced occurrence of hallucinations.

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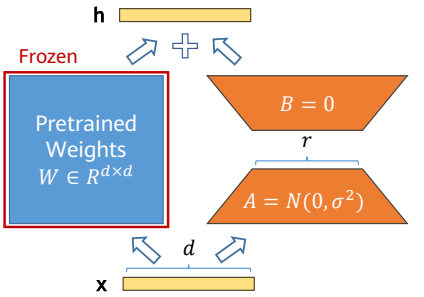
Parameter-Efficient Fine-Tuning Methods

- When fine-tuning a model, one can choose between Full Fine-tuning or Parameter-efficient Fine-tuning (PEFT).
- As language models grow larger, the resource and cost required for Full Fine-tuning become increasingly high. On the other hand, a single consumer-grade computational card often struggles to handle the Full Fine-tuning of large language models.
- Therefore, Parameter-efficient Fine-tuning (PEFT) is generally adopted, which involves fine-tuning only a small number of additional model parameters while keeping most pre-trained parameters fixed. This approach significantly reduces computational and storage costs, and the most advanced PEFT techniques can achieve performance comparable to that of Full Fine-tuning.

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Parameter-Efficient Fine-Tuning Method: LoRA

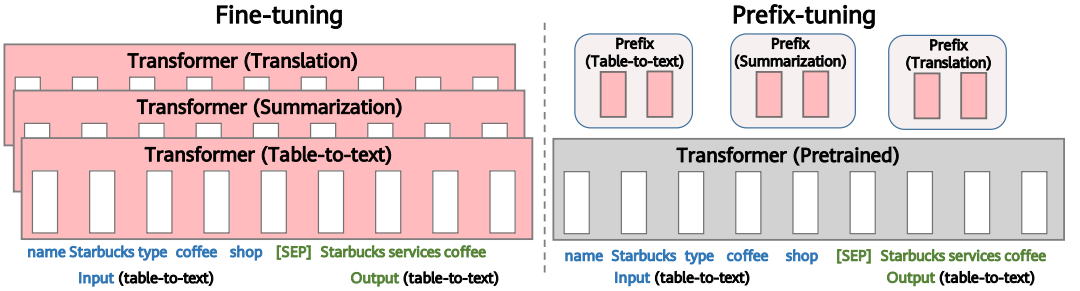
- LoRA was first published by Microsoft on arXiv in 2021, with the related technology being open-sourced simultaneously. It has since been widely applied in fine-tuning large language models.
- Basic Idea of LoRA: LoRA does not directly train the original model. Instead, it fixes the parameters of the original model and attempts to optimize the rank decomposition matrix that changes in the original model.
- Advantages of LoRA:
  - **Generalizable to full fine-tuning, offering high flexibility.**
  - **Pluggable, with no additional inference latency.**
  - In fine-tuning GPT-3 (175B), the number of training parameters is reduced by four orders of magnitude while maintaining comparable accuracy.



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Parameter-Efficient Fine-Tuning Method: Prefix-Tuning

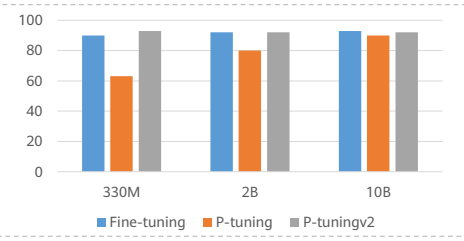
- Prefix-tuning is a lightweight fine-tuning method for generative tasks. It adds a trainable embedding layer to the original model, which is task-specific and referred to as the "prefix." This embedding layer is optimized during training to better understand the intent of the prompt words.
- Compared to traditional fine-tuning, prefix-tuning only optimizes the prefix. Therefore, it requires storing only one large Transformer model and an embedding layer for the known task-specific prefix, resulting in very low overhead for each additional task. This provides an automatic mechanism to improve model performance through prompts.



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Parameter-Efficient Fine-Tuning Method: P-tuning

- Prompt tuning (P-tuning) is a simple fine-tuning method, which is a simplified version of Prefix Tuning. It does not require adjusting model parameters but instead selects a subset of existing parameters as learnable parameters to create each prompt prefix.
- P-tuning V2 is an optimized and adapted implementation of P-tuning, with one of the most significant improvements being the application of continuous prompts to every layer of the pre-trained model, not just the input layer.



Experiments show that by fine-tuning only 0.1% of the parameters, P-tuning achieves performance comparable to fine-tuning on language models ranging from 330M to 10B parameters.

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Comparison of Mainstream Large Model Fine-Tuning Techniques

Fine-tuning on Original Model Parameters

- Full fine-tuning has the highest theoretical upper limit, but it requires the most data, hardware resources, and training time. It is prone to overfitting when the dataset is not large enough.
- Partial fine-tuning is less efficient, updating only some parameters.
- Layer-wise fine-tuning performs poorly.

Parameter Efficient Fine Tuning (PEFT)

- LoRA does not introduce additional inference latency, reduces memory and storage resource consumption, and achieves convergence similar to training the original model, making it the most widely used method.
- Adapter Tuning adds adapter layers, introducing extra computation and resulting in inference delay.
- Prefix Tuning is difficult to optimize, and its performance varies non-monotonically with the scale of trainable parameters.
- P-Tuning retains a portion of the sequence length in the prefix, which can reduce the input sequence length for downstream tasks.

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Huggingface PEFT (1)

- Huggingface PEFT is an open-source model fine-tuning toolkit that includes the parameter-efficient fine-tuning methods mentioned in the previous section. It is designed to effectively adapt pre-trained language models to various downstream applications without needing to fine-tune all parameters of the model, significantly reducing computational and storage costs.
- Based on HuggingFace's Transformers library, PEFT provides a rich set of plugins and extensions, making model fine-tuning simpler and more efficient. PEFT supports multiple mainstream model architectures such as BERT, RoBERTa, and GPT, and can seamlessly integrate with HuggingFace's Model Hub, allowing users to quickly access and use pre-trained models.

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Huggingface PEFT (2)

- Huggingface PEFT supports the following four types of model fine-tuning methods:
  - LoRA
  - Prefix Tuning
  - P-Tuning / P-Tuningv2
  - Prompt Tuning
- In addition to basic model tuning capabilities, PEFT also offers a rich set of plugins and extensions to help users better meet their needs. For example, PEFT supports custom loss functions, custom evaluation metrics, custom data loaders, and more, making model tuning more flexible and efficient.

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Huggingface PEFT (3)

- PEFT Installation and Simple Usage Example:

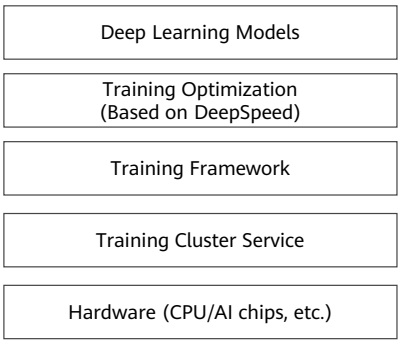
```
# To fine-tune the model using PEFT, you first need to install the PEFT and Transformers libraries. You can install them using the following commands:
pip install transformers peft


from transformers import AutoTokenizer, AutoModelForSequenceClassification
from peft import PEFT
# Load the pre-trained model and tokenizer
tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')
model = AutoModelForSequenceClassification.from_pretrained('bert-base-uncased', num_labels=2)
# Create a PEFT instance
peft = PEFT(model, tokenizer)
# Load training and validation data
train_data = ...
val_data = ...
# Configure fine-tuning parameters
config = {
    'learning_rate': 3e-5,
    'epochs': 3,
    'batch_size': 16,
    'metric_for_best_model': 'accuracy'
}
# Perform model fine-tuning
peft.fit(train_data, validation_data=val_data, **config)
# Save the best model
peft.save_pretrained('best_model')
```


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DeepSpeed (1)

- On the other hand, when fine-tuning large models on multi-computing-card and multi-node clusters, the training efficiency of the cluster system is often low. The efficiency of a single node in the cluster typically does not reach 50% of the maximum efficiency that the node machine can achieve.
- DeepSpeed is an open-source deep learning optimization library developed by Microsoft, designed to improve the efficiency and scalability of training large-scale models. It accelerates training through various techniques, including model parallelism, gradient accumulation, dynamic precision scaling, and mixed precision training. DeepSpeed also provides auxiliary tools such as distributed training management, memory optimization, and model compression to help developers better manage and optimize large-scale deep learning training tasks.



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DeepSpeed (2)

- DeepSpeed mainly consists of three parts:
  - **APIs:** Provides user-friendly API interfaces, allowing model training and inference to be performed by simply calling a few functions. The most important one is Initialize, which is used to initialize the engine and configure training parameters and optimization techniques through its arguments. Configuration parameters are usually stored in a config.json file.
  - **Runtime:** The runtime component is the core component for managing, executing, and optimizing performance in DeepSpeed. It handles tasks such as deploying training jobs to distributed devices, data partitioning, model parallelism, system optimization, fine-tuning, fault detection, checkpoint saving and loading, etc. This component is implemented in Python.
  - **Ops:** The underlying kernels are implemented in C++ to optimize computation and communication, including ultrafast transformer kernels, fused LAN kernels, and custom deals.

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Large Model Compression

- After a model is trained, before deploying it as a service for inference, it is generally necessary to compress and quantize the model to reduce the number of parameters while ensuring that the model's accuracy meets the required standards. This process helps to decrease the consumption of computing and storage resources while improving the model's inference speed.
- Model quantization can be categorized based on different stages: post-training quantization, quantization-aware fine-tuning, and quantization during training. It can also be classified based on different formulas: linear quantization and nonlinear quantization.
- For large model compression and quantization, you can utilize compression tools or prioritize selecting quantized versions of open-source large models during model selection.

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Summary of LLM Quantized Versions

- When major companies launched their own large model series, they anticipated that larger models would require more computing power for deployment. Therefore, they provided quantized versions of the models, which are more suitable for deployment on device-side and edge-side environments.

Company	Model	Release Date	Parameter Count
Google	PaLM2-Gecko	May 2023	-
	Gemini Nano	December 2023	1.8B/3.25B
	Gemma	February 2024	2B/7B
Microsoft	Phi-2	December 2023	2.7B
Modelbest & Tsinghua	MiniCPM	February 2024	2.4B
Alibaba	Qwen1.5	February 2024	0.5B/1.8B/4B/7B/14B/72B
Meta	Llama 2	July 2023	7B/13B/70B
	MobileLLM	February 2024	<1B

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Comparison of Quantized Model Performance

Task	Dataset	LLaMA-7B	TinyLLaMA-1.1B	Phi2-2.7B	Qwen-1.8B
Exam	C-Eval	32.50	27.85	31.86	**53.60**
	CMMLU	31.80	24.64	32.18	**52.12**
	MMLU	46.80	25.75	**58.49**	46.43
	AGI-Eval	21.80	18.54	28.51	**35.83**
Knowledge	BoolQ	74.90	56.06	**77.40**	64.31
	AX-b	53.50	45.47	43.57	**57.79**
Reasoning	PIQA	78.30	70.62	**78.89**	73.83
	CSL	55.60	53.12	50	**55.62**
Understanding	EPRSTMT	46.20	46.25	46.25	**88.12**
	XSum	19.70	**20.15**	13.66	20.03
	LCSTS	9.10	13.97	12.02	**14.87**
Overall	Average	42.75	36.58	42.98	**51.14**

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Large Model Inference Deployment

- After quantization and compression, large models can be deployed for inference. Depending on the business inference scenarios and the conditions of software and hardware equipment, there are mainly three deployment methods: Cloud Deployment, Edge/Device-side Deployment, and Device-cloud Collaborative Deployment.



Cloud Deployment



Edge/Device-Side Deployment



Device-Cloud Collaborative Deployment

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Introduction to Inference Deployment Methods

- Cloud Deployment:
  - Cloud computing platforms offer a large amount of computing resources, meeting high-performance computing needs and are suitable for deploying models with a large number of parameters. However, due to bandwidth limitations of web pages or APIs, they cannot meet the requirements for real-time inference of large volumes of data such as video streams. It is more suitable for scenarios involving small batches of data like text and images.
- Edge/Device-Side Deployment:
  - This type of deployment is mainly suitable for scenarios that require high mobility and real-time performance. However, due to the limited computing power and power consumption of the devices, models generally need to be significantly quantized and compressed, or lightweight versions of large models are directly chosen. As a result, the performance of edge/device-side deployments is usually weaker than that of cloud deployments.
- Device-Cloud Collaborative Deployment:
  - Device-cloud collaboration offloads AI computing workloads by enabling devices and the cloud to work together. Data can be collected and preprocessed on the device side before being sent to the cloud for inference, or real-time inference can be performed on the device side while the cloud is used for model iteration. A typical scenario is the voice assistant on a smartphone.

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Comparison of Inference Deployment Methods

- The following provides a comparison of the performance, cost, and applicable scenarios of three mainstream inference deployment methods.

Category	Cloud-side Deployment	Device-side Deployment	Device-Cloud Collaborative Deployment
Model Size	Large	Small	Smaller
Computing Resources	High computing power	Limited computing power	High computing power
Data Transmission	Small data such as images, text, etc.	Unrestricted	Small data such as images and text
Real-time Inference	Limited by bandwidth and data transmission volume	Good	Limited by bandwidth and data transmission volume
Deployment Cost	High, requiring a commercial closed-loop for cloud-side computing power	Low, users purchase device-side computing power once and for all	Relatively high, balancing both device-side and cloud-side considerations
Application Scenarios	Online translation, text-to-image generation, dialogue systems	Facial recognition at turnstiles, intelligent sorting by robotic arms	Mobile voice assistants

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Quiz

1. (Multiple-Choice Question) Which of the following toolkits can be used for developing large models?
   A. Huggingface PEFT
   B. DeepSpeed
   C. OpenCV
   D. OpenGL

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- Answer: AB

Summary

- This chapter first introduces the overall process of using large models in enterprise scenarios, including scenario analysis, data preparation, model selection, training and optimization, inference, and deployment. It then provides detailed descriptions of the main industry methods and related development tools for each step in the process.

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Thank you.

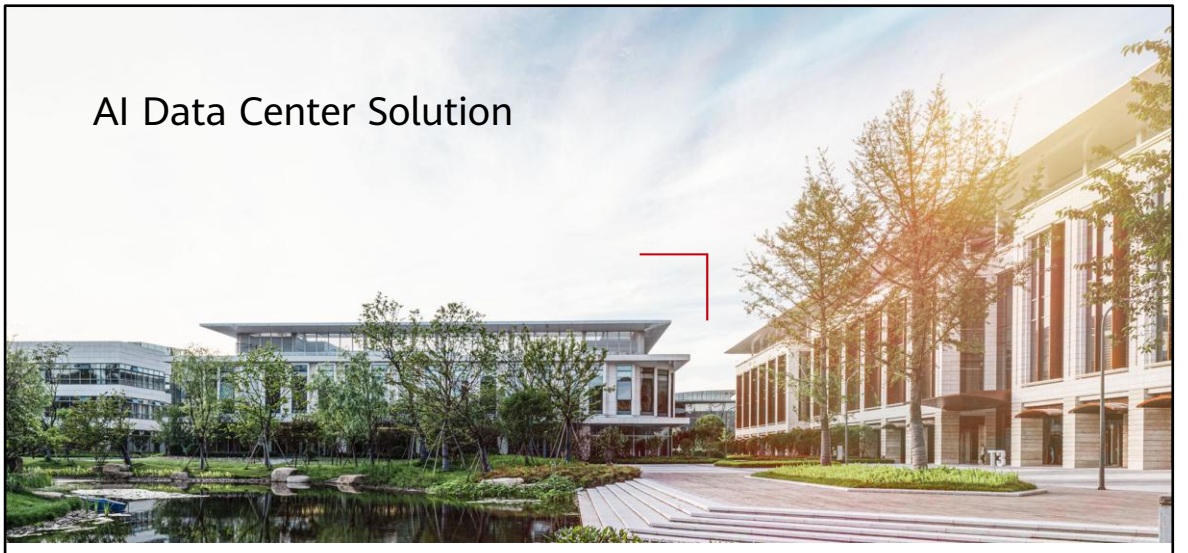
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每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

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AI Data Center Solution



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Foreword

- The 21st century is the era of artificial intelligence. All walks of life around the world are vigorously moving towards digital intelligence transformation. Learning, mastering AI, and actively exploring AI technology will become the trend of the times. The use of AI technologies is inseparable from the "AI Data Center" as a physical foundation. This chapter focuses on the physical foundation of the AI era.
- AI data center construction needs to be planned from the perspectives of Computing, Storage, Network, and Facility. This chapter describes the key technologies of AI data centers, such as computing, storage, network, and facility. Let's start with facility and enter the AI data center world.

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Objectives

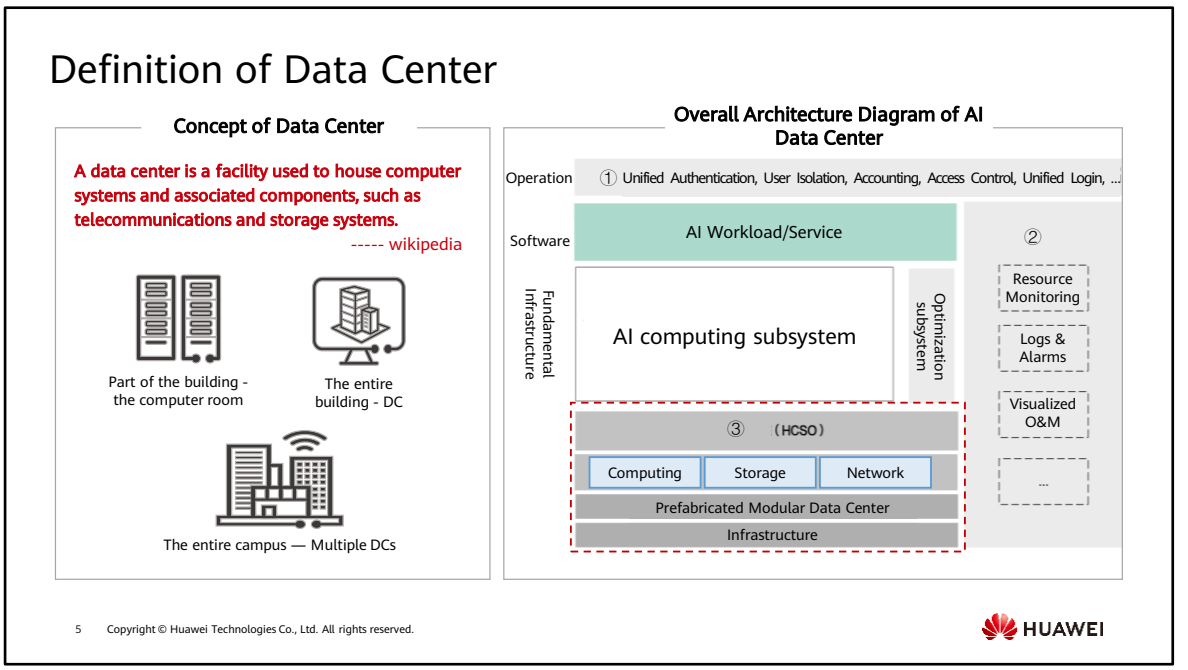
- On completion of this course, you will be able to:
  - Understand the concept of AI data center layering, the characteristics of facilities, and related requirements.
  - Describe AI data center computational capability solutions.
  - Understand the network requirements of large models and AI data center network solutions.
  - Understand key technologies for AI data center networks.
  - Understand the storage requirements of large models and AI data center storage solutions.
  - Describe key storage features and requirements.

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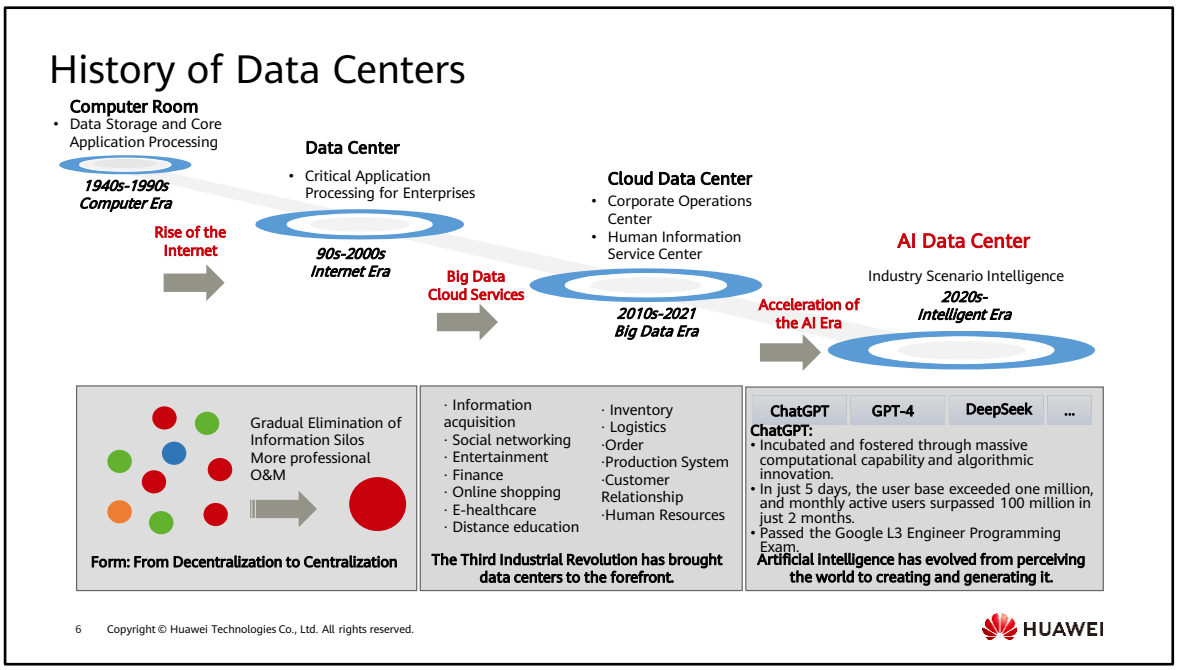
1. AI Data Center Overview
   - Definition and Common Metrics of Data Center Facilities
   - Overall AI Data Center Solution
2. AI Data Center Computational Capability Solutions
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4. AI Data Center Storage Solutions and Key Features
5. DeepSeek All-in-One Solution

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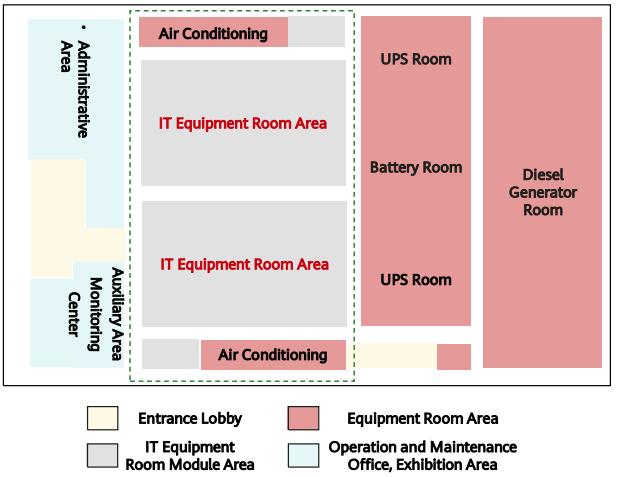
- ① **Operation platform:** A customized secondary operation platform that primarily serves operation administrators and tenants (users). Users can complete functions such as registration, login, AI service activation, ticket submission, and billing through this platform. Operation administrators can use this platform to perform product and pricing entry, order management, customer management, permission management, billing management, ticket management, announcement management, operation analysis, and operation dashboard management.
- ② **O&M platform:** The L2 and above cluster computing centers support continuous remote O&M. SRE O&M administrators perform remote O&M through the unified O&M platform CloudScope of Huawei Cloud. The energy infrastructure is maintained by local O&M administrators of the computing center through the local O&M platform.
- ③ **HCSO computational capability service foundation:** It centrally manages all physical hardware resources, implements resource pooling, on-demand allocation, and elastic scaling, and provides capabilities such as multi-tenancy and network isolation.
- **Computing subsystem:** A full-stack AI computing infrastructure built based on AI chips.
- **Storage Subsystem:** Storage devices used for storing training data and platform-preconfigured images.
- **Network Subsystem:** Devices used for network interconnection within the computing center.

- **Infrastructure:** Planning, design, and construction of data center facilities, including civil engineering, electrical engineering, and fire protection and power supply systems.



- Here, we highlight the characteristics of data centers across different eras: data rooms, data centers, cloud data centers, and AI data centers.

Data Center Physical Model



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- The physical model of the data center is derived from the relevant standard, GB50174-2017.
- **Computer Room**
  - A building space primarily used for the installation and operation of data processing equipment, including functional areas such as server rooms, network rooms, and storage rooms.
- **Support Area**
  - The area that provides power support and safety assurance for the computer room and auxiliary areas, including: power transformation and distribution room, diesel generator rooms, battery rooms, air conditioning rooms, power equipment rooms, uninterruptible power supply system rooms, fire protection facility rooms, etc.
- **Auxiliary Area**
  - Areas used for the installation, debugging, maintenance, operation monitoring, and management of electronic information equipment and software, including cable entry rooms, testing rooms, central control rooms, fire protection and security control rooms, unpacking areas, spare parts storage, printing rooms, and repair rooms.
- **Administrative Area**
  - Areas used for daily administrative management and for clients to manage their hosted equipment, including offices, lobbies, duty rooms, restrooms, changing rooms, and user workspaces.

Key Indicators for Data Centers (1)

Grid Usage Effectiveness

$$GUE = \frac{IT\ Power\ Distribution\ Capacity}{Total\ Utility\ Power\ Capacity}$$

$$= \frac{Total\ Utility\ Power\ Capacity - Cooling\ and\ Power\ Distribution\ PeakCapacity}{Total\ Utility\ Power\ Capacity}$$

Grid Usage Effectiveness (GUE): GUE refers to the ratio of the power capacity that can be used by planned IT equipment to the total utility power capacity of the data center. The amount of power available for IT systems is calculated by subtracting the peak power consumption of cooling and power supply systems from the total power.

12MW Data Center: for every 1% increase in GUE, an additional 1% of cabinets can be added, generating an extra revenue of 1 million CNY per year.

Space Usage Effectiveness

SUE (Space Usage Effectiveness) is the number of IT cabinets that can be deployed per unit of building area in a data center.

It is positively correlated with the proportion of white space area.

Taking the xxx project as an example: 1000 cabinets, 12kW per cabinet.

White space: 2457 m² 35%	Cooling: 2360 m² 3%	Power supply area: 1095 m² 16%	Others: 1120 m² 16%

Total building area: 7,032 square meters; white space accounts for 35%. Each cabinet occupies an average of **7 m²**.

For every 1% increase SUE, annual revenue can increase by 1 million CNY.

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- Model: Self-built *IDC*, *1500* cabinets *in* Beijing, *8 kW per* cabinet, power supply *2N/10 minutes*, cooling system *N+1*, average load rate of *50%* (including an average rack occupancy rate of *85%*), monthly rent per cabinet is *7,350* CNY/month (including electricity cost), electricity rate is *0.75* CNY/*kWh*, water rate is *3.5* CNY/ton, and the lifecycle is *10* years, excluding land acquisition costs.
- As an operation-focused data center, the benefits of improving these two metrics are obvious.

Key Indicators for Data Centers (2)

PUE (Power Usage Effectiveness)

$$PUE = \frac{Total\ Annual\ Electricity\ Consumption}{IT\ Electricity\ Consumption}$$

A smaller PUE value indicates a greener data center with higher energy efficiency.

- Annual PUE is used to calculate energy consumption.
- Peak PUE, denoted as PUE_{Max}, is used to calculate power intake.

WUE (Water Usage Effectiveness)

$$WUE = \frac{Annual\ Water\ Usage\ (L)}{IT\ Electricity\ Consumption\ (kWh)}$$

A smaller WUE value indicates that the data center consumes less water.

· Annual WUE is used to calculate the amount of water consumed.

· Peak WUE is used to calculate the amount of municipal water resources introduced.

· The WUE value generally ranges between 1.2 to 2 L/kWh. A data center with 1,000 cabinets consumes over 63,000 tons of water annually, which is enough for 300 residents to use for one year.

The total building area of a data center = Number of cabinets × 5~15 m², which is specifically related to factors such as system architecture, density, reliability, and operation and maintenance management requirements.

For example, a data center with 1,000 cabinets requires a building area of 5,000–10,000 m², with the IT cabinet area being 2,500~3,000 m² (number of cabinets × 2.5~3 m²).

$$Total\ Power\ Demand = \sum Number\ of\ cabinets \times Power\ per\ cabinet \times PUE_{Max}$$

For example, a data center with 1,000 cabinets, an average power of 6 kW per cabinet, and an estimated PUE_{Max} of 1.5 (for a Class A data center according to national standards), would require a total power demand of 9,000 kW, necessitating two 10 MVA capacity 10 kV dedicated power lines.

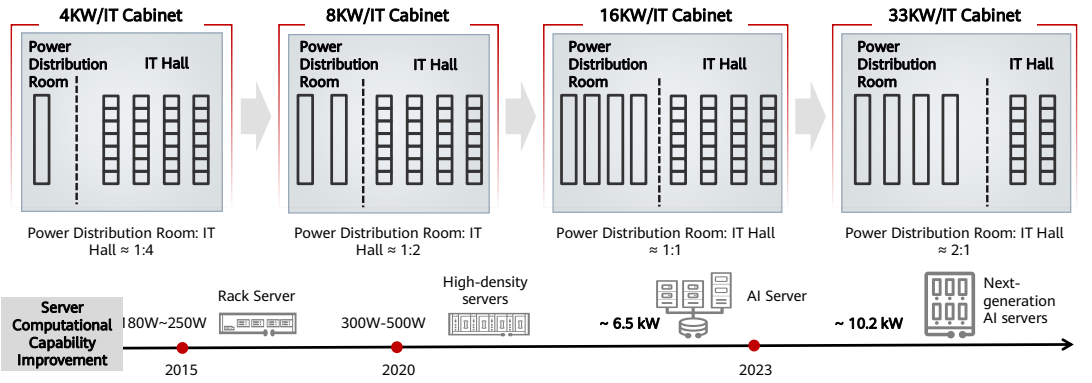
The phased construction of data centers is closely related to investment, as well as factors such as IT scale and development, local resource availability, and market demand.

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- As a new data center, it is not enough to focus solely on PUE; all indicators need to be considered comprehensively.

High-density Power Supply and Distribution Systems

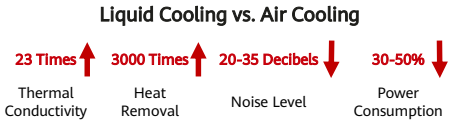


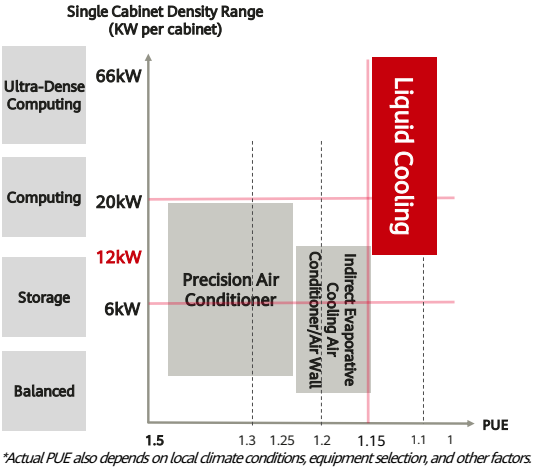
The rapid increase in IT power consumption leads to a growing proportion of space occupied by power supply and distribution systems, resulting in a decreasing "utilization rate" for IT equipment.
High density and space-saving considerations for the power supply and distribution system need to be taken into account.

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The Era of Liquid Cooling

- **Overall:** In the past 10 years, the TDP design has seen a 10% increase in CPU performance and a 12% increase in high-performance computing processors. However, the trend is clearly accelerating, mainly due to two reasons: 1) slowdown in process technology; 2) the process is approaching its limits, leading to a slower decline in power consumption per transistor.
- **AI Computing:** High-performance computing processors consume 700W (in 2023), and an 8-computational-card AI server consumes 10.2kW. The power consumption for 2024 and 2025 models is expected to be 700W and 1000W respectively, with the maximum total power consumption of the entire system estimated at approximately 13kW.
- **General Computing:** CPUs consume 350W (in 2023), and a typical dual-socket server has a total power consumption of 1.3kW. Emerald Rapids and Granite Rapids are expected to have a maximum total power consumption exceeding 1.9kW (in 2024).





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- The full English name of TDP is "Thermal Design Power," which translates directly into Chinese as "heat dissipation design power." It is primarily provided for use by computer system manufacturers, heat sink/fan manufacturers, and case manufacturers, among others, during system design.
- The term "thermal design power" refers to the maximum amount of heat (measured in watts, W) that a chip can dissipate when it is under maximum load. It represents the maximum heat that the computer's cooling system must be capable of dissipating, but it is not the power at which the chip actually dissipates heat.
- Generally, TDP is mainly applied to CPUs. The TDP value of a CPU corresponds to the maximum heat that the final version of the CPU in a series might generate under full load (theoretically, when CPU utilization is 100%). The heat sink must ensure that the processor's temperature remains within the design range even when the processor is operating at its maximum TDP.
- Note: Since the core voltage and current of a CPU are constantly changing, the actual power consumption of the CPU (calculated as power P = current I × voltage U) also varies continuously. Therefore, the TDP value is not equivalent to the actual power consumption of the CPU, nor is there any arithmetic relationship between them.
- For example, the Pentium E2160 has a TDP of 65W, but its average power consumption during actual operation is only 19W.

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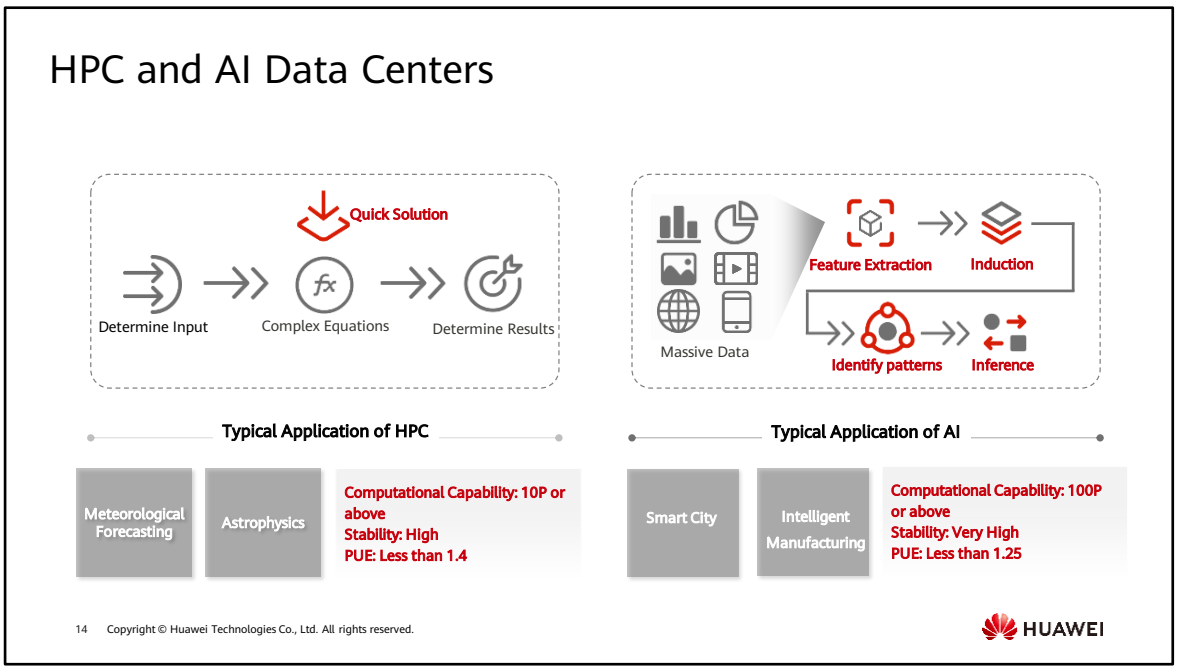
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Types of Data Center

Infrastructure Layer				Application Layer
Comparison	**Edge Data Center**	**AI Data Center**	**HPC**	**Cloud Computing**
	Basic Computing Power	**AI Computing Power**	**HPC Power**	
Definition	Edge data centers are a new type of infrastructure deployed at the network edge, building service platforms close to users to provide resources such as storage, computing, and networking. They move some critical business applications to the network edge to reduce bandwidth and latency losses caused by network transmission and multi-level forwarding.	**Based on a converged architecture computi system and using data as a resource, these centers can leverage powerful computing power to drive AI models for deep data processing, continuously generating various intelligent computing services. These servic are then supplied to organizations and individuals via the cloud.**	**The most complex scientific computing problems in academia an industry.**	Cloud computing is an internet-based computing method that provides shared software, hardware resources, and information to computers and other devices on demand. It is mainly divided into public cloud and private cloud.
Application Scenarios	Smart manufacturing, smart city, live streaming games	Autonomous driving, smart finance, smart healthcare	Astronomy, meteorology, biology, and other scientific research fields	Commercial Hardware/Software Leasing
Computational Precision	Half-precision and single-precision floating-point operations	Half-precision and single-precision floating-point operations	Single-precision and double-precision floating-point operations	/
Chip	Select the appropriate computing chip based on application scenarios and IT platforms	AI Chip	CPU, High-Performance Computing Processor	CPU, High-Performance Computing Processor
Features	Provides edge computing services based on edge-cloud collaboration.	Intelligent training and inference. Depends on computing power, algorithms, and data.	High-speed network MPI parallel computing Mathematical library	SaaS, PaaS, IaaS service models Virtualization, high availability, scalability, on-demand services, security
Calculation Method	Distributed Computing	Distributed Computing	Parallel Computing	Distributed Computing
Infrastructure	Distributed IDC, operator MEC, customer-built nodes, industrial gateways, home gateways, and other edge devices and corresponding infrastructure environments.	Computing, storage, networking	Computing clusters	Computing, storage, networking
Typical Projects	xxx Mine 5G Edge Data Center	xxx AI Computing Center	xx HPC Center	Huawei Cloud

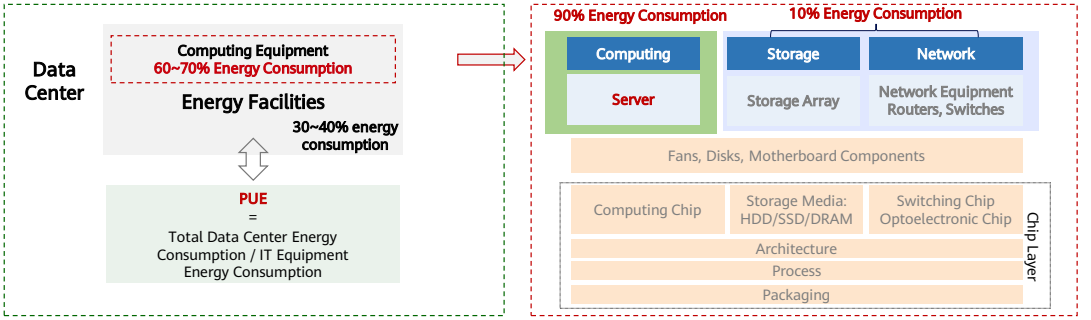
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- High Performance Computing (HPC) is a type of computer system that connects a large number of servers through high-speed networks to form a computing cluster. Together with high-performance storage, it is used to solve the most complex scientific computing problems in both scientific research and industrial applications (the three major paradigms in scientific research: theoretical science, experimental science, and computational science). HPC systems feature extremely fast computing speeds, large storage capacities, and high communication bandwidth.
- AI computing is based on a converged architecture computing system and uses data as its primary resource. It leverages powerful computing capabilities to drive AI models to deeply process data, continuously generating various intelligent services, which are then provided to organizations and individuals via cloud services over the network. Supercomputing centers and AI data centers are both subcategories of data centers, with the main differences lying in the types of chips, servers, and computational capability requirements. Supercomputing centers and AI data centers have higher demands for server computational capability, cooling systems, stability, and PUE (Power Usage Effectiveness).

Typical Energy Consumption Distribution in AI Data Centers

- Computing equipment is the source of computational capability, integrating computing, storage, and network devices. It uses servers and chips as core components to process and output data. The energy consumption of computing equipment accounts for 60–70% of the total energy consumption of a data center.
- According to McKinsey data, servers in commercial and enterprise data centers rarely exceed 60% utilization rates, with up to 30% of servers remaining in a "dormant" state, continuously consuming electricity without providing useful information services.

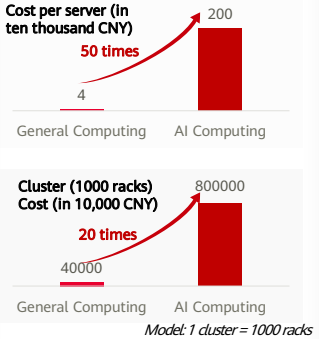


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Safety and Reliability Requirements

Clusters are costly in terms of deployment, failure and depreciation

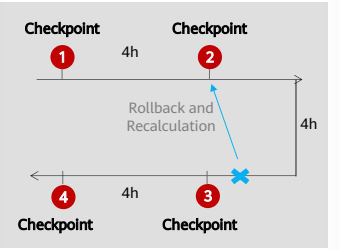
IT costs are 20 times higher, and depreciation during downtime is also 20 times higher.



Model: 1 cluster = 1000 racks (10,000 general-purpose servers / 4,000 AI servers)

Cluster parallel computing leads to high costs for fault rollback.

AI is parallel computing; resumable training after breakpoint requires cluster rollback.



Taking the xx large model as an example: a cluster of 5xxx cards, **checkpoint** interval of 4 hours, results in a loss of over 410,000 CNY per interruption. The customer's SLA compensation amount is even higher.

The fault radius has increased by five times.

AI computing scenarios have higher density and larger fault domains, exacerbating risks.

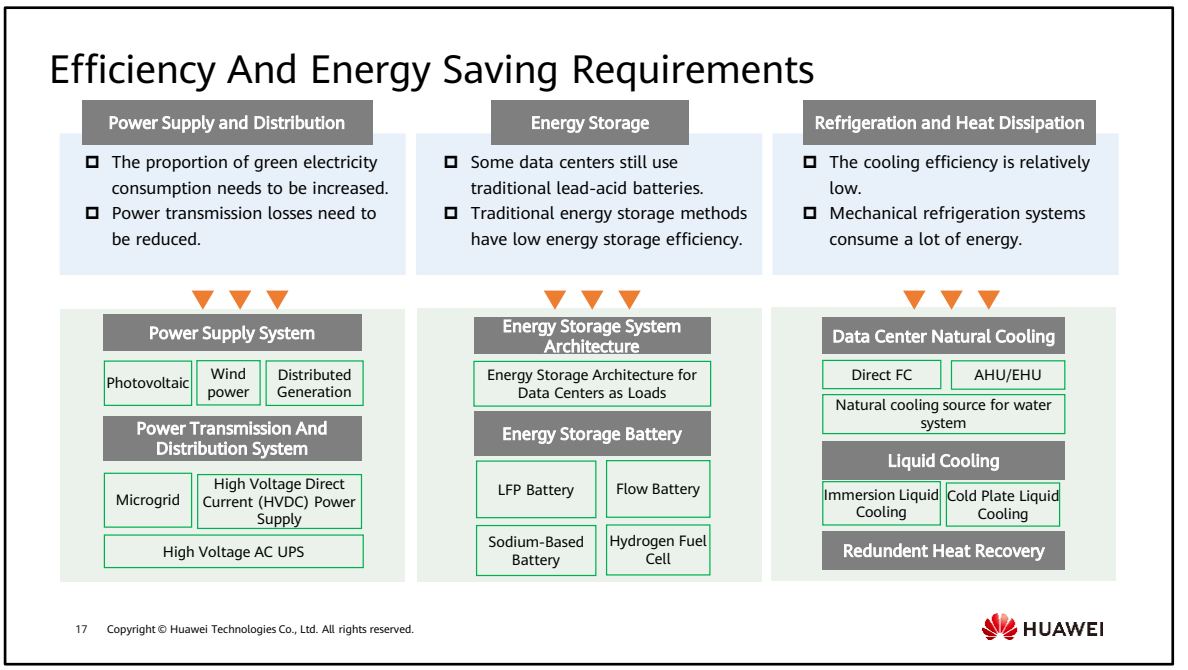
General Computing

Business	General Business
Campus	X Buildings, 10 MW level
IT	General server cluster
Single cabinet	IT cabinet: 8KW/R
Module	0.25 MW

AI Computing

Business	Large Models, Trillion-Parameter Models
Campus	X Clusters, 100X MW Level
IT	32K/64K/128K
Single Cabinet	IT Cabinet: 40KW/R
Module	1.2MW

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- Compared to traditional data centers, "AI data centers" are more environmentally friendly and energy-efficient, offer a higher degree of customization, and feature more advanced design concepts. They have larger capacities, strong scalability, low network latency, and high speed. These characteristics also determine that we need to consider the related requirements for supporting facilities.

Typical AI DC Solution: Facility Architecture (L1)

Elastic Deployment: "Scale on Demand"

Overall Design: Adopt a **Liquid Cooling Micro-module Solution**, or use **cluster deployment**, supporting flexible expansion.

- Ceiling Height: The clear height under beams is recommended to be ≥4.5m
- Raised floor height is recommended to be ≥750mm
- Aisle spacing: Front aisle ≥1.5m, Rear aisle ≥1.5m
- Data Center Floor Load Capacity: ≥1000 kg/m²

Flexible Power Supply, "Adaptive and Flexible"

Power Supply and Distribution Design:

- Adopts **a 2N power distribution architecture with dual UPS systems**, combined with **prefabricated integrated power modules** for full-chain integration, shortening the delivery period. The liquid-cooling cabinets also support **a power supply solution of one mains utility power and one UPS/HVDC**.
- The end-point uses **an intelligent busway** solution to achieve flexible power supply for individual racks, enabling flexible allocation and improving SUE.

Cold Plate Liquid Cooling Room

Power Supply System

Cooling System

Others

Elastic Cooling: "Plug-and-Play"

HVAC Design: Cold Plate Liquid Cooling + Centralized CDU + In-Row Air Conditioning for Supplementary Cooling Solution

- **Cooling Source**: For liquid cooling, closed cooling towers or open cooling towers with plate heat exchangers are used. The primary side inlet water temperature should be **≤35 °C**, and a temperature difference of **8 °C** is recommended for design.
- **End-point**: CDU (Cooling Distribution Unit) solutions are preferred, using prefabricated stainless steel secondary piping. A closed hot aisle configuration is adopted, and the air conditioning supply air temperature is recommended to be ≥25° C.



Cluster Deployment Mode

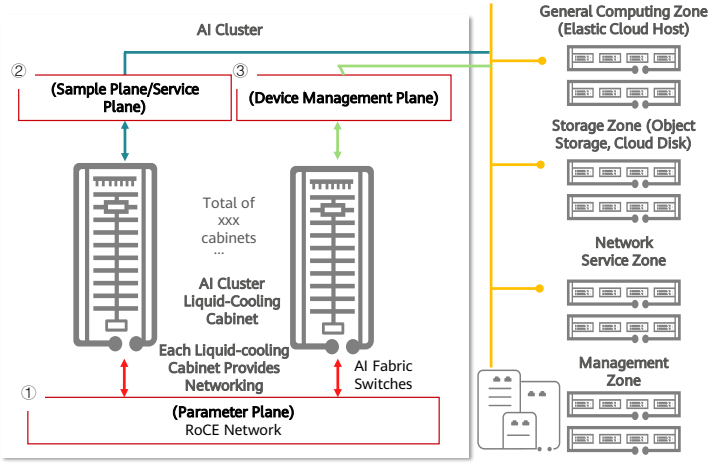
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- Currently, typical AI computing solutions for data centers can be deployed in two main modes: using micro-module form, and directly adopting cluster deployment.

Typical AI DC Solution: L2 Infrastructure Diagram

- AI Data Center AI Cluster



[AI Cluster Liquid Cooling Cabinet Solution]
AI clusters are delivered in high-density and integrated cabinets. Each liquid-cooling cabinet provides ultimate AI computational capability.
[Network solution design]
The AI cluster network consists of three planes: Parameter Plane, Sample/Service Plane, and Device Management Plane.

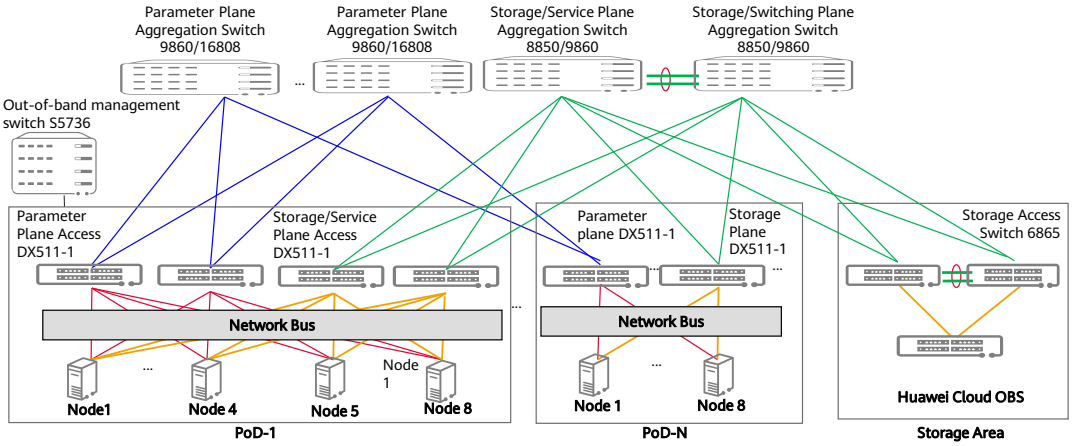
① **Parameter plane:** Servers in each liquid-cooling cabinet are connected to the access switch in the cabinet and converged by the aggregation switch outside the cabinet, implementing high-bandwidth, low-latency, and high-throughput cluster communication for AI training tasks.
② **Sample plane/service plane:** Accesses the sample plane/service plane through access switches. It is used to access storage systems with high-speed and high-bandwidth interconnection and schedule and manage system services.
③ **Device Management Plane:** Access is completed through access switches, primarily used for device management and operation.

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- **Parameter Plane Network:** Used for parameter exchange during distributed training in intelligent clusters. It requires a network with high throughput and low latency, and is deployed as a high-bandwidth intelligent lossless network.
- **Service Plane Network:** Used for system service scheduling and management. It is typically deployed as a TCP/IP lossy network.
- **Sample Plane Network:** Mainly used for high-speed, high-bandwidth interconnection of storage systems accessing the storage area. It is deployed as a TCP/IP lossy network and is integrated with the service plane network.
- **Out-of-band Management Network:** Used for out-of-band management and operation of cluster devices. It is deployed as a TCP/IP lossy network.

Example: Networking for Business and Storage Planes

Network Principle for Parameter Plane: Leaf-Spine architecture is adopted **with a convergence ratio of 1:1**. Leaf switches are DX511 (two per cabinet), with external spine switches 9860 (supporting up to 16 cabinets) or 16808 (supporting up to 144 cabinets).



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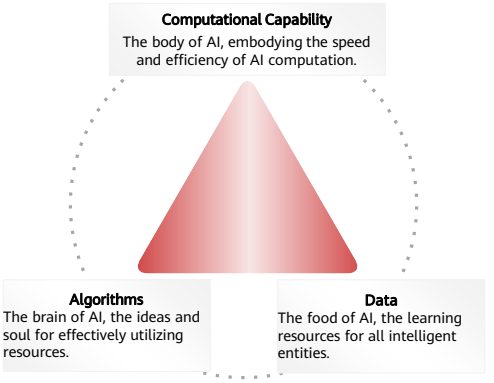
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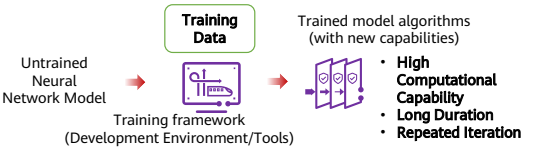
Learning from "Training" and Applying in " Inference "

Computational capability, algorithms, and data form the foundation of AI.

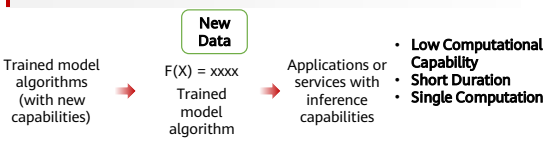


Basic concepts of AI training and inference.

AI Training Learning new capabilities from existing data to form algorithms (models)



AI inference Deploy the trained model into an application or service

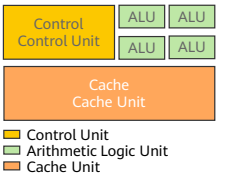


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General Computing vs. AI Computing

General-purpose Computing

Computational capability provided by **CPU**



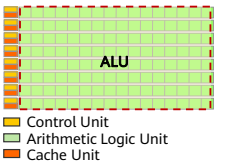
- **Suitable for complex logical operations, such as those used in most general-purpose software.**
- **More than 70% of transistors are used to build cache and control units**, with computing cores ranging from a few to dozens.

General applications

Office work, databases, numerical calculations (such as weather forecasting, fluid simulation, electromagnetic simulation), etc.

AI Computing

Computational capability provided by **AI chips**



- **Suitable for simple logic, compute-intensive, and high-concurrency tasks.**
- **More than 70% of transistors are used to build computing units**, with thousands or even tens of thousands of computing cores.

Specific applications

- Image Recognition: license plate recognition, action recognition, object detection, perimeter detection, etc.
- Natural Language Processing: machine translation, speech recognition, text generation, sentiment analysis, etc.
- Search and recommendation, assisted driving, trend prediction, etc.

ALU: Arithmetic Logic Unit, Arithmetic Logic Unit

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AI Chips (1)

- **From a technical architecture perspective, AI chips can be broadly categorized into the following types:**
  - **CPU (Central Processing Unit):** It is a very large-scale integrated circuit and serves as the computing Core and Control Unit of a computer. Its primary functions include interpreting computer instructions and processing data in computer software.
  - **ASIC (Application Specific Integrated Circuit):** This type of integrated circuit product is designed for a single specific purpose.
  - **FPGA (Field Programmable Gate Array):** Originally designed to achieve semi-custom chip functionality, FPGAs allow their hardware structure to be flexibly configured and changed in real-time according to needs.

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AI Chips (2)

- **From a business application perspective, AI chips can be divided into two types: Training and Inference.**
  - Training typically requires a large amount of data input or employs unsupervised learning methods such as reinforcement learning to train a complex deep neural network model. This process involves massive training data and intricate deep neural network structures, resulting in enormous computational demands. It requires processors with high performance in terms of computational capability, precision, and scalability.
  - Inference refers to using the trained model to "infer" various conclusions based on new data. Although the computational load for inference is much lighter compared to training, it still involves a significant amount of matrix operations. In the inference phase, chips with different technical architectures have substantial application value.

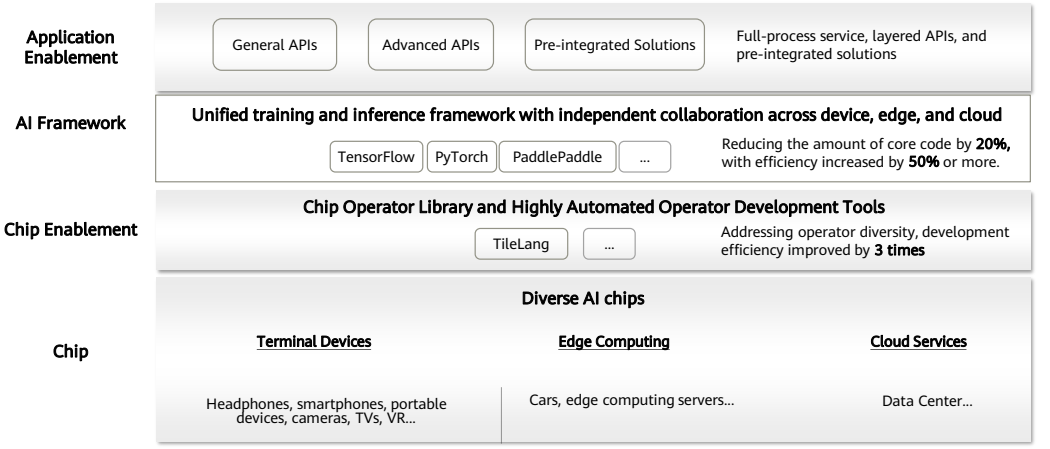
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Background of the AI Chip Industry

- **FPGA-based semi-customized AI chips:** Utilize reconfigurable FPGA chips for semi-customized AI chip design, represent the best choice.
- **Fully customized AI chips:** Designed specifically for deep learning algorithms are developed entirely using ASIC design methods, optimizing performance, power consumption, and area metrics for deep learning algorithms. A typical example of such chips is the Cambricon deep learning processor chip developed by the Institute of Computing Technology at the Chinese Academy of Sciences.
- **Brain-inspired computing chips:** Designed not only to accelerate deep learning algorithms but also aim to develop new computer architectures at the fundamental structural and even device levels. A representative example of this type of chip is IBM's TrueNorth chip.

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AI Full-Stack Solution



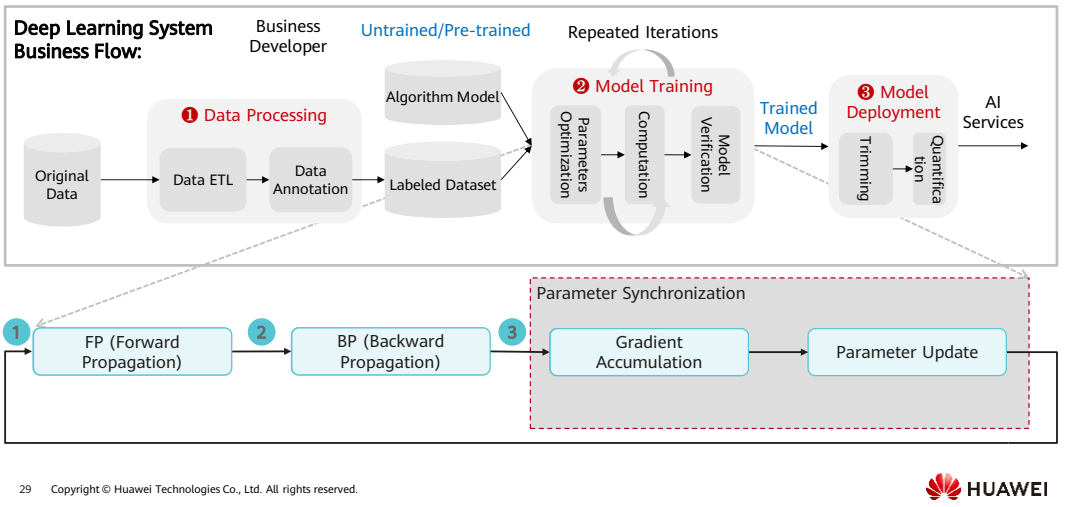
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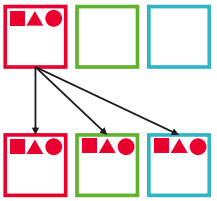
Basic Workflow of Artificial Intelligence



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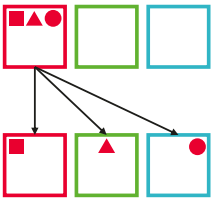
- Machine Learning Training Objective: To obtain model parameters W that minimize the error between the model's output values and the actual sample values.
- Forward Propagation: Calculate the activation values at each layer and ultimately obtain the output value. The activation value an at each layer is obtained by performing a matrix multiplication of the previous layer's activation value an-1 and the current layer's parameter Wn: $an = Wnan$-1.
- Backward Propagation: Based on the output obtained from forward propagation and the error between the model's output and the actual sample values, calculate the gradient ΔWn for each layer. This gradient is derived from the gradient of the next layer ΔWn+1 and the current layer's activation value an.
- Practical Operation: Continuously iterate the computation until $Wnew$ and $Wold$ are sufficiently close, indicating that the model has converged.

Collective Communication for Large Models: Basic Operations



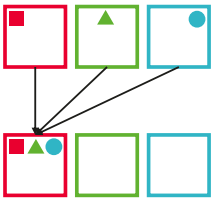
Broadcast

- Sending the entire data from one host to all others.



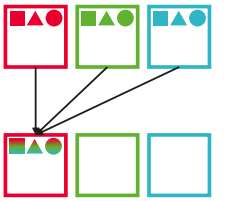
Scatter

- One host splits the data into N parts and sends each part to different hosts.
- The difference between Scatter and Broadcast lies in whether the receiving host receives the same complete data or just a part of the data. Therefore, the network traffic also differs.



Gather

- Multiple hosts send **different parts** of the data to the same host.
- Scatter and Gather are inverse operations of each other.



Reduce

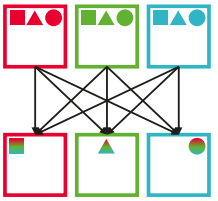
- Multiple hosts send data to the same host. Then, the destination host performs certain operations, such as summing, finding the maximum value, the average, etc.
- The difference between Reduce and Gather lies in whether the destination receives the same or different parts of the data.

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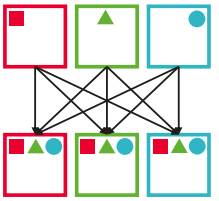
- Note 1: In the figure, boxes of different colors represent different hosts. Solid shapes represent data, and the color of the data indicates which host it originated from.
- Note 2: Complete data consists of squares, triangles, and circles. After splitting, the data consists of only one shape. Mathematically, complete data is a vector/matrix, and split data is the vector/matrix divided into multiple parts.

Collective Communication of Large Models: Combined Operations



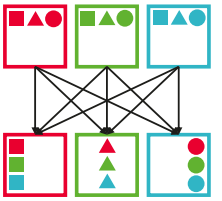
ReduceScatter

- Each host splits the data and sends different parts to different hosts. After receiving the data, the destination hosts perform certain operations, such as summing, finding the maximum value, calculating the average, etc.



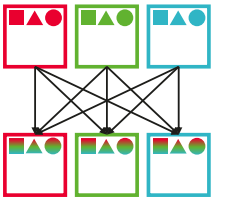
AllGather

- Multiple hosts send **different parts** of the data to all hosts.
- AllGather = Gather + Broadcast



AlltoAll

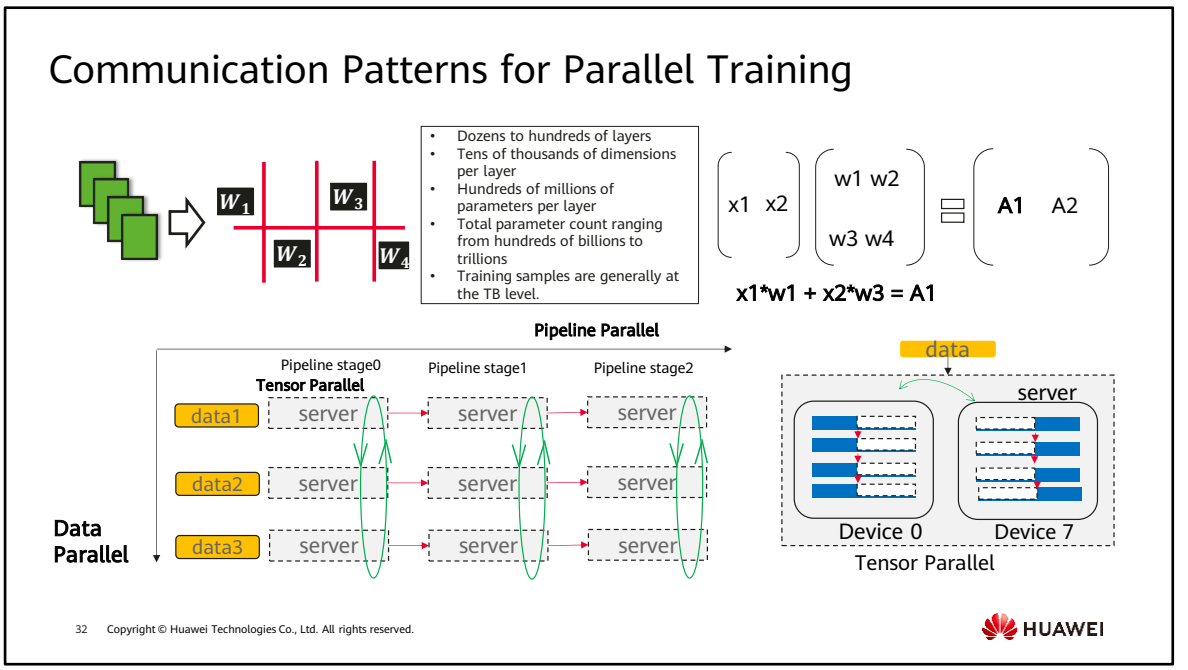
- Each host splits the data and sends different parts to different hosts.
- AlltoAll = All hosts scatter data, but note that there are restrictions on the destination hosts, such as the red receiving squares, green receiving triangles, and blue receiving circles in the figure.
- There is no difference in communication traffic between ReduceScatter and AlltoAll; the only difference is whether the destination hosts perform reduction operations.



AllReduce

- Multiple hosts send the data **same part** to all hosts. Then, all destination hosts perform certain operations, such as summing, finding the maximum value, the average, etc.
- AllReduce = Reduce + Broadcast (PS architecture)
- AllReduce = ReduceScatter + AllGather (Ring architecture)

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- Data Parallelism: Synchronization of model parameter gradients, primarily using AllReduce.
- Pipeline Parallelism: Intermediate computation results (activation values, layer parameter gradients) are passed to adjacent nodes through point-to-point communication.
- Tensor Parallelism: Intermediate computation results and parameters are distributed across multiple nodes where tensors reside, with communication mainly using AllReduce.

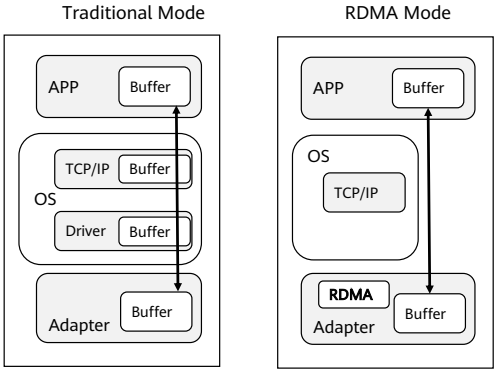
Communication Load Estimation for Model Training

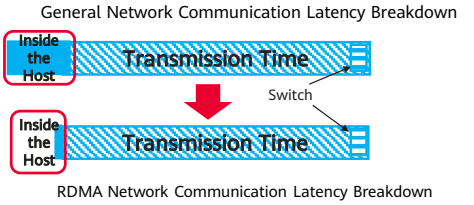
Parallelism Method	Communication Operation	Communication Load (Single Computational Card)	Number of Computational Cards	Computation Time	Cluster Requirements
Tensor Parallelism	Allreduce	Hundred GB	Single machine with 8 cards	Second-level	High-speed internal interconnection
Pipeline Parallelism	P2P	GB level	2 cards between machines	Second-level	High single-node bandwidth
Data Parallelism	Allreduce	GB-level	All cards, divided into multiple groups	Ten-second level	High-throughput Allreduce
Expert Parallelism	All2All	Hundred GB	Hundred of Cards	Second-Level	Overall cluster throughput

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Network Requirement for Model Training: RDMA is Essential

- RDMA (Remote Direct Memory Access) is a method for transferring data across buffers, between application software on two servers.
- The RDMA protocol stack is offloaded to the network interface card, bypassing the kernel, which reduces the processing latency of the protocol stack to approximately 10 ms.





RDMA network, latency within the host has close to zero overhead

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- RoCE: Remote Direct Memory Access over Converged Ethernet

Network Requirement for Model Training: High Bandwidth is Fundamental

- Large model training is bandwidth-sensitive, the data transferring time mainly depends on effective bandwidth.
- The time spent on network switching is not significant, and switching latency has little impact on model training efficiency. Thus, static latency is not a concern.

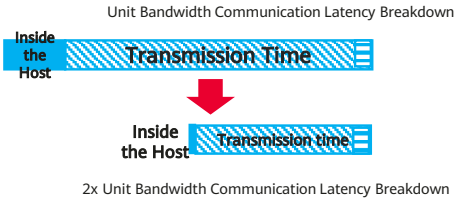
Higher access bandwidth benefits system expansibility

Access bandwidth increased by 16 times: from 100 Gbps to 1.6 Tbps.

- Allreduce communication time reduced by 14 times
- Communication time percentage decreased from 35%->3.7%



Increasing access bandwidth can reduce data transfer time almost linearly

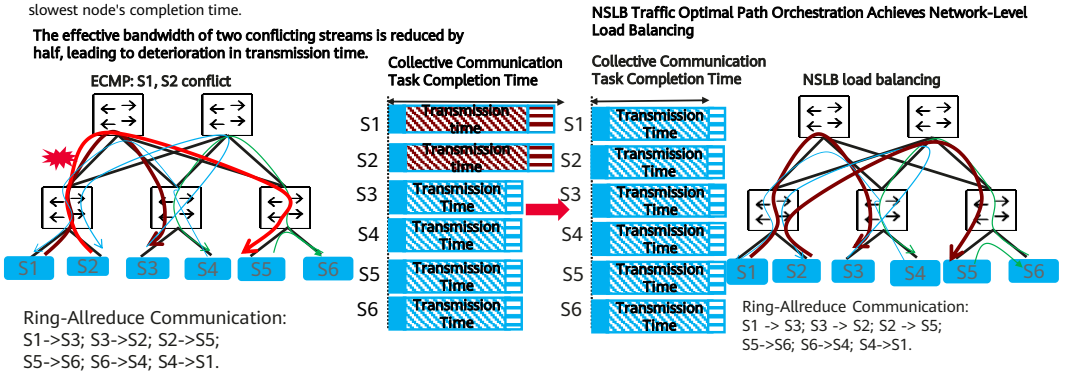


- Increasing access bandwidth directly accelerates data transferring.

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Network Characteristics Requirement for Model Training: Load Balancing is Key

- In large model training systems, the most critical aspect of the network is ensuring effective communication bandwidth. Load balancing has the greatest impact on effective bandwidth and is a key factor.
- In a non-blocking network, if the load is unbalanced, it leads to a reduction in the effective bandwidth of conflicting flows (for N conflicting flows, the effective bandwidth drops to 1/N), worsening the transmission time of these flows. The completion time for collective operations is determined by the slowest node's completion time.



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- NSLB: Network Service Load Balance, based on Huawei's high-precision telemetry capability of switching chips, collects traffic matrices as input for routing algorithms to control the forwarding paths of AI traffic, preventing load imbalance and improving AI training efficiency.
- In the Ring algorithm scenario, when running a single computing task, using the NSLB technology compared to the typical ECMP load balancing technology, the network achieves 100% load balancing, with an average link utilization of 34%, which is 35% higher than that of ECMP. The AI training set performance is improved by up to 113.41%.
- In the Ring algorithm scenario, when running two computing tasks simultaneously, using the NSLB technology compared to the typical ECMP load balancing technology, the network achieves 100% load balancing, with an average link utilization of 29%, which is 15.6% higher than that of ECMP. The AI training set performance is improved by up to 57.29%.
- In the Tree algorithm scenario, when running a single computing task, using the NSLB technology compared to the typical ECMP load balancing technology, the network achieves 100% load balancing, with an average link utilization of 13.8%, which is 1% higher than that of ECMP. The AI training set performance is improved by up to 6.50%.
- In the Tree algorithm scenario, when running two computing tasks, using NSLB technology compared to the typical ECMP load balancing technology, the network achieves 100% load balancing, with an average link utilization of 14%, which is 10.5% higher than that of ECMP. The AI training set performance is

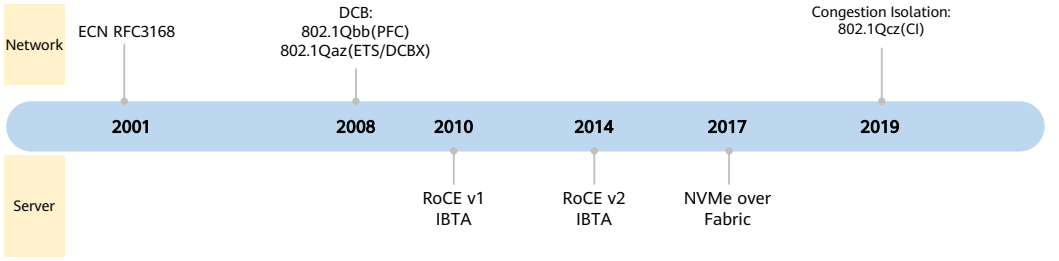
improved by up to 15.81%.

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Development of Lossless Ethernet Protocols



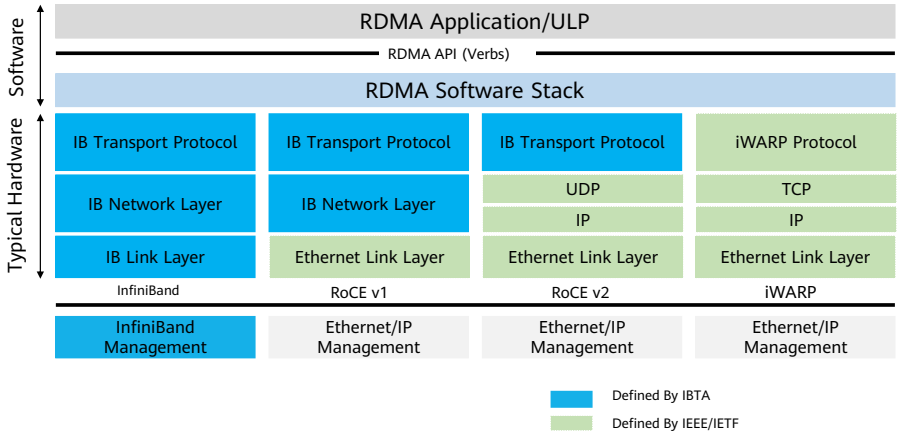
- Lossless Ethernet was first proposed in 2008. The two main foundational protocols are **Priority-based Flow Control (PFC) and Explicit Congestion Notification (ECN).**
- The widespread adoption of high-performance distributed computing, high-performance storage RDMA, and the application of converged Ethernet have led to the extensive use of lossless protocols such as ECN and PFC, thus giving rise to the concept of lossless Ethernet.
- In addition to network-side technologies, lossless Ethernet also involves features on the server NIC side, including RoCE (RDMA over Converged Ethernet) for supporting RDMA protocol transmission over Ethernet, and **NoF (NVMe over Fabric)** for supporting NVMe protocol transmission over Ethernet.

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- In converged Ethernet, the RoCE technology enables Ethernet to carry the RDMA protocol originally used in the IB network, thus meeting the high-speed I/O access requirements between servers. In addition, FCoE (Fiber Channel over Ethernet) is used to carry the iSCSI storage transmission protocol over the Ethernet network.
- ECN was initially defined in RFC 3168 to identify network congestion. PFC is defined in 802.1Qbb and is similar to the IEEE 802.3x PAUSE flow control mechanism. However, PFC supports independent operation based on each priority level, enabling precise control and preventing interference between different priorities. 802.1Q continues to evolve, and 802.1Qcz (CI), a new protocol standard led by Huawei, was defined to address the head-of-line blocking (HOLB) issue in PFC.

RDMA Bearer Network



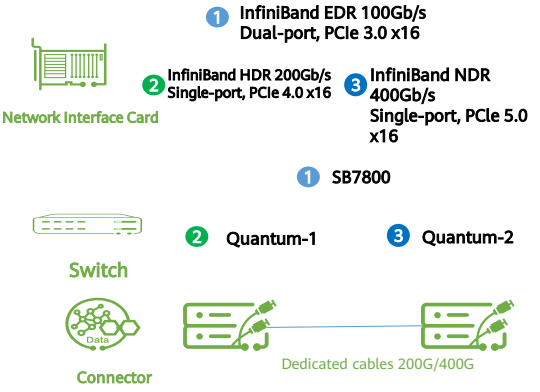
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- Similarities: Both use the same RDMA transport layer standard, defined by the same standards organization, employ the same programming interfaces, and share the same RDMA application ecosystem.
- Differences: IB has a shorter packet header within the same subnet and a slightly higher payload. RoCE benefits from the advantages of Ethernet and IP: large-scale networks, network virtualization, and network operations and maintenance.

IB

- IB is a high-performance, low-latency computing and storage networking technology that utilizes silicon photonics and fiber optic communication for high-speed interconnections, enabling fast and reliable data transmission.
- The IB network has the following characteristics:
  - IB originated as an interconnection technology between servers in high-performance computing systems, offering the advantage of large bandwidth. It enables the fastest possible data packet transfer between servers. IB employs a credit-based flow control system, where source nodes send data packets only after obtaining credit permits.
  - IB supports the RDMA (Remote Direct Memory Access) kernel bypass mechanism, allowing applications to directly read data from the network interface card, significantly reducing protocol stack latency. This also allows the receiver to directly read data from the sender's memory, greatly reducing CPU load.

The key components of an IB network include the subnet manager, IB NIC, IB switch, and IB cables.



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- Restrictions on the IB network: The IB network must be interconnected with devices that support the IB protocol. The existing Ethernet network is not compatible with the existing Ethernet network. A single ecological chain of closed vendors is weak. To prevent vendor lock-in, dedicated IB NICs, IB switches, and cables need to be introduced. The infrastructure network construction cost is high.

RoCE

- RoCE is a network protocol that allows RDMA (Remote Direct Memory Access) to be performed over Ethernet. It offers advantages such as high performance, low latency, low CPU usage, low cost, and high compatibility with existing DC Ethernet networks, making it widely used in fields requiring high-speed data transmission and low latency.
- RoCE networks have the following characteristics:
  - The RoCE protocol combines the high performance of RDMA technology with the openness of Ethernet technology, providing a network with low latency and high bandwidth.
  - RoCE can utilize existing Ethernet infrastructure without the need to set up dedicated network equipment, thereby reducing network deployment and maintenance costs.
  - RoCE can support higher-level network protocols such as TCP/IP and UDP/IP, making it compatible with existing network environments.

There are many manufacturers that support RoCE NICs.

ConnectX-5 25GbE Dual-port SFP28, PCIe Gen 3.0 x 8



ConnectX-6 100GbE Dual-port QSFP56, PCIe 4.0 x16



ConnectX-740OGbE Single-port OSFP, PCIe 5.0 x16



Currently, most switches support the RoCE protocol. Mainstream data center switch manufacturers in China include Huawei and H3C. Below are some models of Huawei data center switches:

CE6885-48YS8CO ((25GE)



CE8855-32CQ4B0 (100/200GE)

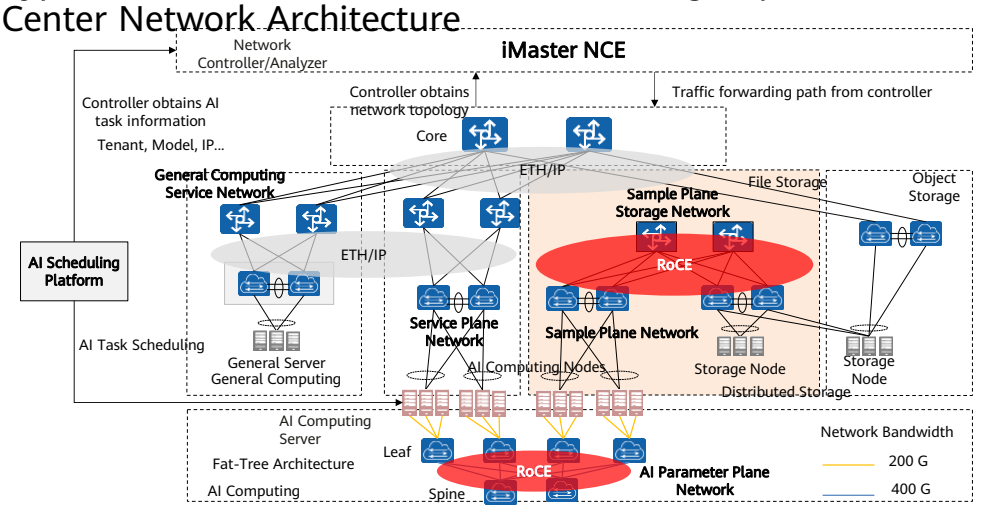


CE8850-64CQ-El (100GE)



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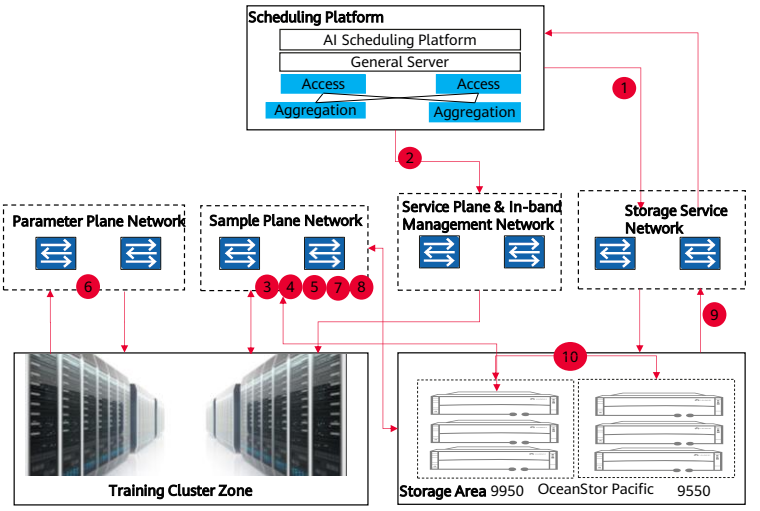
Typical AI Data Center Overall Solution: High-Speed Lossless Data Center Network Architecture



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- The physical network of the AI computing cluster area is designed with different planes:
  - Parameter plane network: A two-layer leaf-spine architecture is used. Eight 200GE network ports are connected to 200GE leaf switches to facilitate parameter exchange during distributed training across multiple machines.
  - Distributed storage network: Two 100GE ports configured on each AI server are connected to 100GE leaf switches, mainly for accessing high-speed, high-bandwidth storage systems in the storage area.
  - Service plane network: Two 25GE/10GE ports configured on each AI server are connected to 25GE leaf switches, mainly for system service scheduling and management.
  - Out-of-band management plane network: One GE network port on each AI server is connected to a GE switch, which is then connected to the management zone network, mainly for out-of-band management and operation of cluster devices.
- The service plane network is typically deployed using traditional TCP. The storage/parameter plane network is used for parameter exchange between compute nodes and requires a high-bandwidth, lossless network.

Cluster Training Workflow (1)



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Cluster Training Workflow (2)

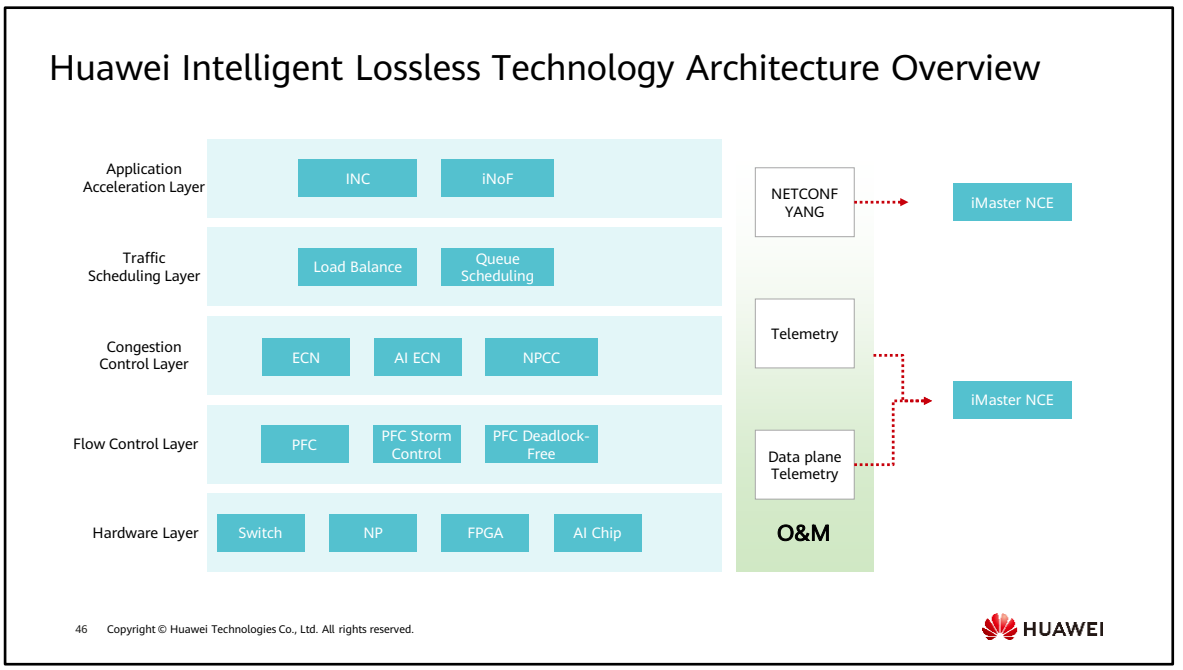
	Business Process Description
1	Import dataset and train model into the storage system through storage network.
2	The AI scheduling platform distributes training tasks through the service plane & in-band management network.
3	The training task images are loaded onto the AI computing nodes via the sample plane network (training images).
4	AI computing nodes load AI models (checkpoints) through the sample plane network.
5	AI computing nodes read training datasets (training data) through the sample plane network.
6	During AI training, model parameter synchronization is completed through the parameter plane network.
7	CheckPoint files generated during AI training are written to the storage system via the sample plane network.
8	The completed AI model is written to the storage system via the sample plane network.
9	Training results (models) are exported through the storage service network.
10	Checkpoints are written/read, and data is read and saved. Data is obtained from the 9950 device via the sample plane network. If the data is on the 9550 device, it is pulled to the 9950 device through the storage backend network and then fetched to the compute node via the sample plane network.

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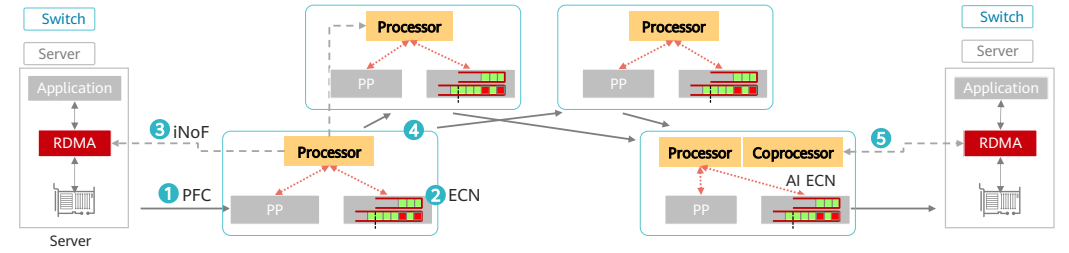
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- Flow Control Technologies
  - PFC: Priority-based Flow Control, a protocol defined by IEEE 802.1Qbb, primarily addresses the issue of frame loss caused by congestion.
  - PFC storm control: PFC storm control mainly resolves the problem of network traffic interruption caused by PFC storms, also known as PFC deadlock detection.
  - PFC deadlock prevention: PFC deadlock prevention identifies and eliminates the necessary conditions for circular buffer dependencies, thereby resolving PFC deadlocks and improving network reliability.
- Congestion Control
  - Static ECN: Explicit Congestion Notification (ECN) is an end-to-end network congestion notification mechanism that allows the network to avoid packet loss when congestion occurs. It is defined in RFC 3168 (2001).
  - AI ECN: The ECN threshold is dynamically adjusted using the iLossless intelligent algorithm to achieve maximum bandwidth and minimum latency.
  - NPCC: Network Proactive Congestion Control (NPCC) actively controls the transmission rate of each flow based on the congestion status of the device itself. It is mainly used in long-distance lossless networks.
  - iQCN: Intelligent Quantized Congestion Notification (iQCN) is used to control the latency of RoCE traffic in scenarios where TCP and RoCE traffic coexist.

Huawei Intelligent Lossless Technology Application Overview



1 Flow Control
- PFC
- PFC head resistance
- PFC deadlock
- Deadlock prevention
- Deadlock detection
- Long-distance flow control

2 Congestion control
- DCQCN
- ECN
- AI ECN
- NPCC
- ECN OVERLAY

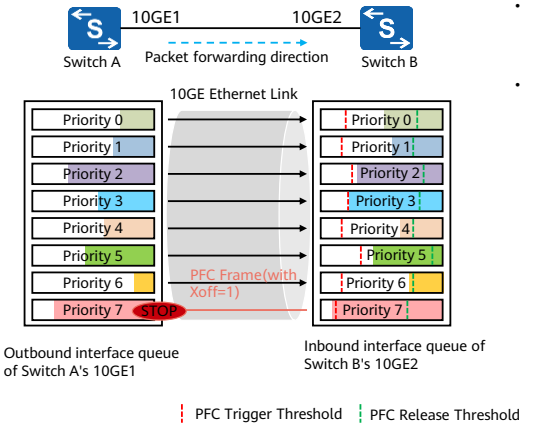
3 NoF
- NoF
- iNoF Plug-and-Play
- Hardzone Domain Isolation
- iNoF Configuration Synchronization
- iNoF Fast Failover

4 Scheduling Technology
- DLB Dynamic Load Balancing
- DPP Large and Small Flows

5 Integrated network computing
- On-Net Computing
- Topology awareness

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Flow Control Technology: PFC (Priority-based Flow Control)

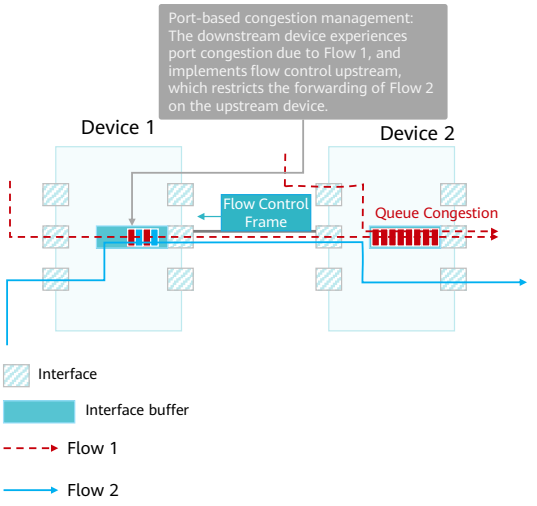


- Definition of Flow Control: By managing the transmission rate between two nodes, flow control aims to prevent the sender from filling up the receiver's buffer, thus addressing the issue of a fast sender and slow receiver.
- Definition of PFC: Priority-based Flow Control (IEEE 802.1Qbb), which is a pr iority-based traffic control mechanism.
  - When the buffer of a specific priority queue at the receiving end exceeds the PFC Trigger Threshold, the receiving end sends a PFC frame(with Xoff=1) to the transmitting end.
  - Upon receiving the PFC frame(with Xoff=1), the transmitting end stops sending traffic to the receiving end.
  - When the buffer of the specific priority queue at the receiving end returns to the PFC Release Threshold , the receiving end sends a PFC Frame(with Xoff=0) to the transmitting end.
  - Upon receiving the PFC Frame(with Xoff=0), the transmitting end resumes sending traffic to the receiving end.

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- PFC is a queue-based flow control mechanism that allows the creation of eight virtual lanes on an Ethernet link. Each virtual lane is assigned a priority level, enabling the independent pause and restart of any one of these virtual lanes while allowing traffic on other virtual lanes to pass through uninterrupted. This method enables the network to provide zero-loss service for individual virtual links, allowing them to coexist with other types of traffic on the same interface.
- To achieve lossless forwarding, the queue depth should be sufficient and the congestion threshold should not be too high, ensuring that packets can be buffered in the queue before PFC Trigger takes effect, thus preventing packet loss due to queue overflow.
- PFC propagates Trigger from the congestion point (the destination) back to the source node in the network, suppressing the rate at which the source sends data. Therefore, the sending end (server) must also support PFC functionality.

Flow Control: HOLB Head-of-Line Blocking Issue



Interface

Interface buffer

Flow 1

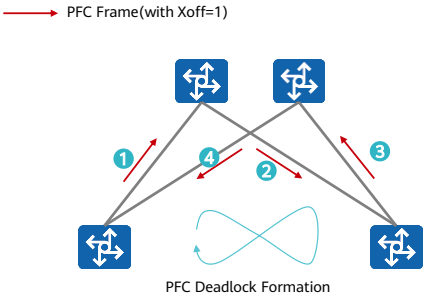
Flow 2

HOL Blocking Problem

- Current Ethernet flow control is based on ports rather than individual flows, which can cause congested flows to block normal flows. This is known as the head-of-line (HOL) blocking issue, which is one of the core issues in high-performance interconnection.
- For example, if Flow 1 causes congestion at Device 2, it generates PFC Trigger on Device 1, causing Flow 2 to be blocked and unable to be forwarded.

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PFC Deadlock



PFC Deadlock Formation

Description of PFC Deadlock

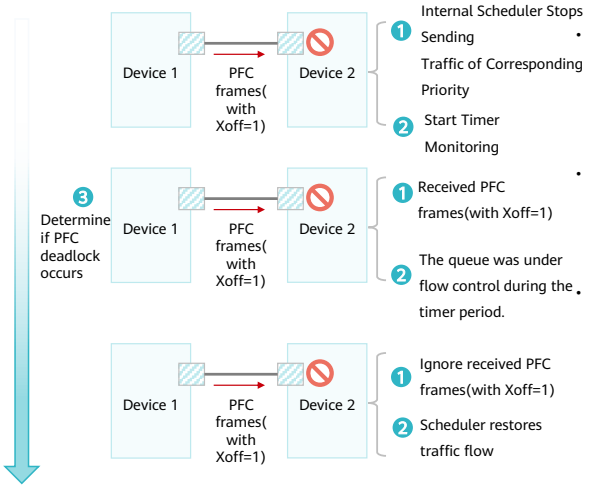
- PFC Deadlock refers to a network condition where multiple switches experience congestion simultaneously due to loops or other reasons, causing the buffer usage on each port to exceed the threshold. These switches then wait for each other to release resources, resulting in a permanent blockage of data flows on all switches involved.

Impact of PFC Deadlock

- Once a deadlock occurs, all switches on the path where the deadlock is generated will stop forwarding traffic, leading to service interruption.

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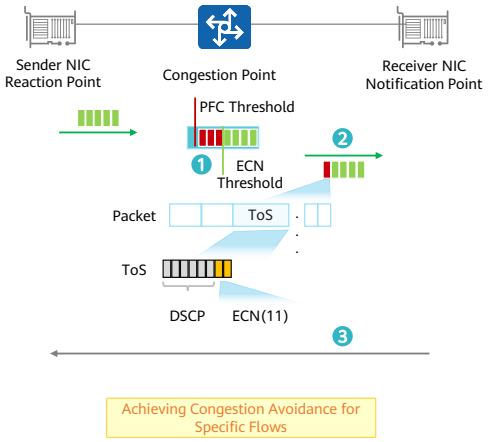
PFC Deadlock Detection



- After receiving the PFC frames(with Xoff=1) sent by Device 1, the internal scheduler of Device 2 will stop sending traffic from the corresponding priority queue and start a timer. It will then begin detecting the PFC frames(with Xoff=1) received in the queue according to the set deadlock detection and precision parameters.
- If a queue remains in the PFC-XOFF state (i.e., under flow control) for the entire duration of the set PFC deadlock detection time, it is considered that a PFC deadlock has occurred, and the PFC deadlock recovery process needs to be initiated.
- During the PFC deadlock recovery process, PFC frames(with Xoff=1) received by the port will be ignored. The internal scheduler will resume sending traffic from the corresponding priority queue, or it may choose to discard the traffic from the corresponding priority queue. After the recovery period, the normal PFC flow control mechanism will be restored. If a deadlock is still detected during the next deadlock detection cycle, a new round of deadlock recovery will be initiated.

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Congestion Control Technologies: ECN and DCQCN



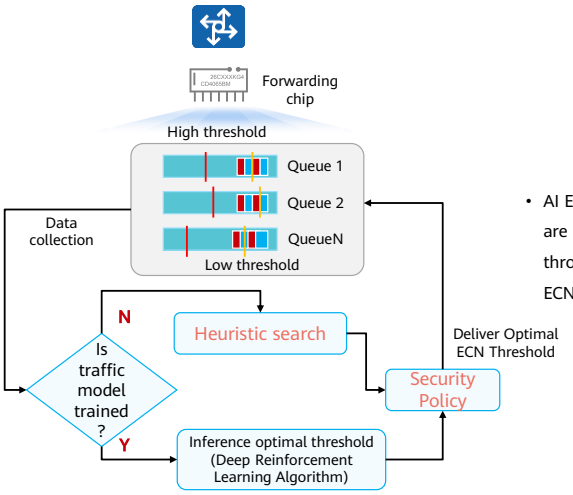
- DCQCN: Data Center Quantized Congestion Notification, a congestion control algorithm for RoCEv2.
- DCQCN can be divided into three parts:
  - Congestion Point Algorithm: Intermediate switches check the current congestion status. When the queue depth exceeds a threshold, they mark packets with ECN and then forward them to the next hop.
  - Notification Point Algorithm: This is performed by the receiver's NIC, which is mainly responsible for notifying the sender of congestion information. RoCEv2 introduces CNP (Congestion Notification Packets) control messages for congestion notification. The receiver's NIC checks the ECN flag of each received packet; if the CN flag is set, it sends a CNP to the sender.
  - Reaction Point Algorithm: Completed by the sender's NIC, responsible for adjusting the sending rate to avoid congestion.

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- When the number of packets in the queue at a congestion point (CP) exceeds the Explicit Congestion Notification (ECN) threshold, the CP marks the packets with an ECN congestion flag (setting the ECN field to 11). The ECN congestion flag uses the last two bits of the ToS field in the IP packet header, and their meanings are as follows:
  - 00 – Transmission does not support ECN, non-ECT.
  - 10 – Transmission supports ECN, ECT(0).
  - 01 – Transmission supports ECN, ECT(1).
  - 11 – Congestion has occurred, CE.
- After receiving a packet carrying the ECN congestion flag, the sink end server (NP, Notification Point) sends a CNP (Congestion Notification Packet) to the source end server (RP, Reaction Point). Upon receiving the CNP, the source end server reduces its packet transmission rate.

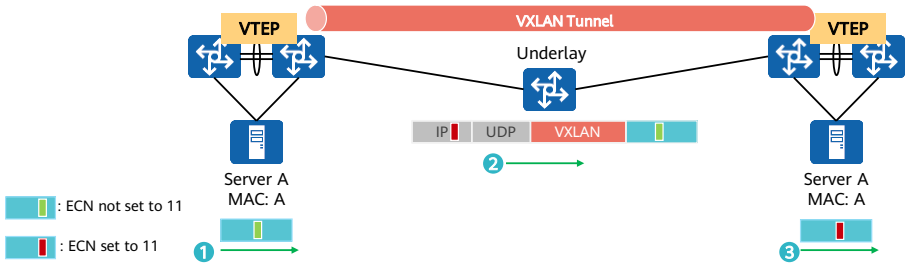
AI ECN adaptively adjusts the ECN threshold



- AI ECN: By analyzing traffic load and composition, AI algorithms are used to train ECN parameters to ensure maximum throughput and minimum latency, and adaptively adjust the ECN threshold.

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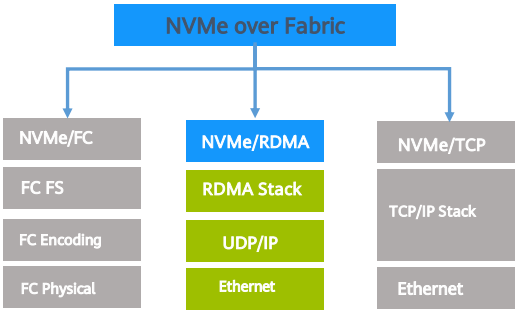
AI ECN Adaptation for VXLAN Scenarios



- Supports ECN marking on the outer IP header of network devices in VXLAN scenarios when congestion occurs on VTEP nodes or underlay nodes. The ECN-marked packets are then sent to the egress VTEP node, which maps the ECN field in the overlay encapsulated IP header to the ECN field in the original packet, restores the user packet, and forwards it to the target server on the user side. This feature enables ECN support in VXLAN scenarios.

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NoF Technology: NVMe over Fabric



- With the continuous development of modern information technology, there is an increasing demand for higher storage performance. In particular, the rapid advancement of flash memory has led to faster storage media speeds, making traditional SCSI protocols difficult to keep up with the fast development of flash drives. There is an urgent need for a new protocol to enable high-performance communication between drives and hosts, which led to the emergence of the NVMe protocol. As the NVMe protocol was rapidly adopted in flash drives, the demand for end-to-end NVMe protocols became increasingly strong. In 2017, the NVMe over Fabric protocol based on the RDMA transport layer was defined and quickly extended to FC and TCP transport layers, thus achieving end-to-end NVMe protocols across the entire path, further improving performance and reducing latency.

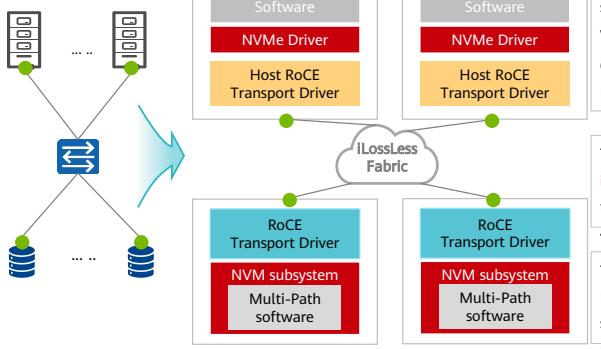
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- In comparison, the Fibre Channel (FC) network that originally carried storage services has become a bottleneck for current storage networks in terms of both bandwidth and latency. To address this, a new generation of storage systems requires a faster and higher-quality network. As a result, storage and networking have undergone deep architectural and protocol-level restructuring, leading to the emergence of NVMe over Fabrics (NVMe-oF), a new generation of storage networking technology.
- NVMe-oF applies the NVMe protocol to the front end of server hosts, serving as the channel connecting storage arrays to front-end hosts, and can replace the SCSI protocol in SAN networks end-to-end.

NVMe over RoCE

- NVMe over RoCE is a type of NVMe over fabric (based on RDMA). It is optimized in terms of performance, cost, network management, and technology development, and is gradually becoming the best application of NVMe over fabric.



NVMe Driver

NVMe over RoCE requires the host's Linux system version to support NVMe over Fabrics Host Driver. The supported kernel version is Linux 4.8.

Currently, mainstream operating systems on the market include Red Hat 8.0/8.1/8.2, CentOS 8.0/8.1/8.2, etc.

Host RoCE Transport Driver

The iNIC that supports NVMe over RoCE smart network interface card requires the host's Linux system version and NIC to support RoCEv2. The kernel version supporting RoCE is Linux 4.7 or later.

RoCE Transport Driver

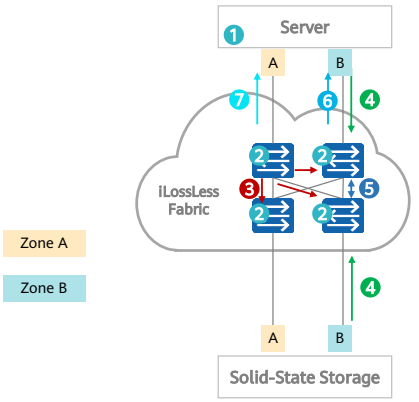
To support NVMe over RoCE, the storage system must support NVMe over Fabrics and RoCEv2. Currently, mainstream storage systems on the market include HUAWEI Storage, Pure Storage, NetApp, etc.

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- SmartNIC: As the interface rates of data center networks have increased (10G > 40G > 100G > 400G) and storage media have advanced, the industry has proposed SmartNICs based on programmable chips (such as NP) to improve the network performance of server-side operating systems, reduce network processing latency on the server side, and support converged Ethernet. These SmartNICs can support protocols such as IB, RoCE, RDMA, and NoF, and directly offload and process network layer information at the NIC level. This reduces the processing burden on the server-side operating system and provides efficient network I/O for servers.

NoF+ Storage Network Working Mechanism



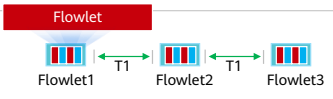
1. Server-side SNSD & UltraPath Plugin Installation
2. Address Planning, Zone Planning (Memory Management), VLAN Planning, Load Configuration, Enable INOF
3. Zone Configuration Data Synchronization
4. Host Server & Storage Login and Registration
5. Real-time synchronization of login status information
6. Notify the Host Server and automatically establish a connection.
7. Failover notification for UltraPath in case of failure on the storage network.

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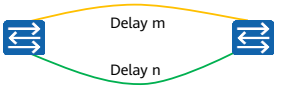
Scheduling Technique: Flowlet & DLB: A Subflow-based Load Balancing Scheme

DLB

A burst of packets with the same flow characteristics but separated by time intervals is called a "Flowlet," such as a TCP burst.



T2 = |m – n|, if T1 < T2, then DLB can be performed.



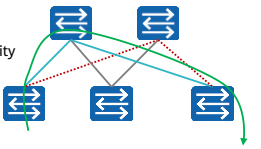
- Flowlet-based load balancing divides the burst traffic of large flows into smaller granularities of Flowlets. It selects paths based on Flowlets and detects the quality of paths across the entire network in real time, choosing the path with the best quality to achieve finer-grained and more balanced load distribution.
  - When the time interval between packets within a Flowlet is less than the delay difference between paths, the same path will still be selected to ensure that the packets within the Flowlet are not out of order.
  - When the time interval between two Flowlets is greater than the delay difference between paths, a less congested link can be selected again without causing packet out of order.

Real-time Scheduling

: Burst Traffic

: Good Link Quality

: Relatively Poor Link Quality



- Real-time perception of the entire network quality, selecting the path with the best quality for traffic load.

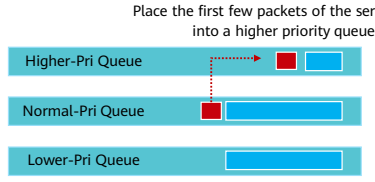
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- DLB: Dynamic Load Balancing.
- Link and Path Quality Measurement
  - Local Link Quality: The switch's forwarding chip quantifies the quality level based on information such as buffer length and bandwidth utilization of each interface.
  - Network-wide Path Quality: The switch's forwarding chip detects the congestion status of each node on all paths in real time through packet probing and quantifies the quality level.

DPP: Solving Elephant and Mice Flow Issues

- Elephant and Mice Flows: Packets entering the queue must be dequeued in order. When large and small flows are mixed, congestion occurs, and large flows can fill up the queue. As a result, smaller flows may experience packet loss because they cannot enter the queue. Even if small flows can enter the queue, the long queue length caused by large flows increases the queuing delay for small flows.



Dynamic Packet Prioritization: Dynamically raises the scheduling priority of delay-sensitive small flows, allowing them to be sent preferentially without being affected by the congestion caused by large flows, thereby ensuring the performance of distributed applications.

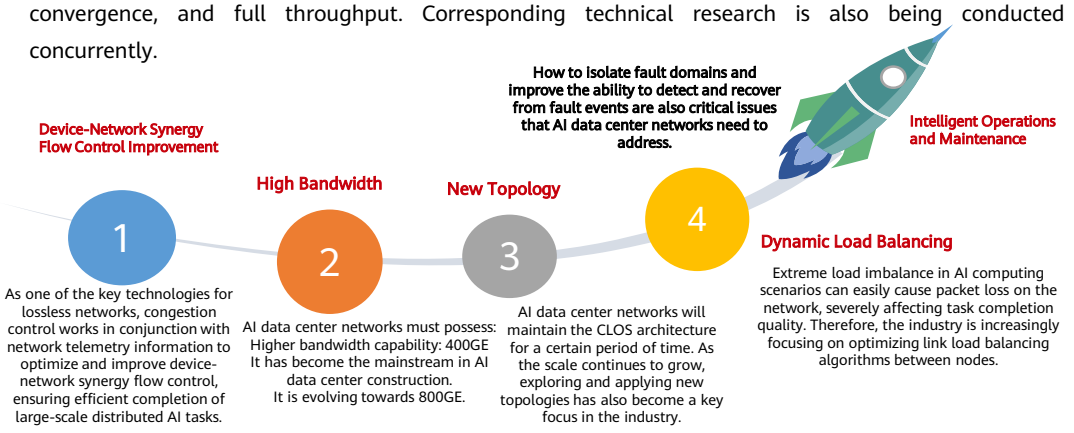
DPP (Dynamic Packet Prioritization)

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- Basic Principles:
  - Packet information in the queue is recorded in flow table entries, and elephant flows are identified based on the flow table entries using elephant flow identification parameters.
  - For identified elephant flows, they are forwarded via a low-priority queue; for mice flows, they are still forwarded in the original priority queue.
  - Subsequent packets entering the queue are processed as follows: if they belong to an identified elephant flow, they are forwarded to a low-priority queue; if they do not belong to an identified elephant flow, the above steps are repeated for processing.
  - In addition, since it takes some time to identify elephant flows, if priority scheduling is started only after the identification is successful, it may affect the mice flows that are blocked before the identification. Therefore, the device tracks each service flow and places the initial packets of each flow into a higher-priority queue to ensure timely scheduling of mice flows.

Technology Evolution Trends

- AI data center networks are evolving towards zero packet loss, high bandwidth, large scale, non-convergence, and full throughput. Corresponding technical research is also being conducted concurrently.



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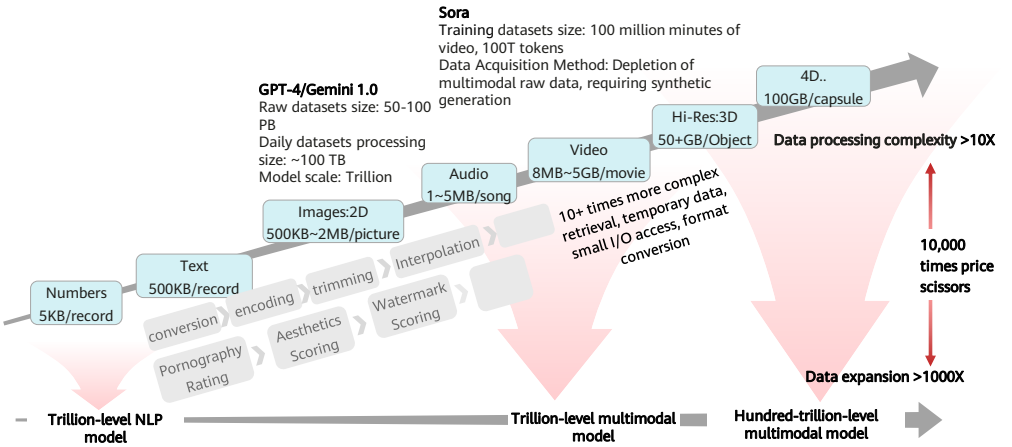
- The CLOS architecture is a multi-stage switching network topology used to build high-performance, highly reliable data center networks. Typically, the CLOS architecture consists of three layers: the input layer, the middle layer, and the output layer. The CLOS switching architecture can achieve strict non-blocking, re-arrangeable, and scalable performance.

Contents

1. AI Data Center Overview
2. AI Data Center Computational Capability Solutions
3. AI Data Center Network Solutions and Key Technologies
4. **AI Data Center Storage Solutions and Key Features**
5. DeepSeek All-in-One Solution

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Complexity of AI Data Processing

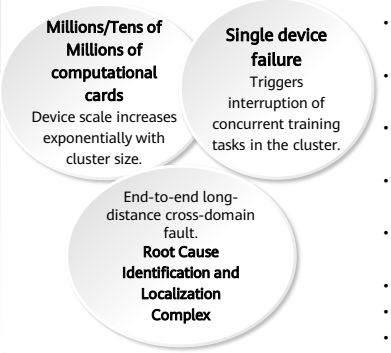


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Challenges of Storage in AI Scenarios 1: Low Cluster Availability

- AI training clusters range from 10,000 to 100,000 computational cards, but available computational capability cannot scale linearly.

The failure rate of ultra-large-scale computing clusters is increasing. This significantly reduces the efficiency of large model training.



- xPU Card Failure
- xPU Memory Failure
- Hard Disk Failure
- Hard disk sub-health
- Optical module failure
- Link failure
- Driver failure
- ...

- During training of a large model, the longest continuous training period is 2.5 days.
- The average failure rate for a 10,000-card cluster is 2 times per day (a 1,000-card cluster experiences 2.4 failures per week), with industry average troubleshooting times ranging from 0.5 to 30 days.

Source: Public data from the internet

Main constraints preventing linear growth of computational capability:

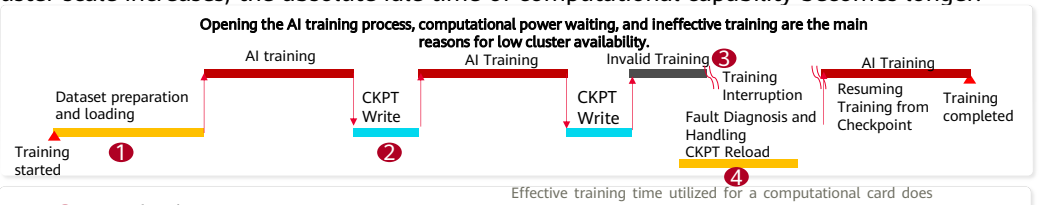
- Increased complexity of task scheduling and reduced efficiency of cluster communication.
- Difficulty in ensuring the reliability of infrastructure such as computing, networking, and storage.
- **Challenges in data synchronization and consistency between nodes have increased.**

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Storage Challenge in AI Scenarios 2: Restricted Storage Cluster Performance

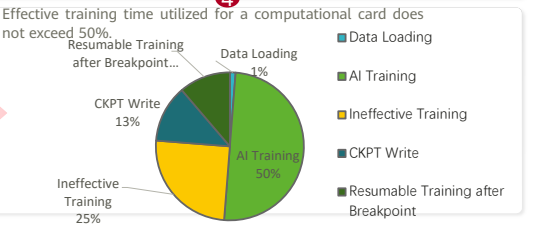
- Dataset loading, CKPT read/write, and fault recovery take more than half of the time. As the cluster scale increases, the absolute idle time of computational capability becomes longer.

AI training process



AI Training Time Analysis

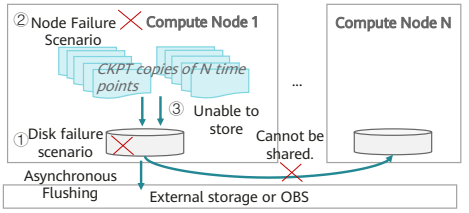
1. **Hour-level** PB-level training dataset loading takes a long time.
2. **0.5-hour** CKPT write time
3. **~2 hours** of ineffective training time
4. **Hour-level** breakpoint/failure recovery time is long



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Challenges of Storage in AI Scenarios 3: Existing Local Disk/Cache Acceleration Solutions Struggle to Meet the Strong Consistency Requirements of Trillion-Level Multimodal Training Data

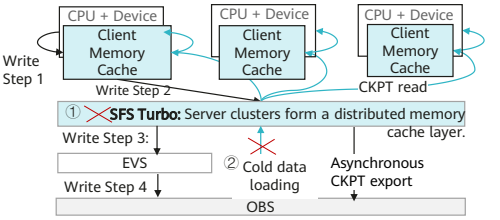
Local Disk Acceleration Solution



1. CheckPoint files are first written from memory to local disks, and then asynchronously flushed to external storage or OBS.
2. Since **local disks cannot be shared**, **disk failures or node failures will cause the entire cluster to lose consistency, making it impossible for the cluster to roll back to the latest durable checkpoint**(wasting N iterations of training time)

Zidong Taichu:"In large-scale cluster scenarios, disk failures and node failures are common occurrences. A single disk failure causing the entire training cluster to wait has a more severe impact than the frequent failures of current optical modules."

Distributed Server Cache Acceleration Solution



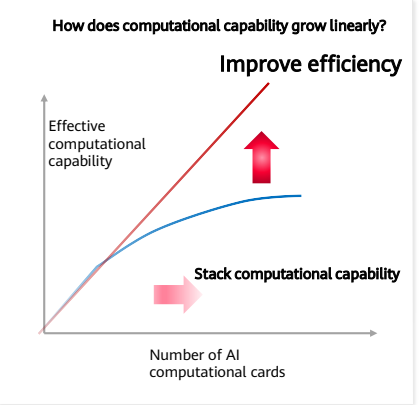
1. CheckPoint files are first written from the training server to the cache server, and then asynchronously flushed to OBS cold storage.
2. The distributed **cache layer lacks reliability guarantees** (no replicas or EC redundancy). When a node failure occurs in the cache layer, files at the latest time point are lost, and the cluster needs to roll back N iterations.

A supercomputing center reported: "Once a node failure occurs in the cache layer, the entire cluster has to roll back and wait, and the performance of reloading data from OBS cold storage is extremely slow, further extending the cluster's waiting time to hours.

Strong Consistency: At any moment, any node, all users query the latest successfully updated data.

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Requirements for AI Data Storage from “Computational Capability" to "Efficiency Improvement"



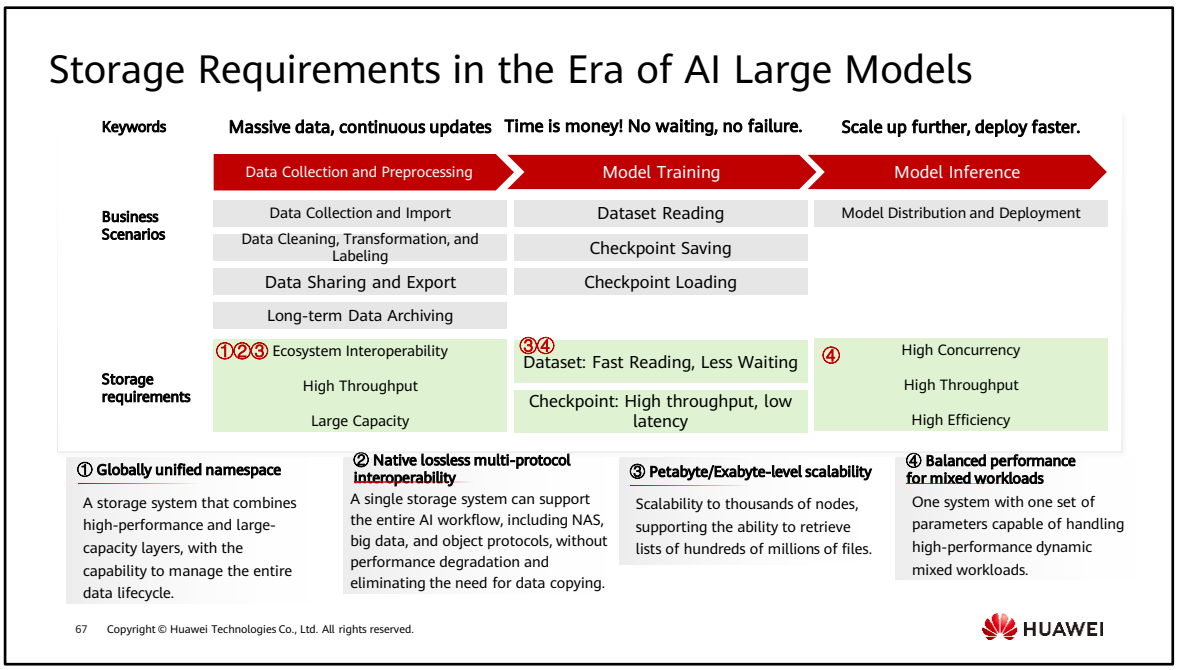
1 Properly configure storage cluster performance to improve computing availability
Switching from stacking computational capability to optimizing full-stack AI training process. The computational capability availability reaches 60% or higher.

2 Choose high-performance, reliable professional external storage
Abandon solutions such as local server disks and cache acceleration to reduce unexpected interruptions in training tasks and improve the availability of computational capability in AI clusters.

3 Build AI-ready data infrastructure
Prepare high-quality industry knowledge and enterprise data assets in advance, establish a data lake, and actively engage in AI model practices.

4 Strengthen data security protection for AI large models
Back up critical enterprise data assets to ensure they can be restored at any time, and strengthen end-to-end ransomware protection systems.

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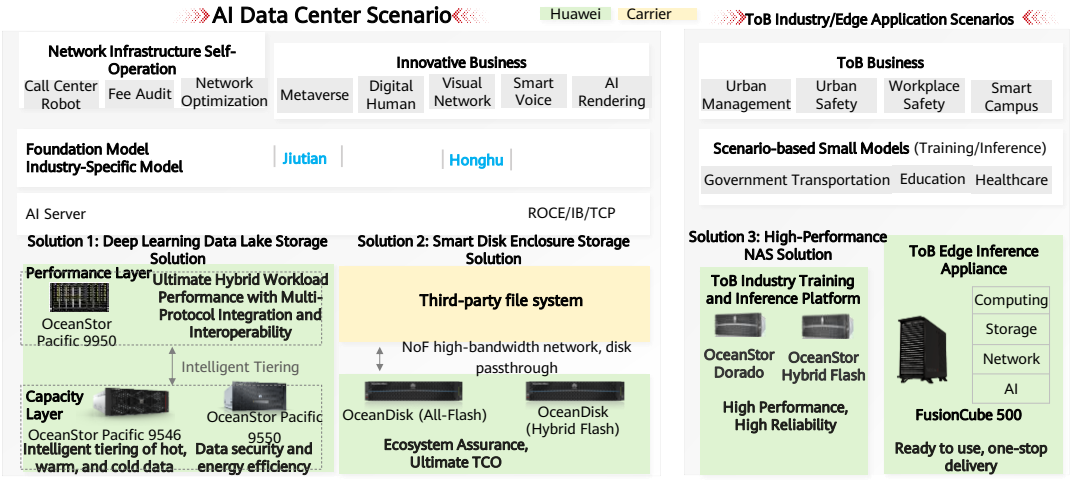


- What kind of storage architecture is the best choice for the era of large AI models?
- The first key feature (global unified namespace): A storage system that integrates both high-performance and large-capacity layers and presents a unified namespace to the outside world, with the capability of managing the entire data lifecycle. ① It allows specifying the placement policy for data when it is first written, enabling different types of data to be stored on different storage layers. ② It supports a variety of data tiering and migration policies, combining access frequency and time-based migration strategies. ③ It has data prefetching capabilities, allowing users to configure preheating policies for specified datasets via commands or APIs to accelerate the cold start speed of planned tasks.
- The second key feature (native lossless multi-protocol interoperability): A storage system that can support the entire AI workflow and simultaneously support protocols required by the AI toolchain, such as NAS, big data, object, and parallel clients. These protocols must be semantically lossless and meet the same ecosystem compatibility requirements as native protocols. In addition, the system must have the capability to efficiently transfer data required for collaboration across different AI processes. At each stage, the toolchain based on different protocol ecosystems should be able to access the same data and metadata. Collaboration between different stages should achieve zero data copying and zero format conversion, allowing the output of one stage to directly serve as the input for the next stage, thus achieving zero wait time for collaborative AI processes.
- The third key feature (PB/EB-level scalability): The system must support

horizontal scalability with thousands of nodes. The system architecture should adopt a fully symmetric design without independent metadata service nodes. As the number of storage nodes increases, the system bandwidth and metadata access capabilities should be able to grow linearly. It should also provide efficient access to hundreds of millions of file lists during the shuffle phase of each epoch in AI training.

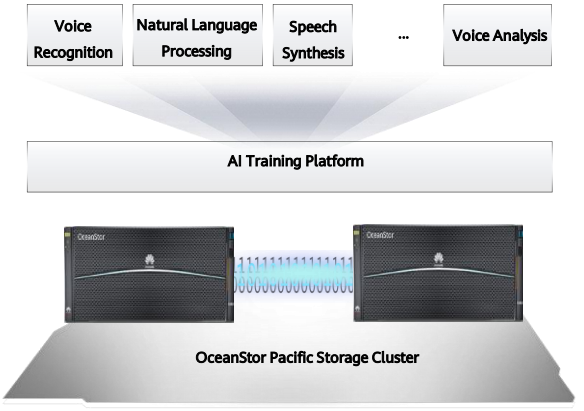
- The fourth key feature (balanced performance for mixed workloads): A single system with a single set of parameters should be capable of handling high-performance dynamic mixed workloads. During the data import phase, both large and small files should be written simultaneously. In the data preprocessing phase, large and small files should be read in batches and processed to generate a massive number of small files. During the model training phase, these small files should be read randomly in batches. When generating checkpoints, the system should support high-bandwidth writes. In the model deployment phase, even when multiple devices concurrently read the same model file, the cluster's aggregated throughput bandwidth should still increase linearly as the number of deployment devices grows.

Typical AI Computing Storage Construction Scenarios



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Huawei OceanStor Pacific collaborates with XX to build a precise and efficient AI training platform



Efficiency improved by 45%

One storage system achieves both high bandwidth and high OPS, improving AI model development efficiency by 45%.

Better cost efficiency

Customizable tiering policies for hot and cold data achieve a balance between storage system performance and capacity; lossless multi-protocol interoperability and zero-copy data transfer accelerate AI model development end-to-end.

Flexible expansion up to EB level

A single cluster can scale up to 4096 nodes, supporting the growth of massive AI training data in the future. The fully symmetric architecture ensures that the storage system's performance improves nearly linearly with increased capacity.

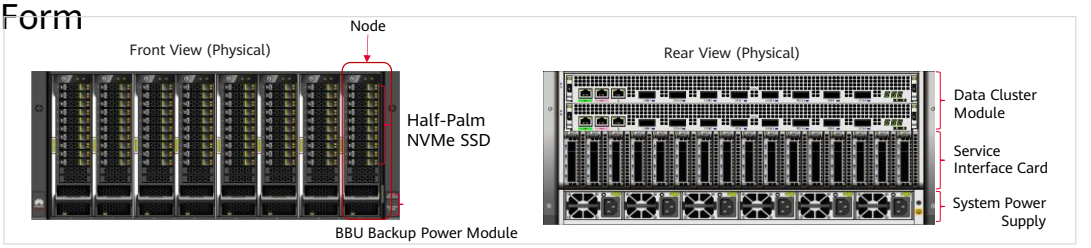
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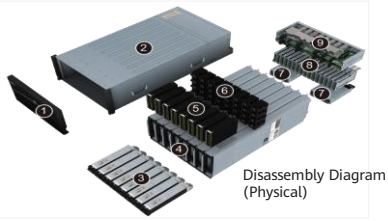
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- Project Background: iFLYTEK is a leading AI enterprise in China, with a national engineering laboratory for speech and language and a national key laboratory for cognitive intelligence. The company has a large-scale computing cluster for AI model development and a wide variety of AI models. The AI training platform has extremely high requirements for storage performance. This project is to procure a complete HPDA solution to meet daily AI training tasks, including enterprise-level distributed storage hardware and an enterprise-level parallel file system.
- Customer Pain Points: Performance Issues: AI training tasks are mainly single-machine multi-computational cards tasks, accounting for about 90% of the workload. The storage system requires high I/O read and write performance, and a single user may concurrently execute multiple tasks. The I/O model is dominated by small files, which are frequently read and written. The current system has low performance for accessing small files, which affects the training cycle and duration. Data needs to be migrated multiple times: from data collection, data preprocessing, model training, to model archiving and backup, the same data is copied multiple times, resulting in low efficiency and high costs.
- Huawei Solution: In this phase of the project, OceanStor Pacific 9540 is used as the data foundation for iFLYTEK's AI training platform (for internal use). In the second phase, the project plans to upgrade to DPC over TCP/IP to further improve the overall system performance.
- Customer benefits: efficiency improved by 45%: A set of storage devices achieves high bandwidth and high OPS, and AI model development efficiency improved by 45%. Cost-effective: Customized tiering policies for hot and cold data balance

between storage system performance and capacity. Lossless multi-protocol interworking, zero data copy, and end-to-end AI model development acceleration; 2.5-fold performance improvement: A single cluster can be expanded to 4096 nodes, supporting the growth of massive AI training data in the future. The fully symmetric architecture ensures that the storage system performance increases linearly as the capacity increases.

OceanStor Pacific 9950 5U High-Density All-Flash Product Physical Form



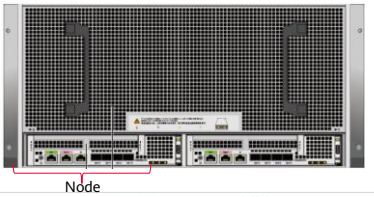


1	Front Panel Cover	2	System Chassis
3	BBU	4	Node
5	Cache Disk Module (Half-Palm NVMe SSD)	6	Fan Module
7	Power Module	8	Interface Card Module
9	Data Cluster Module	10	

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OceanStor Pacific 9550 5U High-Density Hybrid Flash Product Physical Form

Front View (Physical)



Rear View (Physical)





Disassembly Diagram (Physical)

1	Front panel cover	2	Hard disk tray
3	System Chassis	4	Main Storage Hard Disk Module (HDD)
5	Fan Module	6	Power Module
7	Cascade Module	8	Cache Disk Module (Half-Palm NVMe SSD)
9	BBU Module	10	Node

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AI Data Center AI Storage Platform Planning Recommendations

Suggested Ratio of Compute to Storage:

100P computational capability (40 servers * 8 computational cards): Compute-to-storage ratio ≈ 20:1, All-flash to HDD ratio ≈ 2:3

Total Capacity (4.3PB)	=	SSD (1.7PB)	+	HDD (2.6PB)

- **SSD (1.7PB): 32 nodes** OceanStor Pacific 9950 * 10 * 7.68TB NVMe SSD
- **HDD (2.6PB): 5 nodes** OceanStor Pacific 9546 * 60 * 14TB HDD

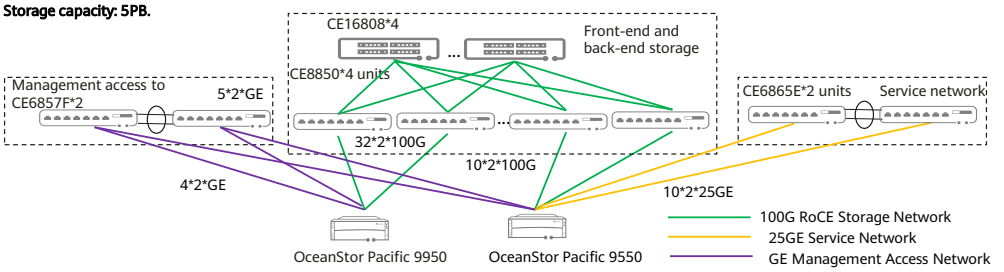
Performance Requirement Suggestions:

100P computational capability : Small file IOPS: > 600,000 IOPS, Bandwidth: > 93 GB/s

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Storage Solution Example

- OceanStor Pacific 9950: 4 chassis with 32 nodes; OceanStor Pacific 9550: 5 chassis with 10 nodes.
- Front-end and back-end networks: 4 x 8850 access switches and 4 x 16808 aggregation switches are used. The uplink and downlink convergence ratio is 1:1.
- Management network: A total of 4 CE6857F switches are deployed in the storage area. Storage nodes connect to the CE6857F switches via optical-electrical conversion modules to meet the access requirements of all storage nodes in phases I and II. Each 9950 chassis has 2 GE ports configured in bond1 mode; each 9550 node has 1 GE port, which is not bonded.
- Service network: A total of 2 CE6865E switches are deployed in the storage area to meet the access requirements of all 9550 storage nodes in phases I and II (reusing the two switches from the resource pool in phase I).
- 9950: One disk pool with 32 nodes (EC 14+2), paired with clustered 8850 TOR switches. 9550: One disk pool with 10 nodes (EC 8+2).
- **Storage capacity: 5PB.**



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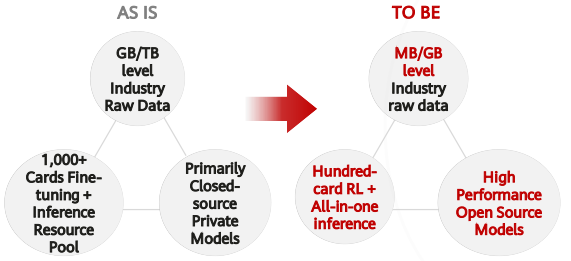
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1. AI Data Center Overview
2. AI Data Center Computational Capability Solutions
3. AI Data Center Network Solutions and Key Technologies
4. AI Data Center Storage Solutions and Key Features
5. **DeepSeek All-in-One Solution**

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DeepSeek accelerates the implementation of large models, quickly meeting industry needs, making all-in-one machines the best choice for small and medium-sized enterprises

DeepSeek's technology for implementing large models: Significantly reduces the requirements and barriers for data, computational capability, and model acquisition.



Balancing safety, efficiency, and personalization
All-in-one machines are the best choice for enterprises deploying DeepSeek.

1 Data Privacy and Security

- Users have full control over data access, usage, and sharing permissions, fundamentally eliminating the risk of data leakage.

2 Rapid Deployment and Ready-to-Use

- Utilizing standardized hardware architecture and software integration, users can quickly start AI projects without complex installation and configuration.

3 Personalization and Flexible Customization

- Users can customize the DeepSeek model capabilities by integrating their own RAG knowledge bases based on business procedures, industry terminology systems, and personalized needs.

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Quiz

1. (Multiple-choice Question) Which of the following are advantages of RoCE networks?
   A. Open Ethernet standard
   B. Low operation and maintenance costs
   C. Intelligent operation and maintenance
   D. High transmission rate

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- Answer: ABCD

Summary

- This chapter begins with the concept of AI data center layering. It first introduces the characteristics and related requirements of AI data center facilities. With the support of large AI models, the power consumption and cooling solutions for AI data centers have entered a new phase. Next, it describes the typical structure and networking solutions of AI clusters.
- The second part presents the computational capability solutions for AI data centers, focusing on AI computational cards.
- The third part discusses the network requirements of large models and the network solutions for AI data centers, with an emphasis on PFC (Priority Flow Control) and ECN (Explicit Congestion Notification). It also introduces key technologies and trends in AI data center networks.
- Finally, it covers the storage requirements of large models and the storage solutions for AI data centers.

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Abbreviations

- IoT: Internet of Things, which is a network that connects all ordinary objects capable of performing independent functions through the Internet and traditional telecommunication networks.
- FPGA: Field Programmable Gate Array, a semi-custom circuit in the field of Application-specific Integrated Circuits (ASIC). It is an advanced product developed based on programmable devices such as PAL, GAL, and EPLD, addressing the shortcomings of custom circuits and overcoming the limitation of the number of logic gates in traditional programmable devices.
- ASIC: Application-specific Integrated Circuit, which is an integrated circuit designed for a specific purpose. It is an IC designed and manufactured according to the requirements of a particular user or for the needs of a specific electronic system.
- ALU: Arithmetic Logic Unit.
- CLB: Configurable Logic Block.
- RAID: Redundant Array of Independent Disks, a technology that combines multiple independent hard disks (physical drives) into a single drive group (logical drive) in different ways to provide higher storage performance and data redundancy compared to a single hard disk.

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Abbreviations

- ROCE: RDMA over Converged Ethernet, a network protocol that allows Remote Direct Memory Access (RDMA) over Ethernet.
- IB: InfiniBand, a switched fabric I/O technology designed to establish a single connection link between remote memory, networks, and servers through a central infrastructure (central InfiniBand switch). The central InfiniBand switch directs traffic, and its tightly integrated design significantly improves system performance, reliability, and efficiency, alleviating data traffic congestion among hardware devices.
- DCQCN: Data Center Quantized Congestion Notification, a congestion control algorithm for RoCEv2.
- DLB: Dynamic Load Balancing.
- CLOS Architecture: Named after Charles Clos, a researcher at Bell Labs, this architecture is a multi-level switching network topology designed for building high-performance, highly reliable data center networks. Typically, the CLOS architecture consists of three layers: input layer, intermediate layer, and output layer. The CLOS switching architecture is strictly non-blocking, re-arrangeable, and scalable.
- DPP: Dynamic Packet Prioritization, dynamically increases the scheduling priority of delay-sensitive small flows to ensure they are sent first without being affected by congestion caused by large flows, thereby guaranteeing the performance of distributed applications.

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把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

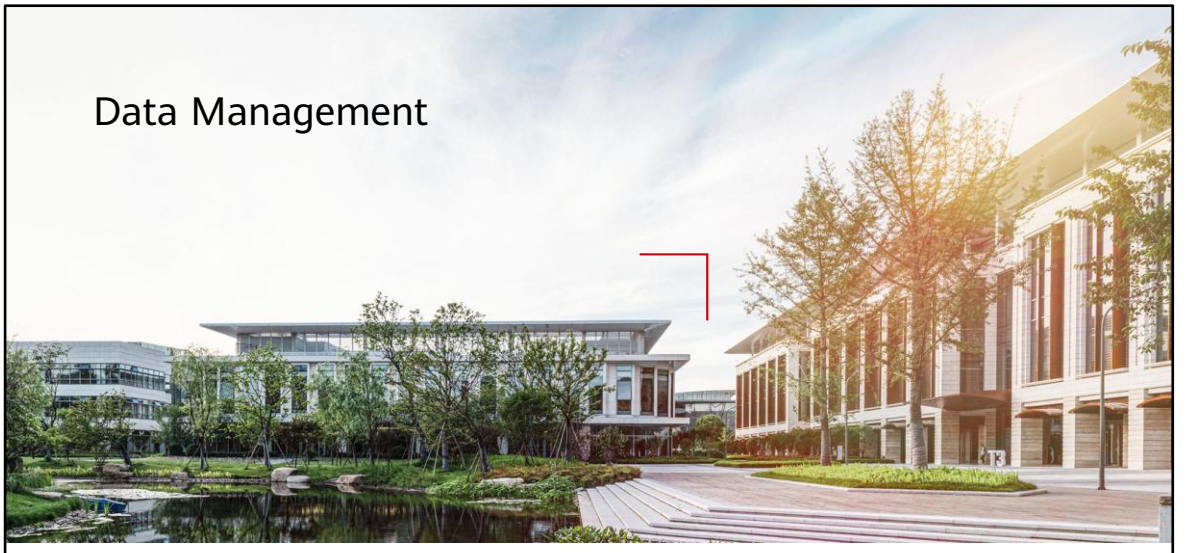
Bring digital to every person, home, and organization for a fully connected, intelligent world.

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Data Management



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Foreword

- This chapter mainly discusses the development and evolution of datasets, standard tools, and training data types in the process of deep learning.

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Objectives

- On completion of this course, you will be able to:
  - Understand the differences between training datasets required for small models and large models
  - Understand commonly used datasets and Labeling tools
  - Understand the evolution of dataset structures

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1. **Differences between Large and Small Model Datasets**
2. Dataset Labeling Tools
3. Data Structures in Deep Learning

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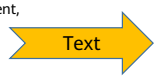
Data: The Spiritual Food for Large Models

- Training large models requires large-scale, high-quality, multimodal datasets, which are typically collected from various fields and multiple data sources. These data can come in various forms such as text, images, audio, and video.





End-to-End Model Production Line
Provides an end-to-end toolchain for model development, training, and inference, with seamless collaboration among DataOps, MLOps, and DevOps, improving development efficiency by 50%.









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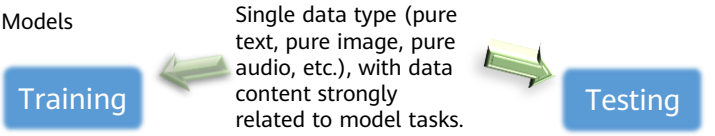
Differences between Large Model Datasets and Traditional Model Datasets

- Datasets required for traditional models:
  - Types of Dataset: Training Sets and Test Sets.
  - Professional Datasets are used for specialized scenarios, with relatively domain-specific content.
  - The datasets size is relatively small, occupying less storage space.
- Datasets required for large models:
  - Types of Dataset: Pre-training Data, Supervised Fine-tuning, Reward Modeling, and Reinforcement Learning. Each stage requires different types of data.
  - A wide variety of data is needed for pre-training.
  - The amount of data required is substantial, requiring significant storage space.

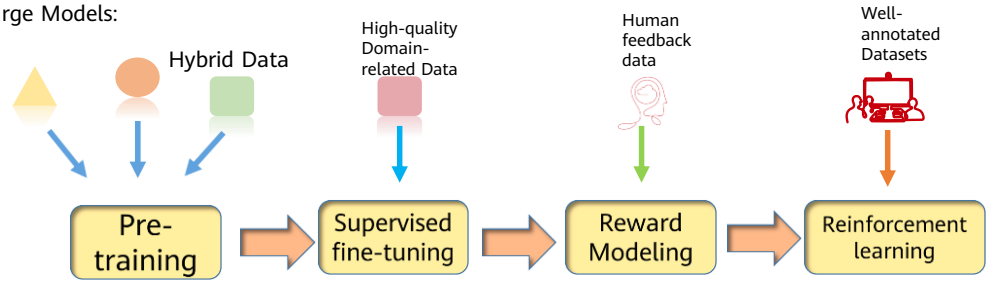
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Differences in Dataset Requirements Across Different Training Phases

- Traditional Models



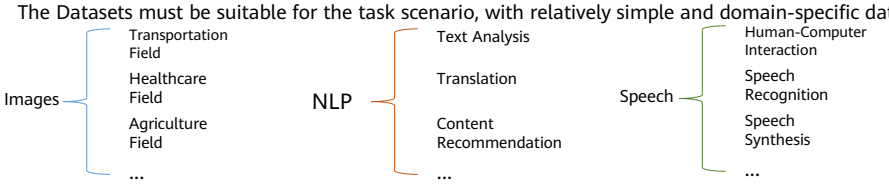
- Large Models:



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Datasets required by traditional models

- The Datasets must be suitable for the task scenario, with relatively simple and domain-specific data content.



- Datasets from different domains:

  ImageNet: Approximately 1,500,000 images, about 150GB after compression.

  COCO: 330,000 images, 80 object categories, 5 captions per image, and 250,000 human portraits with key points, about 25GB after compression.

  IMDB Movie Review Dataset: The training and test sets each contain 25,000 highly polarized movie reviews, approximately 80 MB.

  Wikipedia Corpus: This dataset is a collection of full-text Wikipedia articles, containing nearly 1.9 billion words from over 4 million articles, approximately 20 MB.

  LibriSpeech: This dataset includes a large corpus of about 1,000 hours of English speech, sourced from audiobooks in the LibriVox project, approximately 60 GB.

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Multimodal Dataset

- In traditional deep learning model training, a single type of data is typically used. For example, training NLP models requires corpus data; training object detection models requires image or video data...
- Large models use multimodal data, employing both images and corpus data during training to enable the model to generate images from text.

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GPT Pre-training Dataset

- In 2022, OpenAI released ChatGPT, which sparked a revolution in the AI field, bringing the GPT series models back into the spotlight. However, ChatGPT and the new GPT-4 have not been open-sourced.
- Based on the token data disclosed by OpenAI, we can compile the datasets used for training GPT-3:
  - **Common Crawl Dataset:** This is a web crawling dataset spanning over eight years, containing raw web page data, metadata extraction, and text extraction. As of version 2023-40, the entire dataset occupies approximately 135 TiB of storage space. During training, OpenAI filtered this dataset and extracted high-quality data to train the model. The selected data is about 570 GB, accounting for roughly 60% of GPT-3's training data.
  - **WebText2:** This is an internal OpenAI corpus created by crawling web pages, emphasizing document quality. It accounts for about 22% of the training data.
  - **Books:** Story-based books consist of two main categories—fiction and non-fiction—and are primarily used to train models on storytelling and response capabilities, accounting for approximately 16%.
  - **Wikipedia:** A free, multilingual, collaborative online encyclopedia, mostly consisting of expository text with rigorous content, covering multiple fields, accounting for about 3%.

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LLaMA Pre-training Dataset

- An open-source large language model. It is highly capable and can perform functions such as common-sense reasoning, reading comprehension, joke explanation, code generation, and more. Similarly, its pre-training dataset comes from a wide range of sources and is very large in scale.

Dataset	Sample Ratio	Epochs	Disk Size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

LLaMA2 Training Data

- Sample Ratio: The proportion of this type of data in the total training samples.
- Epochs: The number of training rounds used for each data item.
- Disk Size: The amount of disk space required to store this type of dataset.

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ChatGLM-6B

- ChatGLM-6B is an open-source chat language model that supports both Chinese and English. It uses similar technology to ChatGPT and has been optimized for Chinese question-answering and chat tasks.
- Although ChatGLM has not disclosed the specific datasets used, it has been trained on approximately 1 trillion tokens of bilingual (Chinese and English) data, supplemented by techniques such as supervised fine-tuning, self-supervised learning, and reinforcement learning with human feedback. With 6.2 billion parameters, ChatGLM-6B is capable of generating responses that are quite aligned with human preferences.

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GLM-130B – Journey of a Trillion-Parameter Model

- GLM-130B is an open-source, bilingual (Chinese and English) bidirectional dense model with 130 billion parameters, using a general language model architecture.

- As of July 3, 2022, GLM-130B has been trained on 400 billion text tokens.
  - Chinese: 200 billion tokens
  - English: 200 billion tokens

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Data Labeling Tools

- In fields such as machine learning and natural language processing, the training of most models requires the use of large amounts of data. This data can be divided into two types: Labeled Datasets and Unlabeled Datasets.
- Labeled datasets are suitable for training supervised learning algorithms. These datasets have already been manually or automatically labeled with different categories or tags. For example, in an image classification problem, a labeled dataset might consist of tens of thousands of images, each labeled with its corresponding category (e.g., "cat," "dog"). Labeled datasets can be used to train supervised learning algorithms, such as classification, regression, and other models.
- Data labeling refers to the process of using specific software tools to classify and label data content, such as images, audio, text, and videos, with standard feature labels (e.g., classifying images into categories like "cat," "dog," "bird," etc.), making it easier for machine learning algorithms to understand and process quickly.

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Data Labeling Tool in NLP - NLTK

- NLTK (Natural Language Toolkit) provides an easy-to-use interface for over 50 corpora and lexical resources (such as WordNet) and also offers a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning.
- NLTK is a well-known natural language processing library for Python, which comes with its own corpus and includes features such as part-of-speech tagging, tokenization, and more.
- Breaking long sentences into meaningful components:

```
>>> import nltk
>>> sentence = "hello,,world"
>>> tokens = nltk.word_tokenize(sentence)
>>> tokens
['hello', ',', ',world']
```


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Labeling Tools in the CV Field

- LabelImg is a visual image labeling tool. It is written in Python and uses Qt for its graphical interface. Annotations are saved as XML files in the PASCAL VOC format (used by ImageNet). In addition, it also supports the YOLO format. Datasets required for object detection networks such as Faster R-CNN, YOLO, and SSD can be annotated using this tool to label objects in images.
- Many datasets provide pre-annotated training datasets, for example:
  - COCO Dataset
  - ImageNet Dataset
  - ……

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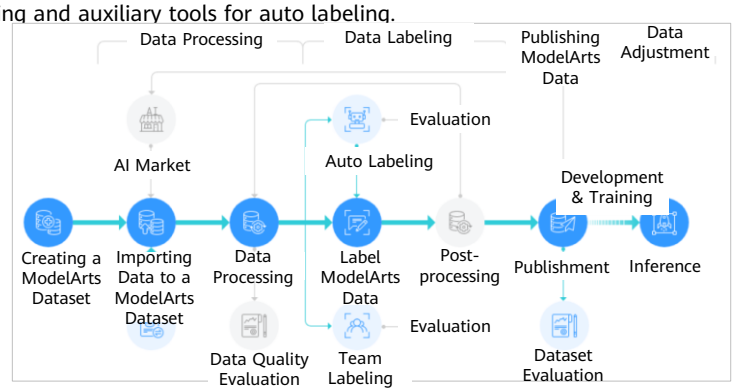
Praat Phonetics Software

- Praat, originally named Praat: Doing Phonetics by Computer, is a cross-platform, multi-functional professional software for phonetics.
- The main functions of Praat include the acquisition, analysis, and labeling of digitized speech signals, as well as performing various processing tasks such as transformation and filtering.

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ModelArts Data Management

- The ModelArts data framework includes data collection, data filtering, data labeling, and dataset version management functions. It supports both automated and semi-automated data filtering, as well as auto labeling and auxiliary tools for auto labeling.



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ModelArts Data Labeling - Manual Labeling

- Since model training requires a large amount of labeled data, data needs to be labeled before model training. ModelArts supports manual labeling for different types of datasets.
  - **Image Classification**: Label the category of an image.
  - **Object Detection**: Label the object categories and their locations in an image.
  - **Text Classification**: Label the category of a text.
  - **Named Entity**: Label the entity segment in a text. Examples include "time" and "location."
  - **Text Triplet**: Label the entities in a text and the relationships between them.
  - **Sound Classification**: Label the category to which an audio belongs.
  - **Speech Content**: Label the speech content within an audio.
  - **Speech Segmentation**: Label the category and speech content of a specific audio segment.
  - **Video Labeling**: Label the categories and locations of objects in video frames.



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ModelArts Data Labeling - Auto Labeling

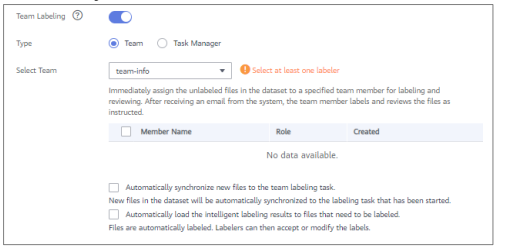
- In addition to manual labeling, ModelArts also provides auto labeling that allows to quickly complete data labeling, saving more than 70% of the labeling time. Auto labeling refers to the process of learning and training based on the labels and images at the current labeling stage, using existing data in the system.
  - Currently, auto labeling function only supports "Image Classification" and "Object Detection".
  - When starting auto labeling, the labeling job must contain at least two types of labels, and each label must have at least five labeled images. There must be unlabeled images available, and it is also required that no auto labeling tasks are currently running in the system.

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ModelArts Data Labeling - Team Labeling

- Data labeling tasks are usually completed by a single person. However, when dealing with large datasets, multiple people are required to complet the task. ModelArts provides team labeling feature that allows a labeling team of multiple people to label the same dataset.
- Currently, team labeling feature only supports the following types: Image Classification, Object Detection, Text Classification, Named Entity Recognition.
- The team labeling feature is managed on a team basis. When enabled, you must specify a team, which can have multiple members.
- An account can add up to 10 teams.

Create Team Labeling Job



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Baidu EasyData Data Labeling

- EasyData is an intelligent data service platform launched by Baidu Brain, providing comprehensive data related services that cater to the needs of AI development, including data collection, data cleaning, and data labeling.
- Data Labeling
  - Pre-configured rich labeling templates
    - Image: Image classification, object detection, image segmentation
    - Text: Text classification, short text similarity analysis, sentiment analysis, text entity extraction
    - Audio: Sound classification
    - Video: Video classification
  - The platform also supports intelligent labeling and multi-user labeling.

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Huggingface

- Huggingface is an open-source community and company dedicated to advancing Natural Language Processing (NLP) technologies and tools. The platform is committed to providing the latest technologies, models, and tools for various NLP tasks, as well as offering developers a convenient way to use, fine-tune, and deploy these technologies.
- Huggingface offers a variety of datasets:
  - Multimodal
  - Computer Vision
  - Natural Language Processing
  - Audio
  - Tabular
  - Reinforcement Learning

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- Huggingface is an open-source community and company dedicated to advancing natural language processing (NLP) technologies and tools. They are committed to providing the latest techniques, models, and tools for various NLP tasks, as well as offering developers convenient ways to use, fine-tune, and deploy these technologies. Huggingface's contributions to the NLP field have been widely recognized, making it an important resource for many developers and researchers. In addition to natural language processing, they also support multimodal tasks such as image and audio processing, and the community offers a vast collection of pre-trained models and datasets.

Transformers Library

- The Transformers library focuses on providing pre-trained models and related tools used in various Natural Language Processing (NLP) tasks.
- The goal of this library is to enable developers to easily use and fine-tune pre-trained NLP models to solve a wide range of text processing tasks, such as text classification, named entity recognition, text generation, and more.
- The Transformers library offers Tokenizer module. It provides the functionality of tokenizers, preprocesses data, and facilitates the conversion between text and token sequences.

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Structured Data

- Structured data is data that uses predefined and expected formats. It is typically presented in tabular form, where rows and columns clearly define data attributes. Due to its quantitative nature, computers can efficiently process structured data.
- For example, a structured customer data table containing columns such as name, address, and phone number can provide information like the total number of customers and the region with the most customers.
- Examples of structured data: Excel files, SQL databases, product catalogs...
- Advantages:
  - Easy to use: Structured data can be quickly understood and accessed. Operations such as updating and modifying structured data are very simple. It is highly storage-efficient because fixed-length storage units can be allocated for data values.
  - Highly scalable: Structured data scales algorithmically. As the size of datasets increases, storage and processing capabilities can be expanded. Modern systems that handle structured data can scale to thousands of TB of data.
  - Easy to analyze: Structured Query Language (SQL) can be used to generate reports and to modify and maintain data. Structured data is also useful for big data analytics.
- Disadvantages:
  - Limited usage, lack of flexibility, and difficulty in scaling.

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Unstructured Data

- Unstructured data exists in an absolutely raw form and generally refers to information that has not been modeled or organized in a predefined manner.
- Unstructured data can include text files, images, audio, video, etc., such as social media posts, chats, satellite images, IoT sensor data, emails, and presentations...

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Semi-structured data

- Semi-structured data refers to data that is captured or formatted in a non-traditional way. It does not follow the tabular data model or the format of relational databases because it lacks a fixed schema. However, the data is not entirely raw or unstructured; instead, it contains some structured elements, such as tags and organizational metadata, which make it easier to analyze. The advantage of semi-structured data is that it is more flexible and easier to scale compared to structured data.
- An example of data in a semi-structured format is a delimited file. It contains elements that can break down the data into separate hierarchies. Similarly, in digital photographs, the image itself has no predefined structure, but possesses certain structural attributes that make it semi-structured. For instance, when taking a photo with a smartphone, it will have some structured attributes such as geolocation, device ID, and date-time stamps. After saving them, labels (e.g., "pet" or "dog") can be assigned to the image to provide additional structure.
- Semi-structured data lies between structured and unstructured data. Due to the lack of a specific relational or tabular data model, semi-structured data cannot be considered fully structured. Compared to unstructured data, semi-structured data is considered easier to obtain information and insights from. However, it does not possess the same level of information integrity or reliance on predefined data models as structured data. Nevertheless, it does contain metadata that can be analyzed, such as tags and other markers.

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Quiz

1. (Multiple-choice Question) Which of the following options belong to unstructured data?
   A. Voice
   B. Images
   C. Text
   D. JSON

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- Answer:ABC

Summary

- This chapter mainly introduces the differences between training datasets for small models and large models, and provides a detailed explanation of the data structures used in deep learning, especially during the training of large models.

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Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

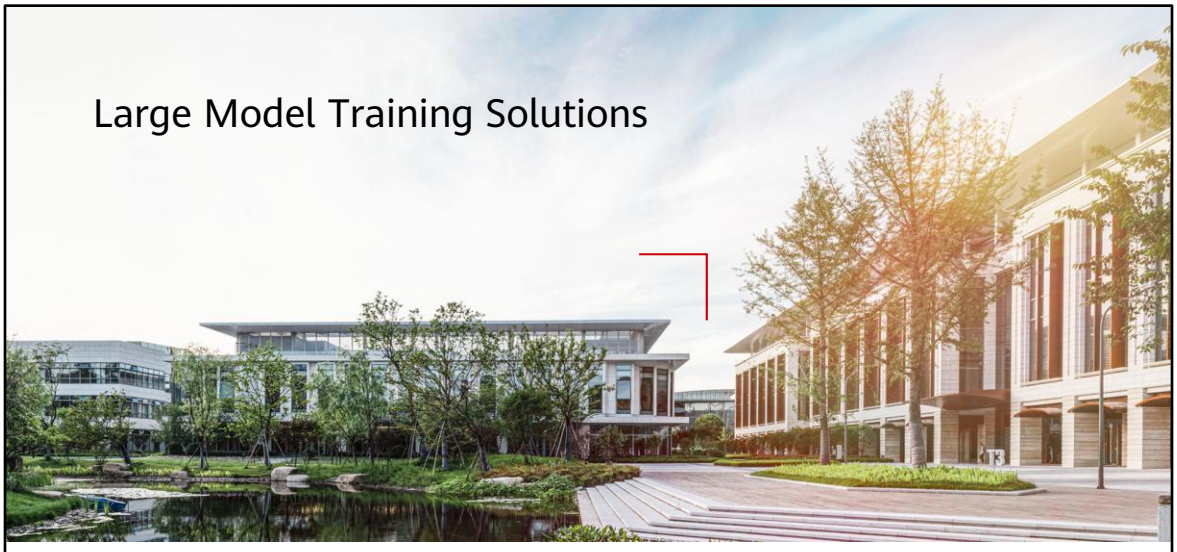
Bring digital to every person, home, and organization for a fully connected, intelligent world.

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Large Model Training Solutions



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Foreword

- Enterprises are facing unprecedented systems engineering challenges during the implementation of large models. From TB-level memory management to complex parallel computing architectures, from high training costs to scarce high-quality data, each phase tests the comprehensive strength of technical teams. This chapter describes the toolchains and policies used during large model training.

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Objectives

- Upon completion of this course, you will be able to understand:
  - Features of large model training
  - TensorFlow and PyTorch development frameworks
  - Large model training solutions

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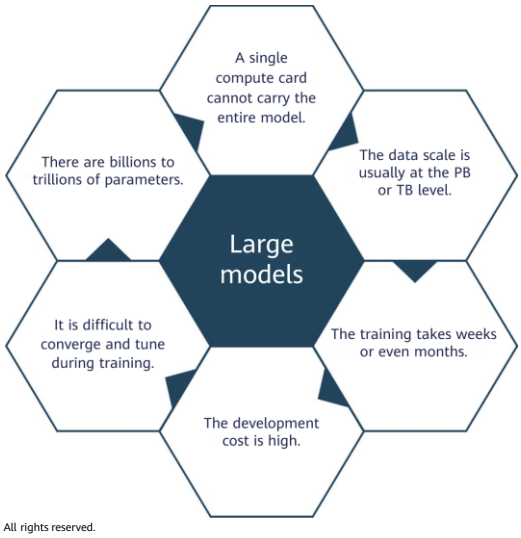
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2. Large Model Training Solutions
3. Large Model Fine-Tuning Solutions

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Features of Large Model Training



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Parameter Quantities of Different Large Models

Model Name	Number of Parameters
Qwen3-32B	32B
Qwen3-235B-A22B	235B/22B
GLM-4.5 (MoE)	355B/32B
DeepSeek-V3 (MoE)	671B/37B
Llama 3.1-70B	70B
Llama 4 Maverick	400B/17B
Falcon	40B
Kimi K2-1T	1T/32B

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Requirements for Memory (Video RAM) Resources

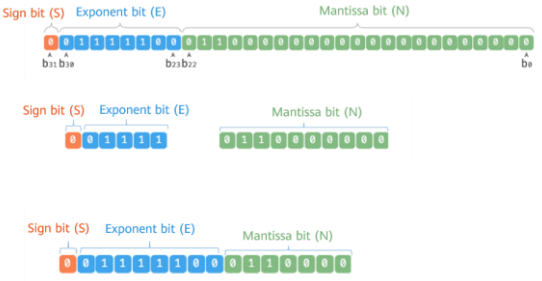
- Large models' requirements on the video RAM are one of the core challenges for deploying and running large models. The video RAM requirements depend not only on the model parameter scale, but also are closely related to the accuracy, inference/training scenarios, and optimization technologies.
  - Take a model with 7B parameters as an example. The following table lists the required memory (video RAM) capacity in the inference scenario.

Data Type	Number of Bytes Occupied by Each Parameter	(Theoretical) Video RAM
FP32	4	28 GB
FP16	2	14 GB
BF16	2	14 GB
INT8	1	7 GB

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Data Precision

- A float32 (FP32) number generally occupies 4 bytes, that is, 32 bits in total. It consists of 1 sign bit, 8 exponent bits, and 23 mantissa bits.
- FP16 is a 16-bit binary floating-point number representation method. It uses 16 bits to encode and store data, including 1 sign bit, 5 exponent bits, and 10 mantissa bits.
- Both FP32 and BF16 have 8 exponent bits. However, the decimal places (mantissa bits) of BF16 are reduced to 7. This design enables BF16 to maintain a certain level of precision while greatly lowering storage requirements and computing costs.
- INT8 and INT4 are signed integer types, each of which has 1 sign bit. The range of data that can be represented by them is small. The two types are usually used for model quantization.



- FP32 parameters can provide high model accuracy, but result in high computing and storage costs. Reducing the bit width causes a decrease in model accuracy.

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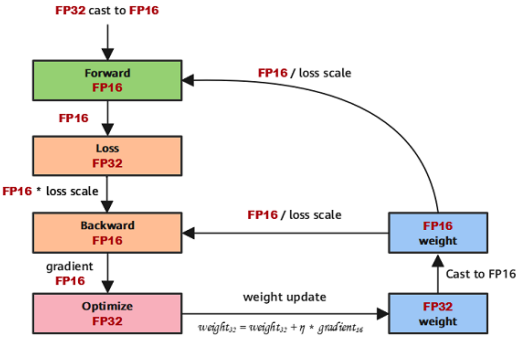
- FP32 is a data type and a single-precision floating-point data format used in deep learning. It is used to represent 32-bit floating-point numbers. In computers, both numbers and characters need to be stored in binary form, and floating-point numbers are binary numbers that can represent decimals. An FP32 number occupies 4 bytes, that is, 32 bits in total. It consists of 1 sign bit, 8 exponent bits, and 23 mantissa bits.
- FP16, a half-precision floating-point format, is a 16-bit binary floating-point number representation method widely used in fields such as computer graphics and deep learning. FP16 uses 2 bytes (16 bits) for encoding and storage, including 1 sign bit, 5 exponent bits, and 10 mantissa bits. This representation method enables FP16 to demonstrate obvious advantages in storage and transmission. It can store a large amount of data in a small storage space.
- Both FP32 and BF16 have 8 exponent bits. However, the decimal places (mantissa bits) of BF16 are reduced to 7. This design enables BF16 to maintain a certain level of precision while greatly lowering storage requirements and computing costs. Although the precision of BF16 is slightly lower than that of FP32, this precision tradeoff is acceptable in many deep learning applications, especially in computer vision and speech scenarios. BF16 is almost equivalent to FP32 in the value range. This helps avoid the overflow and underflow issues that are prone to occur when FP16 is used. However, the mantissa precision of BF16 is low, meaning that BF16 may be less accurate than FP16 in some cases. However,

as deep learning models are usually much more sensitive to exponent bits than to mantissa bits, BF16 can provide prediction precision close to that of FP32 in applications. In terms of performance, both BF16 and FP16 have high throughput and can significantly lower memory requirements. The formats can be selected based on specific application requirements and hardware support.

- The storage and computing requirements of FP16 are halved compared with those of FP32, and the precision loss may be about 2%. In INT8 quantization, the model size is reduced by 75%, the inference speed is improved by 2–4 times, but the error increases.

Mixed Precision Training

- Mixed precision training refers to a computation policy in which different numerical precisions are used for different computation operations of neural networks during training. Some neural network computations are insensitive to the numerical precision. In this case, using a low precision can significantly accelerate training (such as conv and matmul). However, in some neural network computations, the input and output values differ greatly, and high precision is required to ensure the correctness of the result (such as log and softmax).



FP16 and FP32 are used as examples.

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- Parameters are stored in FP32 format.
- During forward propagation, if there are FP16 operators, the operator inputs and parameters need to be cast from FP32 to FP16 for computation.
- Set the loss layer to FP32 for computation.
- During backpropagation, the value is multiplied by the loss scale first to avoid underflow caused by an extremely small backward gradient.
- FP16 parameters are involved in gradient computation, and the result is cast back to FP32.
- The value is divided by the loss scale to restore the enlarged gradient.
- The optimizer checks whether the gradient overflows. If yes, the optimizer skips the update. If no, the optimizer updates the original parameters using FP32.

Factors Affecting Memory (Video RAM) Resources of Large Models

Factor	Impact Mechanism	Example
Model parameter scale	The number of parameters directly determines the video RAM usage. Each time 1 billion parameter are added, the required video RAM space increases by 4 GB when FP32 precision is used.	28 GB video RAM is required to store a 7B model in FP32 precision.
Data precision	FP32 (4 bytes) > FP16/BF16 (2 bytes) > INT8 (1 byte). A lower precision indicates that the required video RAM space is smaller.	After INT8 quantization, the video RAM space required by the 7B model is reduced from 28 GB to 7 GB.
Inference/Training	Gradients and optimizer states need to be stored during model training. The video RAM space required for training is 2–3 times that required for inference.	The video RAM space required for training a 7B model may exceed 80 GB in FP32 precision.
Intermediate activations	The intermediate results (such as the attention matrix of the Transformer) generated during forward propagation occupy extra video RAM.	During the inference of a 7B model, the intermediate activations may occupy 10–20 GB.
Optimization technology	Technologies such as mixed precision training, gradient checkpointing, and tensor parallelism can significantly lower the demand for the video RAM.	The demand for the video RAM can be reduced by 50% by using FP16 mixed precision training.

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Requirements on the Training Mode - Distributed Parallelism

- During model training, the parameters and gradients of the model need to be stored. This requires a memory space of 2 Ψ + 2 Ψ. During FP32 mixed precision training, Adam needs an FP32 model copy, momentums, and variances to store the optimizer states of the model. This requires a memory space of 4 Ψ + 4 Ψ + 4 Ψ. In summary, 16 ΨB memory is required for training. That is, 24 GB memory is required for training a GPT-2 model.

	Category	Size (Ψ indicates the number of model parameters.)	Total Size
Content that must be stored	Parameters (FP32)	4Ψ	12Ψ
	Momentums (FP32)	4Ψ	
	Variances (FP32)	4Ψ	
Intermediate values	Parameters (FP16)	2Ψ	4Ψ
	Gradients (FP16)	2Ψ	
Total			16Ψ

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Training Data Requirements of Large Models

- Large models generally have hundreds of millions to trillions of parameters. To effectively adjust these parameters so that models can accurately execute specific tasks (such as image recognition and natural language processing), a large amount of training data is required to provide sufficient "signals" to guide the learning process.

Number of Parameters	Number of Tokens (1 trillion tokens are about 2,000 GB, depending on the number of bytes)
DeepSeek-V3/R1	14.8 T
LLaMA 33B	1.4 T
LLaMA3	15 T
Falcon 40B	1 T

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Computing Resource Requirements of Large Model Training

$$\text{Training time} \approx \frac{\text{Coefficient} \times \text{Number of tokens} \times \text{Number of model parameters}}{\text{Number of computational cards} \times \text{Peak FLOPS of computational cards} \times \text{computational card utilization}}$$

The coefficient is 6 or 8.
For each token and each model parameter, two floating-point operations need to be performed.
If the activation recomputation technology is used to reduce the video RAM usage of intermediate activations, an additional forward propagation is required. Therefore, the sum of the coefficients of forward propagation, backpropagation, and activation recomputation is 4 (1 + 2 + 1). If the activation recomputation technology is not used, the sum of the coefficients is 3 (1 + 2).
Therefore, in one training iteration, 8 (2 × 4) or 6 (2 × 3) floating-point operations need to be performed.
The computational card utilization ranges from 0.3 to 0.55.

Parameter Quantity (P) (B)	Training Phase	Data Volume (T) (Billion Tokens)	Number of Cards (n)	Training Duration (Days)
175 (e.g., GPT-3)	Pre-training	3,500	8,192	49
	Secondary training	100	2,048	5.5
65 (e.g., LLaMA)	Pre-training	1,300	2,048	27
	Secondary training	100	512	8
13 (e.g., LLaMA)	Pre-training	1,000	256	34
	Secondary training	100	128	7

Take the GPT-3 model with 175 billion parameters and 3,000 billion tokens as an example. The time required for training this model on 1,024 cards with the computing power of 312 TFLOPS is as follows:

$$\frac{8 \times (300 \times 10^9) \times (175 \times 10^9)}{1024 \times (312 \times 10^{12}) \times 0.45} \approx 2921340\ seconds \approx 34 days$$

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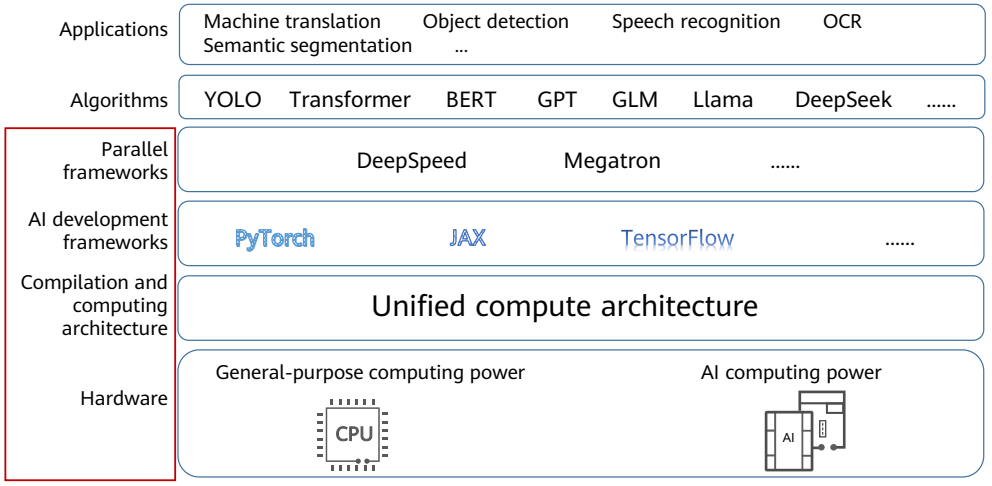
- Utilization: For an 80 GB card with a theoretical computing power of 312 TFLOPS, Megatron can use the tensor parallelism and pipeline parallelism technologies to achieve a utilization of 51.4%, that is, 160 TFLOPS per second.

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AI System Architecture



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Solution Overview

AI training needs to consider the development framework, parallel training policy, and computing hardware configuration.



Development framework

Select a proper deep learning framework and toolchain to quickly build AI training tasks.



Parallel training

Select a proper parallel training policy to push the computing limit of a single card and implement ultra-large-scale model training

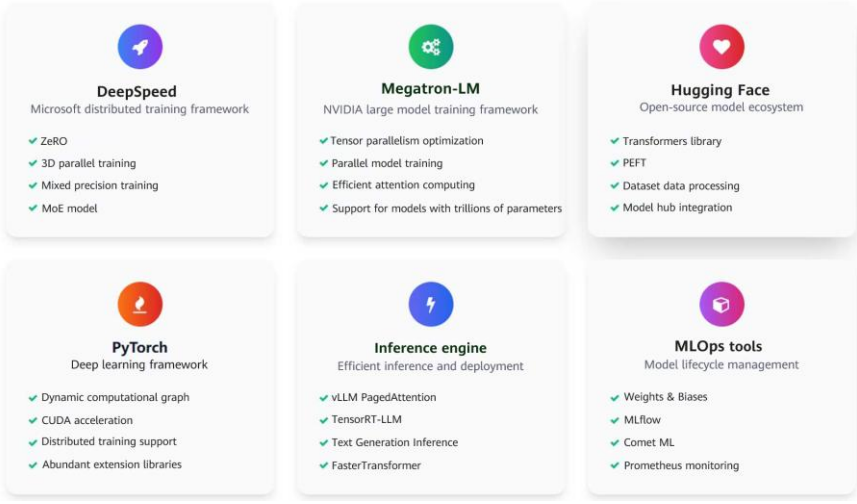


Computing hardware

Configure high-performance computing clusters and optimize the storage and network architectures to ensure training efficiency and stability.

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Open-Source toolchains



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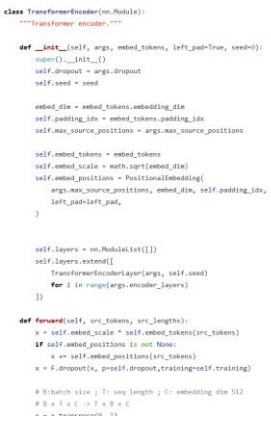
Challenge: Algorithm Implementation

Deep learning algorithms

Llama, DeepSeek, GLM, GPT, Stable Diffusion, and more

Code implementation

Based on AI development frameworks



```
class TransformerEncoder(nn.Module):
    """Transformer encoder."""

    def __init__(self, args, embed_tokens, left_pad=True, seed=0):
        super().__init__()
        self.dropout = args.dropout
        self.seed = seed

        embed_dim = embed_tokens.embedding_dim
        self.padding_idx = embed_tokens.padding_idx
        self.max_source_positions = args.max_source_positions

        self.embed_tokens = embed_tokens
        self.embed_scale = math.sqrt(embed_dim)
        self.embed_positions = PositionalEmbedding(
            args.max_source_positions, embed_dim, self.padding_idx,
            left_pad=left_pad,
        )


        self.layers = nn.ModuleList([])
        self.layers.extend([
            TransformerEncoderLayer(args, self.seed)
            for i in range(args.encoder_layers)
        ])

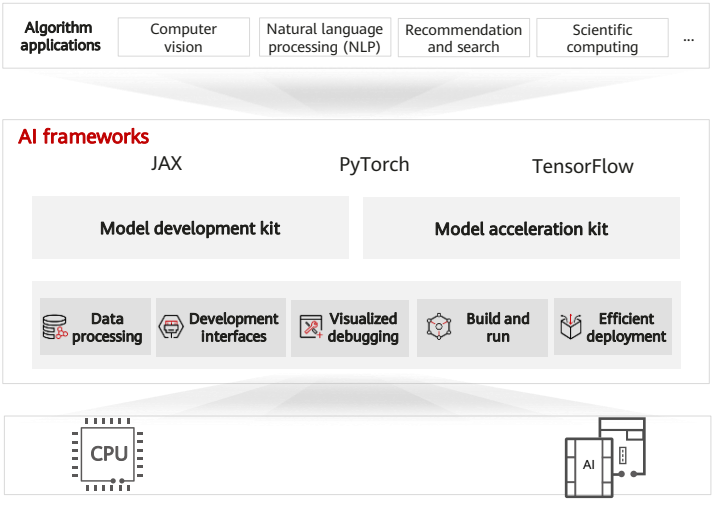
    def forward(self, src_tokens, src_lengths):
        x = self.embed_scale * self.embed_tokens(src_tokens)
        if self.embed_positions is not None:
            x += self.embed_positions(src_tokens)
        x = F.dropout(x, p=self.dropout,training=self.training)

        # B:batch size ; T: seq length ; C: embedding dim 512
        # B x T x C -> T x B x C
```

- Algorithm development efficiency
- Hardware platform compatibility
- Operator types and qualities
- Toolchain
- Multi-device deployment

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Functions of AI Development Frameworks



01. Data preprocessing

Preprocess data (data conversion and augmentation) in different formats, such as audio, video, image, and text.

02. Development interfaces

Provide unified interfaces for developers throughout the AI development process to accelerate scientific research and development.

03. Debugging and tuning

Provide visualized debugging and tuning capabilities for developers to view and trace the model training process.

04. Compilation and execution

Compile user code and implement high-performance training through full-stack hardware-software synergy.

05. Inference deployment

Implement model delivery and computing scheduling, support quick deployment by industry customers, and maximize the computing power of AI chips.

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- Deep learning frameworks lower the entry barrier for AI development. Developers do not need to compile code starting from complex neural networks and backpropagation algorithms. Instead, they can use existing models to configure parameters as required, and the model parameters can be automatically trained. In addition, developers can add custom network layers to existing models, and select the required classifiers and optimization algorithms.

PyTorch

- PyTorch is a machine learning computing framework released by Meta. Its predecessor, Torch, was a scientific computing framework with extensive support for machine learning algorithms and functioned as a tensor operation library similar to NumPy. While Torch was highly flexible, its adoption was limited due to its use of Lua, a less mainstream programming language. This led to the rise of PyTorch, which is based on Python.
- PyTorch is one of the mainstream deep learning frameworks.

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Advantages of PyTorch

Advantage Category	Core Feature	Description	Differentiated Value
Dynamic computational graph	Building computational graphs during runtime	Control flows such as **if** and **for** are supported to facilitate the debugging of complex models (such as variable-length sequence processing).	The dynamic computational graph is more flexible than the static graph of TensorFlow, and is suitable for fast iteration in the research phase.
Usability	Native support for Python and simplified API design	NumPy-like operation interfaces enable model definition, training, and inference to be completed using only 10 lines of code.	The learning curve is smooth. Developers can focus on algorithm innovation instead of framework syntax.
Community ecosystem	Open-source and active	PyTorch has been starred more than 800,000 times on GitHub. The pre-trained model libraries are provided by head organizations such as Hugging Face.	Comprehensive third-party toolchains (such as TorchVision and Lightning) are available, forming a closed-loop ecosystem comprising frameworks, libraries, and models.
Production deployment	TorchScript static graph conversion	PyTorch supports model serialization and C++ export, and is compatible with TensorRT optimization.	The flexibility of dynamic graphs and the performance of static graphs are balanced. The solution can be deployed on the cloud or edge.
Research field adaptation	Automatic differentiation using Autograd	Dynamic graphs support high-order operations (such as neural architecture search) and transparent gradient computation.	Compared with the static graph framework, PyTorch is more suitable for exploring cutting-edge algorithms (such as GAN and reinforcement learning).
Cross-platform support	Compatibility with multiple operating systems and hardware	PyTorch supports the Windows, Linux, and macOS operating systems, and adapts to mainstream computing hardware.	PyTorch provides the mobile deployment solution (PyTorch Mobile) to cover all device scenarios.
Scalability	Modular design	Custom operators and loss functions can inherit from nn.Module, C++ extension is supported.	The dynamic graph extension of PyTorch is more intuitive than the static graph extension of TensorFlow.

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PyTorch Community Ecosystem

- Extensive pre-trained model libraries: PyTorch is deeply integrated into libraries such as Hugging Face Transformers to provide a large number of open-source models (such as GPT and BERT) and support quick fine-tuning and deployment.
- Strong community support: Developers can obtain a large number of tutorials, tools, and pre-training weights through channels such as GitHub and forums, lowering the development barrier.
- Industrial-grade toolchains: PyTorch can be used with optimization libraries such as DeepSpeed and FairScale to further improve training efficiency (such as ZeRO and 3D parallelism).

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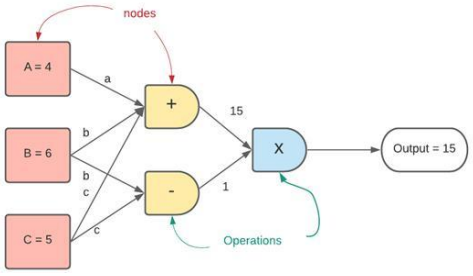
TensorFlow (1)

What is TensorFlow?

An open-source deep learning framework developed by Google. It uses the computational graph model for numerical computing.

Development history:

- It was open-sourced by Google in 2014
- It adopted static graphs in the early stage (1.x era).
- Version 2.x supporting dynamic graphs was released in 2019.
- Its ecosystem is continuously improved through continuous updates.



Main application fields

Computer vision, NLP, speech recognition, and recommendation system

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TensorFlow (2)

Core features:
- Support for the static image mode and dynamic image mode
- In-depth integration with Keras and high usability
- TensorBoard visualization tool
- Cross-platform deployment capability

Production tools

TensorFlow Lite	TensorFlow.js	TensorFlow Serving	TFX
Deployed on mobile devices	Running on the web client	Deployed on servers	MLOps pipeline

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JAX

Design philosophy
- Open-sourced by Google in 2018
- Focus on high-performance numerical computing
- Functional programming style
- Scientific research-level flexibility + Production-level performance

Core technologies
- Optimization using the accelerated linear algebra (XLA) compiler
- Automatic differentiation (Autograd)
- Vectorization (vmap)
- Parallel computing (pmap)

API features

NumPy-compatible APIs

Low learning cost

@jax.jit
def func(x):
Just-in-time compilation
Ultimate performance

Automatic differentiation

Powerful tool for scientific research

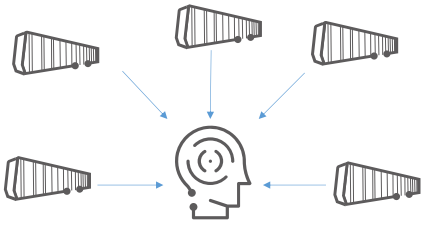
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Challenge: Multi-device Collaborative Training



Multiple compute nodes collaborate to train a model.

- How do I train a model on multiple compute nodes?
- How do I update parameters?
- How do I maximize the utilization of training resources?
- What can I do if the device memory or video RAM is insufficient?

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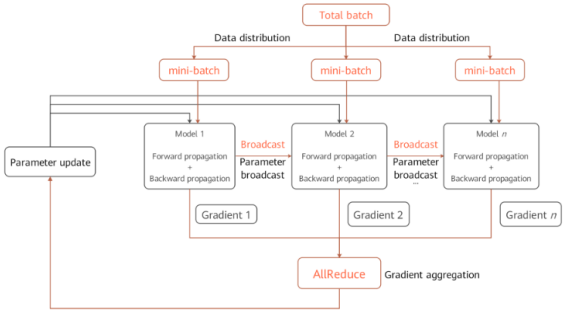
Distributed Model Training

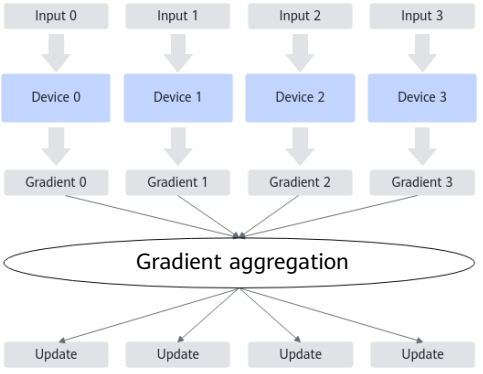
- During the training of a large model, a large amount of memory is required to store parameters such as intermediate activations and weights. A model with tens of billions of parameters cannot be trained on a single compute node. As a result, model training is inefficient or even impossible in specific cases. To solve this issue, multi-device or multi-node distributed training is required.
- The core idea of distributed parallelism is to distribute computing and storage to different devices. The main paradigms for training large-scale deep learning models are as follows:
  - Data parallelism (DP)
  - Model parallelism (MP)
  - Hybrid parallelism (HP)

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Data Parallelism

- In data parallelism, each computing device has a complete copy of the neural network model. During iteration, each computing device is allocated only a subset of a batch of data samples, and performs forward propagation and backpropagation of the network model based on data in the subset.





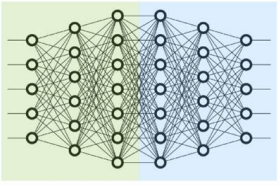
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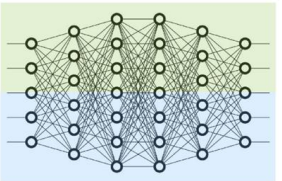
- Data parallelism divides training data into multiple batches and allocates them to different devices for parallel processing. Each computational card processes data of different batches in parallel, and then the results are combined. In data parallelism, each computational card has its own memory and computing resources and can process data independently. However, each computational card stores a complete copy of model weights, posing high requirements on the video RAM. In other words, the model is replicated on each worker so that each worker has a complete copy of the model. The input dataset is sharded, and a small batch of training data is split among multiple workers. The workers periodically aggregate their gradients to ensure that all workers have a consistent weight version. For a large-scale model that cannot be stored on a single worker, data parallelism can be used on small shards in the model.

Model Parallelism (1)

- Model parallelism is mainly used to solve the issue of insufficient memory on a single node. It can be classified into the following types from the perspective of computational graph partitioning:
  - Intra-layer parallelism (tensor parallelism): Parameters at the same layer are allocated to different devices.
  - Inter-layer parallelism (pipeline parallelism): The model is divided by layer, and different layers are placed in different devices.



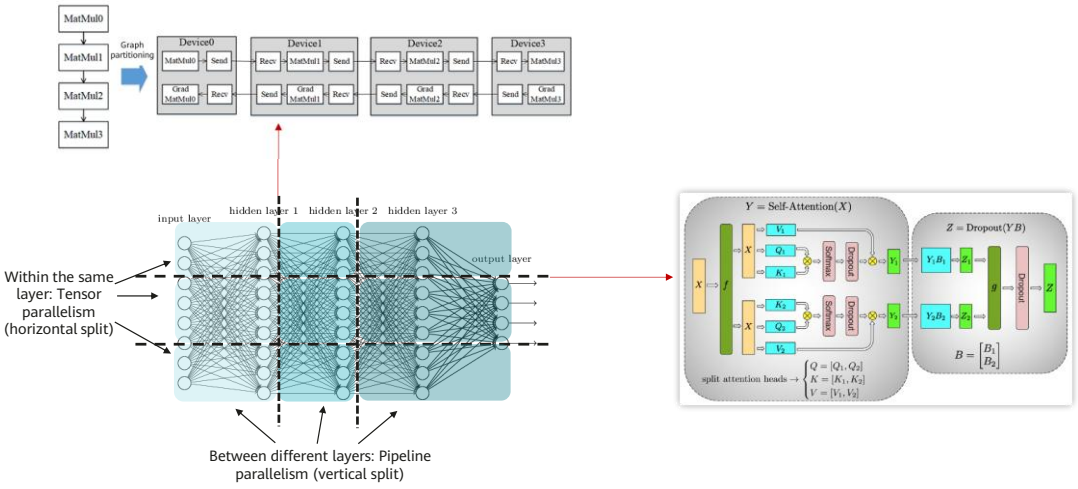
Inter-layer parallelism



Intra-layer parallelism

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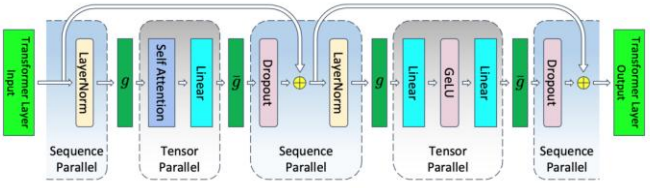
Model Parallelism (2)



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Sequence Parallelism

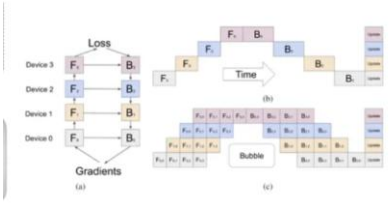
- Sequence parallelism is a parallel computing mode that performs column split on an input sequence. It can further improve computing efficiency based on tensor parallelism.
  - Input split: An input sequence with a length of **L** is split into **N** parts, and each device processes a subsequence with a length of **L**/**N**.
  - Independent computing: In operations such as LayerNorm and Dropout, each device independently processes its own subsequence.
  - All-Gather operation: All subsequences need to be concatenated into a complete sequence by using All-Gather before the sequence is sent to modules that require global information, such as the self-attention module.
  - Global computing: The complete sequence is used for computing in the self-attention module.
  - Reduce-Scatter operation: After the computing is complete, the result is split back to each device through Reduce-Scatter.



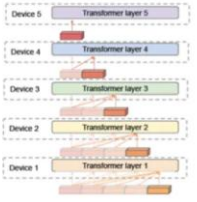
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Hybrid Parallelism

- Hybrid parallelism is a combination of multiple parallelism policies, such as data parallelism, pipeline parallelism, and tensor parallelism. By combining different parallelism policies, hybrid parallelism can fully leverage the advantages of various parallelism policies to maximize computing performance and efficiency.



GPipe
Data parallelism + Pipeline parallelism



TeraPipe
Token parallelism +
Pipeline parallelism

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Recomputation (1)

In the standard backpropagation algorithm, the activations of all layers need to be computed and saved during forward propagation.

Forward Propagation: Input → Layer 1 → Layer 2 → ... → Layer N → Output

↓ Save A1 ↓ Save A2 ↓ Save AN

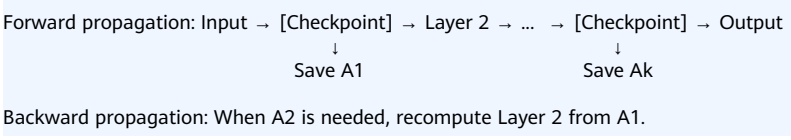
- The memory consumption is proportional to the model depth.
- Each layer of Transformer has a large number of temporary variables.
- Attention matrix computation produces a huge number of intermediate results.

- Recomputation is a policy of trading time for space. In forward propagation, intermediate results (on which backpropagation depends) are not saved. Instead, they are recomputed during backpropagation, lowering the demand for video RAM during model training.
- The recomputation technology allows larger-scale models and longer sequences to be computed, especially in memory-constrained environments. However, it may increase computational overheads because some intermediate results need to be recomputed.

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Recomputation (2)

- After recomputation optimization:
  - Forward propagation phase: Only key checkpoints are saved.
  - Backpropagation phase: When the activation of a layer is required, recomputation starts from the latest checkpoint.



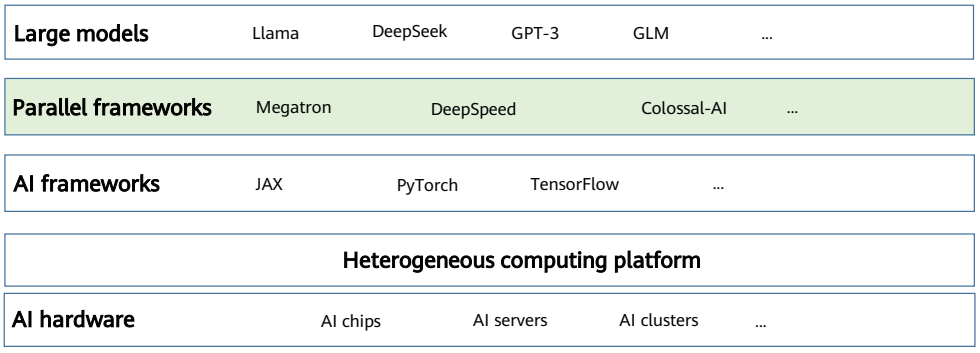
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- Checkpoint selection: Select the optimal layer as the checkpoint.
- Recomputation scheduling: Recomputation is efficiently scheduled during backpropagation.
- Memory management: Unnecessary activations are promptly released.

Software Architecture of Deep Learning Models

- The training acceleration libraries for large models are located between the model training frameworks and models, and are used to improve training and inference.



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- Position of DeepSpeed in the software architecture of deep learning models

Introduction to DeepSpeed

- DeepSpeed is an open-source deep learning optimization library developed by Microsoft to **improve the training efficiency and scalability of large-scale models**. It uses diverse technical means to accelerate training, including model parallelism, gradient accumulation, dynamic precision scaling, and local mixed-precision training.
- DeepSpeed also provides auxiliary tools for distributed training management, memory optimization, and model compression to help developers better manage and optimize large-scale deep learning training tasks.
- DeepSpeed has a powerful ecosystem and natively supports diverse computing devices.

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- DeepSpeed is a deep learning optimization software suite that makes distributed training and inference simple, efficient, and effective. What can it do? Its functions are as follows: Be used for the training/inference of dense or sparse models with billions or trillions of parameters; achieve excellent system throughput and effectively extend to thousands of computational cards; perform training/inference on resource-constrained systems; perform inference with unparalleled low latency and high throughput; implement extreme compression at low costs to achieve unparalleled inference latency and model downsizing.

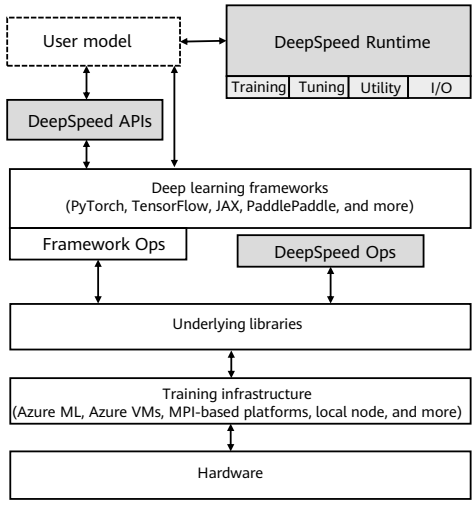
Software Architecture of DeepSpeed (1)

- DeepSpeed consists of three parts:
  - APIs: Easy-to-use APIs are provided. Only a few APIs need to be called for model training and inference. The most important API is the initialize API that is used to initialize the engine and configure training parameters and optimization technologies. Generally, configuration parameters are stored in the **config.json** file.
  - Runtime: The runtime component is DeepSpeed's core component for management, execution, and performance optimization. For example, it can be used to deploy training tasks to distributed devices, save and load checkpoints, and implement data partitioning, model partitioning, system optimization, fine-tuning, and fault detection. This component is implemented using Python.
  - Ops: It is an underlying kernel component implemented using C++. It is used to optimize computing and communication, and provides a range of underlying operations.

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Software Architecture of DeepSpeed (2)

- Architecture advantages:
  - Training and inference can be separately optimized on the training framework.
  - This architecture can better utilize the entire ecosystem than tightly coupled architectures. In addition, it is easier to maintain than deeply integrated architectures.
  - This architecture is irrelevant to infrastructure, allowing users to choose the platforms they prefer, such as Azure ML and Azure VMs.



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- Why is this architecture used?
  - Training and inference can be separately optimized on the training framework.
  - This architecture can better utilize the entire ecosystem than tightly coupled architectures. In addition, it is easier to maintain than deeply integrated architectures.
  - This architecture is irrelevant to infrastructure, allowing users to choose the platforms they prefer, such as Azure ML and Azure VMs.

ZeRO (1)

- Zero redundancy optimizer (ZeRO) is a video RAM optimization technology proposed by Microsoft in its DeepSpeed framework. Its core philosophy is to eliminate memory redundancy through distributed storage and on-demand loading of model states, thereby supporting ultra-large-scale model training with limited hardware resources.
- It gradually optimizes the video RAM usage in the following three phases:
  - ZeRO-1: Optimizer state sharding
    - Objective: Distribute optimizer parameters (such as Adam's momentum and variance) to multiple devices.
    - Implementation: Each device stores only part of the optimizer states, but retains all model parameters and gradients.
    - Effect: The video RAM usage is reduced by about 8 times (for the Adam optimizer). This phase is applicable to medium-scale models.

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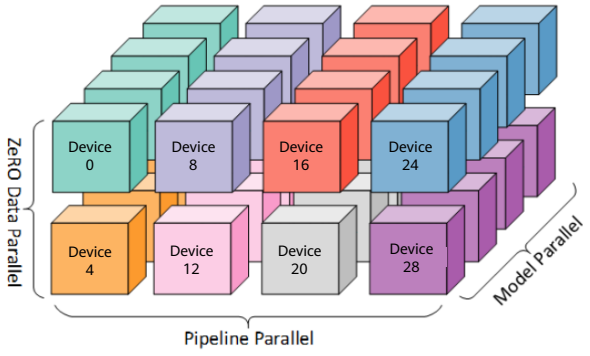
- ZeRO is a new memory optimization technology for large-scale distributed deep learning. It can train deep learning models with 100 billion parameters on a cluster at a speed that is three to five times the optimal system throughput. It also provides a clear path for training models with trillions of parameters, demonstrating an unprecedented leap in deep learning system technologies. As a part of DeepSpeed, ZeRO is used to improve the efficiency of the video RAM and computing.
- ZeRO can overcome the limitations of data parallelism and model parallelism, and achieve the advantages of both parallelism policies. Memory redundancy is eliminated by partitioning model state parameters, gradients, and optimizer states between data parallelism processes, instead of replicating them. Dynamic communication scheduling is used during training to share necessary states between distributed devices in order to maintain the computation granularity and communication load of data parallelism.

ZeRO (2)

- ZeRO-2: Gradient sharding
  - Objective: Further disperse gradients based on ZeRO-1.
  - Implementation: Gradients are synchronized to the corresponding devices through the reduce operation instead of the global all-reduce operation.
  - Effect: The video RAM usage is further reduced, and larger models can be trained.
- ZeRO-3: Model parameter sharding
  - Objective: Eliminate redundancy and shard all model states (parameters, gradients, and optimizer states).
  - Implementation: During forward propagation, only the parameters of the current layer are loaded. After backpropagation, the parameters are updated synchronously.
  - Effect: The video RAM usage is minimized, but the communication overhead increases. Therefore, an efficient communication framework is required.

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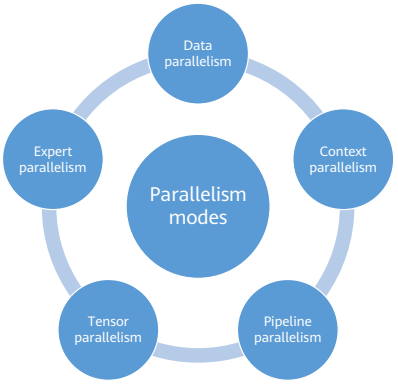
3D Parallelism



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Megatron

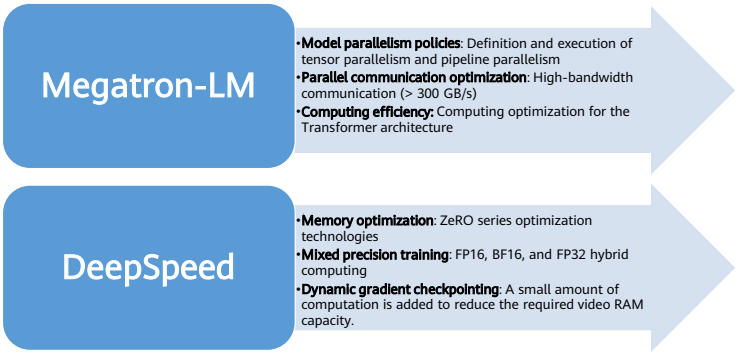
- Megatron-LM (Megatron for short) is a toolbox launched in 2019 to specifically train ultra-large models.
- Megatron is a basic operating system. It is a general-purpose training framework for both language models and domain-specific large models (such as those in image and biomedicine fields). Its core "underlying module" is called Megatron-Core, which is similar to the chip and system kernel of a mobile phone and provides the most basic parallel computing capability.



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Megatron-DeepSpeed

- Megatron-DeepSpeed integrates the core advantages of Megatron-LM and DeepSpeed to create a large model training solution that achieves the ultimate performance and high memory efficiency.



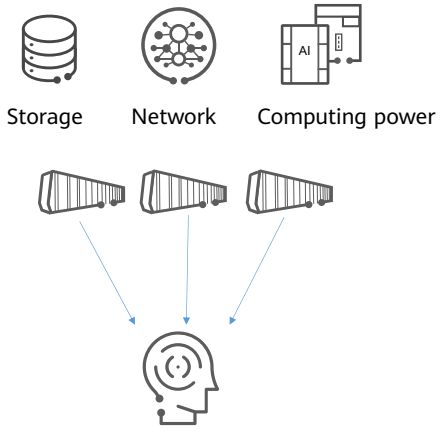
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1. Basic Features of Large Model Training
2. **Large Model Training Solutions**
   - Training Solution Overview
   - AI Development Frameworks
   - Distributed Parallelism Policies
   - Unified Compute Architecture
3. Large Model Fine-Tuning Solutions

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Challenge - Computing Efficiency



Training a model with hundreds of billions of parameters generally requires a cluster consisting of thousands of computing chips. During training, the following issues may occur:

- Parameter synchronization - bandwidth throttling
- Heterogeneous hardware
- Low operator affinity (hardware adaptation)
- Long operator scheduling time
- ...

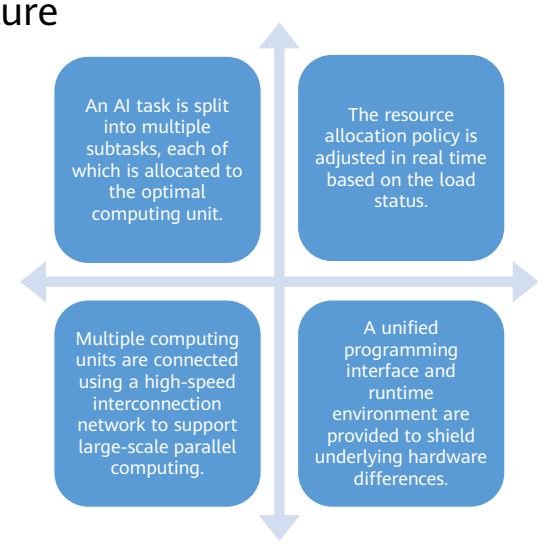
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- Currently, large models are generally trained on a multi-node server based on a solution that integrates computing (computing chip), storage, and network.
- Parameter synchronization: In multi-node training, gradient synchronization takes 30–40% of the training time. For example, when a model with trillions of parameters is trained, the single synchronization latency can be within seconds.
- Bandwidth throttling: Cross-node data transmission is one of the major bottlenecks.
- Heterogeneous hardware: The computing speed varies depending on the computing performance of different nodes, resulting in waiting time.

Unified Compute Architecture

- A unified compute architecture tightly couples different types of computing units and storage resources to form a unified computing entity, thereby implementing efficient collaboration and intelligent scheduling of resources.



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Computing Power Optimization

- Fully unleashing hardware computing power: The parallel computing capability of the AI chip is maximized.
- Direct connection to the physical layer: A vast amount of data is allocated to multiple threads for processing, and multiple cores of the AI chip are called to implement parallel computing.
- Operator optimization and computing efficiency.
- Efficient memory management: The data transmission between the CPU and device is optimized to reduce latency and improve scheduling efficiency.
- Computing efficiency optimization: Technologies such as operator optimization and memory management are used to improve performance.

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   - Large Model Fine-Tuning Algorithms
   - Open-Source Tools for Fine-Tuning Large Models

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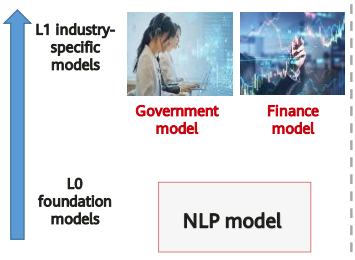
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Types of Fine-Tuning for Large Models

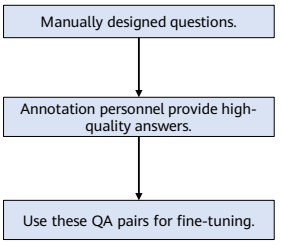
Prerequisites for fine-tuning a large model:

1. The large model is still underfitting.
2. The distribution of model pre-training data is inconsistent with the distribution of specific task data.

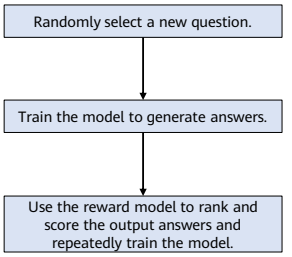
Continual pre-training:
Improves the knowledge and capability of the model in a specific domain by adding domain data.



Supervised fine-tuning (SFT):
Stimulates the large model's capability to understand and answer various questions in a specific domain.



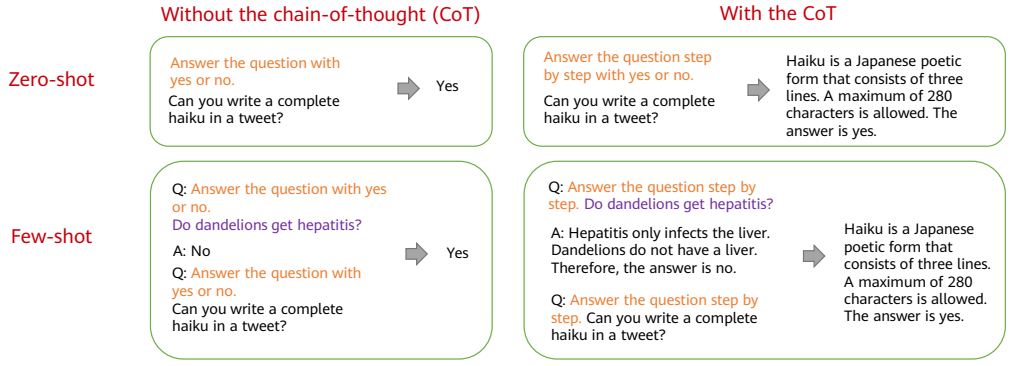
Reinforcement learning from human feedback (RLHF)/Reward model/Reinforcement learning: Aligns the answers of large models with human preferences.



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Instruction Fine-Tuning for Large Models - SFT

- Instruction fine-tuning refers to the process of constructing instances of the instruction format and then fine-tuning large language models (LLMs) in a supervised manner. The instruction format generally includes the task description, a pair of input and output, and examples (optional).



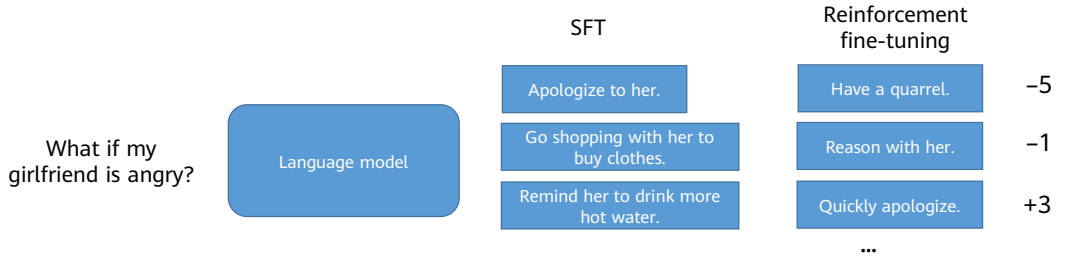
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- SFT can be simply considered as instruction fine-tuning.
- Instruction fine-tuning is a process of further training LLMs on a dataset consisting of (instruction, output) pairs. The instruction refers to the human instruction, and the output refers to the expected output that follows the instruction. This process helps bridge the gap between the next-word prediction target of the LLM and the target of enabling the LLM to follow human instructions.
- Instruction fine-tuning can be considered as a special form of SFT. However, their goals are different. SFT is a process of fine-tuning a pre-trained model using labeled data so that the model can better execute specific tasks. Instruction fine-tuning is a process of further training LLMs on a dataset consisting of (instruction, output) pairs to enhance the capabilities and controllability of the LLMs. The particularity of instruction fine-tuning lies in the structure of its dataset, that is, pairs of human instructions and expected outputs. This structure enables instruction fine-tuning to focus on making models understand and follow human instructions.
- In general, as a special form of SFT, instruction fine-tuning focuses on enhancing the capabilities and controllability of LLMs by understanding and following human instructions. Although their goals and approaches are similar, the special data structure and task focus of instruction fine-tuning make it a unique subset of SFT.

Alignment of Large Models - RLHF

- Data that humans prefer is used for training:
  - Honesty: Is it false or misleading information?
  - Harmlessness: Does it cause physical or mental harm to people or the environment?
  - Helpfulness: Does it solve the user's task?



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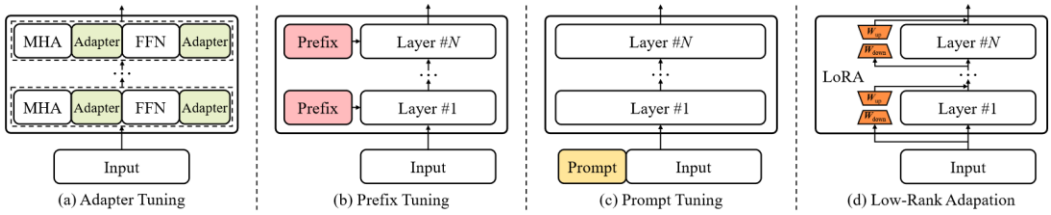
Efficient Fine-Tuning Technologies

- Efficient fine-tuning technologies can be roughly classified into three categories: adding extra parameters (addition-based), selecting some parameters for update (selection-based), and introducing reparameterization (reparameterization-based). The addition-based technology can be further classified into two sub-categories: adapter-like method and soft prompts.
  - Examples of the addition-based technology: Prefix tuning, prompt tuning, adapter tuning, and their variants.
  - Examples of the selection-based technology: BitFit.
  - Examples of the reparameterization-based technology: LoRA, AdaLoRA, and QLoRA.

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Common Fine-Tuning Methods for Large Models

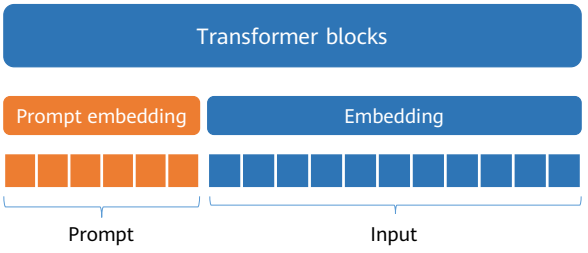
- There are diverse fine-tuning methods for large models. The core idea is to minimize the modification to the parameters of the pre-train model in order to improve the model capabilities in specific domains or on specific downstream tasks. The following methods can be used for instruction fine-tuning.



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Fine-Tuning Methods for Large Models - Prompt Tuning

- The purpose of prompt tuning is to train a small model with a small number of parameters for each specific task without changing the parameters of the foundation model. These small models are called on demand when specific tasks are executed.
- The basic principle of prompt tuning is to add some special tokens of a specific length before the input sequence **X** to increase the probability of generating the expected sequence.
- Prompt tuning in performed in the embedding phase. To be specific, X = [x(1), x(2), ..., x(m)] is changed to X' = [x'(1), x'(2), ..., x'(k); x(1), x(2), ..., x(m)], and Y = WX'.



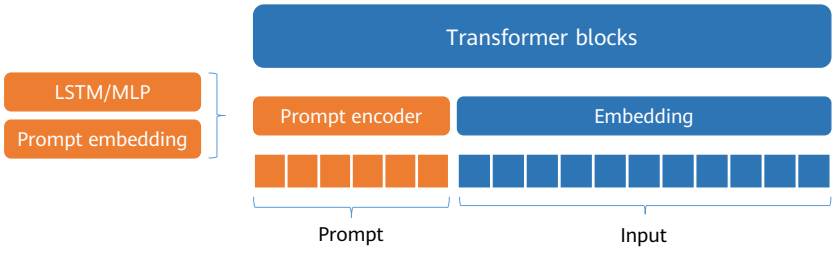
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- Prompt tuning is an efficient fine-tuning method in which only the representation layers of new prompts are trained and other model parameters are fixed.
- The new prompts can be classified into hard prompts and soft prompts.
- Soft prompts refer to broad or vague prompts that allow models to have a higher degree of freedom in generating results. They are generally used to inspire models to generate creative results.
- Hard prompts are specific and clear prompts that require models to generate accurate results based on the given information. They are generally used for tasks that require models to provide accurate answers.
- In parameter-efficient fine-tuning (PEFT), soft prompts are the text that is randomly initialized, whereas hard prompts are the text that needs to be specifically configured.

Fine-Tuning Methods for Large Models - P-Tuning

- Based on prompt tuning, P-Tuning accelerates model convergence by adding the long short-term memory (LSTM) or multilayer perceptron (MLP) encoding module.
- P-Tuning v2 adds prompt tokens to each layer so that more learnable parameters and prompts in deeper structures directly affect model prediction, improving model flexibility and efficiency.

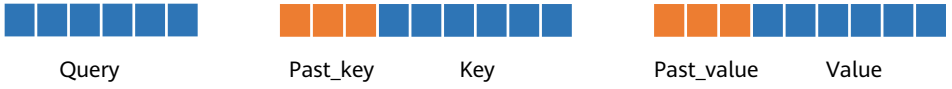


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- Principle of P-Tuning: Based on prompt tuning, P-Tuning further encodes and computes the prompt part to accelerate convergence.

Fine-Tuning Methods for Large Models - Prefix Tuning

- Prefix tuning is inspired by the prompt engineering practice that adding proper conditions to the prompt context without changing the large model can guide the large model to perform better.
- The purpose of prefix tuning is similar to that of prompt tuning. The only difference is in the implementation.
  - Prompt tuning adds specific tokens before the input sequence **X** in the embedding phase.
  - Prefix tuning adds specific prefixes to the encoder and decoder networks of the Transformer. To be specific, **W** in Y = WX is changed to W' = [W(p); W], and Y = W'X.
  - Prefix tuning ensures that the foundation model remains unchanged. During inference, some parameters are added before **W** as required.



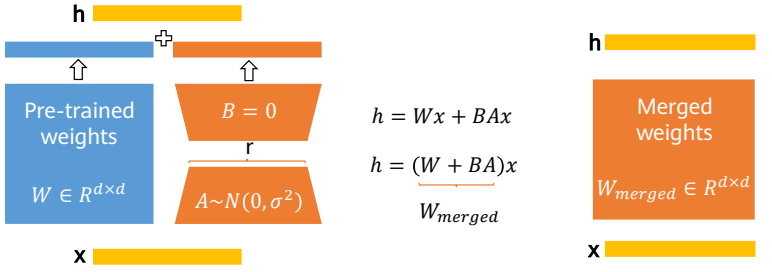
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- Prefix tuning embeds trainable parameters, that is, the prefix, into the entire model.
- Prefix tuning places multiple prompt vectors before the key matrix and value matrix of each multi-head attention.
- **past_key_values**: Historical computation results of keys and values in the Transformer model. They are used for decoding acceleration of generative models. The decoding logic is to predict a new token each time based on the historical input, add the new token to the input, and then predict the next token. This process involves a large number of repeated computations. Therefore, the computation results of keys and values can be cached as **past_key_values** and serve as the input to the next computation. This technology is also called KV cache.
- In prefix tuning, the learnable part, also called the prefix, is placed at each layer of the model in the form of **past_key_values**.

Fine-Tuning Methods for Large Models - LoRA

- LoRA is another technical route that is totally different from prompt tuning and prefix tuning.
- Assumption behind LoRA: Existing LLMs are over-parameterized, and there is a low-dimensional essential model behind over-parameterized large models.
- Basic idea of LoRA: LoRA does not directly train the original model. Instead, it fixes the parameters of the original model and attempts to optimize the original model's rank factorization matrix that is changed after fine-tuning.



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- The blue part in the figure refers to the pre-trained model parameters. LoRA adds structures A and B next to the pre-trained model structure. The parameters of structures A and B are initialized to Gaussian distribution and **0**, respectively. Therefore, the parameter added at the beginning of training is **0**. The input dimension of structure A and the output dimension of structure B are the same as the input and output dimensions of the original model, respectively. The output dimension of structure A and the input dimension of structure B are far less than the input and output dimensions of the original model. This is the reflection of low-rank (similar to the ResNet structure). In this way, the parameters to be trained can be greatly reduced.
- The basic idea of LoRA includes the following steps:
  - First, to adapt to a specific downstream task, train a specific model and change Y = WX to Y = (W + ΔW)X, where **ΔW** is the fine-tuning result.
  - Then, perform low-dimensional decomposition on **ΔW**: ΔW = AB (**ΔW** has m × n dimensions, **A** has m × r dimensions, **B** has r × n dimensions, and **r** is the low dimension in the assumption mentioned earlier).
  - Next, train A and B using specific training data to obtain **ΔW**. In the inference process, directly add **ΔW** to **W**. There is no additional cost.
  - In addition, if LoRA needs to adapt to different scenarios, the switching can be performed through simple matrix addition: (W + ΔW) – ΔW + ΔW'.

- During training, only the parameters of A and B are updated, and the pre-trained model parameters remain unchanged. During inference, reparameterization can be used to combine AB and W. In this way, no extra computation is introduced during inference. For different downstream tasks, only A and B need to be trained based on the pre-trained model. This accelerates the training of large models.

Comparison of Mainstream Fine-Tuning Technologies for Large Models

- LoRA does not introduce additional inference latency and reduces the consumption of memory and storage resources. It converges the model to the performance level of the original pre-trained model and is the most widely used method.
- Adapter tuning adds the adapter layer, which increases the computation workload and inference delay.
- Prefix tuning is difficult to optimize, as its performance varies non-monotonically with the number of trainable parameters.
- P-tuning preserves part of the sequence length, thereby reducing the input sequence length of downstream tasks.

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Hugging Face Transformers

- Hugging Face Transformers is a pre-trained algorithm library based on the Transformer model structure.
  - It supports the training and inference of mainstream pre-trained NLP models.
  - It uses the pipeline to quickly construct various NLP tasks, such as question answering, text classification, and image classification tasks.
- Libraries related to Hugging Face Transformers:
  - Tokenizer: Provides different types of tokenizers, vocabulary construction methods, and data conversion methods to convert text data into data that can be processed by neural networks.
  - Datasets: Integrates various types of public datasets for users to load and use and provides common data loading and processing methods.
  - Evaluate: Provides common evaluation functions to evaluate the capabilities of a single model and compare the capabilities of different models.
  - PEFT: Provides common parameter-efficient fine-tuning (PEFT) methods in the industry, such as LoRA and P-Tuning.
  - Optimum: A model inference acceleration toolkit that supports diverse types of backends, and allows users to select backends based on hardware devices.
  - Accelerate: Uses short code to implement distributed parallel computing and FP16 half-precision computing in multi-node multi-device and single-node multi-device scenarios.
  - Gradio: A simple and easy-to-use demonstration system setup tool that provides simplified front-end pages for large model applications.

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Hugging Face PEFT

- PEFT is an open-source PEFT library of Hugging Face. It provides the latest PEFT technology and can be seamlessly integrated with Transformers and **Accelerate**.
- The project URL is as follows: https://github.com/huggingface/peft. The PEFT library can easily convert common Hugging Face models into models that support lightweight fine-tuning.
- Currently, the following PEFT methods are supported:
  - LoRA
  - Prefix tuning
  - P-Tuning
  - Prompt tuning
  - AdaLoRA

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Hugging Face TRL

- Transformer Reinforcement Learning (TRL) is a reinforcement learning framework developed by Hugging Face for task-oriented dialog systems (TODSs).
- The TRL library focuses on constructing and optimizing interactive agents and uses reinforcement learning to improve their performance in multi-turn dialogue. The TRL project aims to simplify the development of TODSs and enable researchers and developers to quickly experiment with different reinforcement learning algorithms and apply them to real-world dialog systems.
- Project URL: https://github.com/huggingface/trl

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DeepSpeed-Chat

- DeepSpeed-Chat is the latest tool released by Microsoft for training ChatGPT-like models in a simple and efficient manner. It is designed based on Microsoft's large model training tool DeepSpeed. This tool has the following features:
  - Complete code for training ChatGPT-like models: The code covers the download of the pre-trained model and data, as well as the InstructGPT training process and testing.
  - Models of different scales: The number of model parameters ranges from 1.3 billion to 66 billion. Models suitable for beginners and commercial deployment are available.
  - Efficient training: The latest technologies, such as ZeRO and LoRA, are used to improve the training process and speed.
  - Inference APIs: Easy-to-use inference APIs are provided to facilitate dialog-based interactive tests.
- Project URL:
  - https://github.com/microsoft/DeepSpeedExamples/tree/master/applications/DeepSpeed-Chat

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LLaMA Factory

- LLaMA Factory is an easy-to-use and efficient platform for training and fine-tuning LLMs. With LLaMA Factory, you can fine-tune hundreds of pre-trained models locally without writing any code.
  - Model types: LLaMA, LLaVA, Mistral, Mixtral-MoE, Qwen, Yi, Gemma, Baichuan, ChatGLM, Phi, and more.
  - Training approaches: (incremental) pre-training, (multimodal) SFT, reward model training, proximal policy optimization (PPO) training, direct preference optimization (DPO) training, Kahneman-Tversky optimization (KTO) training, odds ratio preference optimization (ORPO) training, and more.
  - Operational precision: 16-bit full fine-tuning, frozen fine-tuning, LoRA fine-tuning, and AQLM/AWQ/GPTQ/LLM.int8/HQQ/EETQ-based 2/3/4/5/6/8-bit QLoRA fine-tuning.

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Summary

- This chapter describes the large model training solutions, including the full-process development tools and kits provided for users to help them train and fine-tune models more efficiently.

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Quiz

1. (Multiple-answer question) Which of the following can be used for LLM training and fine-tuning?
   A. AI computational cards
   B. DeepSeek
   C. DeepSpeed
   D. LLaMA Factory

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- Answer: ACD

Quiz

2. (Multiple-answer question) Which of the following tools can provide distributed training policies for large models?

   A. CPU

   B. DeepSpeed

   C. Megatron

   D. LoRA

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- Answer: BC

Thank you.

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每个组织，构建万物互联的智能世界。

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Model Inference and Deployment Solutions



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Foreword

- AI is deeply integrated into people's daily life and has yield unusually brilliant results in diverse fields. However, the value of AI is not limited to model training. If models cannot be efficiently deployed in application scenarios to implement accurate inference, it is difficult to create concrete value. Therefore, developers need to select proper deployment and inference modes and ensure stable running and high performance of models.
- This chapter describes the AI model inference and deployment solutions, development trends of large model deployment, and common algorithms for large model inference and optimization.

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Objectives

- Upon completion of this course, you will be able to:
  - Understand the inference metrics of large models.
  - Understand the optimization methods of large model inference.
  - Describe different AI model deployment solutions.
  - Master common optimization methods of large model deployment.

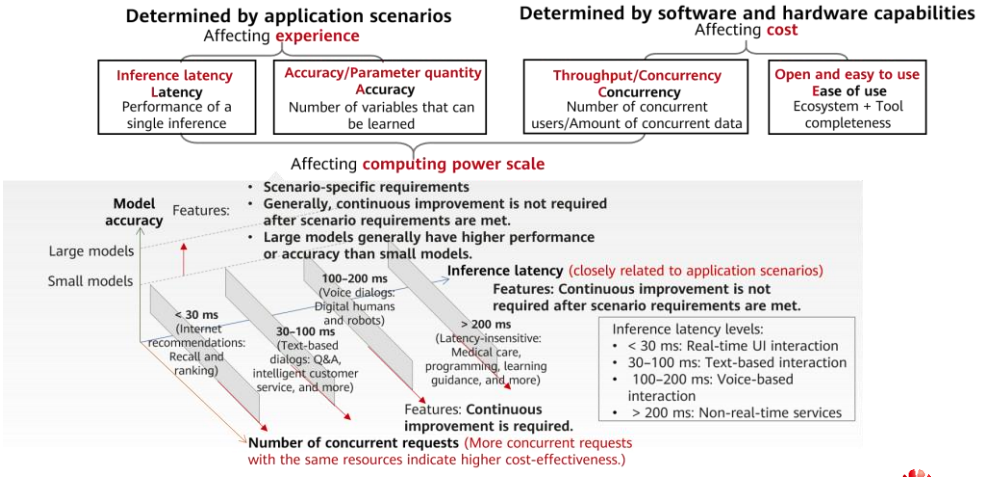
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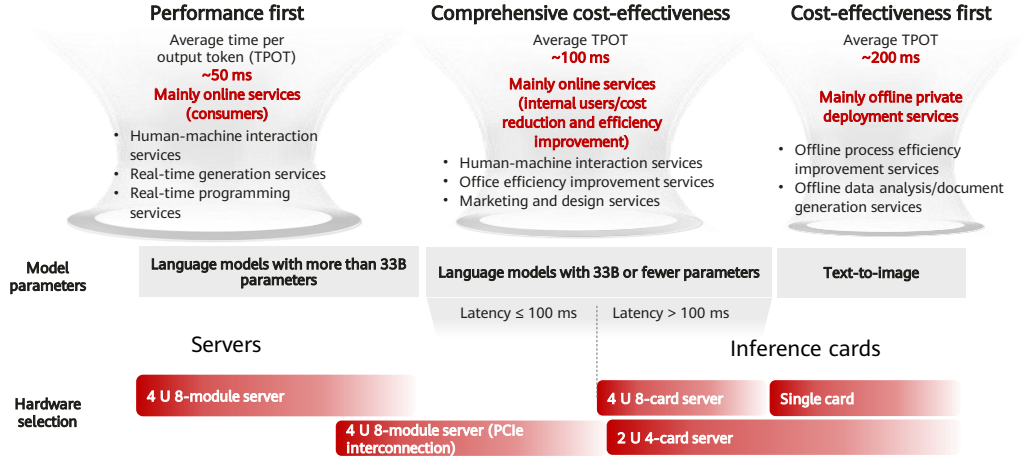
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Building an Inference Metric System That Balances Experience and Cost



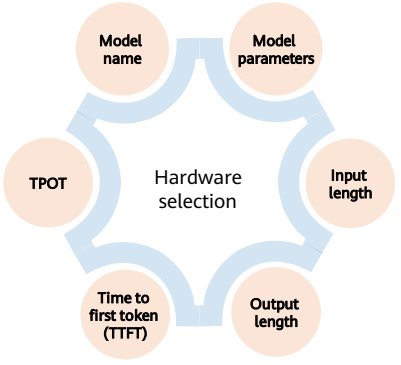
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Selecting Large Model Inference and Deployment Modes Based on Experience Objectives (1)

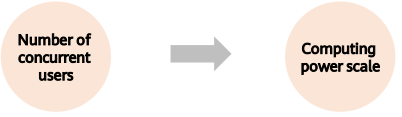


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Selecting Large Model Inference and Deployment Modes Based on Experience Objectives (2)



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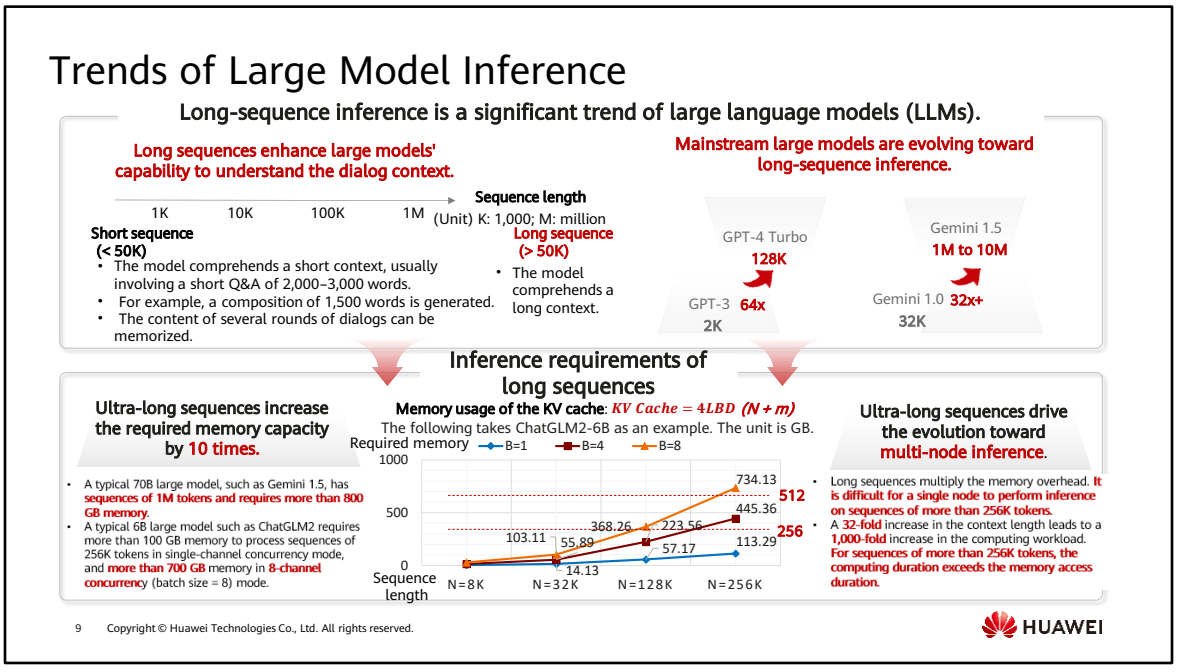
For example, according to performance tests, in the office scenario, the LLaMa2-7B model is used to support 120 concurrent users. When the latency is less than 200 ms and inference cards are used to provide computing power, 10 cards are required to achieve the average peak value.

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- **L** indicates the number of network layers. **B** indicates the batch size. **D** indicates the feature dimension. **N** indicates the length of the input sequence. **m** indicates the length of the output sequence.

Challenges of Large Model Inference (1)

Latency and response time

In scenarios where chatbots and intelligent assistants are used, models are expected to have low latency and short waiting time.

Memory usage and model size

The video RAM or memory is insufficient for excessively large models or extremely long sequences.

Scalability and throughput

In application scenarios, model inference systems may face different request loads. Therefore, the systems must have excellent scalability and high throughput, and be capable of integrating computing and storage resources.

Hardware compatibility and acceleration

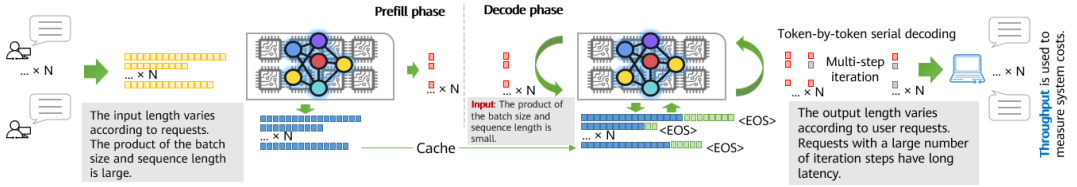
It is critical to effectively utilize hardware resources. Hardware-aware algorithms need to be designed and optimized to maximize the potential of underlying hardware.

Trade-off between accuracy and efficiency

The optimization of LLM inference may involve a trade-off between efficiency and accuracy.

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Challenges of Large Model Inference (2)



Larger model scales, longer sequences, and autoregressive decoding pose challenges in memory access and computing power utilization.

If the model scale is large, the memory capacity and memory access are bottlenecks.

- **Single-device inference cannot be performed due to insufficient memory capacity.**
  - **Model parameters:** 10 billion or 100 billion model parameters exceed the memory capacity of a single device.
  - **KV cache**: As the batch size and sequence length increase, more memory is required.
- **The memory access bandwidth hinders the attainment of the 50 ms/token inference latency.**
  - Multi-channel concurrency alleviates bandwidth pressure and improves throughput, but increases latency.
  - As the sequence length increases, KV cache memory access becomes a bottleneck, and the inference latency increases exponentially.

The computing power utilization of autoregressive decoding is low, and it is difficult to achieve both low latency and high throughput.

- **Inference in the prefill and decode phases differs greatly, making it difficult to fully utilize computing resources.**
  - **Request-based scheduling results in a large amount of idle computing power.** The lengths of sequences in the same batch differ greatly.
  - **It is difficult to batch requests in different phases.** The inputs and KV cache dimensions in the prefill and decode phases differ greatly.
- **In the decode phase, the computing power utilization of token-by-token serial decoding is low.**
  - General matrix-vector multiplication (GEMV) is mainly used for serial decoding, and the arithmetic intensity is low.
  - The KV cache memory access volume increases with the sequence length, and the proportion of the Attention module's computing workload in the entire model inference process increases accordingly.

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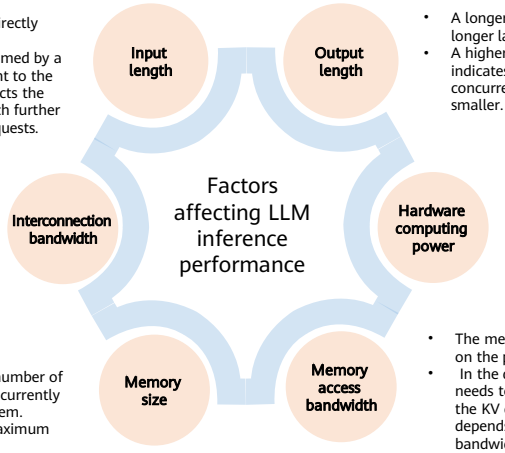
- Before performing optimization, you need to know what causes slow model computing.
  - First, large models need to process long inputs and outputs (generally the window size is greater than 4K). Because the current token and all previous tokens need to participate in the attention computing, the size of the matrix to be computed by the model increases as the values of **seq_len** and **n** increase.
  - Second, the generation of each token depends on all the previous computations. Tokens can only be generated one by one, and concurrent computing is not supported.
- Simply speaking, the inference bottleneck is that attention computing has a computational complexity of O(N^2) and cannot be performed concurrently.

Factors Affecting LLM Inference Performance

- The duration of the prefill phase is directly affected.
- In the decode phase, the time consumed by a single token is constant and irrelevant to the input length. However, this time affects the memory usage of the KV cache, which further affects the number of concurrent requests.

- If a single card is insufficient for deploying an instance, the instance needs to be deployed across cards. In this case, the inference computing result needs to be communicated between cards. Therefore, a smaller interconnection bandwidth results in longer TTFT.

- The memory size determines the number of inference requests that can be concurrently handled by the LLM inference system. Therefore, it greatly affects the maximum number of concurrent requests.



- A longer output results in more iterations and longer latency in the decode phase.
- A higher memory usage of the KV cache indicates that the maximum number of concurrent requests supported by the memory is smaller.

- The hardware computing capability directly affects the TTFT, inter-token latency (ITL), and throughput, and slightly affects the TPOT.

- The memory access bandwidth has little impact on the prefill phase.
- In the decode phase, data in the KV cache needs to be extracted, calculated, and written to the KV cache again. Therefore, the TPOT depends heavily on the memory access bandwidth.

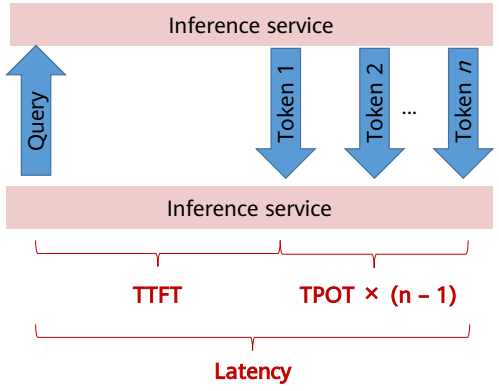
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Performance Metrics of LLM Inference: Latency



TTFT/Prefill latency

Duration from the time when a request is received to the time when the first token is generated

The TTFT greatly affects user experience of online interactive applications such as knowledge Q&A and chat.

TPOT/Decode latency

Average time for a large model to generate a token through incremental inference after the first token is generated

The TPOT has great impact on scenarios with long output tokens, such as code generation applications.

Latency

Total duration from the time when data is sent to the time when a system response is received. Generally, three metrics are concerned: the average latency, P95, and maximum latency.

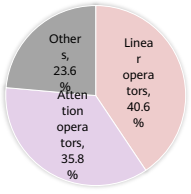
Latency = TTFT + TPOT × (Total number of generated tokens – 1)

The latency indicators concerned vary depending on scenarios. For example, the troubleshooting system focuses on the maximum latency, whereas online customer service focuses on the P95 latency.

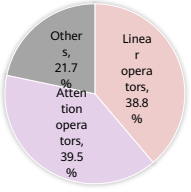
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Analysis of the LLM Inference Latency: Where Has the Time Gone?

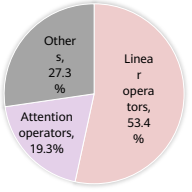
In the inference computing of LLMs, the main operators are **attention operators and linear operators**. The execution time of the two types of operators accounts for more than **75%** of the total operator execution time. In addition, many fine-grained operators, such as RMSNorm and SiLU, are distributed in computational graphs during LLM inference.



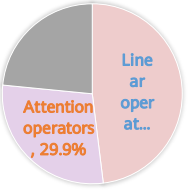
Llama2-7B
Text length: 128



Llama2-7B
Text length: 2,048



Baichuan2-13B
Text length: 128

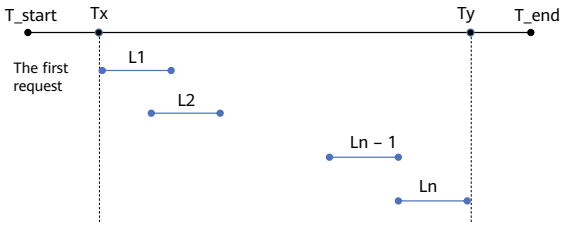


Baichuan2-13B
Text length: 2,048

A survey on efficient inference for large language models. https://arxiv.org/abs/2404.14294

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Performance Metrics of LLM Inference: Throughput



TPS = total_output_tokens / (Tx – Ty)

QPS = n/(Ty – Tx)

Throughput

Amount of data processed by the system in a unit of time. This metric affects the computing cost for generating each token.

Tokens per second (TPS) = Total number of tokens/Processing time (s)

Or

Query per second (QPS) = Total number of requests/Processing time (s)

Maximum number of concurrent requests

Maximum number of requests that can be concurrently processed by the system. It is an instantaneous metric.

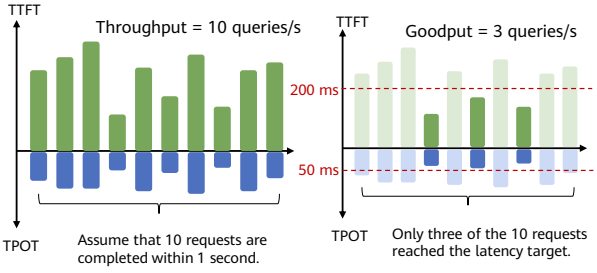
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Performance Metrics of LLM Inference: Goodput

Goodput

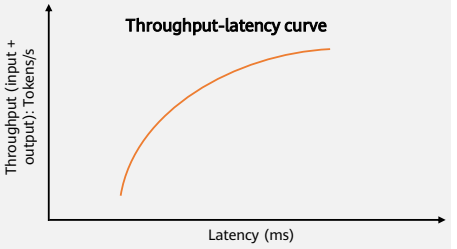
Throughput that meets the latency requirements. Goodput is a comprehensive metric that balances both latency and throughput.

High throughput ≠ High goodput



The latency and throughput are balanced by adjusting the batch size.

If the batch size is small, its increase makes the throughput increase faster than the latency. However, when the batch size exceeds a specific value, the doubling of the batch size only increases the latency but does not increase the throughput.



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Accuracy Metrics of LLM Inference

Absolute accuracy: Task score on the downstream dataset

Examples:

- Text generation
  - Prefill phase: BoolQ, C-Eval (few-shot), and MMLU (few-shot)
  - Decode phase: TruthfulQA
- Code generation: HumanEval and HumanEval-X
- Long sequences: LongBench (5K to 15K), L-Eval (3K to 200K)

Relative accuracy: Logits consistency

- Generally, **greedy search** is used for post-processing, and randomness should not be added to post-processing.
- If the Kullback-Leibler divergence of the probability distributions of two groups of logits (for example, the logits before and after model migration) is close to **0** after softmax, the two groups of probability distributions are statistically equivalent.
- When the model output token is inconsistent with the theoretical output token, the KL divergence of the probability distributions of the output logits corresponding to the first inconsistent token needs to be checked. If the KL divergence is close to **0**, it is considered that the probability distributions of the two inconsistent output tokens are equivalent and the model sampling probability is not affected.

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Objectives and Optimization Approaches of Large Model Inference Acceleration

- **Improving throughput:** From the perspective of the LLM service system, a higher throughput indicates higher resource utilization and lower costs of the system.
- **Reducing latency**: From the perspective of users, a lower latency indicates a smoother user experience.

Computing acceleration: Accelerating model computing

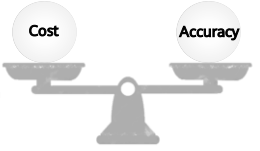
The model structure remains unchanged. The algorithms and hardware utilization are improved to improve the computing efficiency without compromising the model output quality.

- Compute-side optimization
  - Reduce unnecessary or repeated computations.
  - Maximize hardware concurrency.
- Memory-side optimization
  - Accelerate the memory I/O speed or reduce unnecessary memory read/write operations.

Model compression: Reducing the model computing workload

The model structure is changed to reduce specific computations, ultimately accelerating inference and reducing resource consumption. However, the model output quality may be compromised.

- Quantization
- Pruning
- Distillation
- Low-rank factorization
- Simplified Attention
- Speculative inference

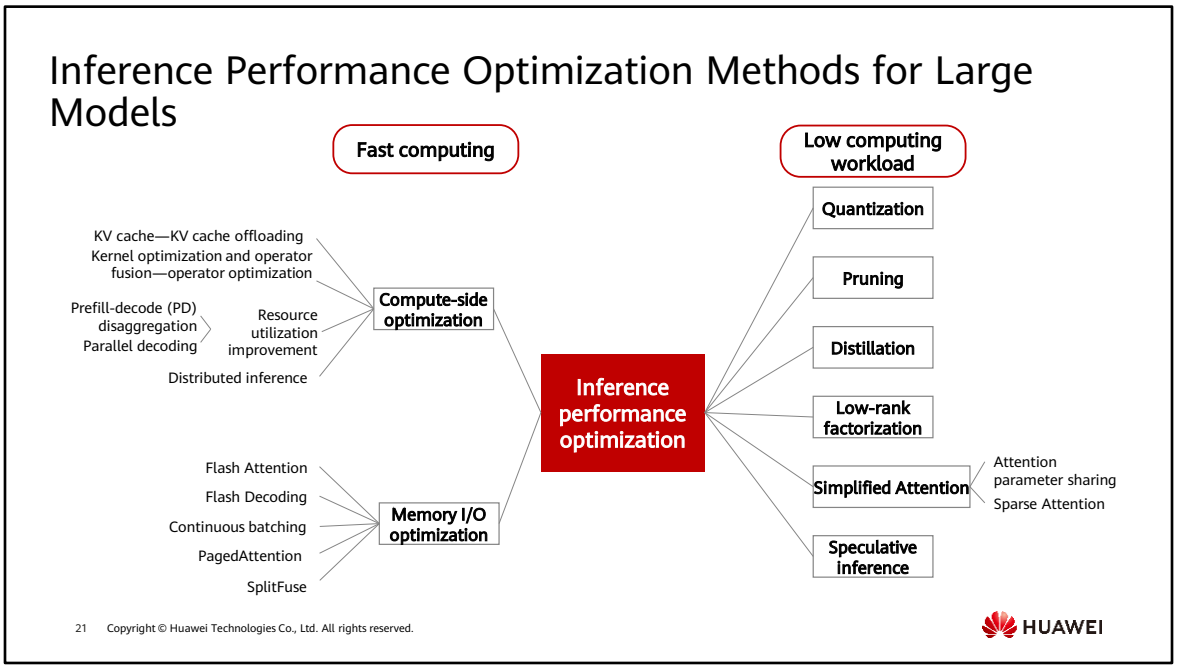


It is critical to strike a balance between cost and accuracy.

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- Acceleration optimization has two objectives: lower latency and higher throughput. Latency refers to the time for returning a single request, and throughput refers to the total number of requests processed within a specific period of time. Sometimes, the two objectives cannot be achieved at the same time. For example, if a model is divided into multiple small models for distributed computing (tensor parallelism), the speed of a single request can be improved by several times (the latency is lower). However, due to the communication and aggregation costs, the system consumes more resources to process a single request, resulting in a lower throughput.
- In this chapter, inference optimization technologies are classified into two types: computing acceleration and model compression.
  - Computing acceleration: The algorithms and hardware utilization are improved to improve computing efficiency without compromising the model output quality. In essence, model computing is accelerated.
  - In model compression, the model structure is changed to reduce specific computations (for example, through sparse attention) or reduce the computing precision (for example, through quantization). In this way, inference is faster and resource consumption is lower. However, the model output quality may be compromised. The essence of model compression is to reduce the model computing workload.



- The essence of any computing is that CPU/AI chips execute a series of instructions. Given a fixed model structure, we can perform the following three types of optimizations:
  - Reducing the number of instructions to be executed: That is, reduce unnecessary or repeated computations.
  - Maximizing the hardware concurrency: Enable a single instruction to process multiple pieces of data at a time, that is, single instruction multiple data (SIMD). Alternatively, use the multi-core mechanism of CPU and AI chips to simultaneously execute multiple instructions.
  - Improving the memory I/O speed: Use the cache locality to improve the memory's speed of reading instructions, or reduce unnecessary memory read/write operations.
- The first two types are called compute-side optimization, and the last type is called memory I/O optimization.

KV Cache

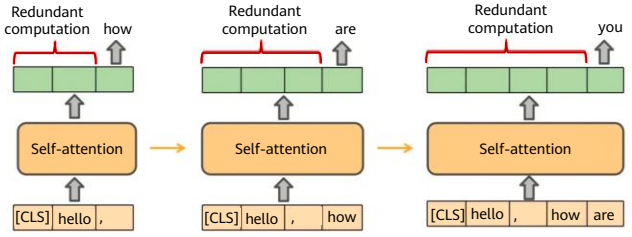
- Background
  - In a traditional Transformer model, each time a new token is generated, the Attention of all previous tokens needs to be recomputed, resulting in a large number of redundant computations.
  - When the model is applied to infinite input streams, a longer context causes more memory to be occupied by the matrix. That is, when the sequence length exceeds the Attention window size set during pre-training, the performance deteriorates.
- KV cache technology
  - The KV cache technology optimizes the inference process by caching the keys and values used in the Attention mechanism.
- Technical benefits
  - Higher inference efficiency: The KV cache technology caches the key and value vectors to avoid repeated computation each time a new token is generated, significantly improving inference efficiency.
  - Lower computing costs: The KV cache technology reduces the model computing costs by avoiding repeated computation, especially in the case of processing long-sequence text.
  - Improved resource utilization: The KV cache technology lower the requirements on hardware resources and improves resource utilization by reducing the computing workload.
  - Lower video RAM usage: The KV cache technology can reduce the video RAM usage by means of quantization, sparsification, paging, and Attention mechanism optimization. In this way, large models can be effectively used even when video RAM resources are limited.

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- At the self-attention layer of the Transformer model, the model calculates the key and value vectors of each token for a given input sequence. The values of these vectors remain unchanged in the sequence generation process. Therefore, by caching these vectors, repeated computation can be avoided each time a new token is generated. Instead, only the query vector of the new token needs to be calculated, and self-attention computation is performed using the cached key and value vectors.
- The KV cache is the simplest and most direct optimization method. Generally, the default implementation of models has the KV cache, and you only need to set **use_cache** to **True** in the configuration.
- This technology is implemented in the decode phase to trade space for time.

Video RAM Optimization—KV Cache

- Inference process of a generative model: After an input text is provided, the model outputs an answer (length: **N**). Inference is performed **N** times in this process. That is, a GPT model outputs only one token per inference step. The output token is concatenated with the input tokens to serve as the input for the next inference step. This process repeats until a terminator is encountered.



- The KV cache technology saves reusable key-value vector results to avoid repeated computation.
- One of the most commonly used technologies for optimizing the inference performance of large models is KV cache. This technology can improve inference performance by trading space for time without compromising computation precision. Currently, mainstream LLM inference frameworks support and enable this function by default.

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KV Cache Computation Process

- After the KV cache is introduced, the inference process is divided into the following two phases:
  - Prefill phase: refers to the process of computing the first output token. During the computation, the key cache and value cache need to be computed and saved for each Transformer layer. The FLOPs are the same as those when the KV cache is disabled. This phase involves a large number of general matrix multiply (GEMM) operations and is compute-bound.
  - Decode phase: refers to the process of computing the second output token to the last token. In this phase, the KV cache stores the historical key-value results. In each round of inference, the cache can be read, and the new key and value computed in the current round are added to the cache. The general matrix-vector multiplication (GEMV) operation is performed, the FLOPs decrease, and the inference is faster than that in the prefill phase. This phase is memory-bound.

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Disadvantages of the KV Cache

- The KV cache trades space for time. When the input sequence is extremely long, a large number of keys and values need to be cached, occupying a large amount of video RAM space.
- Assume that the length of the input sequence is **s**, the length of the output sequence is **n**, and the KV cache is stored in FP16 format. The following uses GPT-3 (175B) as an example to compare the video RAM capacity occupied by the KV cache and that occupied by model parameters. The video RAM capacity occupied by the weights of the GPT-3 model is about 350 GB (FP16), the number of layers is 96, and the number of dimensions is 12,888.

Batch Size	s + n	KV Cache (GB)	KV Cache/Weight
4	4,096	75.5	0.22
16	4,096	302	0.86
64	4,096	1,208	3.45

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Improvements to the KV Cache

- KV cache sharing: Multi-query attention (MQA) and grouped query attention (GQA)
  - MQA is a variant of multi-head attention. The main difference is that different attention heads in MQA share a set of keys and values, and each header retains only query parameters. As there is only one key-value matrix, the video RAM usage is significantly reduced and the video RAM is more efficient.
  - GQA is a compromise between multi-head attention and MQA. It groups query heads and shares a key head and a value head in each group.
- Window optimization
- Quantization and sparsification
- Storage and computing optimization: PagedAttention

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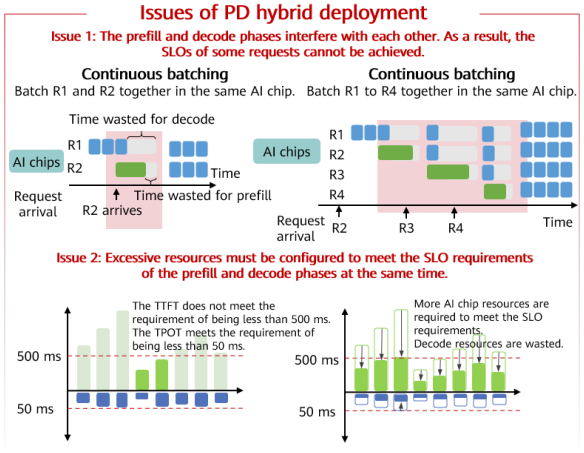
- Because MQA changes the structure of the Attention mechanism, the model needs to support MQA from the beginning of training.
- Although the KV cache is important, it occupies and wastes a large amount of space. PagedAttention is introduced to solve the waste issue. The size of the KV cache depends on the value of **seqlen**. However, the sequence length varies depending on the batch. After all, people enter questions of different lengths, and models generate answers of different lengths. The KV cache is applied for based on the value of **max seq len**. As a result, a large amount of video RAM space is wasted in existing decoder inference systems.
- PagedAttention divides the KV cache of each sequence into multiple blocks. Each block contains a fixed number of labeled keys and values. During attention computing, the PagedAttention kernel efficiently identifies and obtains these blocks, and accelerates computing in a parallel manner.

Kernel Optimization and Operator Fusion

- Background
  - Traditional computing methods face performance bottlenecks, especially when deep learning models are executed on hardware such as AI chips. Memory access latency and underutilization of computing resources become key factors that limit performance improvement.
- Kernel optimization and operator fusion
  - Each kernel call causes additional overheads, for example, communication between the AI chip and the CPU, and memory copy. Therefore, all attention computing can be implemented in the same kernel to avoid additional overheads. Attention-specific optimization can also be implemented in the kernel. Mainstream inference libraries have efficient kernel implementation.
  - Operator fusion is a method that combines multiple computing operations into a single computing unit to reduce the number of memory access times and improve computing efficiency.
- Technical benefits
  - Improved computing efficiency: Kernel optimization and operator fusion reduce memory access overheads and improve computing efficiency.
  - Enhanced parallelism: If the hardware capacity cannot be fully utilized when a single kernel is used to execute operators without data dependencies, it is conducive to implement kernel parallelism through operator fusion to enhance parallelism.
  - Improved hardware utilization: Operator nodes in computational graphs are orchestrated in parallel to improve the overall computing parallelism. In particular, when there are parallel branch nodes on the network, the parallel acceleration effect is excellent.

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PD Disaggregation Deployment



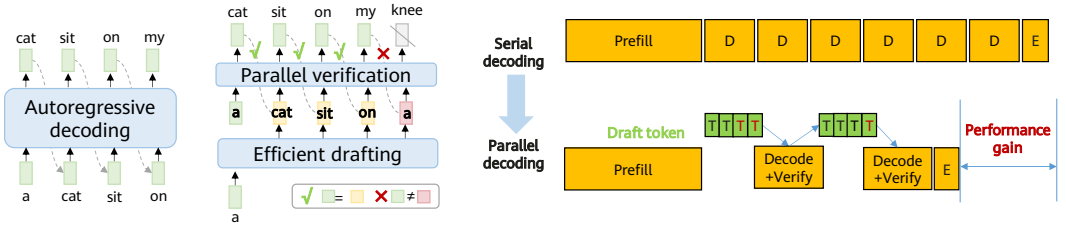
- **Background**
  - **The prefill and decode phases interfere with each other during inference by large models.** In the prefill phase, the decode phase waits. In the decode phase, the prefill latency increases.
  - **It is difficult for the prefill and decode phases to achieve the SLO at the same time.** The prefill phase is compute-intensive, whereas the decode phase is memory- and bandwidth-intensive. To meet the Service Level Objective (SLO) requirements (fulfillment rate of the TTFT and TPOT) at the same time, excessive resources must be configured.
- **PD disaggregation deployment solution**
  - Prefill and decode are deployed on different nodes.
  - Prefill and decode transfer the KV cache through the RDMA over Converged Ethernet (RoCE) network.
  - Prefill and decode are separately scheduled and elastically scaled.
- **Benefits of PD disaggregation deployment**
  - In a project, PD disaggregation deployment improves the resource utilization by 30% compared with PD hybrid deployment.
  - The prefill and decode nodes can be flexibly configured, making the SLO more controllable.

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- In the hybrid deployment scenario, it is difficult to coordinate resources required by the prefill and decode phases.
- SLO is a key indicator used to measure service performance and reliability. It is generally used to ensure that services meet user expectations and requirements. In scenarios such as natural language processing or LLM inference, the SLO can specifically refer to the fulfillment rate of the TTFT and TPOT.
- During decoding, prefilling cannot be performed. As a result, the resource utilization is low and computing power resources are wasted.

Parallel Decoding

- Parallel decoding:
  - Multiple draft tokens are obtained in a simple and quick manner (main differences between different parallel decoding algorithms).
  - The large model performs parallel verification on the draft tokens and retains the part that meets the output conditions.
- The operational intensity is enhanced to decode multiple tokens within a short period of time to accelerate inference.



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- Parallel decoding enables fast prediction and parallel verification. The computing power utilization of decoding is improved and the memory access is reduced.
- In the autoregressive decoding mode of LLMs, token-by-token serial decoding needs to be performed. The model weight and KV cache need to be read each time decoding is performed. The decoding process is limited by the memory bandwidth, resulting in underutilized computing resources.
- **D** indicates decode, and **E** indicated end.

Distributed Inference

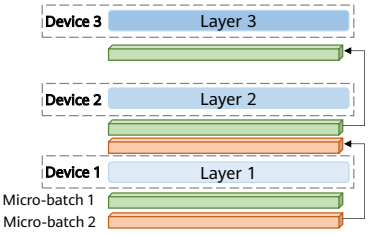
- Core issues: Single-device inference cannot meet the requirements on the storage (insufficient video memory capacity) and speed (low latency).
- Application scenarios: When there are a large number of model parameters, TP splitting is performed to reduce the video RAM usage of the weight on a single device. In this way, more video RAM is reserved for the KV cache to support more requests and meet the cost-effectiveness requirements.
- Distributed inference solution: The distributed inference solution is similar to the distributed training solution. The mainstream distributed parallelism modes are as follows:
  - Data parallelism: A model is deployed on multiple AI chips, and each AI chip contains a complete model. The dataset is divided into multiple parts, and each AI chip is responsible for the inference of a part of data.
  - Pipeline parallelism: A model is split vertically, and each AI chip contains only some layers of the model. After data is calculated on an AI chip, the output is sent to the next AI chip for subsequent computing.
  - Tensor parallelism: A model is split horizontally. Each layer of the model is split and placed on different AI chips. The computing of each layer requires the collaboration of multiple AI chips.

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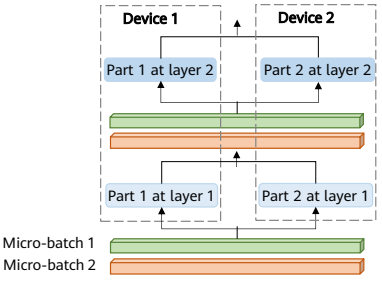
- Communication is traded for computing.
- The computing time of a single device is short.
- Performance ranking: data parallelism > tensor parallelism > pipeline parallelism. The communication load of data parallelism is the lowest.

Pipeline Parallelism and Tensor Parallelism

- Generally, pipeline parallelism is a last resort when the memory of one AI chip is unable to store the entire model. Layers still run in sequence, but this cannot accelerate model computing. Data parallelism and tensor parallelism can accelerate model inference. Data parallelism in the inference phase is simple because inference results do not need to be aggregated to update parameters, which is required during training. Therefore, you only need to deploy the model on multiple AI chips for inference.



Pipeline parallelism



Tensor parallelism

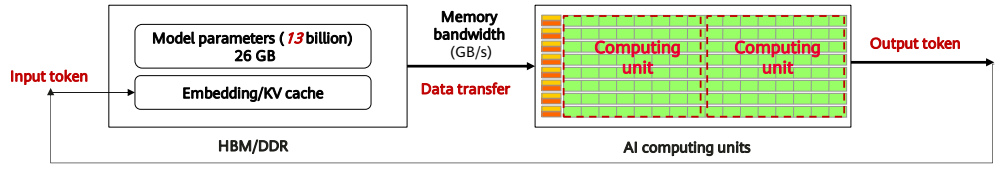
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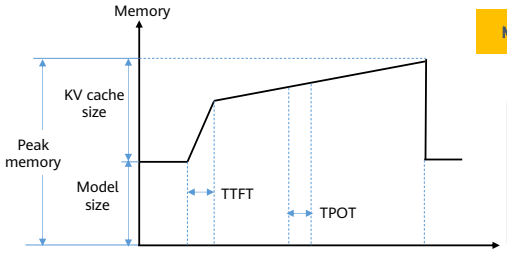
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- Layer-level splitting
- Pipeline parallelism: Inter-layer splitting
- Tensor parallelism: Intra-layer splitting

Resources Required for Inference Must Meet Storage and Transfer Requirements

- Storage: There is sufficient device memory for storing the model weight, KV cache, and system overheads.
- Transfer: Data of the required volume (number of model parameters + KV cache size) can be transferred on the prerequisite that the TPS and latency requirements are met.





Model inference space = Model weight space + Total KV cache space + Others

Number of parameters × 2

$(s + n) \times h \times 2(FP16) \times 2(K \& V) \times l \times b$

- s = seq-length
- n = Number of tokens generated at a time
- h = hidden-layer size (key/value vector dimension)
- l = layer
- b = batch size

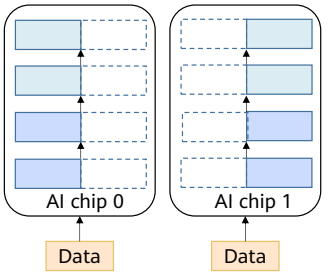
For example, for a 13B model, the weight requires a space of about 26 GB. Assuming that the KV cache requires a space of about 1.6 GB, the total memory usage is about 30 GB. If the TPOT is 50 ms, the effective memory bandwidth must be greater than 600 GB/s.

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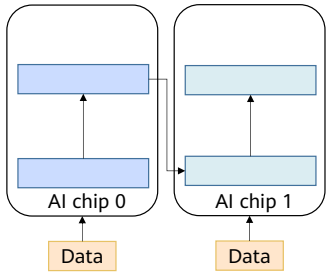
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- Storage: The memory space of a single card needs to be greater than the model inference space. If the model inference space is greater than the memory capacity of a single card, hardware supporting multi-card high-speed interconnection is needed.
- Transfer: The product of Memory bandwidth × Utilization × Latency must be greater than the model inference space.
- hidden-layer size = num_kv_head × head_dim

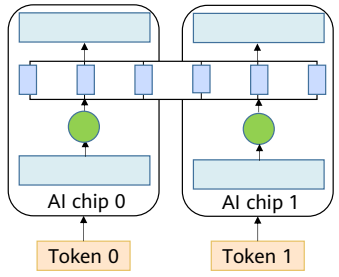
Performing Multi-card Parallel Inference When Model Inference Space Exceeds Memory Capacity of a Single Card



Tensor parallelism:
Matrices are split and a single operation is split to different AI chips for execution.



Pipeline parallelism:
Different model layers are executed on different AI chips.



Expert parallelism:
Different expert networks in the expert layer of the model are executed on different AI chips.

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- Mixture of experts (MoE): One layer of a large model is split into multiple expert networks with the same structure. In each iteration, some experts are activated for computing based on samples to train ultra-large-scale sparse models.

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2. Optimization Methods of Large Model Inference
3. **Inference Systems and Frameworks**
   - Inference Systems
   - Inference Frameworks
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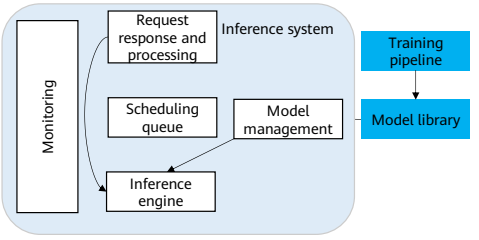
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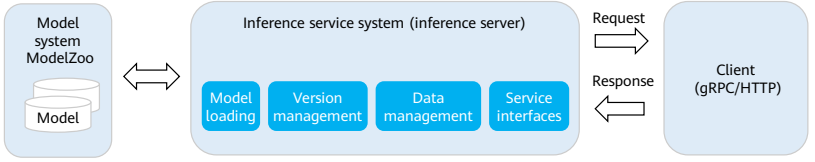
What Is an Inference System?

An inference system consists of the inference engine, inference management services, and model system. It is an AI system used to deploy models and execute inference and prediction tasks.

Components

- **Model system**: A trained model is saved in a file system. As the performance of the trained model continuously improves, models of new versions may be generated. They are stored in the file system and managed using specific model version management protocols.
- **Management services:**
  - Data management: The batch size of input data is dynamically optimized, and model outputs are processed.
  - Version management: Model versions are managed. New versions can be brought online and models can be rolled back to earlier versions.
- **Inference engine:**
  - Service interfaces: Service interfaces (such as HTTP and gRPC) are provided for clients to invoke.
  - Model loading: Models are deployed and brought online through the service system. The inference system loads models to the memory first.



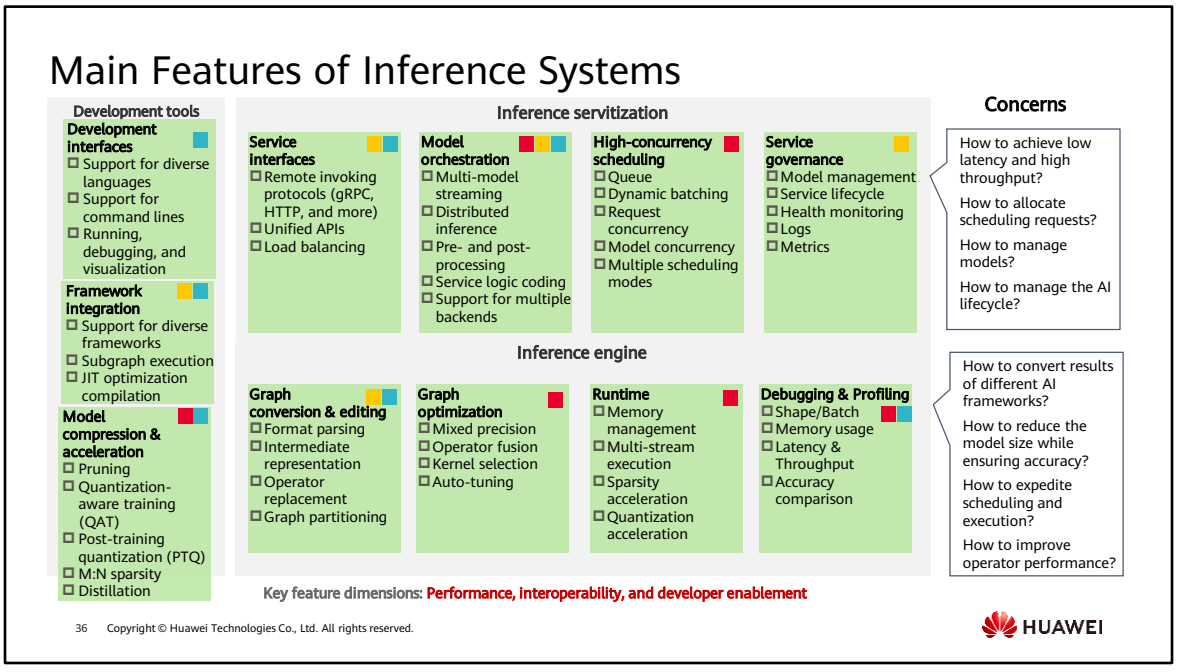


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- The preceding figure shows the architecture and process of an inference service system. First, a trained model is stored in a file system. As the training performance is optimized, multiple model versions may be generated. These models are properly stored in the file system according to the predefined version management rules.
- Then, these models are deployed and brought online through the service system. During the rollout, the inference system loads the model to the memory and manages model versions, ensuring that new versions can be quickly rolled out and models can be quickly rolled back to earlier versions. In addition, the system dynamically optimizes the batch size of input data to meet different processing requirements. The inference system also provides diverse service interfaces (such as HTTP and gRPC) for clients to invoke.
- You can use these interfaces to continuously send requests to the inference service system and receive responses. In addition to providing services for users, the inference system can function as a microservice and be invoked by other microservices in the data center to play its specific role in the entire request handling process.
- The inference system is not limited to using data centers as its core service mode. It also needs to fully adapt to and serve mobile devices at the edge. When discussing the strategy of the entire inference service, we need to consider not only the efficient inference capability of data centers but also the inference

requirements of edge devices. To sum up, the inference system involves complex systems engineering, and its complexity is reflected in that it needs to consider multiple factors.



- Interoperability dimension: Inference systems interact with deep learning frameworks and other modules in data centers.
- R&D enablement dimension: Development interfaces, compression acceleration tools, debugging, profile, visualization, flexibility, scalability, and composability.
- The inference engine is responsible for delivering high performance. Servitization is used to implement interoperability. Development tools are used for enablement.

Relationships Between Inference System Concepts

Vehicle system	VS	Inference system
Automotive engine	VS	Inference engine
Driving service	VS	Inference service



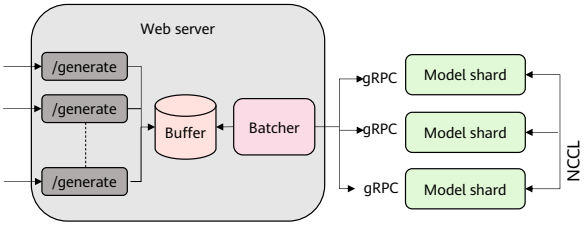
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Hugging Face TGI Framework (1)



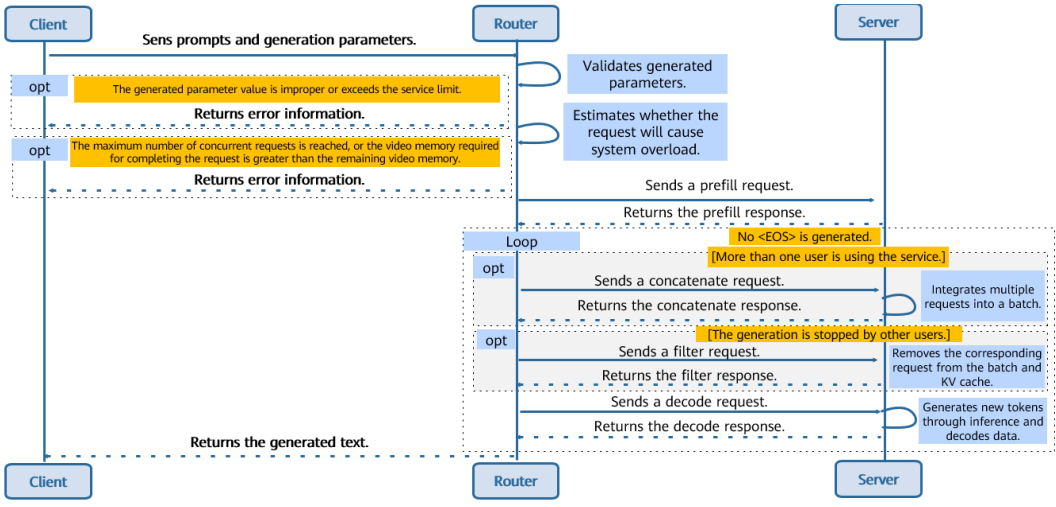
Service logic process

1. After multiple clients simultaneously request the **/generate** service of the web server, the server integrates these requests into a batch in the Buffer component.
2. The requests are forwarded to the inference engine of the AI chip through gRPC for computing and generation.
3. The requests are distributed to multiple model shards, which communicate with each other through NCCL. This is because the video memory capacity is limited or multiple AI chips are required for distributed inference to improve computing efficiency.

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- The service logic process of Hugging Face TGI is mainly based on the router and server.

Hugging Face TGI Framework (2)



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- The four interactions (prefill, decode, concatenate, and filter) between the router and the server are the core service logic of large model inference.

Key Features of Hugging Face TGI

- Supports inference with tensor parallelism.
- Supports continuous batching of incoming requests to improve the overall throughput.
- Optimizes the Transformers code used for inference on mainstream model architectures using FlashAttention and PagedAttention.
- Performs quantization with bitsandbytes (LLM.int8()) and GPT-Q.
- Provides built-in service evaluation, which can monitor server loads and learns about the server performance.
- Easily runs your own models or models from any Hugging Face repository.
- Easily generates text by providing custom prompts to guide the model output.
- Uses OpenTelemetry and Prometheus metrics for distributed tracing.

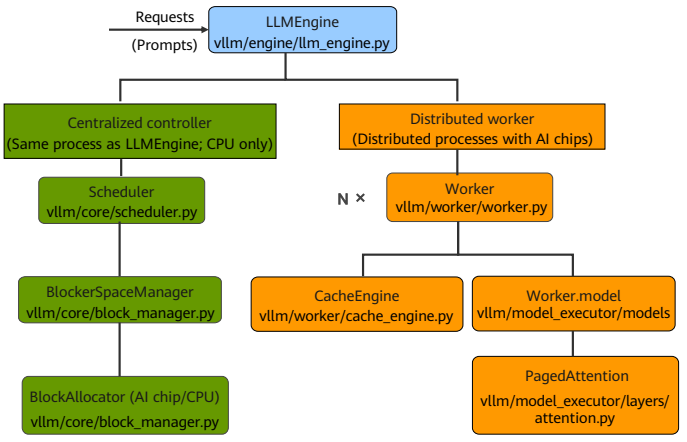
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Introduction to vLLM

- vLLM is an open-source framework for accelerating inference of large models. It efficiently manages tensors cached in attention through PagedAttention, achieving 14 to 24 times higher throughput than Hugging Face Transformers.
- PagedAttention is the core technology of vLLM and solves the memory bottleneck issue in LLM services. In the autoregressive decoding process, traditional attention algorithms need to store the attention key and value tensors of all input tokens in the memory of AI chips to generate the next token. These cached key and value tensors are usually referred to as KV caches.

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Overall vLLM Architecture



LLMEngine can be divided into two parts.

Centralized controller:

- The scheduler determines the data to be sent to models for inference in each inference phase, and allocates physical KV cache blocks to the models.
- The scheduler maintains BlockSpaceManager, which manages BlockAllocator (the class that participates in physical block allocation).

Distributed workers:

- Distributed workers are distributed systems. They load the model to be used to each card, perform inference on the data sent by the controller, and return related results.
- Distributed workers specify the method of controlling workers.

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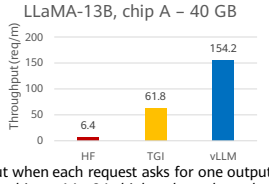
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- BlockAllocators are classified into two types, which manage physical blocks on the AI chips and CPUs, respectively.
- Distributed worker: Each worker can be regarded as an AI chip. Currently, vLLM supports TP/PP inference for models that cannot be installed on a single device.
- Distributed workers: They are marked in orange blocks in the figure. According to the vLLM source code, it is more appropriate to write them as executors. It is the management and control center of all workers. It specifies the method for managing and controlling these workers and is responsible for initializing the distributed environment. Currently, the following methods are supported: 1. **cpu_executor** (rarely used): This method is applicable when CPUs are used for inference. 2. **AI chip_executor**: This method can be used in the case of a single card (world_size = 1). 3. **ray_AI chip_executor**: It is an executor implemented using the distributed computing framework ray and applies to the multi-device environment. Workers refer to AI chips in terms of hardware, and worker instances (each process on the AI chip maintains its own worker instance) in terms of code. Each worker instance manages and controls two important instances: CacheEngine and Worker.model. The CacheEngine manages and controls the KV cache physical blocks on AI chips/CPUs. (The block manager of the scheduler only allocates physical block IDs, whereas the CacheEngine manages data in physical blocks based on the ID allocation result.) According to the vLLM code, it is more appropriate to replace Worker.model with

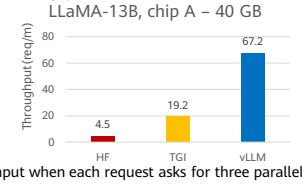
model_runner. The Worker.model loads models and performs inference. The logic related to PagedAttention is maintained in the code associated with the Worker.model instance.

Features of the vLLM Framework

- The core of vLLM is the PagedAttention algorithm. Other key technologies of vLLM are as follows:
  - KV cache video RAM optimization (INT4/8 quantization of the KV cache)
  - Continuous batching
  - Optimization of the unified computing device architecture kernel for AI chips
- Flexibility and ease of use
  - Seamless integration with Hugging Face models
  - High-throughput services that support decoding algorithms such as parallel sampling and beam search
  - Support for tensor parallelism in distributed inference
  - Support for streaming output
  - Compatibility with OpenAI interface services



Serving throughput when each request asks for one output completion. vLLM achieves 14x–24x higher throughput than HF and 2.2x–2.5x higher throughput than TGI.



Serving throughput when each request asks for three parallel output completions. vLLM achieves 8.5x–15x higher throughput than HF and 3.3x–3.5x higher throughput than TGI.

Performance: vLLM is far ahead of the HF (without LLM-accelerated inference) baseline and TGI frameworks.

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- PagedAttention is used to effectively manage the KV cache.
- Continuous batching, rather than static batching, of the transferred requests
- Support for inference with tensor parallelism
- Support for streaming output
- Compatibility with OpenAI interface services
- Seamless integration with Hugging Face models

Feature Comparison of Inference Frameworks

Framework	vLLM	TGI
Vendor	Community open-source project	Hugging Face
Code repository	https://github.com/vllm-project/vllm (29.6k stars)	https://github.com/huggingface/text-generation-inference (9k stars)
Features	• SOTA throughput • Efficient memory management (**near optimal**) • Optimized, unified computing device architecture for AI chips	• Safetensors • Watermarking
Multimodal support	√ (LLaVA, MobileVLM, Obsidian, and more)	

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- In terms of inference acceleration performance, vLLM provides faster acceleration, more stable video RAM usage, and a larger batch size than TGI.
- All tested inference acceleration methods cause the inference performance to deteriorate. Tasks with stricter requirements on the results (the result has only one answer) have more severe deterioration.
- PagedAttention is a technology used to improve LLM inference efficiency. It manages the cached tensors in the attention mechanism by page. Its working principles include memory paging, efficient access, and dynamic adjustment. It improves memory utilization and computing efficiency, and has high flexibility and scalability. It can be used in scenarios such as LLM inference, distributed inference, and resource-constrained environments.
- Continuous batching is a technology that improves the efficiency of computing and data processing. It continuously collects tasks and stores them in the buffer. When the threshold is reached or a specific period of time elapses, batch processing operations are triggered, including task collection, batch processing triggering, execution, and result return. It improves efficiency and resource utilization, and has strong adaptability. It can be used in scenarios such as real-time services, data processing, and model inference to reduce task startup costs and waiting time. It is capable of adapting to different task types and workloads.
- Medusa is a large model inference acceleration technology proposed by

institutions such as Princeton University. It introduces multiple "Medusa heads" on the last hidden state of the original language model to predict subsequent tokens in parallel. In addition, it uses the tree-based attention mechanism to process multiple candidate sequences. There are two training policies: Medusa-1 (only the decoding heads of the original model are trained) and Medusa-2 (the decoding heads and the original model are trained together). This technology can significantly accelerate inference and improve the inference speed by 2.2–3.6 times. It is easy to integrate and flexible, enabling large models to be used in scenarios such as real-time interaction.

- Chunked prefilling + batching in the decode phase is a technology that improves the inference efficiency of large models. It includes chunked prefilling, decoding, and batching operations. Chunked prefilling divides input data into small chunks and pre-processes them. Decoding is performed to generate the output result. Batching can be performed in the prefilling and decoding phases to utilize parallel computing.

Feature Comparison of Inference Frameworks

Framework	vLLM	TGI
Memory offload	√	
Continuous batching	√	√
Flash Attention	√	√
Speculative inference by large and small models	√	
Medusa		√
Chunked prefilling + Batching in the decode phase	√	
Automatic prefix caching	Experimental feature	
Multi-LoRA support	Experimental feature	√
Quantization support	• GPTQ • AWQ • SqueezeLLM • FP8 KV cache	• Bitsandbytes • GPTQ • EETQ • AWQ

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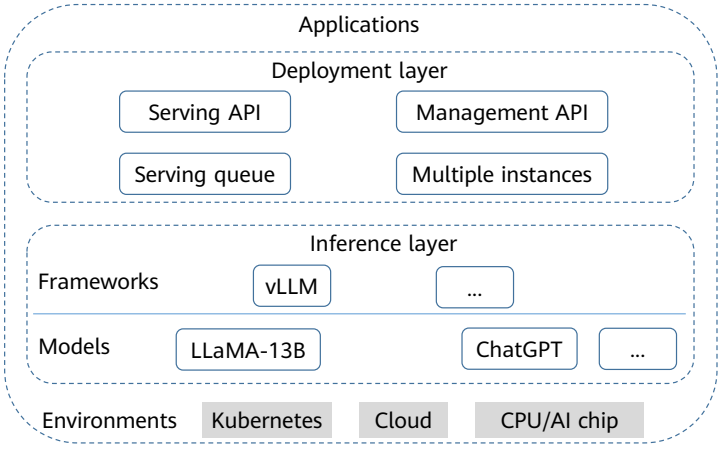
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AI Model Deployment Architecture



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- The model deployment layer connects to the lower-level inference layer and provides model service capabilities for upper-level applications. The framework at this layer is called the model deployment framework. The model deployment framework is based on the inference framework and provides capabilities such as auto scaling, observability, and multiple model instances, inference frameworks, service protocols, and clouds.

Optimization Technologies and Tools During Model Deployment

- In the model deployment process, some optimization methods and technologies can be used to improve the model performance, reliability, and security.
  - The model compression technology can reduce the model size, computing resource consumption, and storage resource usage.
  - The model acceleration technology can accelerate inference.
  - The containerization technology can implement quick model deployment and version control.
- Some useful tools can be used to assist in model deployment tasks.
  - Deployment services provided by open-source frameworks such as TensorFlow Serving and ONNX Runtime.
  - Containerization technologies such as Docker and Kubernetes. They can be used to quickly deploy and manage models.
  - AI computing resources provided by cloud service providers.

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Deployment Solutions: Cloud-based Deployment

- In cloud-based deployment, a model is deployed on cloud-based computing resources. Users send requests to the cloud server by accessing web pages or calling APIs. The cloud server processes the received requests and returns results.

Advantages	Disadvantages
• Higher computing capability: The cloud computing platform usually has a large number of computing resources and high scalability to meet the high-performance computing requirements of AI models. • Higher resource utilization: Cloud-based deployment enables resource sharing and proper allocation, improving resource utilization. • Lower O&M costs: The cloud computing platform maintains and manages hardware and software, reducing O&M costs. • There are no strict restrictions on the power consumption, temperature, and model size. Models can be easily protected on the cloud. This deployment solution is applicable to large-scale data processing and complex computing.	• The inference service is highly dependent on the network, and the data transmission cost is high. • Data privacy issues may occur. • It is difficult to customize models.

- Examples: Intelligent recommendation, automatic review, online translation, AI filter, ChatGPT, and Midjourney.

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Deployment Solutions: Edge/On-device Deployment

- Edge/On-device deployment mainly applies to embedded devices. Models are packaged and encapsulated into software development kits (SDKs) and integrated into embedded devices. Data processing and model inference are performed on devices.

Advantages	Disadvantages
• Low latency, high reliability, and good privacy protection • Small device size and low power consumption • Applicable to scenarios that require high real-time performance	• The power consumption and heat generation are strictly restricted. The model size must be less than the device memory size. • The hardware computing power is low. • Data is scattered and difficult to train. • Models are more vulnerable to attacks at the edge. • There are diverse platforms, and no universal solution is available.

- Examples: Optical character recognition (OCR) function on mobile phones, automatic smile capture, and facial recognition gate.

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Deployment Solutions: Device-Cloud Synergy Becomes the Mainstream

- Device-cloud synergy: Devices and the cloud collaborate to distribute AI computing workloads.
- Based on the workload distribution mode, there are different cloud-device hybrid modes:
  - Hybrid AI with devices as the focus. Devices function as anchors for real-time inference. The cloud is used to distribute and process tasks that cannot be fully executed by devices. It is also used for model tuning iterations.
  - Hybrid AI that features collaborative processing by the cloud and devices. Devices collect data and send it to the cloud, and the cloud invokes model capabilities to perform inference.

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- On-device AI can understand personal data in devices, and accurately predict user behavior and provide assistance based on such data. Cloud-based AI has unparalleled performance advantages. It provides more precise information based on devices, and can answer and assist users in a more intelligent and accurate manner.

Characteristics of Different Model Deployment Solutions

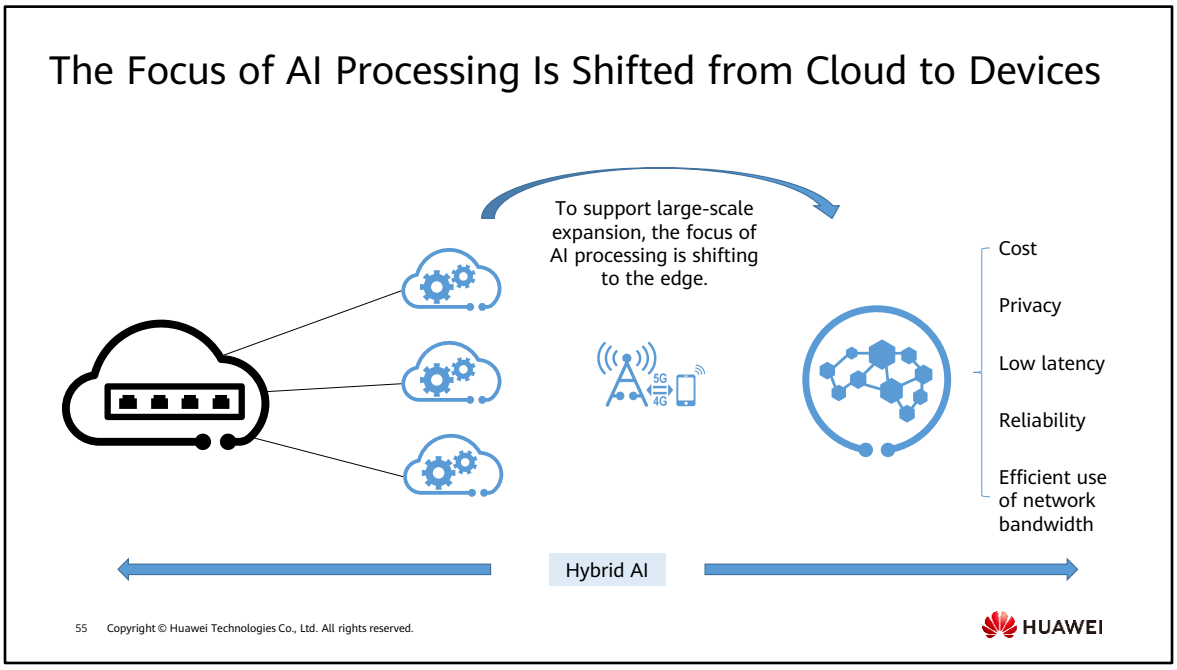
	Cloud-based Deployment	On-device Deployment	Device-Cloud Synergy Deployment
Computing power	Robust computing power and elastic scalability	Limited computing power and poor scalability	Robust computing power and elastic scalability
Latency	Network latency + Cloud-side computing latency	Device-side computing latency	Device-side computing latency + Network latency + Cloud-side computing latency
Network dependency	Strong dependency	Weak dependency	Strong dependency
Transmission data	Raw inputs, such as prompts or multimodal data such as images and videos	None	Raw inputs, intermediate results, or intermediate computation data
Device-side power consumption	Low; insensitive	< 10 W; extremely sensitive	< 10 W; moderately sensitive
System architecture	Open and highly centralized	Closed and decentralized	No standard architecture
Ecological diversity	High; high standardization; xPU	Relatively high; diverse chip architectures and many SoCs	Relatively high. Mainstream frameworks basically support both the cloud and devices.
R&D cost	Low. Complete auxiliary devices are available and the portability is high.	Relatively high. Auxiliary devices are incomplete and the portability is limited.	Relatively high. Both the device and cloud sides need to be considered.
Inference cost	High. The cloud-side computing power cost needs to be recovered through business value realization or borne by service providers.	Low. The device-side computing power is bought out by users at a time.	Medium. The computing power to be provided by devices is low.

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Deployment Modes: Service-based Deployment and SDK-based Deployment

- Service-based deployment is mainly used to deploy the central server on the cloud. Generally, the training engine library is used to provide the inference service. RESTful APIs or gRPC APIs are provided for users to easily invoke models for prediction and inference.
- SDK-based deployment is mainly used to deploy embedded devices. Languages such as C++ and Python are used to implement efficient engine libraries used for pre-processing, post-processing, and inference. Developers can integrate SDKs into embedded devices to implement local model inference and real-time response.

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- Device-side AI deployment helps significantly reduce computing power costs and power consumption, and streamline the business models of AI monetization. As the number of daily active users and usage frequency increase, the cost and energy consumption will increase significantly if only cloud-based computing power is used to support model inference. In terms of AI running on devices, take Stable Diffusion, which can run on mobile phones under Qualcomm's full-stack optimization, as an example. The cost generated each time is small or can be ignored. This effectively reduces the cost of application vendors and promotes the implementation of software application business models.
- Device-side AI deployment reduces issues regarding privacy leakage, data security, and latency by keeping sensitive user information locally. In a data theft attack, multiple outputs of the target model are used to obtain the data distribution used in the training process. When an attacker knows the specific data used in the large model training process, data privacy is damaged. Device-side AI deployment stores personal information locally, which can fundamentally avoid data security issues. In addition, experience measured by metrics such as latency is important to end users. When the access traffic of a cloud-based large model reaches the peak, feedback delay or even denial of service (DoS) may occur.

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Evaluation Metrics for Measuring the Model Inference Speed

- TTFT: Time required for the model to generate the first output after a user initiates a query.
- TPOT: Time required to generate an output token for each user who queries the system. This metric is related to each user's perception of the model speed. For example, a TPOT of 100 ms/token indicates that each device can process 10 tokens per second or about 450 words per minute. This is far faster than the reading speed of ordinary people.
- Latency: Total time required for the model to generate a complete response for a user. The overall response latency can be calculated using the preceding two metrics: Latency = (TTFT) + (TPOT) × (Number of tokens to be generated).
- Throughput: Number of output tokens that the inference server can generate per second for all user requests.

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- In real-time interaction, low-latency response is important. However, it is unimportant in offline workloads.

Factors Affecting the Model Inference Speed

- FLOPs: Total addition and multiplication operations of the model.
- MAC: Memory access cost.
- Parallelism degree: A higher parallelism degree leads to faster model inference.
- Computing platforms: AI chips, CPUs, and more.

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Model Quantization

- Model quantization refers to the process of converting floating-point numbers of a neural network into fixed-point numbers.
  - Low-precision models: The model weight format is FP16 (half-precision floating-point) or INT8 (8-bit fixed-point integer).
  - Normal-precision models: The model weight format is FP32 (32-bit single-precision floating-point).
  - Mixed-precision models: Both FP32 and FP16 weight formats are used in the model. FP16 reduces the required memory capacity by half, but some parameters or operators must use the FP32 format to maintain accuracy.
- Advantages of model quantization:
  - Reducing the model size: For example, 8-bit integer quantization can reduce the model size by 75%.
  - Reducing the storage space required: This is of great significance when the storage space at the edge is insufficient.
  - Reducing memory consumption: Smaller models require less memory.
  - Accelerating inference: The time for accessing FP32 floating-point numbers once can be used to access INT8 integers for four times. The integer arithmetic is faster than the floating-point arithmetic.
  - Reducing device power consumption: Lower memory usage and faster inference result in lower power consumption of devices.
  - Supporting microprocessors: For 8-bit microprocessors, the floating-point arithmetic is slow when the power consumption is low, and 8-bit quantization is required.

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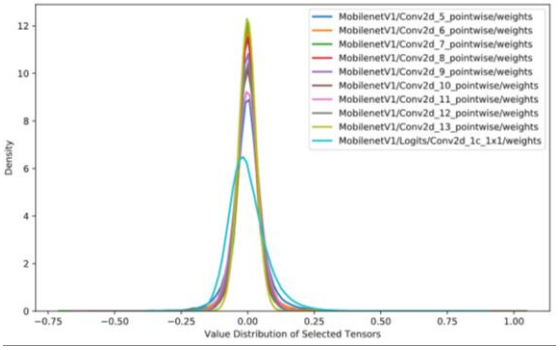
- Currently, low precision generally refers to INT8.
- CPUs can achieve faster computing by using INT8.
- Some hardware accelerators, such as DSP and AI chips, support only INT8.
- Generally, only INT8-quantized models are used in the industry. For example, mobile model inference frameworks such as NCNN and TNN support INT8-quantized models and the inference functions of quantized models.

Weight Distribution in a Neural Network

Range of FP32: $[(2-2^{23})\text{x}2^{127},\ (2^{23}-2)\text{x}2^{127}]$

Range of INT8: $[-128, 127]$

Generally, the weight distribution range of a neural network is narrow and quite close to zero.



Weight distribution of the 10-layer MobileNetV1

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Precision Loss Caused by Quantization

Model	fp32 Accuracy	PTQ best Calibration	PTQ best Accuracy	PTQ best Relative	QAT Accuracy	QAT Relative
MobileNet v1	71.88	99.9%	70.39	-2.07%	72.07	0.26%
MobileNet v2	71.88	99.99%	71.14	-1.03%	71.56	-0.45%
ResNet50 v1.5	76.16	Entropy	76.05	-0.14%	76.85	0.91%
ResNet152 v1.5	78.32	Entropy	78.21	-0.14%	78.61	0.37%
Inception v3	77.34	Entropy	77.54	0.26%	78.43	1.41%
Inception v4	79.71	99.99%	79.63	-0.10%	80.14	0.54%
ResNeXt50	77.61	Entropy	77.46	-0.19%	77.67	0.08%
ResNeXt101	79.30	99.999%	79.17	-0.16%	79.01	-0.37%
EfficientNet b0	76.85	Entropy	72.06	-6.23%	76.95	0.13%
EfficientNet b3	81.61	99.99%	80.28	-1.63%	81.07	-0.66%
Faster R-CNN	36.95	Entropy	36.82	-0.35%	36.76	-0.51%
Mask R-CNN	37.89	99.9999%	37.80	-0.24%	37.75	-0.37%
Retinanet	39.30	99.999%	39.19	-0.28%	39.25	-0.13%
FCN	63.70	Entropy	64.00	0.47%	64.10	0.63%
DeepLabV3	67.40	99.999%	67.50	0.15%	67.50	0.15%
GNMT	24.27	Entropy	24.53	1.07%	24.38	0.45%
Transformer	28.27	99.99%	27.71	-1.98%	28.21	-0.21%
Jasper	96.09	Entropy	96.11	0.02%	96.10	0.01%
BERT Large	91.01	99.999%	90.20	-0.89%	90.67	-0.37%

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- Summary of PTQ and QAT: PTQ best reports the best accuracy and corresponding calibration for each model. QAT reports accuracy after fine-tuning starting from the best PTQ model.

Model Quantization Classification

- By quantization solution:
  - L1: Data-free quantization solutions. Floating-point parameters are directly converted into quantized numbers without using any calibration set. Such solutions are easy to use, but generally cause great precision loss.
  - L2: Calibration solutions. Based on the calibration set, a small amount of actual data is input for statistical analysis. Many chip vendors, such as TensorRT, Qualcomm, HiSilicon, Horizon, and Cambricon, provide this function.
  - L3: Fine-tuning solutions. Based on training fine-tuning, the quantization error is simulated and modeled during training, and the weights are adjusted to facilitate quantization. The advantage is that the precision is improved. The disadvantage is that the model training code needs to be modified and the development period is long.
- By quantization phase:
  - PTQ
  - Quantization-aware fine-tuning (QAF)
  - QAT
- By quantization formula:
  - Linear quantization (used as an example in this chapter)
  - Non-linear quantization

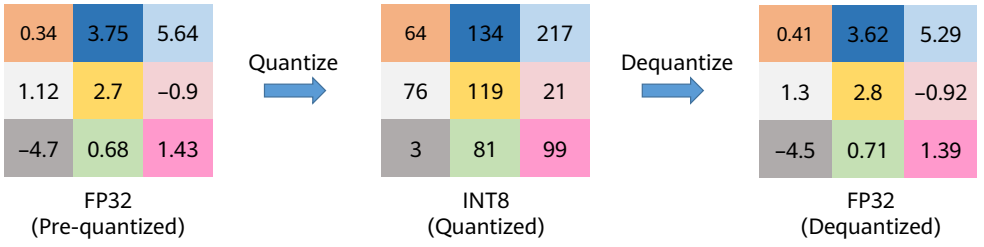
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- Data-free solutions: Qualcomm's paper *Data-Free Quantization Through Weight Equalization and Bias Correction* has proved that high precision can be achieved without using any calibration set.

Model Quantization Principles

- The principle of model quantization is to establish an effective data mapping relationship between fixed-point numbers and floating-point numbers.
- There are two important processes: quantization and dequantization.
  - Quantization converts floating-point real numbers to integers (FP32 → INT8).
  - Dequantization converts integers to floating-point real numbers (INT8 → FP32).



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- INT8 can also be replaced with UINT8, but the value ranges of the two data formats are different.
- For operators do not support fixed-point computation, dequantization is required.

Example of Quantization Algorithms for Large Models: QLoRA

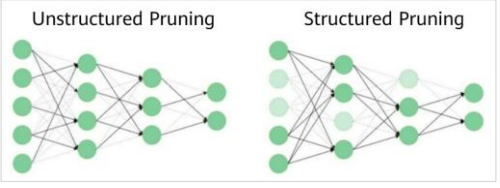
- QLoRA combines two methods: model quantization and low-rank adaptation (LoRA). LoRA is a parameter-efficient fine-tuning (PEFT) technology. QLoRA enables a 65B large model to be fine-tuned on a single 48 GB AI chip. The QLoRA quantization method (the backend is provided by the bitsandbytes library) is also the official model quantization implementation of Transformers.
- Innovative points of quantization:
  - QLoRA uses a new NormalFloat (NF) data type that is theoretically optimal for normal distribution weights. In addition, the NF type helps mitigate the impact of outliers.
    - The lattice points of NF4 are truncated based on the quantile of normal distribution. They are sparse at both ends and dense in the middle. The lattice point distribution is consistent with the data distribution.
  - Double quantization: The quantized scale data is further quantized using the symmetric quantization algorithm.
    - QLoRA divides every 64 parameters into a block, that is, **block_size** is **64**. The scale data is calculated for each block. The quantized scale data is usually stored in FP32 format. When there are a large number of blocks, the video RAM usage of the scale data cannot be ignored. Therefore, QLoRA further quantizes the scales to the FP8 format, and the block size of double quantization is 256. In this way, the video RAM usage is further reduced.

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- The quantization process is basically the same as that described earlier.
- Before double quantization, an additional video RAM space of 0.5 bits (32/64) is required for the quantization of each parameter. After double quantization, only an additional video RAM space of 0.127 bits (8/64 + 32/(64 × 256)) is required for the quantization of each parameter.

Model Pruning

- Model pruning is also called model sparsification. Different from model quantization that compresses each weight parameter, model pruning attempts to "delete" some weight parameters. The principle of model pruning is to remove "unimportant" model weights in order to reduce the model parameter scale and computing workload without compromising model accuracy.
- Pruning can be classified into unstructured pruning and structured pruning. The main difference between them lies in the pruning target and the generated network structure. Structured pruning prunes connections or hierarchical structures based on specific rules while retaining the overall network structure. Unstructured pruning prunes individual parameters and results in irregular sparse structures.



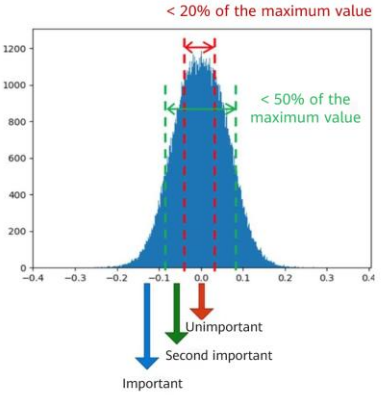
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- https://users.soe.ucsc.edu/~pang/200/f18/papers/2018/5784-learning-both-weights-and-connections-for-efficient-neural-network.pdf

General Procedure for Model Pruning

- Training the initial model: First, an initial large model needs to be trained to reach the required performance level.
- Accurately evaluating the importance of parameters: The functions of each model parameter are analyzed based on the absolute values and gradient information of weights, providing strong support for optimization decision-making.
- Pruning: Unimportant parameters or connections are pruned based on the evaluation result. The pruning can be structured or unstructured.
- Correction and fine-tuning: After pruning, correction and fine-tuning need to be performed to ensure that the model performance does not deteriorate significantly.



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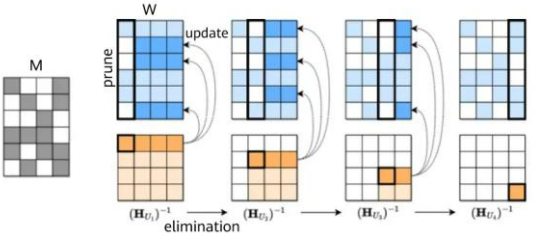
Unstructured Pruning

- Unstructured sparsity is also referred to as fine-grained sparsity. It prunes a single weight value or the weight values of an entire row or column in a weight matrix. The pruned new weight matrix becomes a sparse matrix (the pruned values are set to **0**).
- Fine-grained pruning causes irregular computing features and is not friendly to data access and large-scale parallel computing in computing devices.

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Typical Algorithms for Unstructured Pruning of Large Models: SparseGPT

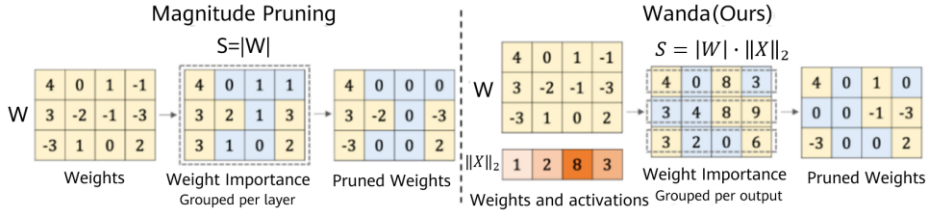
- The SparseGPT method proposed in 2023 can run a model with hundreds of billions of parameters on a single AI chip for several hours while maintaining high model accuracy. It can also prune the model to a sparsity level of 50% to 60% without greatly degrading performance.
- Given a fixed pruning mask **M**, the corresponding weights in each column of the weight matrix **W** are incrementally pruned using the Hessian inverse sequence. In addition, the remaining weights (on the right side of the pruned columns) in the same rows as the pruned weights are updated. To be specific, the weights on the right (marked in dark blue) of the pruned weights are updated to compensate for pruning errors, whereas the unpruned weights (marked in light blue) are not updated.



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Typical Algorithms for Unstructured Pruning of Large Models: Wanda

- A new pruning metric is introduced. The importance of each weight is determined by the product of its magnitude and the norm activated by the corresponding input. Using this new metric, the Wanda algorithm obtains sparse LLMs by locally comparing weights within each output of the linear layer and deleting weights of lower priorities.



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Structured Pruning

- The basic pruning unit of structured sparsity is one or more channels of a convolution kernel or weight matrix. Because structured pruning does not change the sparse degree of the weight matrix, most existing computing platforms and frameworks support this method.
- Structured sparsity is also referred to as coarse-grained sparsity or block sparsity in many references.

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Typical Algorithms for Structured Pruning of Large Models: LLM-Pruner

- Philosophy: Pruning + A small amount of data + A small amount of training = Efficient LLM compression.
- LLM-Pruner involves three stages: discovery, estimation, and recovery.
  (1) Discovery stage: This step focuses on identifying the interdependent structures in the LLM. Such structures need to be removed at the same time to ensure the structure accuracy after pruning. (Coupling inside the MLP, coupling inside the MHA, and channel coupling in the entire network)
  (2) Estimation stage: Once the coupling structures are grouped, the second step is to estimate the contribution of each group to the overall model performance, and to determine the groups to be pruned.
  (3) Recovery stage: This step involves fast post-training to mitigate possible performance degradation caused by structure deletion. (Efficient fine-tuning policies such as LoRA are used.)

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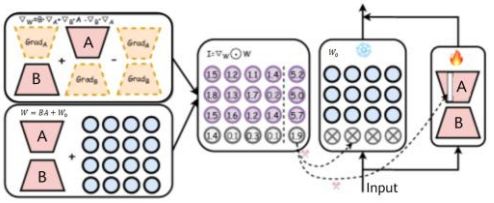
Low-Rank Factorization

- Low-rank factorization aims to approximate a given weight matrix by factorizing it into two or more matrices of smaller dimensions. Its core idea is to find a factorization of a large weight matrix W in order to obtain two matrices U and V, where $W \approx UV$. U is an $m \times k$ matrix, V is a $k \times n$ matrix, and **k** is far less than **m** and **n**. The product of U and V approximates the original weight matrix, thereby greatly reducing the parameter scale and computing overhead.

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Typical Algorithm for Low-Rank Factorization of Large Models: LoRAPrune

- This algorithm combines low-rank factorization and pruning.
- It introduces a novel weight importance criterion called the LoRA-guided criterion for large-scale pre-trained models (LPMs) and can work seamlessly with LoRA. With the gradient of LoRA, the importance of the pre-trained weights can be effectively estimated without calculating the gradient.
- Based on the proposed criterion, LoRAPrune is introduced. It is a new framework that combines PEFT and pruning. Because gradients can be efficiently approximated and weights can be updated through LoRA, LoRAPrune allows iterative structured pruning. In this way, LoRAPrune can achieve a high compression rate while ensuring that the pruned model can be seamlessly integrated with LoRA weights.
- Pruning experiments on the LLaMA model show that LoRAPrune can efficiently perform structured pruning with up to 65 billion weights on an AI chip. In addition, the pruning result of LoRAPrune is significantly better than that of other pruning methods.

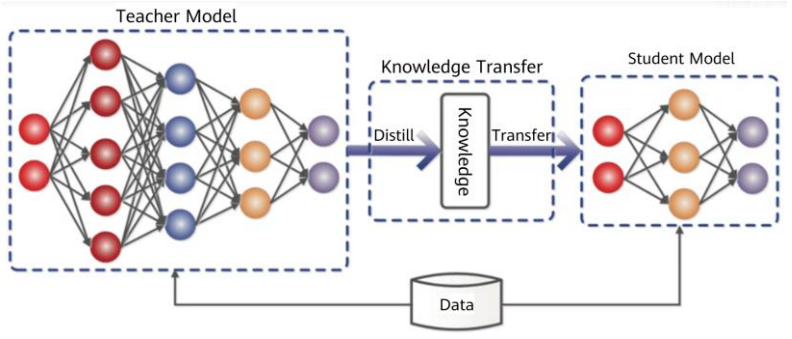


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- LoRAPrune requires only 52.6% of the memory overhead required by LLM-Pruner, while achieving lower perplexity on WikiText2 and PTB datasets.
- https://ar5iv.labs.arxiv.org/html/2305.18403

Knowledge Distillation

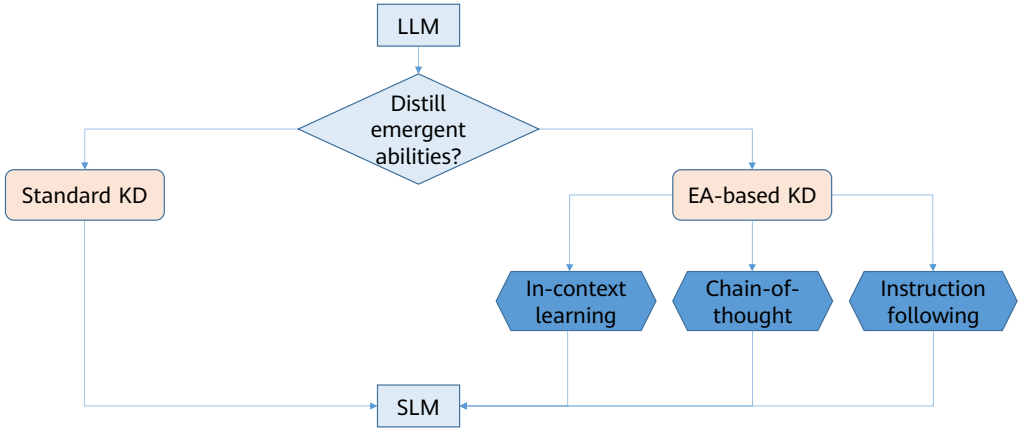
- Knowledge distillation (KD) refers to the process of transferring the knowledge learned by a complex model or ensemble model (teacher) to another lightweight model (student) to make the model lightweight (easy to deploy) and minimize performance loss.
- KD uses the teacher-student model, in which the teacher outputs knowledge and the student receives knowledge.



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- The knowledge of a large teacher model is extracted and condensed into a small student model. Simply put, a large teacher neural network teaches its knowledge to a small student network. This involves a knowledge transfer process, that is, knowledge is transferred from the teacher network to the student network. Generally, the teacher network is oversized, and it teaches knowledge to the small student network.

KD Approaches for Large Models

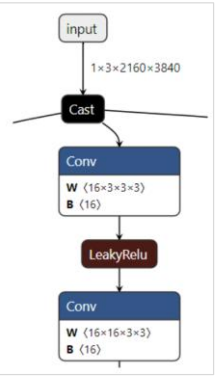


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- There are not many achievements regarding this research. Here is a brief introduction.
- In-context learning (ICL) distillation: This approach transfers the ICL, few-shot learning, and language modeling capabilities of LLMs to SLMs. It combines the objective of ICL with those of traditional language modeling. ICL distillation is explored under two few-shot learning paradigms: Meta-ICT and Multitask-ICT.
- [Lion et al., 2023] The adaptability of LLMs is leveraged to improve the performance of student models. LLMs identify and generate "difficult" instructions, which can then be used to enhance the capabilities of student models.
- EA is short for emergent abilities.
- SLM is short for small language model.

Operator Compilation Optimization

- Computational graph optimization: It aims to perform equivalent combined transformation on the computational graph to reduce the read and write operations of operators and improve efficiency. The most common method is operator fusion.
  - For example, Conv and ReLU can be fused into one operator to reduce I/O access and bandwidth overheads, and save computing units.
  - In the implementation of Transformer, the multiplication of the Q, K, and V matrices can be fused into one operator.
- Data layout optimization: In the TensorFlow framework, the input format is NHWC. In the PyTorch framework, the input format is NCHW. An optimization direction is to design better data layouts to reduce the access to stored data. In terms of hardware, some hardware has optimal performance in a specific storage mode. Therefore, optimization can be performed based on the read and write characteristics of the hardware.



Computational graph

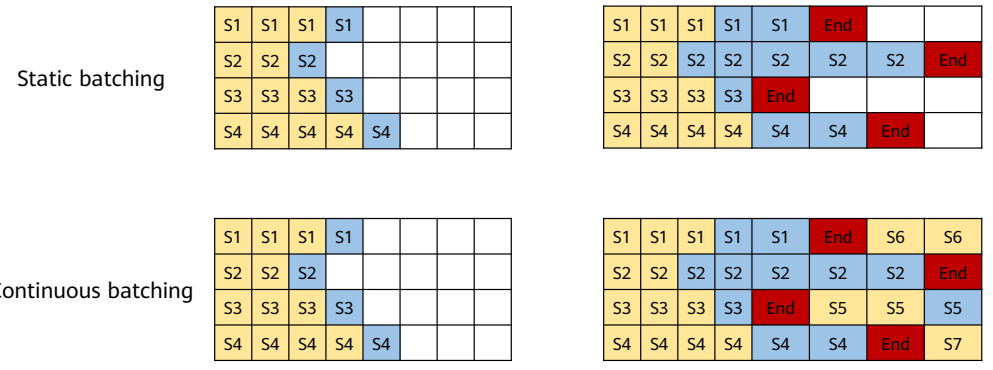
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- Inference can be understood as the process of data from the start point to the end point of a computational graph.
- Computational graph optimization can be summarized as operator disassembly, aggregation, and reconstruction to achieve better performance on hardware devices.

Continuous Batching

- Continuous batching is a powerful technology for improving the LLM throughput.



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- Continuous batching is a memory optimization technology that does not require model weight modification.
- Static batching: In the first round of iteration (left), each sequence generates a token (blue) based on the prompt (yellow). After several rounds of iterations (right), the completed sequences are of different sizes because they go through different rounds of iterations. When the iteration ends, an end of sequence (EOS) token (red) is generated. Although sequence 3 is completed after two iterations, in static batching, the AI chip will complete this sequence at last.
- Seven sequences are completed using continuous batching. The left figure shows the batch after a single iteration, and the right figure shows the batch after multiple iterations. Once an EOS token is generated for a sequence, new sequences (that is, S5, S6, and S7) are inserted at its location. This achieves higher AI chip utilization because AI chips do not need to wait for all sequences to complete before processing a new one.
- Real-world models are more complex than this simplified model. Computing is required in the pre-fill phase, and the computing mode is different from that in the generation phase. Therefore, the computing in the pre-fill phase cannot be easily batched with token generation.

Quiz

1. (Multiple-answer question) Which of the following factors affect the model inference speed?
   A. FLOPs: Total addition and multiplication operations of the model.
   B. MAC: Memory access cost.
   C. Parallelism degree: A higher parallelism degree leads to faster model inference.
   D. Computing platform

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- Answer: ABCD

Summary

- This chapter focuses on AI model inference and deployment. It first describes the factors that affect model inference and the inference optimization methods. Then, it introduces inference systems and related tools. Finally, it describes the common optimization methods for large model deployment from the perspectives of model compression and computing acceleration.

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Acronyms and Abbreviations

- AI: artificial intelligence
- CPU: central processing unit
- FLOPs: floating-point operations
- GPT: generative pre-trained transformer. It is a large-scale natural language generation model developed by OpenAI based on Transformer.
- MAC: memory access cost

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Thank you.

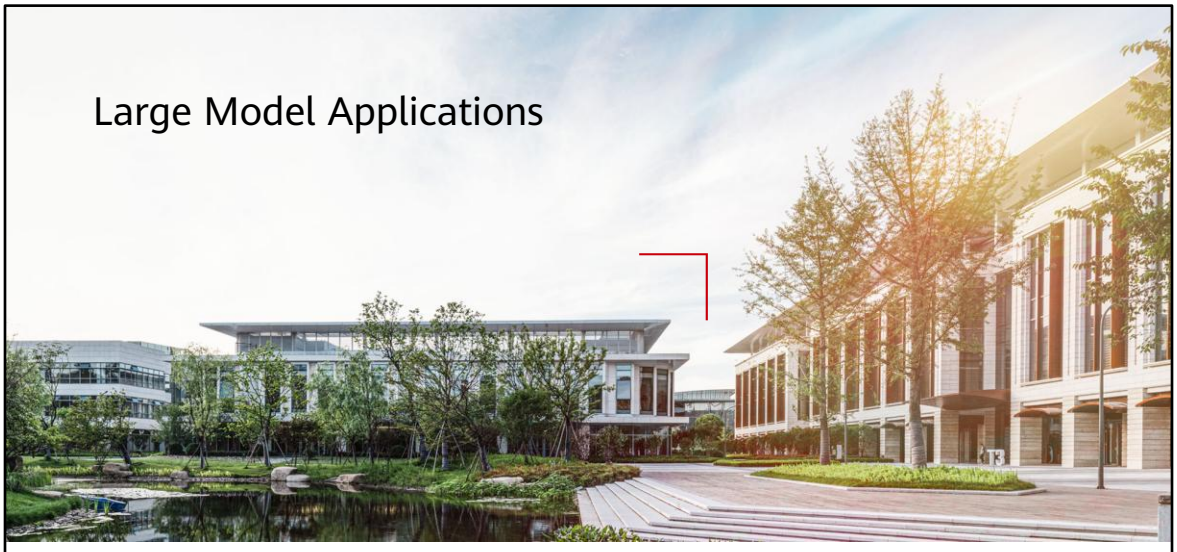
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

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Large Model Applications



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Foreword

- As AI constantly develops, large models, as an important milestone in the AI field, are changing the way people work and live in an unprecedented manner. These models not only understand and generate natural languages, but also demonstrate excellent performance in aspects such as image recognition and voice processing. This chapter focuses on several common application directions of large models: prompt engineering, retrieval-augmented generation (RAG), and agents. It aims to help you understand the directions and development status of large models in real-world applications.

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Objectives

- Upon completion of this course, you will be able to:
  - Be familiar with the basic usage methods of prompt engineering.
  - Be familiar with knowledge bases.
  - Understand agents and their workflow.

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1. **Prompt Engineering**
2. Knowledge Bases
3. Agents

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Prompt Engineering



- Prompt engineering is the art and science of designing and optimizing prompts. It can guide AI models, especially large language models (LLMs), to generate required answers. Carefully designed prompts can provide context, description, and examples for models to help them understand the inputs and generate meaningful answers. Prompt engineering can be understood as providing a road map for AI to guide it toward the direction of the desired output.

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- Prompt engineering refers to the process of optimizing prompts through structured text to guide LLMs to output the expected results.
- Researchers can use prompt engineering to improve LLMs' capabilities of handling complex tasks, such as Q&A and arithmetic inference.
- Developers can use prompt engineering to design and develop powerful engineering technologies for efficiently integrating with LLMs or other ecosystem tools.
- Prompt engineering plays an important role in implementing interaction and interconnection with LLMs and understanding them. It can be leveraged to improve the security of LLMs and enable LLMs. For example, specialized knowledge and external tools can be used to enhance LLM capabilities.

Zero-Shot Prompting

- In zero-shot prompting, users do not provide examples related to task results, but directly prompt LLMs to provide task-related answers.
- LLMs are capable of implementing zero-shot prompting, but the results also depend on the task complexity and the scope of existing knowledge.

```
Q: <Question>?
A:
```


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Example of Zero-Shot Prompting

Prompt:

Some substances in apples may cause allergic reactions, leading to symptoms such as rashes and difficult breathing. In addition, excessive fructose and dietary fiber in apples may cause dyspeptic symptoms such as abdominal distension and diarrhea.
Positive and negative evaluations:

Output:

The description includes the potential negative impact of apples and does not directly mention any positive aspects of apples. The positive/negative evaluation analysis of the description is as follows:

1. **Negative evaluation:**
   - "Some substances in apples may cause allergic reactions": This describes the potential health risks of apples for specific people, and is a clear negative evaluation.
   - "Leading to symptoms such as rashes and difficult breathing": The symptoms of allergic reactions are listed in detail, further emphasizing the potential negative health impact of apples.
   - "Excessive fructose and dietary fiber in apples may cause dyspeptic symptoms such as abdominal distension and diarrhea": Again, the potential side effects of apples are emphasized. Therefore, this is a negative comment on the potential adverse effects of apples.
     In the preceding description, no positive attributes of apples, such as nutritional value or health benefits, are mentioned. Therefore, the description is a negative evaluation.
2. **Positive evaluation:** Although there is no direct positive evaluation in the description, it can be inferred that the mentioned allergens and digestive issues generally do not hinder most people from eating apples because apples are widely recognized as healthy food. However, this positive view is based on general knowledge rather than the description itself.

In conclusion, the description focuses on the potential risks of apples and does not explicitly express any positive features of apples. Therefore, it can be judged as a negative evaluation. It should be noted that, to provide a comprehensive evaluation, positive aspects such as the nutritional and health benefits of apples usually need to be considered.

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- If only simple data judgment is required, the output is complex and contains a large amount of unnecessary information.
- GPT models are autoregressive language models. An autoregressive model predicts the next word based on the current input, combines the prediction result and the input, and uses the combination result as the input of the model to predict the next word. This process repeats.
- GPT-3 researchers found that large-scale data training leads to an interesting behavior called in-context learning (ICL). It can achieve excellent results based on just a few examples of downstream tasks without adjusting model parameters.
- When a pre-trained GPT-3 model is migrated to a new task, the model does not need to be retrained. Instead, only the task description (optional), several examples (task query and corresponding answers, organized in pairs), and the query to be answered by the model need to be provided. If the preceding content is packaged as the input of the model, the model can correctly output an answer corresponding to the last query.

Few-Shot Prompting

- Although LLMs demonstrate amazing zero-shot capabilities, they still perform poorly on complex tasks when using zero-shot settings.
- Few-shot prompting can be used as a technique to enable in-context learning (ICL) and provide demonstrations in prompts to guide models to achieve better performance. The demonstrations serve as the conditions for subsequent examples to guide models to generate the desired response.
- The number of few-shot examples can be selected.

```
Q: <Question>?
A: <Answer>
Q: <Question>?
A: <Answer>
Q: <Question>?
A: <Answer>
Q: <Question>?
A:
```


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- One-shot; few-shot

Example of Few-Shot Prompting

Prompt:

Some substances in apples may cause allergic reactions, leading to symptoms such as rashes and difficult breathing. In addition, excessive fructose and dietary fiber in apples may cause dyspeptic symptoms such as abdominal distension and diarrhea.

Evaluation: Negative

Banana is a kind of fruit that is beneficial to the human body.

Evaluation: Positive

Durian is sweet and delicious, and has high nutritional value.

Evaluation:

Output:

Positive

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What Is CoT?

- Chain-of-thought (CoT) was first proposed in Google's paper *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*.
- CoT is an improved prompting strategy for enhancing the performance of LLMs in complex reasoning tasks, such as arithmetic, common sense, and symbolic reasoning.

Q: "In a bag containing six red balls and four blue balls, if three balls are drawn consecutively, what is the probability of drawing two red balls and one blue ball?"

A: The probability is 0.424.

Q: "In a bag containing six red balls and four blue balls, if three balls are drawn consecutively, what is the probability of drawing two red balls and one blue ball?"

A: **Calculating all possible combinations:** First, we need to calculate the number of all possible combinations of the three balls drawn from the bag. There are 10 balls in total. Therefore, the number of combinations is C(10, 3) = 120.
Calculating the number of combinations that meet the conditions: We need to calculate the number of combinations of two red balls and one blue ball. The number of combinations of red balls is C(6, 2) = 15, and the number of combinations of blue balls is C(4, 1) = 4. Therefore, the number of combinations of two red balls and one blue ball is 60 (15 × 4).
Calculating the probability: There are 120 possible combinations in total, among which 60 combinations meet the conditions. Therefore, the probability is 0.5 (60/120).

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- In 2021, prompt learning evolved through multiple phases, starting with discrete prompt learning (prompt combinations) and advancing to continual prompt learning (continuous spatial representation), ultimately reaching a climax. However, prompt learning based on continuous space also has many limitations, such as resource consumption and training instability. In this year, most researchers agreed that prompt learning would bring a next-generation revolution in the field of natural language processing. However, most research work was related to model training or new language model structures.
- In 2022, the performance of large-scale language models has notably improved. In addition, as the model scale continues to increase, it is easier to prompt models. In particular, breakthroughs have been made in tasks on which models performed poorly. However, large models still need improvement in arithmetic, common sense, and symbolic reasoning. Large models have strong in-context few-shot learning capabilities. However, it is time-consuming to create numerous intermediate steps for supervised fine-tuning. In addition, traditional prompting methods have room for improvement in mathematical calculation and commonsense reasoning. Therefore, it remains a challenge to combine in-context few-shot learning and intermediate steps to improve capabilities such as arithmetic, common sense, and symbolic reasoning.

Features of CoT

- An effective CoT should have the following features:
  - **Logic**: The thinking steps in a CoT should have logical relationships. They should be interconnected to form a complete thinking process.
  - **Comprehensiveness**: A CoT should consider each question as comprehensively and meticulously as possible to ensure that no possible factors or impacts are ignored.
  - **Feasibility**: Each thinking step in a CoT should be feasible, that is, it should be operable and implementable.
  - **Verifiability**: Each thinking step in a CoT should be verifiable, that is, its correctness and validity can be verified based on actual data and facts.

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Composition and Classification of CoT

- A complete prompt that contains the CoT usually consists of three parts: instruction, rationale, and exemplars.
  - Instruction: describes the question and specifies the output format of the large model.
  - Rationale: refers to the intermediate reasoning process of the CoT, including the solution to the question, intermediate reasoning steps, and any external knowledge related to the question.
  - Exemplars: provides the basic format of input-output pairs for the large model in a few-shot manner. Each exemplar contains the question, inference process, and answer.
- Based on whether exemplars are included, CoT can be classified into zero-shot-CoT and few-shot-CoT.
  - Zero-shot-CoT can "wake up" the reasoning capability of a large model by adding a classic line of "Let's think step by step" to the instruction without adding exemplars.
  - Few-shot-CoT describes the problem solving procedure in detail in the exemplars to enable the model to acquire the inference capability through imitation.

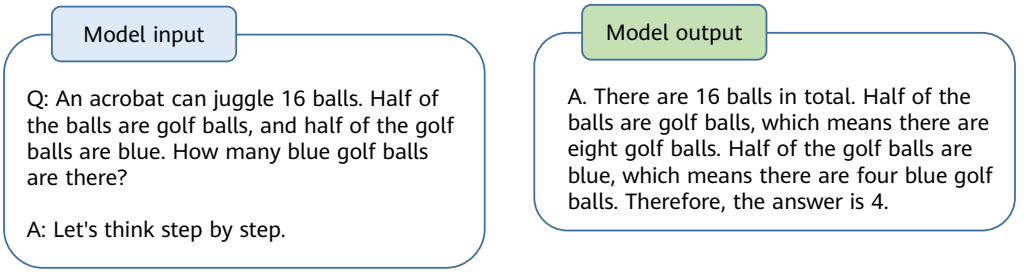
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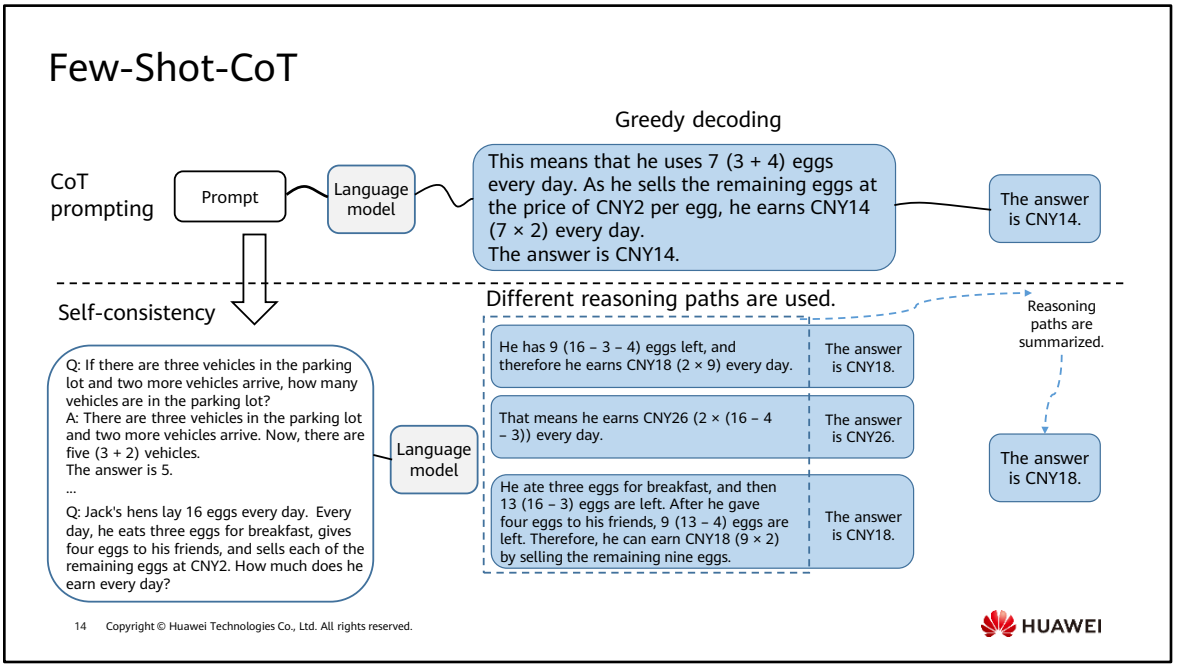
- Currently, there is no widely accepted theory about why the CoT takes effect. However, a series of experiments on the interaction between the CoT and large models have been conducted in many papers. Some interesting phenomena in these experiments may help us understand the working principle of CoT. A small model scale may cause CoT to fail. In simple tasks, CoT does not improve model performance. The CoT performance can be improved by increasing the degree of interconnection between training data. Errors in the exemplars or invalid reasoning steps do not cause the CoT performance to deteriorate. If we summarize and extend these phenomena, we may draw the following conclusions: First, CoT requires the large model to have some fundamental knowledge. If the model is too small, it cannot understand the fundamental atomic knowledge and therefore cannot perform reasoning. Second, CoT can build a bridge between the basic knowledge that it masters to connect known information into a "chain" so that the large model does not deviate during the reasoning process. Finally, the purpose of CoT is to force the model to perform reasoning, instead of teaching the model how to complete reasoning. A large model is capable of reasoning after pre-training. CoT only specifies the model output format so that the model can generate answers step by step.

Zero-Shot-CoT

- The zero-shot-CoT prompting process is a simple zero-shot prompting method developed based on CoT prompting. Researchers have found that by appending the words "Let's think step by step" to the end of a question, an LLM can generate a CoT that answers the question. From this CoT, they can extract more accurate answers.



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- Few-shot CoT: A complex reasoning task can have multiple reasoning paths (that is, problem solving approaches), each of which can lead to the correct answer. Therefore, in the self-consistency decoding process, the greedy decoding policy is abandoned. Instead, sampling is used to select and generate different reasoning paths, and each path corresponds to a final answer.

Application Scenarios of CoT

- CoT prompting is mainly applicable to domains that require complex reasoning, for example, arithmetic, common sense, and symbolic reasoning.
- Multimodal CoT has a great application prospect. In CoT, the multimodality can refer to the multimodal input or output.
- In addition to multimodal CoT, CoT has also been used in fields such as text summarization (SumCoT), open-domain question answering (self-prompting LLMs), machine translation (MAPS), chemistry (ChemCrow), and medicine (Med-PaLM).

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- Multimodal-CoT (MM-CoT) is the first part of the research on multimodal inputs. MM-CoT focuses on embedding CoT by using fine-tuning methods. Languages and images are integrated into a two-phase framework that involves reasoning generation and answer reasoning. In addition, the fine-tuned large model is used to provide the multimodal CoT capability to the input. Based on MM-CoT, the GoT-Input method extracts the mind map generated by the CoT to construct a 3-tuple. Then, the GNN is used to unify the text, image, and CoT to generate the final answer that contains CoT information. VCoT solves the problem of multimodal output. It recursively fills image information by using the "title"of the image to be generated and core identification focuses as the start process of image generation, thereby generating a multimodal output.

Limitations of CoT

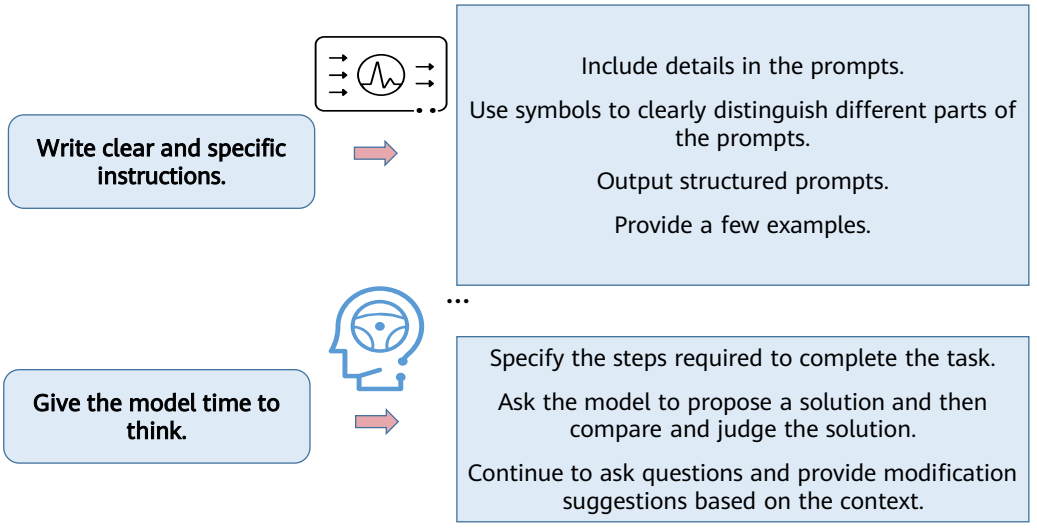
- CoT prompting requires users to provide some reasoning processes in natural languages, which may impose burdens and requirements on users
- CoT prompting may not be applicable to all types of questions, especially those that require expertise or common sense.
- CoT prompting may cause LLMs to rely heavily on the given reasoning process and ignore other possible solutions or perspectives.
- CoT can emerge only when the model scale is large enough.
- CoT prompting cannot solve mathematical problems of the primary school level.

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- Currently, CoT only works in a few fields such as mathematical problems, five commonsense reasoning benchmarks (CommonsenseQA, StrategyQA, Date Understanding, Sports Understanding, and SayCan). Its performance improvement effect on other types of tasks, such as machine translation, needs to be evaluated.

Approaches to Writing Good Prompts

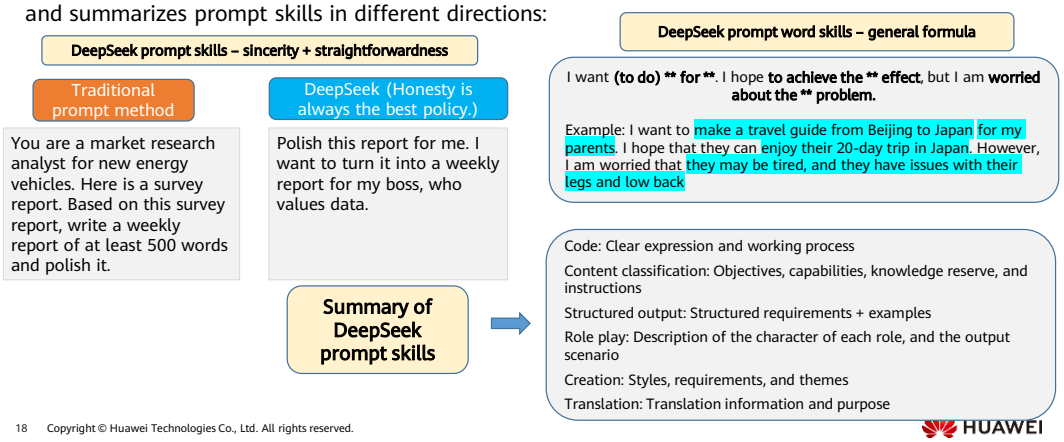


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- Provide as clear and specific instructions as possible to express what you want the model to do. This will guide the model to output content as you expect and reduce the probability of receiving irrelevant or incorrect replies.
- Most large models have the emergent abilities to solve complex problems and cover a wide range of knowledge.
- They can understand and analyze the natural language entered by users, and provide services and help for users in different fields and tasks.
- However, specific skills are needed to communicate with large models.

Prompt Changes

- With the emergence of reasoning models represented by DeepSeek-R1, prompts change accordingly, but the overall principle does not change greatly. The following lists two typical prompt requirements and summarizes prompt skills in different directions:



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1. Prompt Engineering
2. **Knowledge Bases**
   - Concepts
     - RAG
     - LangChain-centric Knowledge Base Generation and Construction
     - Challenges and Development Directions of Knowledge Base Query
3. Agents

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What Is a Knowledge Base?

- A knowledge base is a centralized repository created using knowledge management software. This repository can be used to easily create, organize, search for, and share knowledge, and gather scattered knowledge. Its structure enables employees or customers to find answers to questions or doubts. Nowadays, knowledge bases have become an integral part of any knowledge management system. Based on their functions, knowledge bases can be classified into the following types:
  - Internal knowledge bases for providing services to employees
  - External knowledge bases for providing services to customers

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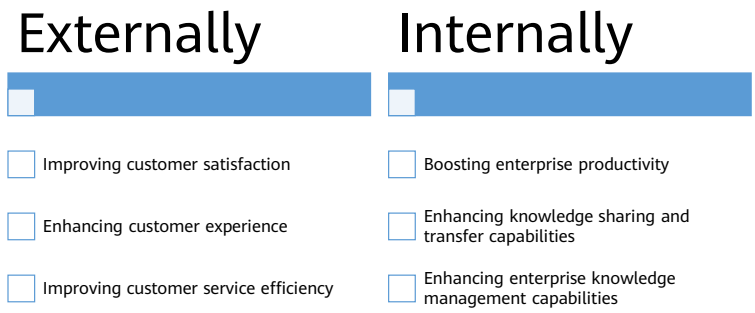
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- Internal knowledge bases:
  - Internal knowledge bases provide services for employees. This means that they do not provide access to people who are not platform users.
  - Internal knowledge bases can be created using anything for internal use. The following lists some examples:
  - Best coding practices of developers
  - Onboarding materials of new employees
  - Content authoring guidelines for content writers
  - Corporate policies
  - Any information that needs to be shared internally
  - An internal knowledge base can even be a cloud disk folder shared only between employees or specific departments.
- External knowledge bases:
  - External knowledge bases are designed to serve customers. They are usually open to all and can be easily found on the Internet.
  - The help center and documentation of any software are external knowledge bases.
  - Take the help center of a digital bank as an example. Its external

knowledge base provides information about application usage methods, transactions, card issues, and personal data. All of this is directly related to customers.

Benefits of Knowledge Bases

- Knowledge bases are widely used in fields such as chatbot systems, search engines, speech recognition, natural language processing (NLP), and machine translation.



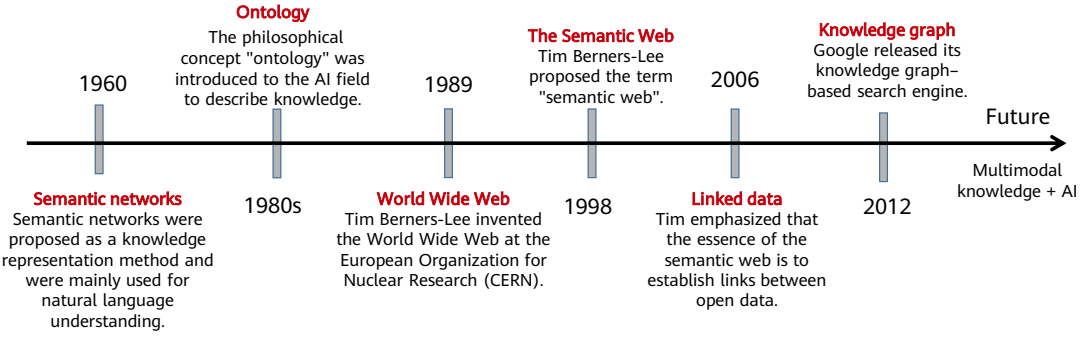
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- Benefits of external knowledge bases to customers:
  - Improving customer satisfaction: According to statistics, more than 60% of users prefer convenient self-service rather than contacting teams by phone, email, or social media. Essentially, customers are allowed to explore the information they are looking for based on their own conditions rather than the company's conditions. With knowledge base software, there is no need to submit work orders to the service desk or wait for "the next available representative" or similar things. This reduces the waiting time during issue resolution and improves customer satisfaction.
  - Enhancing customer experience: In today's fast-paced and interconnected world, most consumers cannot afford to wait for help. When they encounter problems, they want to get answers immediately. In this case, knowledge bases play a key role. They provide customers with extensive, in-depth information and allow them to study specific topics of interest. This essentially provides customers with a better experience because with knowledge bases, they can get more out of the products or services.
  - Improving customer service efficiency: Knowledge bases are available 24/7, allowing customers to solve their problems at their convenience. Powerful knowledge bases can provide accurate information to help customers achieve their goals, regardless of whether they are looking for basic information about a branded product or service, when they start to use

products, or for other purposes.

- Benefits of internal knowledge bases to employees:
  - Boosting enterprise productivity: First, it makes the team more productive and efficient. Because consumers can easily solve their problems through self-service, the company team does not need to spend much time solving relatively simple problems. In turn, the team can focus on more pressing customer issues to improve customer satisfaction.
  - Enhancing the knowledge sharing and transfer capabilities: First, creating a knowledge base requires all department members to share knowledge. Once the team starts to manage knowledge and build a knowledge base, it can be used as an information center database that keeps evolving. The free flow of knowledge keeps all departments "on the same page" when interacting with customers because the knowledge base ensures that they provide the same information.
  - Enhancing the enterprise knowledge management capability: The company team can use the knowledge base as part of its overall knowledge management strategy to enhance the knowledge management capability of the team.

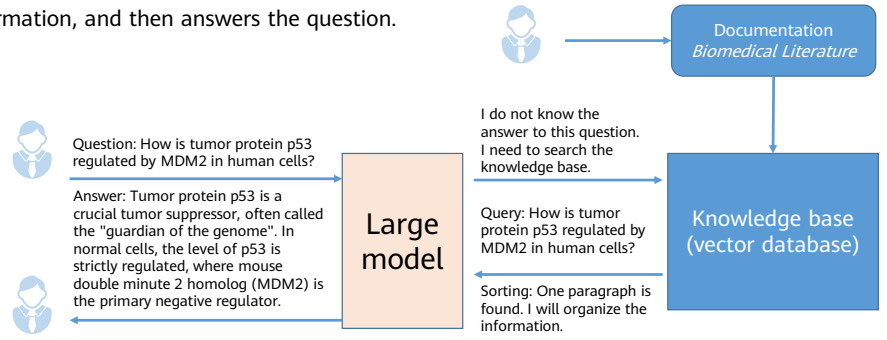
Development History of Knowledge Bases



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Relationships Between Large Models and Knowledge Bases

- The training data of large models is public and universal knowledge. When you ask a large model a question about your company or a professional question in a specific domain, it cannot provide an accurate answer. Knowledge bases serve as external information bases of agents. When an agent does not know the answer to a question, it queries the knowledge base, summarizes the queried information, and then answers the question.



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- We can compare it to the human customer service scenario. When a user asks a product question, the customer service personnel will query the enterprise knowledge base first if they do not know how to answer the question. They organize the queried information and then answer the question. Agents work in the same way. This process is called RAG.
- Large models can be used with knowledge bases to achieve the following synergies:
  - Accurate queries and answers: Knowledge bases provide structured data query interfaces. Large models can use these interfaces to quickly retrieve relevant knowledge and generate accurate answers. This greatly improves the efficiency and accuracy of information retrieval.
  - Knowledge update and maintenance: Large models can continuously learn and update new knowledge. New knowledge can be promptly added to knowledge bases to ensure the timeliness and accuracy of information in knowledge bases.
  - Enhanced application scenarios: Based on the structured information of knowledge bases and the language generation capability of large models, more intelligent and personalized services can be provided in diverse application scenarios such as intelligent Q&A, decision-making assistance, and content recommendation.

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What Is RAG?

- With RAG, an LLM first retrieves relevant information from a large number of documents, and then generates an answer or text based on the retrieved information. RAG helps improve the prediction quality.



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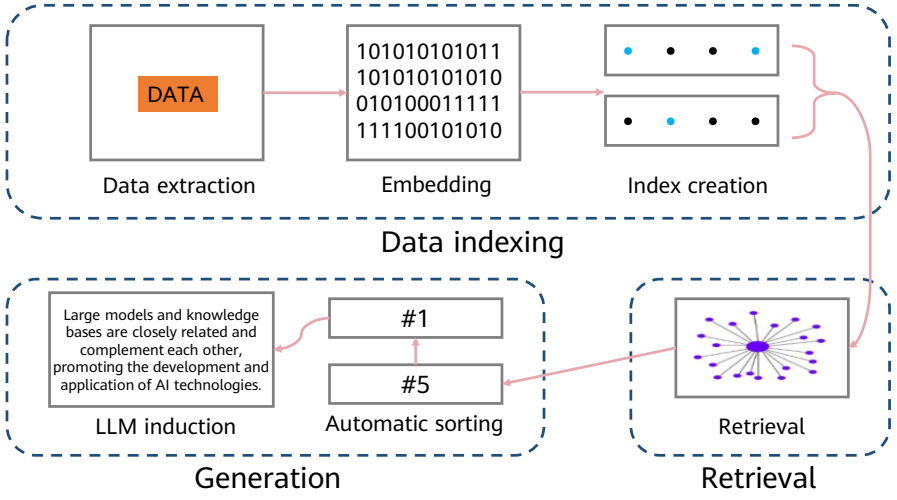
- Although LLMs have strong capabilities, they have the following issues: 1. Hallucination: The underlying principle of text generation by LLMs is probability-based token-by-token generation. Therefore, "plausible-sounding but incorrect answers" are inevitable. Timeliness: A larger LLM scale leads to higher model training costs and longer training periods. If time-sensitive data cannot be used for training, LLMs cannot answer time-sensitive questions such as "Could you please recommend several popular movies?" Data security: General-purpose LLMs do not have internal corporate data or user data. If a company wants to use LLMs on the premise of security, the best way is to store all data locally and complete all computing of the corporate data locally. The online large model implements only the induction function.

Benefits of RAG

- Scalability: The model size and training cost are reduced, and knowledge can be easily expanded.
- Accuracy: By referencing information sources, users can verify the accuracy of answers. This enhances people's trust in the model output.
- Controllability: Knowledge can be updated or customized.
- Interpretability: The retrieved items are used as the reference sources for model prediction.
- Versatility: RAG can be fine-tuned and customized for diverse tasks, including Q&A, text summarization, and dialog systems.
- Timeliness: The latest information can be identified using the retrieval technology. This enables RAG to outperform traditional language models that rely only on training data in terms of the timeliness and accuracy of answers.
- Customization: RAG can provide professional knowledge in diverse fields by indexing the text corpora related to specific fields.
- Security: RAG implements better control over data usage by setting roles and implementing security control in databases. In contrast, fine-tuned models may not be clear enough in terms of managing data access permissions.

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Typical RAG Implementation



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Data Indexing—Data Extraction

- Data acquisition: Data of different formats (PDF, Word, Markdown, database, and API) is obtained from different sources and loaded. The data is processed into the same normal form.
  - Word files: Text elements, such as titles, tables, and paragraphs, can be obtained through parsing. The text segments and their attributes are stored for subsequent segmentation.
  - PDF files: Multiple open-source models are used for collaborative analysis. They detect 10 types of areas: text, title, figure, figure caption, table, table caption, header, footer, reference, and equation. In addition, they unify the optical character recognition (OCR) and text attribute classification tasks.
  - PPT files: PPT files are converted into PDF files and then parsed using the preceding method.
- Data cleansing: The source data is deduplicated, filtered, compressed, and formatted.
- Information extraction: Key information, including the file name, time, chapter title, and image, is extracted from data.

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Data Extraction—Embedding and Index Creation

- Embedding refers to the process of converting text, images, audio, and videos into a vector matrix, that is, a format that can be understood by computers.
- Index creation refers to the process of creating indexes after data embedding and writing the indexes to the database. It can be summarized as the process of importing data to the database.
  - Common tools:
    - FAISS
    - ChromaDB
    - Elasticsearch
    - Milvus

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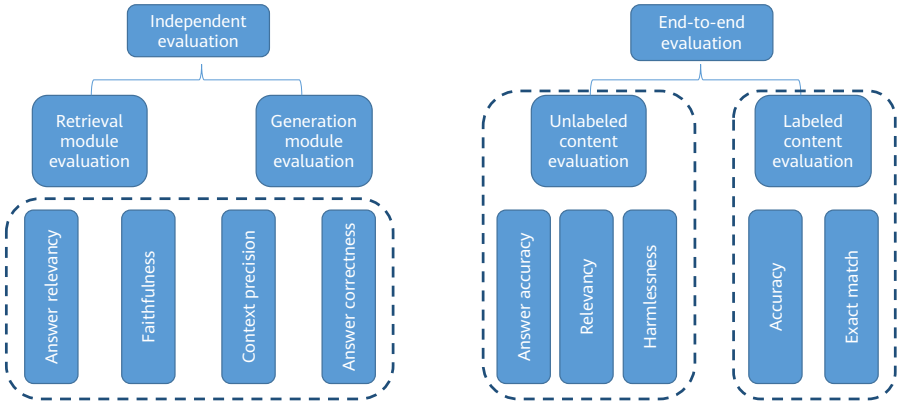
Retrieval

- The major retrieval methods are as follows:
  - Embedding similarity retrieval: Similarity calculation methods include the Euclidean distance, Manhattan distance, and cosine.
  - Re-ranking: Data is adjusted based on factors such as the correlation and matching degree to obtain a ranking that better suits the service scenario.
  - Switching between query methods: This is a query and retrieval mode. The following query methods are used:
    - Subquery: Different query policies can be used based on scenarios, including tree query (step-by-step query from leaf nodes and combination), vector query, and sequential query of chunks.
    - HyDE: It is a method of generating similar or more standard prompt templates.

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RAG Evaluation Methods

- Currently, two methods are used to evaluate the effectiveness of RAG: independent evaluation and end-to-end evaluation.



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- Answer relevancy: This metric is used to evaluate the relevancy between the generated answer and the provided question prompt. If the answer is incomplete or contains redundant information, its score is relatively low. This metric is calculated based on both the questions and answers. The score ranges from 0 to 1. A higher score indicates higher relevancy. Take the following question as an example: What are the main characteristics of a healthy diet? Low-relevancy answer: A healthy diet is important to the overall health. High-relevancy answer: A healthy diet should include various fruits, vegetables, whole wheat foods, lean meat, and dairy products to provide necessary nutrients.
- Faithfulness is an evaluation metric used to check the factual accuracy of the generated answer in the given context. The evaluation process involves a comparison between the answer content and the retrieved context. This metric is also represented by a value between 0 and 1. A larger value means a higher consistency between the answer and the context. Take the following question as an example: What are the main achievements of Madame Curie? Background: Marie Curie (1867–1934) was a pioneering physicist and chemist, the first woman to win a Nobel Prize, and the only person to win Nobel Prize in two different fields. High-faithfulness answer: Marie Curie won Nobel Prize in both physics and chemistry, making her the first woman to make this achievement. Low-faithfulness answer: Marie Curie won Nobel Prize only in the field of physics.
- Context precision is a metric used to evaluate whether all entries related to the benchmark information are correctly ranked in the given context. Ideally, all relevant content should appear at the beginning of the sequence. This evaluation metric is also represented by a score between 0 and 1, and a higher score reflects higher accuracy.

Related metrics include the hit rate, mean reciprocal rank (MRR), normalized discounted cumulative gain (NDCG), and precision.

- Answer correctness is a metric used to measure the matching degree between the generated answer and the benchmark answer. This evaluation metric considers the comparison between the benchmark answer and the generated answer. The score is generally between 0 and 1, and a higher score indicates a higher consistency between the generated answer and the actual answer. The following is an example: Basic fact: The Eiffel Tower was completed in Paris, France, in 1889. High answer correctness: The Eiffel Tower was completed in Paris, France, in 1889. Low answer correctness: The Eiffel Tower was completed in 1889 and stood in London.

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What Is LangChain?

- LangChain is a framework for building LLM–based applications. It provides a range of tools and functions to help developers easily use LLMs (such as GPT-4) to create various complex NLP tasks. It also supports multiple languages, including Python and JavaScript (TypeScript).



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Models

- LangChain model is an abstraction that represents different types of models used in the framework. Models in LangChain are classified into the following types:
  - LLMs: These models take text strings as input and return text strings as output. They are the pillars of numerous language model applications.
  - Chat models: Chat models are supported by language models but have more structured APIs. They take the chat message list as the input and return chat messages. This makes it easy to manage the conversation history and maintain the context.
  - Text embedding models: These models take text as the input and return floating-point lists that represent text embeddings. These embeddings can be used for tasks such as document retrieval, clustering, and similarity comparison.

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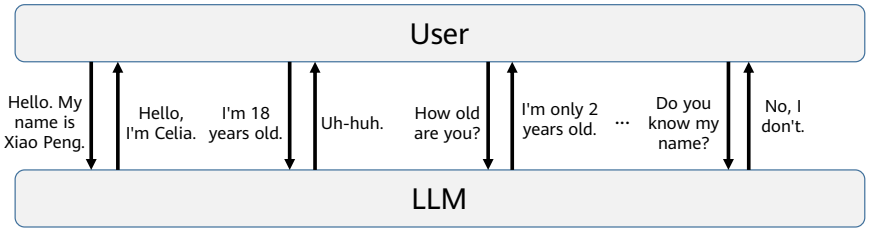
Prompts

- Prompts refer to the inputs required for models to generate content. They include the background knowledge required for models to generate content, the instructions that users expect models to execute, and the formats that the model outputs need to comply with. In LangChain, prompts consist of three parts: prompt templates, output parsers, and example selectors.
- Prompt templates allow you to generate prompts based on the specified template.
- Output parsers are used to parse the LLM text into structured data.
- Example selectors help you build few-shot prompts.

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Memory

- In LangChain, memory refers to the short-term memory of LLMs. Why does it refer to short-term memory? This is because after an LLM is trained (some long-term memory is obtained), its parameters do not change due to user inputs. When a user interacts with a trained LLM, the LLM temporarily memorizes the user input and the generated output to predict the subsequent output. After the model output is complete, the LLM "forgets" the previous user input and the model output. Therefore, such information can only be referred to as the short-term memory of the LLM.

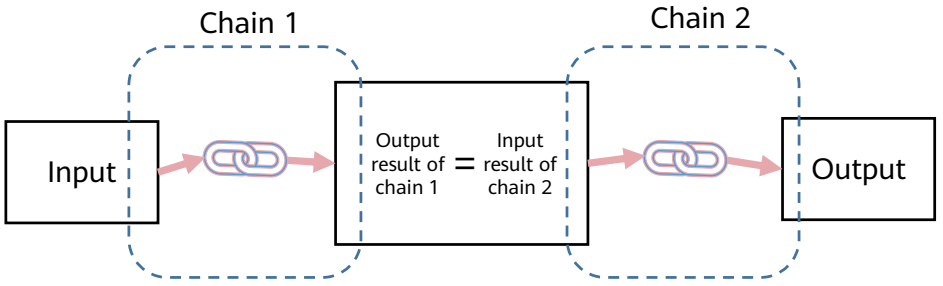


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- LangChain provides the following short-term memory management methods: BufferMemory, BufferWindowMemory, ConversionMemory, and VectorStore-backed Memory.

Chains

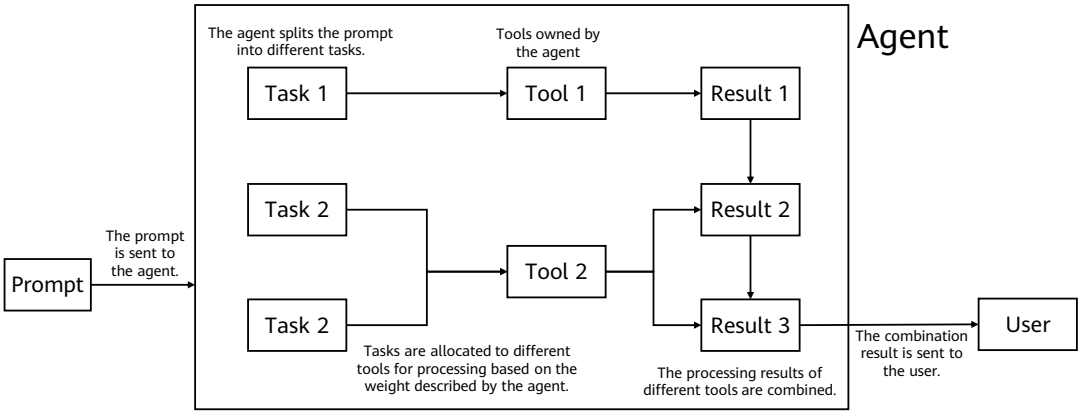
- Chains can be considered as tasks. A chain is a task. Multiple chains can be executed one by one.



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Agents

- Agents can be simply understood as tools that can dynamically select and call chains or existing tools for us.



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Indexes

- Index components provide document processing capabilities for LangChain, including document loading and retrieval. The "document" refers to documents in a broad sense, including not only text content in TXT, EPUB, and PDF formats, but also emails, blockchains, telegrams, Notion, and even video content.
- Index components are classified into the following types:
  - Document loaders
  - Text splitters
  - Vector stores
  - Retrievers

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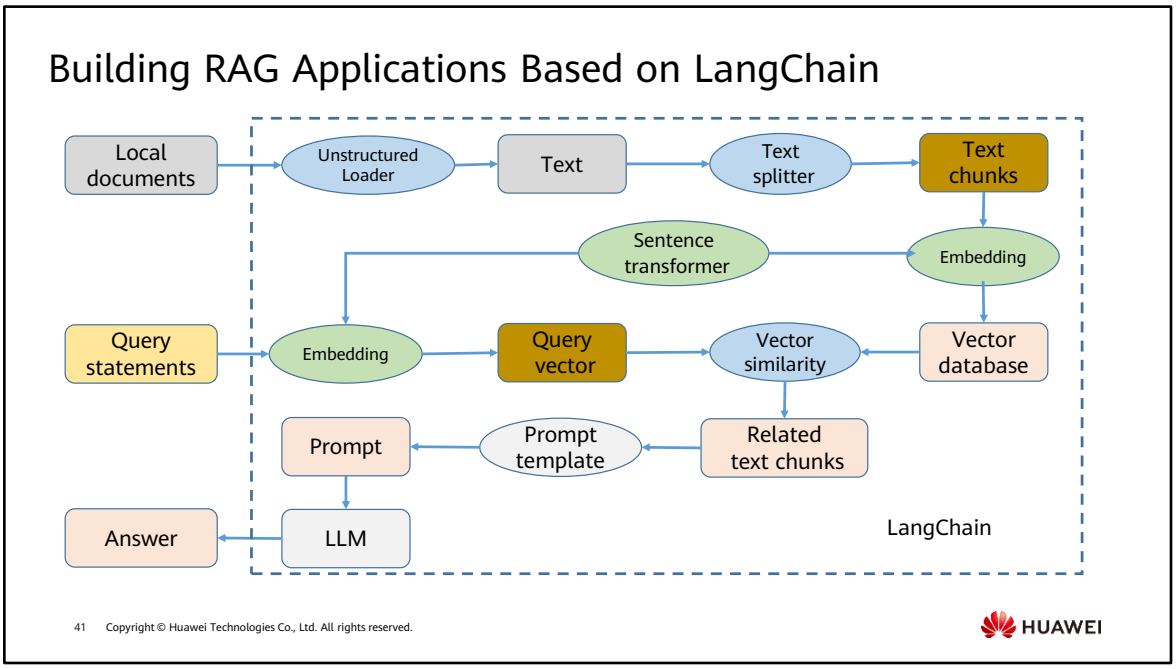
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- Document loaders are mainly based on Unstructured, which is a Python package that can convert various types of files into text.
- Text splitters: Models have restrictions on the length of input characters. Long text needs to be split into multiple small text fragments.
- Vector stores are special databases designed to store vectors created by embeddings and provide functions such as similarity query.
- Retrievers are used by models to query storage data. The retrievers of LangChain have at least one method: **get_relevant_texts**. This method receives the query string and returns a group of documents.

Document-based Q&A in LangChain

- Retrieval Q&A chain:
  - The retrieval Q&A chain is a specific implementation in the LangChain framework, and covers the entire process of RAG. With the retrieval Q&A chain, developers can easily build applications related to Q&A. This chain uses the RAG method and the retrieval module to retrieve relevant information from external knowledge bases and generate answers.

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- To build an RAG application based on LangChain, first, the UnstructuredLoader component is used to load local documents (personal databases are in the form of local documents). This component converts local documents of different formats into plain text. Then, text splitter component splits the extracted plain text into chunks. Next, the open-source word embedding model Sentence Transformer converts text chunks into the vector format, and stores them in the Chroma-based vector database. Sentence Transformer converts each user input into vectors of the same dimension. Then, it performs similarity matching in the vector database to find relevant text chunks, and embeds the relevant text chunks into the written problem template. Finally, InternLM answers the question.

Building a Vector Database

- This process involves source file loading, document chunking, and document embedding.
  - Source file loading: Determine the type of the source file, and select a loader based on the file type.
  - Document chunking: Split the document and manually control chunking.
  - Document embedding: Select the embedding model, implement embedding, and store vectors.

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- This process involves source file loading, document chunking, and document embedding.
  - Source file loading:
    - Determining the type of the source file: Determine the type of the source file by detecting the file name extension or content. The source file may be a text file (.txt), document file (.docx or .pdf), or image file.
    - Selecting a loader: Select a proper loader based on the type of the source file. The loader is used to convert formatted text into unformatted strings to facilitate subsequent processing.
  - Document chunking:
    - Split document: Split the loaded document based on the character string length. A single document usually exceeds the upper limit of the model context. Therefore, the document needs to be divided into smaller chunks.
    - Manual chunking control: You can choose to manually control the chunk length and the length of the overlapping interval to flexibly adapt to different requirements and model requirements.
  - Document embedding:
    - Selecting an embedding model: Select an appropriate embedding model, which converts text data into a high-dimensional vector representation. A

pre-trained word embedding model (such as Word2Vec, GloVe, and BERT) can be used.

- Embedding process: Use the selected model to perform embedding on each document chunk and generate the embedding representation of the document.
- Storing vectors: Store the generated document vectors in the vector database for subsequent semantic retrieval. Chroma can be used as the vector database. It is a lightweight tool that supports semantic retrieval.

Vector Databases

- Vector databases are characterized by efficient storage and retrieval of large-scale vector data. They are mainly used to retrieve images, audio, and text. They use dedicated data structures and algorithms to calculate and query the similarities between vectors.

Vector Database	GitHub Stars	Language	Cloud
Chroma	7,400	Python	✕
Milvus	21,500	Go/Python/C++	
Pinecone	✕	✕	
Qdrant	11,800	Rust	
Typesense	12,900	C++	✕
Weaviate	6,900	Go	

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- In a vector database, data is stored and processed in the form of vectors. Therefore, original non-vector data needs to be converted into vectors. Data embedding refers to the process of converting non-vector data into vector data. Through data embedding, efficient similarity calculation and query can be implemented in the vector database. In addition, vector databases use different retrieval algorithms, such as KD-Tree, VP-Tree, LSH, and inverted index, to accelerate vector similarity search. Algorithms need to be selected and parameters need to be tuned based on specific application scenarios. The algorithm should be selected based on the features and data volume of the datasets, as well as the requirements on query, search accuracy, and efficiency.

File Import

- The following is a simple Python code segment. The **SQL Basic Tutorial.pdf** file can be imported to the database by introducing the document_loaders module of LangChain.

```
from langchain.document_loaders import UnstructuredMarkdownLoader
loader = UnstructuredMarkdownLoader("../data_base/knowledge_db/ SQL Basic
Tutorial.pdf")
pages = loader.load()
...
```


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File Chunking

- LangChain can also be used to chunk text based on parameters such as the character quantity and length.

```
# Import the text splitter.
from langchain.text_splitter import RecursiveCharacterTextSplitter

# Length of a single text chunk in the knowledge base
CHUNK_SIZE = 500

# Overlap length of adjacent texts in the knowledge base
OVERLAP_SIZE = 50
```


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Knowledge Base Embedding (1)

- Embedding, which is common in the NLP field, needs to be performed on the split document.

```
from zhipuai_embedding import ZhipuAIEmbeddings
# embedding = HuggingFaceEmbeddings(model_name="moka-ai/m3e-base") is the method of
introducing a locally deployed model.
embedding = ZhipuAIEmbeddings() # Introduce the online embedding service.

query1 = "Travel"
query2 = "Rocket"
query3 = "Aerospace"
emb1 = embedding.embed_query(query)
```

- The following is an example of the result:

```
The length of the embedding generated for "Travel" is 1024. The first 30 values are as follows:
 [-0.06620011 0.11531917 -0.0707648 -0.06679663 -0.00087593 0.23256738 -0.20461585 -0.07101158 -
0.00645613 0.32598388 -0.01385093 0.1837097 -0.06796395 0.17077696 -0.0064841 -0.17289667
0.03198481 0.12143167 -0.05484311 0.34414008 -0.01926355 0.11278316 -0.12705955 0.04611887 -
0.20642205 -0.2279439 0.06027906 0.11457541 -0.02675474 -0.29482642]
```


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- The online interfaces of large models, such as the interfaces of OpenAI or ChatGLM, are used.
- Locally deployed embedding models such as M3E and Moka can be used. For convenience, the ChatGLM interface services are used, and LangChain is used for interconnection.

Knowledge Base Embedding (2)

- After each document is embedded, the results are stored in the database. Here a lightweight vector database called Chroma is used. It is also integrated in LangChain.

```
from langchain.vectorstores import Chroma
from zhipuai_embedding import ZhipuAIEmbeddings
persist_directory = '../../data_base/vector_db/chroma' # Database persistence path
vectordb = Chroma.from_documents(
    documents=split_docs[:100],
    embedding=embedding,
    persist_directory=persist_directory # Save the persist_directory directory to a disk.
)
```


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Retrieval Embedding

- The following uses the query "What is a database?" as an example to describe the execution process. The top three retrieved items are output.

```
question = "What is a database?"
 docs = vectordb.similarity_search(question,k=3)
```

```
Item 0 found:
1-1 What Is a Database?
Chapter 1 Database and SQL
What Is a Database?
Learning Focus
A database is a collection of data that is stored and processed by a computer and can be accessed efficiently.
The computer system used to manage databases is referred to as a database management system (DBMS).
The DBMS enables multiple users to safely and easily operate a large amount of data.
```

```
Item 1 found:
Chapter 1 Database and SQL
The dataset is referred to as a database (DB). After data such as the name, address, phone number, email address, hobby, and family members is saved in a database, the required information can be quickly obtained at any time.
The computer system used to manage databases is referred to as a database management system (DBMS). System users cannot directly access databases.
```


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Submitting Combined Prompts to LLMs to Generate Answers (1)

- First, LangChain is used to build a prompt and a retrieval chain.

```
from langchain.prompts import PromptTemplate
template = """Answer user questions in a concise and professional manner based on the following known information.
  Known information: {context}
  Question: {question}""" QA_CHAIN_PROMPT = PromptTemplate(input_variables=["context","question"], template=template)

from langchain.chains import RetrievalQA
qa_chain = RetrievalQA.from_chain_type(llm, retriever=vectordb.as_retriever(), return_source_documents=True,chain_type_kwargs={"prompt":QA_CHAIN_PROMPT})
```


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Submitting Combined Prompts to LLMs to Generate Answers (2)

- A user retrieves "What is a database?". After a series of retrieval chains, the following answer is obtained:

```
question = "What is a database?"
 result = qa_chain({"query": question})
 print("Answer generated by the large model based on document knowledge bases:")
 print(result["result"])
```

```
Answer generated by the large model based on document knowledge bases:
"A database is a collection of data that is stored and processed by a computer and can be accessed efficiently. It is usually managed by a relational database management system (RDBMS). The database management system (DBMS) enables multiple users to safely and easily operate a large amount of data. There are many types of databases, including relational databases and text databases. Databases are widely used in scenarios such as bank account management, phonebook, and online shopping. The DBMS can overcome the disadvantages of text files or spreadsheet software in sharing data, operating numerous data formats, and implementing automatic read and write. It improves the efficiency of data management and processing."
```


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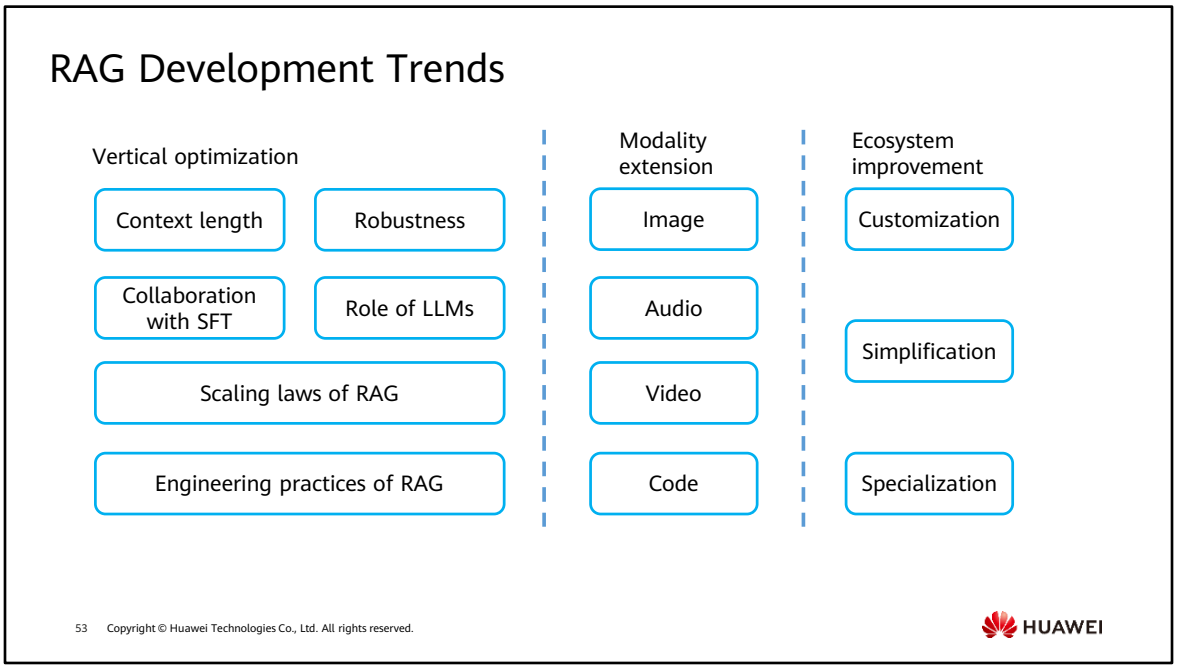
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Challenges to RAG

- The RAG solution is an overall architecture of domain knowledge Q&A. It faces many challenges in application because domain knowledge Q&A has high requirements on the accuracy and completeness of answers. The following lists some major challenges:
  - Improper text segmentation results in inaccurate or incomplete recalled content.
  - As a mainstream retrieval method, vector retrieval is mainly based on similarity judgment. However, differences may be caused by various reasons.
  - The rankings and priorities of multiple search results are improper. As a result, the LLM answers do not focus on the key points.
  - How do LLMs ensure the coherence and consistency of answers?
  - How can LLMs better understand domain knowledge segments?
  - How can high-quality QA pairs be provided to assist in RAG tuning?
  - Multimodal support is an inevitable trend.
  - How can the availability of RAG in a specific field be evaluated?

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- Vertical optimization aims to further address the challenges faced by RAG:
  - Context length: What can I do if the retrieval content exceeds the window limit? If the context window of LLMs is no longer restricted, how should RAG be improved?
  - Robustness: What can be done if incorrect information is retrieved? How to filter and verify the retrieved content? How to improve the anti-toxic and anti-noise capabilities of models?
  - Collaboration with fine-tuning: How to fully utilize RAG and SFT at the same time? How to coordinate and organize them? Should the serial, alternate, or end-to-end mode be used?
  - Scaling law: Do RAG models comply with the scaling law? Does RAG show or in what scenarios does it show inverse scaling law?
  - Role of LLMs: LLMs can be used for retrieval (replacing the retrieved content with the generation of LLMs, or retrieving the memory of LLMs), generation, and evaluation. How to further explore the potential of LLMs in RAG?
  - Engineering practice: How to reduce the retrieval latency of ultra-large-scale corpora? How to ensure that the retrieved content is not disclosed by large models?
- Multimodal expansion of RAG:
  - How to extend the evolving technologies and philosophy of RAG to data of other modalities such as images, audio, videos, and code? This can enhance the tasks of

a single modality and integrate multiple modalities based on the philosophy of RAG.

- The application of RAG is not limited to question answering systems, and its influence is expanding to more fields. Nowadays, diverse tasks such as recommendation systems, information extraction, and report generation are benefiting from the RAG technology. The RAG technology stack is also booming. In addition to the known tools such as LangChain and LlamaIndex, more targeted RAG tools are emerging in the market. For example, RAG can be customized based on the purpose to meet scenario-specific requirements. The usage of RAG can be simplified to further lower the entry threshold. Functions are specialized and gradually oriented to the production environment.

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Introduction: Human Thinking vs. AI



Perception

See, hear, speak, read, write, etc.
Senses of sight, hearing, touch, smell, taste, etc.

Cognition

Induction, reasoning, deduction, calculation, cooperation, sharing, empathy, reflection and improvement, etc.
(Brain-based learning)

Decision-making

Execute plans and solutions.

See, hear, and speak | **Act** | **Think** | **Learn** | **...**

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What Is an AI Agent?

- An AI agent is an intelligent entity that can perceive the environment, make decisions, and take actions. Unlike traditional AI, an AI agent can achieve the specified goals autonomously by reasoning independently and invoking tools as needed.
  - For example, when asked to place a take-out order, an AI agent can directly invoke a food delivery app, select food, and then invoke the payment program to place an order and pay, all without human intervention.
- The concept of agent was first proposed by Minsky in his book *Society of Mind* published in 1986. Minsky believed that certain individuals in society can seek solutions to problems after negotiation. These individuals are called agents. He also believed that agents should be socially interactive and intelligent. This concept was then introduced into the fields of AI and computer science, and quickly became a hot research topic. However, due to data and computing power limitations, it was difficult to implement truly intelligent AI agents.



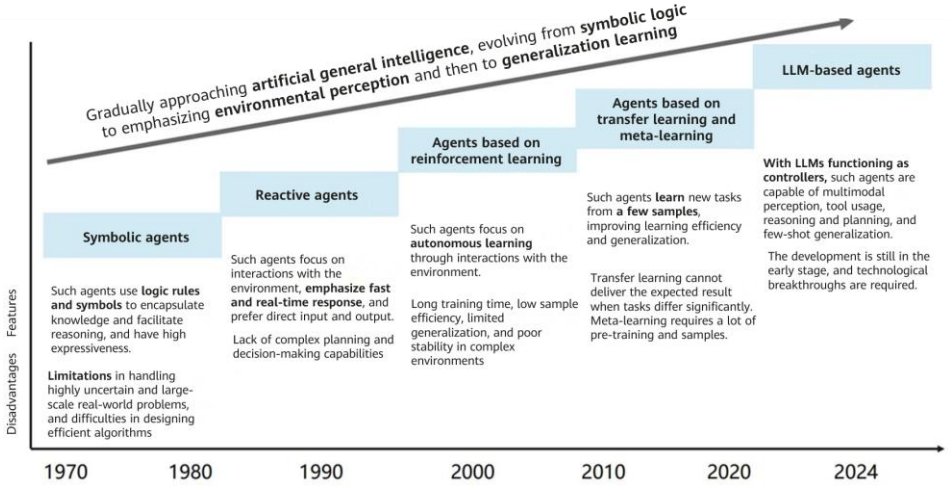
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AI Agent: A Key Step Toward AGI

AI Level (Analogous to Autonomous Driving)	Name	Feature	Example
L1	Tool	Humans complete all work without any explicit AI assistance.	Most software
L2	Chatbot	Humans do most of the work. They ask AI for advice and information, but AI does not directly process tasks.	First-generation ChatGPT
L3	Copilot	Humans and AI collaborate with comparable workloads. AI completes drafts based on the prompts provided by humans. Humans set goals, modify the drafts, and confirm the final outputs.	GitHub Copilot, Midjourney, and Jasper
L4	Agent	AI does most of the work, and humans set goals, provide resources, and supervise results. AI decomposes tasks, selects tools, monitors the progress, and independently ends tasks after the goals are achieved.	AutoGPT
L5	Intelligence	AI independently decomposes the goals set by humans, searches for resources, selects and uses tools, and completes all tasks without human supervision.	AGI and robots

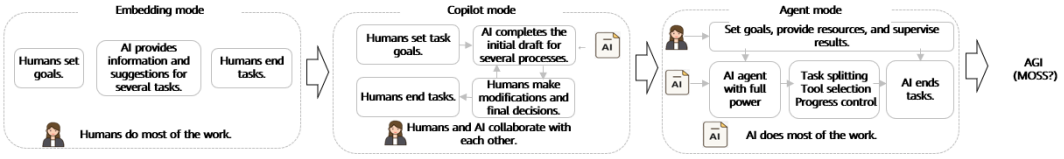
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Development History of AI Agents



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AI Software Evolution Trends



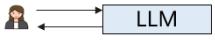
Phase 1: Foundation models (simple task processing)

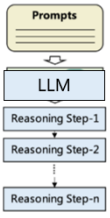
Phase 2: Individual agents (complex task processing)

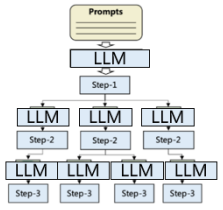
1. Tool invoking capability (RAG based on external databases)
2. Proactive planning capability (CoT, subgoal decomposition, etc.)
3. Memory capability (short-term and long-term memory)

Phase 3: Multiple agents (multi-agent collaboration)

1. Enhanced expertise: Comprehensive and accurate results are generated based on extensive expertise.
2. Robustness and reliability: The multi-agent system can reduce the risk of single points of failure, ensure continuous operation, and reduce errors.



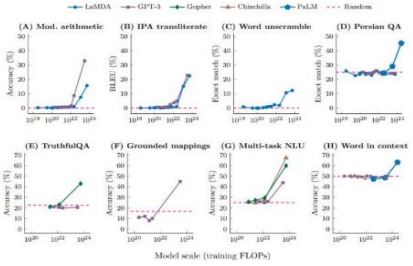




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LLMs Promote AI Agents

- AI agents are expected to interact with their environments in a human-like manner. The powerful capabilities of LLMs make this more attainable. Trained on vast datasets rich in human behavioral patterns, LLMs provide a robust foundation for simulating human-like interactions. In addition, as the model scale increases, LLMs' emergent abilities such as ICL, reasoning, and CoT can simulate the way humans think. With LLMs as the brain, AI agents can decompose complex problems into executable subtasks and implement human-like natural language interaction, which is hard to realize in the past. The rapid development of LLMs boosts the development of AI agents.



Emergent abilities of LLMs Data source: *Emergent Abilities of Large Language Models* (Wei, et al.)

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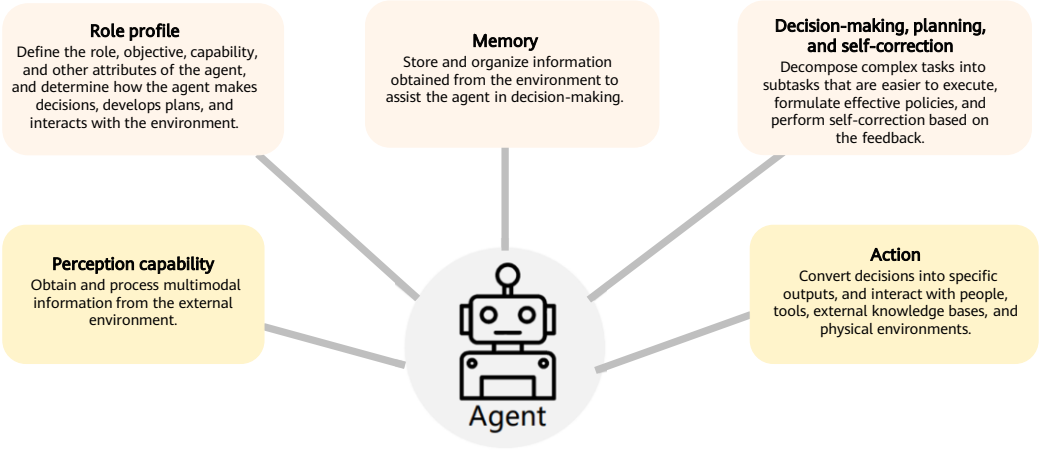
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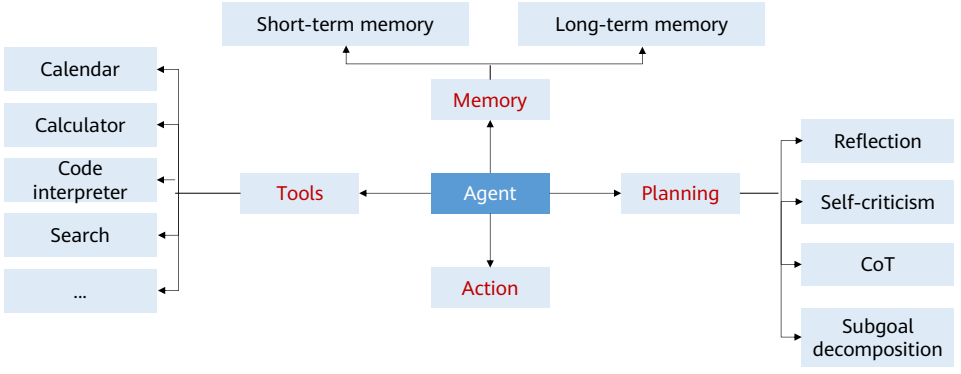
Core Capabilities of AI Agents



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Four Elements of AI Agents

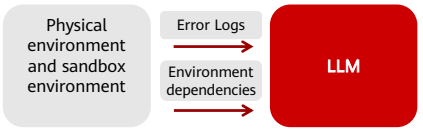
- In June 2023, Lilian Weng, director of applied research at OpenAI, wrote a blog post, arguing that AI agents may be the beginning of a new era. She proposed the basic architecture of agent = LLM + planning skills + memory + tool use. The LLM functions as the agent's brain and provides capabilities such as reasoning and planning in the system.



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Major Modules of Agents

The perception module collects, processes, and interprets various information in the environment.



The planning module determines the scale of the task that can be completed.

- Goal decomposition capability
- State modeling capability
- Reasoning and planning capabilities
- Decision-making and selection capabilities
- Dynamic adjustment capability
- Task scheduling capability

The action module (tool library) determines the tools or devices required by the task.

- Browser
- Chart generation tool
- Time software
- ...

The memory module determines the context length of the task that can be completed.

Short-term memory

GPT-4 Turbo	128k
Llama 2	4k
Claude	200k

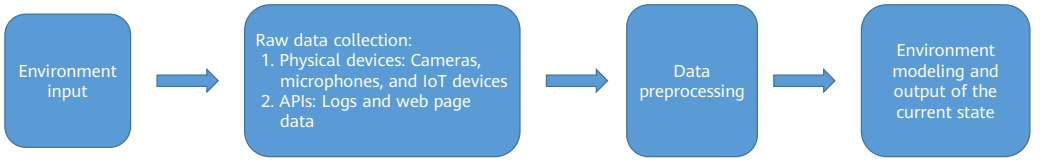
Long-term memory

- Pinecone
- Zilliz
- Milvus

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Perception Module

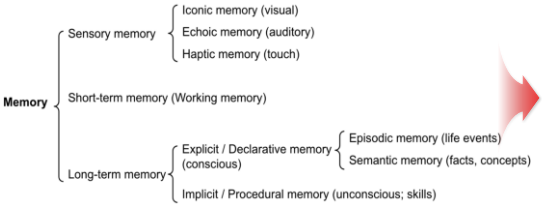
- In the agent system, the environment perception module is a key component for perceiving the external world and understanding the environment state. It receives and processes various inputs from the environment. It is capable of perceiving textual, visual, and auditory inputs, tactile and olfactory data, as well as the ambient temperature and humidity. With these perception capabilities, the agent can better understand its environment and make more rational decisions.
- The following lists some examples:
  - A home service agent perceives user locations, facial features, and voice commands.
  - An autonomous vehicle perceives roads, other vehicles, pedestrians, and traffic lights.
  - A chatbot agent perceives users' voice and emotions to understand user intents.



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Memory Module (1)

- Sensory memory: This is the earliest stage of memory. It lasts only a few seconds and provides the capability of retaining the impression of sensory information (such as visual and auditory data) after the original stimulus ends. Sensory memory includes iconic memory, echoic memory, and haptic memory.
- Short-term memory: It stores information that we are currently aware of, and lasts for about 20 to 30 seconds to perform complex cognitive tasks.
- Long-term memory: It can store information for a long period of time, from several days to decades. Its storage capacity is basically unlimited.



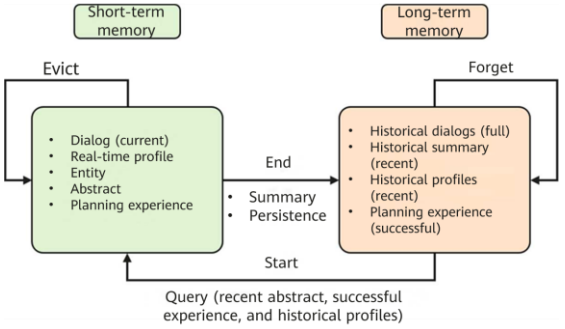
Data source: *Psychology: The Science of Behavior* (Neil R. Carlson)

	Human	Agent
Sensory memory	Visual/Auditory/Haptic input	Text parsing/Multimodal parsing as input
Short-term memory	Memory in the brain	Prompt engineering (restricted by the context length of Transformer)
Long-term memory	Long-term memory area of the brain	Vector database

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Memory Module (2)

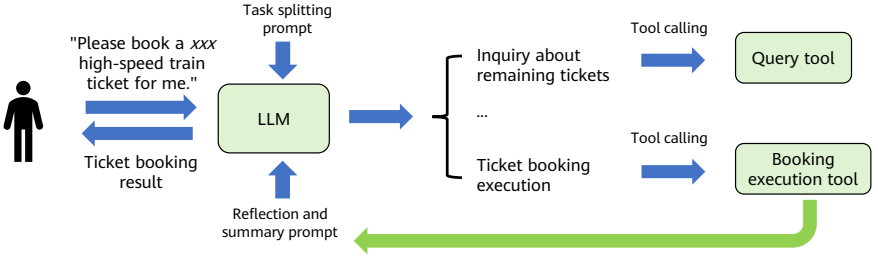
- Application of short-term memory and long-term memory in agents
  - Agent's short-term memory: Contextual information (dialogs and planning experience, such as complete dialog history, entities, abstracts, and profiles) exchanged with the LLM in the current session.
  - Agent's long-term memory: Successful planning experience accumulated (persisted and summarized) through interaction with the LLM, including the profile and summary. Before answering a question, an LLM retrieves relevant memory and reflects on whether the retrieved memory is helpful for the generation of the answer.



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Planning Module (1)

- **Planning**: Refers to the agent's capability of decomposing a complex task into multiple subtasks.

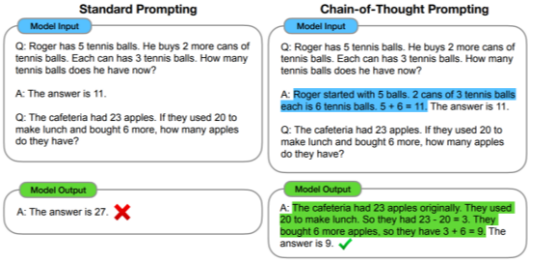


- **Subgoals and decomposition**: Agents can decompose large tasks into smaller, manageable subgoals to effectively process complex tasks.
- **Reflection and refinement**: Agents can make self-criticism and self-reflection on past behavior, learn from mistakes, and make improvements for future steps to improve the quality of final results.

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Planning Module (2)

- CoT:
  - CoT is a common prompting technique that guides large models to decompose tasks. The main method is to provide few-shot examples of task decomposition and use the ICL capability of large models to imitate similar task decomposition and planning.

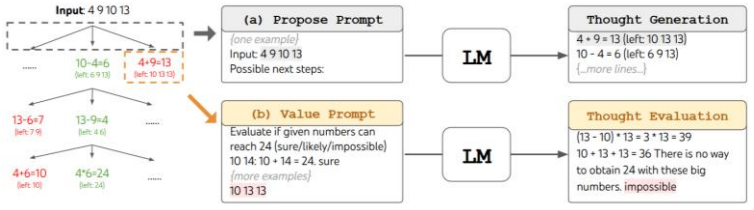


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- https://www.foundingminds.com/chain-of-thought-reasoning-the-magic-behind-the-o1-model/

Planning Module (3)

- Tree of thoughts (ToT):
  - CoT can only perform left-to-right logical reasoning and cannot solve complex problems. ToT, an advanced version of CoT, combines the search method with the LLM and empowers the LLM with the exploration capability. ToT uses depth-first search (DFS) or breadth-first search (BFS) to search for feasible policies, enhancing LLMs' decision-making capability on more complex problems.

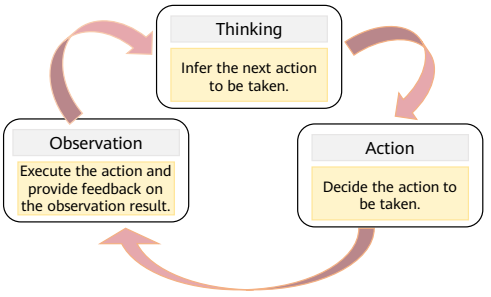


The following takes the 24 puzzle as an example (manipulate four integers through addition, subtraction, multiplication, and division so that the end result is 24):
Input: 4, 9, 10, 13
Step 1:
a. 10 – 4 = 6. The remaining numbers are 6, 10, and 13. This is feasible.
b. 4 + 9 = 13. The remaining numbers are 10, 13, and 13. The three numbers are too large to obtain 24.
Continue the search based on 6, 10, and 13.

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Planning Module (4)

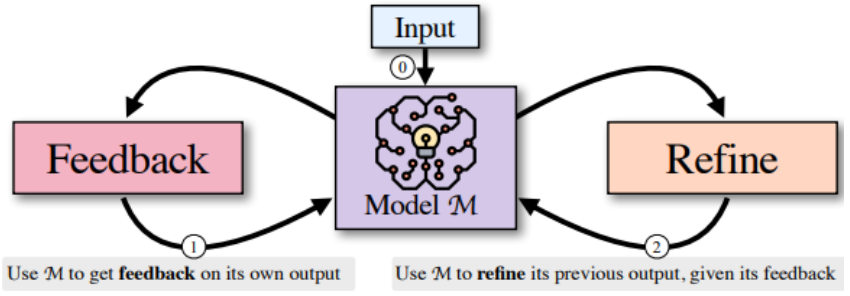
- Reflection and improvement – ReAct:
  - Few-shot prompting guides LLMs to generate reasoning trajectories and task-specific actions.
  - During reasoning, the action plan is continuously created, maintained, and adjusted. In addition, external information can be obtained through tool calling and interaction with external environments, and can be integrated into the dynamic reasoning process.



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Planning Module (5)

- Self-refine is one of the classic methods for agent self-optimization. Its core idea is that the initial output of an LLM is generally not ideal; by feeding the output back to the same LLM, the LLM can generate an improved output. This embodies the fundamental principle of reflection.



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Action Module (1)

Intrinsic behavior

- **Memory retrieval** (long-term memory → working memory): The module retrieves relevant information from the long-term memory and loads it into the working memory.
- **Memory storage** (working memory → long-term memory): The module updates the long-term memory. For example, it stores the reasoning and decision-making processes and results in the episodic memory; the module extracts knowledge updates from the information exchanged with the environment and stores such knowledge in the semantic memory.
- **Short-term memory reasoning and summary**: The module summarizes the latest observation results and action trajectories in the working memory, as well as information retrieved from long-term memory.

External behavior

- Textual or other modal outputs
- Tool use
  - API or function calling
  - Code generation, data analysis using Python code, etc.
- Difficulties:
  - Unified learning paradigm
  - Precise calling of a vast number of tools
  - Tool parameter filling
  - Accurate, error-free code generation
- Embodied action
  - Interaction with the real world

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Action Module (2)

- Tool calling – function calling:
  - Function calling is a key capability that allows an LLM to automatically select and call predefined functions based on the context during a dialog with a user, and incorporate the call results into the generated response. Unlike traditional API calling, function calling is automatically triggered by the LLM based on semantic reasoning without manual programming. The entire process is called by the LLM and is highly flexible.

Step 1: The developer defines the function.

```
{
  "name": "get_weather",
  "description": "Get the current weather of a specific city.",
  "parameters": {
    "type": "object",
    "properties": {
      "location": {
        "type": "string",
        "description": "City name"
      }
    },
    "required": ["location"]
  }
}
```

Step 2: The user enters a request.

"Please check the weather in Hangzhou."

Step 3: The model determines the function to be called.

```
{
  "function_call": {
    "name": "get_weather",
    "arguments": {
      "location": "Hangzhou"
    }
  }
}
```

Step 4: The system calls the corresponding function or API.

The system calls the backend service, such as the API for obtaining weather information, based on the model output.

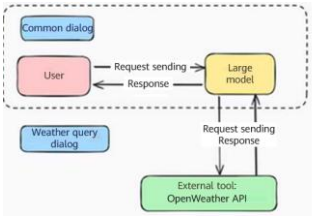
Step 5: The model generates an answer based on the returned result.

"It is sunny in Hangzhou, with a temperature of 28°C."

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Action Module (3)

- Tool calling – Model Context Protocol (MCP):
  - MCP is an open standard protocol that defines standard, secure, and bidirectional connections and communications between AI applications (including LLMs, intelligent assistants, and agents) and external data sources, tools, and services. MCP is used to manage plans, trace execution, and organize contexts, whereas function calling is only an execution mode of a single step in MCP.



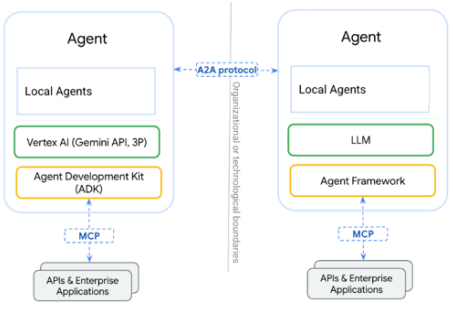
A typical MCP format contains the following elements:

```
{
  "user_input": "Please check the weather in Beijing and recommend what to wear today.",
  "conversation_history": [
    {"role": "user", "content": "....."},
    {"role": "assistant", "content": "....."}
  ],
  "task_state": {
    "step": 1,
    "goal": "Get the weather + dressing advice",
    "tools_used": []
  },
  "available_functions": [
    {"name": "get_weather", "description": "Get the weather of a specific city", ...},
    {"name": "get_dressing_advice", "description": "Recommend what to wear based on the weather", ...}
  ],
  "system_instructions": "Call functions step by step and return the final summary."
}
```


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Action Module (4)

- Agent2Agent (A2A) protocol:
  - The A2A protocol is an open standard designed to enable seamless communication and collaboration between AI agents. It enables agents that are of different frameworks and built by different vendors to communicate with each other in a common language, breaking silos and facilitating interoperability.



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Multi-agent Workflow

- Workflows can effectively help us improve the quality of work results. Take the production in human society as an example. The production capacity of an individual is limited. However, if an organization (such as a company) is established for production, the efficiency can be improved through systematic division of labor and management. In this way, the effect of 1 + 1 > 2 can be achieved.
- The workflow of a multi-agent system (MAS) is a dynamic process in which multiple autonomous agents cooperate with each other to complete a complex task. The core of the workflow lies in distributed decision-making, collaboration, and resource coordination. The multi-agent workflow is applicable to scenarios that require distinct roles, division of labor, dynamic adaptability, and high robustness (such as logistics, autonomous driving, and robot collaboration).



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Design Modes of Multi-agent Workflows

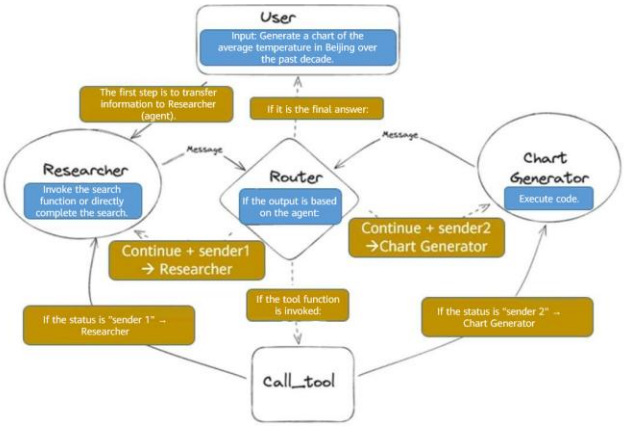
- Tools are terminals that implement functions, including the network interfaces that are called or local software and hardware. Agents rely on these tools to make a difference.
- Defining a workflow is complex. It is necessary to define how to divide responsibilities and plan phases, what different roles should do in each phase, what conditions should be met, and how to execute actions. The following three modes are recommended for designing workflows:



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Multi-agent Collaboration Mode

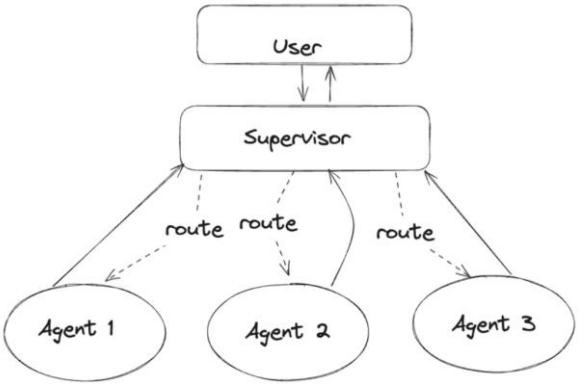
- In this mode, multiple agents are included in a workflow and collaborate to improve the output quality.



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Supervisor Mode

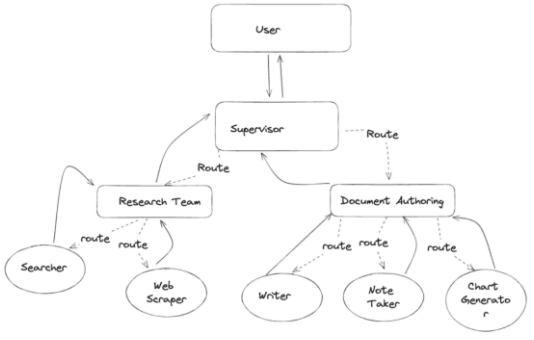
- In this mode, agents work independently and do not share the same register. The supervisor coordinates different agents and determines the final output.



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Hierarchical Team Mode

- Based on the supervisor mode, different agents are classified into multiple teams and agents in the same team can collaborate more efficiently. Finally, the supervisor coordinates multiple teams to generate the final output.

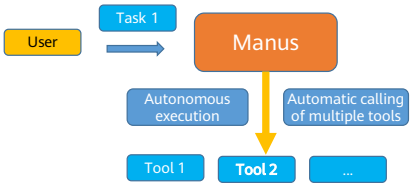


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- Agents work in workflow mode, which may be the major development mode of AI applications in the future. In such a system, the division of labor becomes more and more refined. Each agent can have better capabilities in a specific aspect. This means that a system may need *small models or fine-tuned LLMs in diverse domains*. Due to the refined division of labor, the functions of many agents are increasingly limited. Technical architectures with lower power consumption can be used to implement such agents. Only the agents that require high intelligence need large models that are more intelligent than GPT-4. This is more *environmentally friendly* and can alleviate the shortage of chip and power resources.

Manus

- Manus, launched on March 6, 2025, is a general-purpose AI agent developed by the Chinese team Monica.
- It is designed as an AI system that can autonomously execute complex tasks. Unlike chatbots, Manus can autonomously plan and execute multimodal tasks. It can automatically search for information and perform multi-step actions without continuous manual guidance. It can also crack verification codes, process complex files, and generate structured output such as tables and presentations.



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Quiz

1. (Single-answer question) Which of the following statements about the combination of LangChain and RAG is false?
   A. The data loader loads data from data sources and converts it into a document object.
   B. The text splitter splits a large document object into smaller document objects to facilitate subsequent retrieval and generation.
   C. The text embedder converts text into embeddings to measure the similarities between texts.
   D. The vector store does not need the indexing technology, and can directly store and query embeddings.

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- Answer: D

Summary

- This course introduces three large model–based application instances: prompt engineering, RAG, and agents. Prompt engineering effectively guides large models to generate the expected answers or solutions by elaborately designing the input text. RAG combines the advantages of traditional information retrieval and the powerful language understanding capability of large models to implement more accurate information extraction and content generation. Agents leverage the cognitive capability of large models to enable machines to execute complex tasks, thereby driving automation to a higher level.

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每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

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